

Cross-disciplinary approaches to Information Structure in Niger-Congo languages

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Preface

The eight papers in this volume are a selection of papers presented at the workshop “Cross-Disciplinary Workshop on Information Structure in African Languages,” held at the African Linguistics School 6 in Porto-Novo, Benin.

We take this opportunity to thank all the other participants of the workshop for their lively presentations, which included papers, tutorials, software demonstrations and panel discussions: Enoch Aboh, Akinbiyi Akinlabi, Vicki Carstens, Veneeta Dayal, Didier Demolin, Rebecca Jarvis, Jason Kandybowicz, Nancy Kula, Amani Lusekelo, Simanique Moody, Katie Russell, Hannah Sande, Lotfi Sayahi, John Singler, and Malte Zimmermann.

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Chapter 1

Introduction

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1 Background

A fundamental aspect of human communication relates to managing information that is already known to participants in a discussion, and adding new relevant information to this ongoing discussion. The interaction between these two sides of communication creates a discourse that is in principle limitless. Given the enormous amount of information that participants must process in any given discourse, human languages involve specific devices which allow speakers and hearers to identify what an utterance is about (i.e., the theme or topic), and what new information is contributed about that shared knowledge (i.e., the rheme or focus). In this regard, Halliday (1967), for instance, remarked that speakers divide utterances into hierarchically organized units of information such that what is being commented on, the topic, precedes the comment containing the new information, that is, the focus. In most commonly cited languages as examples (e.g., Indo-European) this organization of discourse, also referred to as *information structure* (IS), interacts with sentence intonation. In Romance and Germanic, for instance, the phrase bearing the most prominent accent in an utterance tends to coincide with some component of IS (i.e., the focus, or the topic).

African languages have much to contribute to the study of information structure. However, the grammatical devices they employ to express discourse elements tend to include specific morphemes or tonemes rather than prominent accent. Grammatical focus markers occur in the grammars of many West African



languages along with topic markers, and other discourse related markers (e.g., Ansre 1966, Bámgbóšé 1966, 1990, Hyman 1979, 2004, Hyman & Magaji 1970, Ameka 1991, Aboh 2004, 2007, Fiedler & Schwarz 2010, Biloa 2013, Titov 2019, among others). There can be overt focus markers, or verbs and nominals can be marked with additional morphology that indirectly signals focus, or lack morphemes. In Bantu languages, verbs are marked for the so-called “conjoint-disjoint” distinction (van der Wal & Hyman 2017), markers that indicate the type of relation with what follows the verb. Conjoint markers indicate a kind of focus interpretation or set of alternatives for the referent, while disjoint markers tend to indicate focus on the predicate or truth value. These can be accompanied by tonal changes as well, and in Rukiga the distinction is only tonal (van der Wal & Asiimwe 2020). In addition, some Bantu languages express information structure through the “augment” on nominals, a prefix preceding the noun class prefix. If the augment is present, it has a referential function, either definite or indefinite. For a recent overview of information structure in Bantu languages, see van der Wal (2025). Finally, Bisang & Sonaiya (2000) argue that different types of copula verbs are used for different kinds of focus in Yoruba. They note that the focus copula [ni] is used in “preconstructed domains”, while the copula [jé] is used in non-preconstructed domains (preconstructed domains express shared knowledge between interlocutors).

Focus and topic can also involve word order changes in African languages, such as left or right dislocation or, a position immediately after the verb (IAV), as well as verb doubling (Aboh 2007, Kandybowicz 2008). What constitutes a focused element and what information it expresses depends on salience, unexpectedness, surprise, comparison with alternatives, or other properties (van der Wal 2016). Similar factors can also determine if a topic element is left or right dislocated. Tone changes can accompany topic or focus. For example, cleft and focus constructions can require different tone on the verb, or there can be a verbal high (H) tone marker that accompanies focus constructions, as in Medumba (Keupdjio 2020). However, as Féry & Ishihara (2016) note “as far as we know, focus does not directly condition the analogous use of obligatory prosodic prominence or prosodic restructuring in Bantu or other African languages.” Focus can determine how syntactic constituents map to phonological phrases, and these phrases can in turn be indicated by vowel duration or tone, but there is no direct encoding of focus via prosodic prominence on the focused element (e.g. Zerbian 2017 on Northern Sotho). Indeed, Kügler & Genzel (2012) actually find a significantly lower pitch on focused constituents in Akan rather than pitch raising or expansion. The Güldemann et al. (2015) survey of information structure in African languages comes to the same conclusion and suggests that a theory of “focus

realization that universally defines prominence in terms of prosodic structure (either phrasing or pitch accent)” must be abandoned.

Nevertheless, phonetics and phonology are important for the study of information structure in African languages. Prosodic phrasing, or the grouping of words into prosodic units marked by tone, vowel lengthening or vowel harmony has long been an area of research and debate. Is prosodic phrasing built off syntactic units directly (the “direct syntax” approach) or does it operate semi-independently (the “indirect syntax” approach)? Some languages provide evidence for direct syntax, with sensitivity to binary branching of syntactic units. Yet others show preferences for binary branching in prosodic units, which may connect only partially to syntactic units. Recent theoretical innovations such as MATCH theory (Selkirk 2011) have been developed in part from data on African languages, such as XiTsonga showing how binary prosodic phrasing is favored.

The phonetic study of intonation is important to understanding the ways in which information structure is expressed. Many African languages are tonal, and information structure employs tone and prosody to express topic, focus, and the grammatical structures of questions, and to track discourse participants, even if it does so in an indirect manner rather than employing prominent accent on the salient element. Intonational tone is overlaid on lexical and grammatical tone, causing tone neutralization, tone shift and pitch range changes. Intonation can also be expressed by vowel lengthening and other prosodic factors. The two volume *Prosodic Typology* books (Jun 2005 and Jun 2014) contain no studies of African languages, except for Egyptian Arabic. Downing & Rialland (2017) explicitly addressed this omission and brought attention to the contributions of African languages to the study of intonation and sentence prosody. Yet only twelve languages are featured in the book from the many hundreds of African languages, and only four of them were non-Bantu languages. Much more research remains to be done.

In addition to word order and focus morphemes, information structure involves tracking referential forms through pronouns, definiteness or modification. Recently mentioned entities are referred to by pronouns or by agreement. Many West African languages make use of logophoric pronouns (Hagège 1974, Clements 1975, Güldemann 2003, Ameka 2017) as well as logophoric verb marking (Curnow 2002), which are used with verbs of speech to track reference to the original speaker. Some Ethiopian languages also use switch reference systems (Treis 2012) to index discourse referents.

Finally, sign languages also employ information structure (Kimmelman & Pfau 2016) and are known for topic and focus constructions, and tracking discourse referents through the use of classifiers, pointing and signing space. The study

of African sign languages has been enriched over the last decade through key dissertations and books on individual sign languages (Nyst 2007, Lutalo-Kiingi 2012, Tano 2017, Morgan 2022). However, information structure in African sign languages is still an unexplored topic.

African languages therefore offer a direct window into the grammar of discourse, and how it impacts the structure of language. Much of the research on African languages specifically has analyzed the morphosyntax of information structure (e.g. edited volumes by Aboh et al. 2007, Fiedler & Schwarz 2010) or the prosody of intonation (Downing & Rialland 2017). Joint work by phonologists and syntacticians can be very fruitful (Cheng & Downing 2012), but is not often undertaken. And yet, information structure crucially involves multiple dimensions of grammar.

2 The current volume

The current volume emerged from a workshop “Cross-disciplinary workshop on information structure in African languages”, funded by the National Science Foundation (BCS 2141420) that was held at the end of the African Linguistics School 6 in Porto-Novo, Benin in 2022. The African Linguistics School is a two-week school that provides instruction from professors from Europe, Africa and North America to graduate students from across Africa, as well as a few from outside Africa. Courses are offered in phonetics, phonology, morphology, syntax, semantics, fieldwork, sociolinguistics and signed languages. A hallmark of the school is the emphasis on the training of new generations of African scholars, and certain previous participants are invited back to teach once they have obtained their PhDs. The workshop featured a number of talks on African languages by students and professors that showcased research on different topics in information structure, seeking to bridge knowledge gaps across linguistic sub-domains (syntax, semantics, pragmatics, phonology/phonetics). The workshop also featured tutorials, a software demo and lively discussion panels. Participants were invited to submit their papers to this volume and all papers were peer-reviewed. The current volume contains eight papers from the workshop, all of which feature authors who are students or recent PhDs. Six of the papers are single-authored papers by African scholars presenting data on their own languages. In addition, two of the editors of the book, Dr. Sampson Korsah and Dr. Abdul-Razak Sulemana, are both ALS alumni and have served as ALS instructors.

The topics addressed in the papers range from the phonetics of polar questions to the semantics of discourse particles, and from verb phrase ellipsis to verb

phrase doubling. They include syntactic analyses of determiner phrase agreement patterns and the semantics of clausal determiners. The syntax and prosody of topic and focus constructions show both commonalities and differences across Niger-Congo languages.

The languages featured in the book are all Niger-Congo languages, and the title of the book reflects this fact. They include two languages of Ghana, Akan (Asante Twi) and Likpakpaanl, the Nigerian language Ìkálè, Ede Chaabe, a language of Benin, two Cameroonian Grassfields languages, Dschang and Ngemba, and the Sudanese Kordofanian language, Tira.

3 Featured papers

Two papers address how focus and topic are structured in two distantly related Niger-Congo languages, Ìkálè and Tira. Both languages employ word order and focus markers to convey topic and focus with both in-situ and ex-situ patterns, although the position of the focus marker differs, being clause initial in Tira and clause-final in Ìkálè. These structures are reinforced through prosodic patterns, fitting well within the typology of prosodic structure observed in other African languages.

In “The syntax of focus in Ìkálè,” Daniel Aremu explores the distribution of the focus particle *rìn* in Ìkálè, a southeastern dialect of Yorùbá. In Ìkálè, the focus particle appears clause finally instead of adjacent to the focused element. Subject focus is ex-situ while non-subject focus may be either in-situ or ex-situ. Word order and resumptive subject pronouns disambiguate which argument is focused. Aremu argues that this poses a problem for syntactic models that adopt the theory of antisymmetry (Kayne 1994), as the Focus Phrase (FocP) would show a different spec-head-complement order than the other phrases to account for the linear order. However, he proposes that an analysis in which the focus head is merged to the right clause-final position provides a more economical and motivated analysis than an alternative that respects antisymmetry.

In “Prosody of Topic and Focus in Tira,” Nina Hagen Kaldhol, Sharon Rose and Mark Simmons show how both types of structures in Tira, a Kordofanian language of Sudan, are conveyed via word order, wherein the topic or focused element occurs phrase-initially, is not case-marked, and triggers agreement on the verb. Other arguments must be post-auxiliary or, if no auxiliary, post-verbal. These structures are similar to those identified in Ìkálè, but the focus particle appears preceding the focused element. In Tira, H tone is used as a diagnostic of the structure in two ways. First, cross-word H tone spreading does not occur between

the topic element and the rest of the sentence, but does occur between elements in the remainder of the sentence. Second, focus constructions are marked by a H tone at the left edge of the verb, in addition to a focus marker and word order changes. Polar questions show the same kind of patterns, but are distinguished from declaratives by raising of non-final H tones in the utterance.

In another paper that examines tone realization, “Intonation and Prosodic Domains in Ede Chaabe,” Tajudeen Mamadou demonstrates the distinction between declaratives, polar questions and exclamatives with respect to final pitch lowering. Like many languages in the Macro-Sudan belt, including Tira, Ede Chaabe marks yes/no questions with what Rialland (2007, 2009) labels “lax prosody”, which includes phrase-final lowering. However, in Ede Chaabe, lowering in polar questions begins earlier than the final syllable, a contrastive difference from exclamatives, wherein the lowering is confined to the final syllable. Mamadou provides acoustic evidence for how extensive this lowering is depending on the overall length of the sentence and the length of the final verb phrase.

Determiner phrases (DPs) can also reflect focus through word order and concord within DPs. In “DP-internal focus affects agree: Insights from Bamileke Ngemba,” Leonel Fongang identifies a pattern in which adnominal possessive pronouns are contrastively focused and appear before the noun instead of following it. The normal noun class concord patterns are disrupted for singular nouns and the noun class agreement prefixes are replaced by glides that signal animacy. Fongang argues that syntactic agreement has failed and the glides represent semantic agreement. He proposes an analysis in which the N POSS order is adnominal, and the focus POSS N order is appositional. Syntactic number agreement fails in the appositional order as it is outside the DP phase and has no Number specification. With plural agreement, however, gender can combine with the specified Number feature and this is visible to the POSS head, resulting in agreement, hence the presence of the regular noun class prefix.

Augustina Owusu’s paper “Definitely a question: Definite markers in Akan ex-situ questions” addresses the semantics of the definite determiner *nó* in Akan, which can function as both a definite marker within determiner phrases, and a clausal determiner, with a core meaning of weak familiarity and existence presupposition. The presence of clause final determiner *nó* in ex-situ wh-questions presupposes familiarity with the existential closure of the question nucleus. In this sense, such questions are comparable to cleft questions in terms of semantics. They do not permit the cancellation or suspension of the existence presupposition, unlike wh-questions without clause-final *nó*. This paper thus contributes to the understanding of cross-category determiners and the function of such determiners in different kinds of wh-questions.

In “The descriptive and syntactic properties of the *ànáá* particle(s)”, Comfort Ahenkorah examines the particle *ànáá* in Akan, which has multiple uses. She argues that this is not one single particle but four distinct particles with different functions: a question marker, a disjunctive marker, a discourse particle and a tag. The particles have distinct semantics, and occupy different positions in the clause, and some can cooccur. The question marker is a clause-typing particle - it marks polar questions and subordinate interrogative clauses. This particle can combine with *ànáá* when it functions as a disjunctive, creating disjunctive questions. *ànáá* also functions as part of a tag clause or as a tag, where it is used in biased questions. The discourse particle use of *ànáá* expresses possibility or the speaker’s uncertainty. Ahenkorah carefully examines the semantics of each of the uses and provides arguments for different positions that they occupy in the left periphery.

Ahenkorah and Owusu’s papers are both semantics papers that show the value of investigating what appears on the surface to be a single word or particle, but turn out to have multiple functions and may actually be different homophonous particles. These careful studies show how fine-grained analysis of discourse and sentential particles illuminate their information structure roles and the intent of the speaker.

Ellipsis, or the absence of some elements of the sentence, relies on the ability of the interlocutor to interpret the meaning based on the context. Samuel Acheampong’s paper “Verb phrase ellipsis in Likpakpaanl” is a novel exploration of ellipsis, a topic that is underexplored in African languages. He shows that Likpakpaanl has a number of modal verbs, but only two of them are able to license verb phrase ellipsis, *ɲmàn* ‘can’ and *bàn* ‘want.’ The former licenses ellipsis of its VP complement and the latter licenses ellipsis of its CP complement. Acheampong analyzes the data within the theory of PF deletion (Merchant 2019) in which there is an ellipsis [E]-feature. He provides further syntactic arguments from antecedent contained deletion and reconstruction effects for this analysis.

In contrast to verb phrase ellipsis, verbs can be doubled in some languages. In “On the clausal syntax of Dschang predicate focus doubling,” Colin Brown and Harold Torrence present an analysis of predicate doubling, which signals contrastive focus in Dschang. They argue that Dschang has two focus positions, one that is used for focus in the left periphery, and another low focus position below TP used for predicate focus doubling. The two positions can cooccur, but verbs cannot appear in the high focus position, and are the only elements in the low focus position. Objects appear between the two verbs, and the rightmost verb copy carries a focus marking suffix. They argue that Dschang provides evidence

for the existence of a low vP peripheral focus position, as previously argued for subjects Belletti (2001, 2004).

In summary, the papers featured in this volume make descriptive and theoretical contributions to the study of information structure in African languages, most particularly those of the vast Niger-Congo phylum. Each paper presents a wealth of novel data that carefully illustrate and illuminate how the languages convey and structure information to aid speakers in discourse. This is managed with the prosody of tone, through raising or lowering pitch, adding high tones, or allowing or preventing cross-word tone spreading. It can be done via word order changes, focus particles, verb doubling and agreement patterns. Discourse particles, clausal determiners and the behavior of modals in verb phrase ellipsis all serve to convey semantic information about the speaker's attitude and intent.

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Chapter 2

The syntax of focus in Ìkálẹ̀

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This paper presents novel data and analysis of the focus constructions in Ìkálẹ̀, a southeastern dialect of Yorùbá. I show that, although many morphosyntactic focus marking African languages have their morphological focus marker(s) in a post-focal, pre-focal position or suffixed to the verb, Ìkálẹ̀ realizes its focus marker at the clause-final position, regardless of the constituent that is fronted for focus purposes. This position of the focus marker poses a challenge to Kayne's (1994) *antisymmetry theory of syntax* which claims that, underlyingly, sentence structure has the order: Spec-Head-Complement, and that the differences found cross-linguistically are due to different movement possibilities. To account for this, I propose two analyses, one in favour of the antisymmetry theory, and another that simply merges the focus head to the right of its complement. Considering the principle of economy, I argue that the latter analysis is preferred due to its simplicity, while showing that the analysis that meets the antisymmetry syntax is uneconomical and unmotivated. In addition to the focus description and analysis, I present data on association with focus and argue that the exclusive focus sensitive particle *nùkàn* 'only' has both an adnominal adjunction and an adverbial/extended verbal projection adjunction (à la Buring & Hartmann 2001).

1 Introduction

Studies on the syntax of focus constructions have been carried out across a wide range of languages and language families. African languages, especially, have shown different, yet interesting, focus realization patterns. From the mechanism employed for focus marking to the asymmetries between subject and non-subject focus, local and non-local focus, and so on, African languages have provided not only interesting data, but also analytical strength to the theory of focus and its interpretation, especially within the cartography framework.



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For a simple definition of *focus*, I adopt Aboh et al.'s (2007: 1) definition where focus is "that part of the clause that provides the most relevant or most salient information in a given discourse situation." In line with Rooth's (1985, 1992) *alternative semantic approach* to focus, a focus constituent invokes alternatives that are contextually salient and relevant to the linguistic discourse (cf. Krifka 2008). In order to make these definitions explicit, consider the examples below. One of the contexts for testing for focus is through a *question-answer pair* (Hamblin 1976). When a question like (1a) is asked, it requires an answer which must be in focus (1b). It is this focus (in this case, banana) that indicates the presence of a set of salient alternatives (which the banana is a part of). Thus, the thing that Adam ate yesterday could have been any of the things in the following set: [banana, egg, rice, avocado, ...]. Importantly, the set of alternatives has to be contextually relevant (cf. Rooth 1992, Beaver & Clark 2008).

- (1) a. What did Adam eat yesterday?
- b. Adam ate A BANANA yesterday.

In fact, (1b) can be used to answer not just (1a), but also questions such as: *What did Adam do yesterday?*; *What did Adam do to the banana yesterday?*; *Who ate a banana yesterday?*; *When did Adam eat a banana?*. The only difference would be the placement of the focal stress. Languages that mark focus in such a way (i.e. using stress) are *intonation focus marking languages*. For example, English and German mark focus prosodically (a.o., Büring & Hartmann 2001, Krifka 2006, 2008, Féry & Ishihara 2009, Rochemont 2013). Apart from prosody, there are languages that mark focus via morphology and/or syntactic reordering. Such focus-marking strategies are dominant among African languages, and there is a large body of research in this regard (see, among many others, Hartmann & Zimmermann 2004, Aboh et al. 2007, Hartmann & Zimmermann 2009, Amaechi & Georgi 2019, Korsah & Murphy 2020, Issah & Smith 2020).

Different questions have been raised regarding focus over the years. Some of these questions are:

1. What are the types of focus found in languages?
2. What are the different focus marking strategies?
3. Do these strategies lead to different interpretations of focus (contrastive vs. new information focus interpretation, for example)? (cf. É. Kiss 1998) and so on.

The syntactic representation of focus has also been of interest for the past decades. Particularly, Rizzi's (1997) split-CP (cartographic) approach has provided different structural positions for different information structural elements. This does not only occur in the left periphery of the entire clause but can also occur in what Belletti (2004) calls the *clause-internal periphery* (cf. Belletti 2009, Belletti & Rizzi 2017). This means that while the left periphery has the FocP in the region above the TP/FinP, the clause-internal periphery has the same projection within the clause – in the region above the little vP or the VP (cf. Belletti 2004, Aboh 2007).

Thus, this paper seeks to contribute to the already existing studies on focus realization in Yorùbá (Owolabi 1981, 1983, Awoyale 1985, Awobuluyi 1987, 1992, Yusuf 1989, Adesola 2005, Ilori 2010). What is new about this paper, however, is that it considers one of the dialects of Yorùbá – Ìkálẹ̀ – and proposes a theoretical analysis for it. Such an endeavour is well motivated considering the focus marking pattern that is observed in the dialect, which is slightly different from that of Standard Yorùbá, as discussed by some of the Yorùbá scholars mentioned above. One major difference is with respect to the surface realization of the focus markers. While the focus marker *ni* immediately follows the ex-situ focus constituent in Standard Yorùbá (2),¹ the focus marker *rín* is realized in the clause-final position in Ìkálẹ̀ (3).² In addition to focus, association with focus has been a relatively well studied phenomenon in African languages (cf. Grubic 2015, Carstens & Zeller 2020). This study attempts a brief description of this phenomenon in Ìkálẹ̀, and proposes a superficial analysis for it. The aim is to show that there is potential work to be done in this area, i.e. among African languages, compared to what has been done for European languages, for example.

- (2) a. Standard Yorùbá
 Kí **ni** Adé pa?
 what FOC A. kill
 ‘What did Adé kill?’
 b. Ìkálẹ̀
 [Èkú]_F **ni** Adé pa.
 rat FOC A. kill
 ‘Adé killed A RAT’

¹The focus marker *ni* is in boldface, while the focus constituent is inserted in an F-marked square bracket [XP]_F (à la Jackendoff 1972). The corresponding focus constituent in the English translation is in uppercase.

²I must note here the existence of a comparative descriptive study on Ìkálẹ̀, Ondo and Igbo by Akintoye (2020).

- (3) a. Nèé Adé rí-i?
who A. see-Q
'Who did Adé see?'
b. [Tolú]_F Adé rí rín.
T. A. see FOC
'Adé saw TOLÚ.'

The rest of the paper is structured as follows. §2 gives a brief background to some relevant aspects of Ìkálẹ̀ grammar. Afterward, §3 begins by introducing the focus marking strategies in Ìkálẹ̀, pointing out the asymmetry between Subject Focus (SF) and Non-Subject Focus (NSF) respectively. Furthermore, arguments in favour of a movement approach, instead of a base-generation approach are provided. §4 briefly discusses the Association with Focus (AwF) phenomenon, with an emphasis on the exclusive Focus Sensitive Particle (FSP) *nùkàn* 'only'. The paper is concluded in §5.

2 A brief background to Ìkálẹ̀

Ìkálẹ̀ is a southeastern dialect of Yorùbá, a Benue-Congo language (cf. Akinkugbe 1976, Adeniyi 2010).³ Going by Akintoye (2020), it is spoken in areas such as Okitipupa, Érinje, Òde-Aye, Òṣòrò, Òde-Ìrèlẹ̀, Igbódigò, Ìkòyò Ọ̀mèn, Àyèkà, Ìgbìsìn-Olòò, Àkótógbò, Àjàgbà, Ìyònsòn and Ìjì-Òsun of Ondo state, Nigeria. The total number of speakers is unknown. However, the dialect is expected to have over 500,000 speakers, according to Wikipedia.

There is a consensus on the fact that the different dialects of Yorùbá make use of the Yorùbá orthography (Bámgbóṣé 1967, Awobuluyi 1978). Ìkálẹ̀ can be said to have three discrete level tones: high, mid, and low. While the high and the low tones are overtly marked, the mid tone is not marked in the orthography of the language. This is in line with the already established tonal system in Standard Yorùbá (cf. Courtenay 1971, La Velle 1974, Akinlabi 1985, Laniran 1988, Connell & Ladd 1990, Laniran 1992, Laniran & Clements 2003). Moreover, apart from minor differences in the pronunciation of some words, and some dialect-specific tonal interactions, there is no difference with regard to the number of tones in the dialect, based on my elicitation. Similarly, there is no observed tonal reflex which is significant to our analysis in this paper.

³Some other dialects include Ègbá, Šakí, Ìlájẹ, Àkúrẹ, Mòbà, Ìjẹ̀sà, Èkiti, Ifẹ, Yàgbà, Ọ̀wórò, Ondó, Ọ̀wò, among others (see, a.o., Adetugbo 1967, Akinkugbe 1976, Adeniyi 2010, Akintoye 2020 for a regional classification of Yorùbá dialects).

The canonical word order of a simple sentence in Ìkálẹ̀ is the same as in Standard Yorùbá, viz: Subject + (Tense Aspect Modal) + Verb + Object (S (TAM) V O). The sentence in (4a) is in the simple past tense (which is morphologically unmarked), and consists of a subject, followed by a verb, and then a direct object. This sentence can be used as an answer to the question *What happened?*. On the other hand, (4b) and (4c), show that the tense and aspect markers in the language are in a preverbal position. Usually, the tense morpheme is followed by the aspect morpheme when they co-occur in a sentence as in (4d).⁴

- (4) a. Adé pa ekún nẹ.
 A. kill rat DEF
 ‘Adé killed the rat.’
 b. Adé áa pa ekún nẹ.
 A. FUT kill rat DEF
 ‘Adé will kill the rat.’
 c. Adé ti pa ekún nẹ.
 A. PFV kill rat DEF
 ‘Adé has killed the rat.’
 d. Adé áa ti pa ekún nẹ.
 A. FUT PFV kill rat DEF
 ‘Adé would have killed the rat.’

Furthermore, there is what is called the High Tone Syllable (HTS) in Yorùbá and its dialects. It is a tonal reflex that occurs on the last syllable of the subject of a sentence (see a.o., Bámgbóṣé 1967, Fresco 1970, Oyelaran 1970, Stalhke 1974, Awobuluyi 1978, Adewòlẹ 1988, Awobuluyi 1992, Bisang & Sonaiya 1999, Ilori 2010).⁵ It becomes obvious when the subject ends in a mid or low tone. For instance, *Ṣògò* originally ends in a mid tone, but in a sentence like (5a), its final syllable has a high tone. Otherwise, a high toned syllable in the form of *ó* is inserted (5b). With the subject ending in a high tone as in (6), the HTS does not become obvious, allowing the insertion of *ó* to be optional. The HTS is said to occupy the head of TP (cf. Ilori 2010).⁶

⁴All the Ìkálẹ̀ data used in this study are from a series of interviews with my language consultants: Mr. Ayomide Akinlalu and Pa Akinlalu. I am grateful to them.

⁵There are controversies around the functional status of the HTS (see, for a brief summary on the different claims about the HTS, Ilori 2010, Akintoye 2020)

⁶See Section 3.1.1 for further discussion on the HTS and resumptive pronouns.

- (5) a. *Şògó pa ekún nè.*
 S.HTS kill rat DEF
 ‘Şògo killed the rat.’
 b. *Şògo ó pa ekún nè.*
 S. HTS kill rat DEF
 ‘Şògo killed the rat.’
- (6) *Bólá (ó) jẹ ejíjẹ nè.*
 B. HTS eat food DEF
 ‘Bólá ate the food.’

The data above show that (a) *Ìkálẹ̀*, just like the standard dialect, is isolating, and (b) there is, at least, a tense phrase (TP) and an aspect phrase (AspP) in the syntactic structure of the dialect. Furthermore, the fact that the dialect linearly shows the subject preceding the tense and/or the aspect morphemes suggests that the subject does not remain in the Spec,vP, but moves to the Spec,TP. Thus, we have the following structure for a simple sentence in *Ìkálẹ̀*:

- (7) $[_{TP} \text{Adé} [_{T'} [_{T} \text{áa}] [_{AspP} \text{ti}] [_{vP} \langle \text{Adé} \rangle [_{v'} [_{v} \text{pa}] [_{VP} [_{V} \langle \text{pa} \rangle] [_{DP} \text{ekún nè}]]]]]]]$

The left periphery of the clause consists of the projection that clause-types a sentence as declarative or interrogative, for example. Traditionally, the CP layer is responsible for clause-typing. However, since Rizzi’s (1997) famous cartographic structure in the left periphery of the clause, the traditional CP has been split into different functional projections that represent distinct information structural notions such as Focus (FocP) and Topic (TopP). The former has been argued to possess the focus feature or interpretation. In a language like English and Italian, the head of the FocP is assumed to be null. In contrast, in languages with morphological focus marking (like Nupe, Dagbani, Igbo, Yorùbá, Gungbe, Guruntum, and many others), the head of the FocP is occupied by an independent morpheme that possesses the focus feature (cf. Yusuf 1989, Rizzi 1997, Aboh 1998, Aboh et al. 2007, Kandybowicz 2008, Hartmann & Zimmermann 2009, Fiedler et al. 2010, Ilori 2010, Amaechi & Georgi 2019, Issah & Smith 2020). Therefore, as we will see in Section 3, the ex-situ focus in the dialect occupies the specifier position of the FocP. Extending our structure in (7) to the complementizer domain, the following structure is proposed.

- (8) $[_{ForceP} \dots [_{FocP} \dots [_{TP} [_{T'} [_{T}] [_{AspP}] [_{vP} [_{v'} [_{v}] [_{VP} [_{V}] [_{DP}]]]]]]]]]]$

In order to account for how the focus feature is interpreted, and how focus movement takes place in the present study, I adopt Pesetsky & Torrego's (2007) *feature sharing* approach to agreement and an EPP-feature. The latter requires that the specifier position of any functional head must not be null. It must be filled with a phrase with an overt head. Pesetsky & Torrego (2007) believe that agreement does not necessarily need to be based on whether a feature is (un)interpretable. In Chomsky's (2000, 2001) agreement approach, within the Minimalist Inquiry (MI) and Derivation by Phase (DbP) framework, an uninterpretable feature must be unvalued, and when it is valued, it must be deleted. Thus, the probe (with an uninterpretable unvalued feature [*uF*]) searches for a goal with an interpretable valued feature [*iF*]. After Agree has taken place, the uninterpretable feature on the probe gets deleted. Pesetsky & Torrego's (2007) feature sharing approach, on the other hand, provides more agreement patterns and/or possibilities. Feature deletion is not a necessary requirement for Agree. In fact, features can be shared between the probe and the goal, i.e. "when Agree applies between a probe feature *F* at a syntactic location α and a goal feature *F* at location β , we propose that the output is a single feature *F* shared by two locations" (Pesetsky & Torrego 2007: 267). The following prose statement is adapted from Pesetsky & Torrego (2007: 268).

(9) **Agree (Feature sharing version)**

- a. An unvalued feature *F* (*a probe*) on a head *H* at syntactic location α (F_α) scans its c-command domain for another instance of *F* (*a goal*) at location β (F_β) with which to agree.
- b. Replace F_α with F_β so that the same feature is present in both locations.

One of the consequences of this approach is that the probe that is valued can serve as the goal for an unvalued higher head *H* with the same feature *F*. This would then be an instance of *F* in three locations, and it can go on. Thus, the Agreement pattern can be iterative. This approach will be important for the proposed analysis in this paper.

3 Focus realization in Ìkálẹ̀

Both ex-situ and in-situ foci exist in Ìkálẹ̀. Ex-situ focus is realized, firstly, when the focused constituent occurs in a position in the left periphery of the clause, and secondly, with the presence of the morphological focus marker *rín* in the clause-final position. In in-situ focus, on the other hand, the focused constituent remains

in its canonical position with no morphological marking. The consequence of this is that the pragmatics of the discussion, whether in form of the Current Question (CQ) or in form of the context of the discussion, would be resorted to, in order to know what is in focus (cf. Roberts 1996, 2004, Beaver & Clark 2008).

In this section, first, I describe the data showing the focalization of both the Subject and Non-subject in Ìkálẹ̀. In the process, I show that there are a few asymmetries between SF and NSF. For instance, while SF cannot occupy an in-situ position, NSF can occur both in-situ and ex-situ. Secondly, I provide arguments for the movement approach of the focalized constituents.

3.1 Subject focus

Subjects are usually topics (cf. Rizzi 1997, Fiedler et al. 2006, Krifka 2008). Thus, in intonation focus-marking languages, the focal stress is realized on the subject in order to focalize it. However, since Ìkálẹ̀ is not an intonation focus-marking language, it employs a different way of distinguishing between a subject topic and a subject focus. When the subject is in focus, it occurs outside the matrix clause.⁷ With a *wh*-question context like (10a), the answer in (10b) is the only one that is felicitous. The focused subject is realized outside the Spec,TP, leaving a third person singular resumptive pronoun (henceforth RP) *ó* behind. The sentence in (10c) is not a felicitous answer to the question in (10a), with the subject in an in-situ position. Thus, SF in Ìkálẹ̀ is only ex-situ.

- (10) a. Nòò jẹ ejíjẹ nẹ?
who eat food DEF
'Who ate the food?'
b. [Adé]_F *(ó) jẹ ejíjẹ nẹ *(rín).
A. RP eat food DEF FOC
'ADÉ ate the food.'
c. *[Adé]_F jẹ ejíjẹ nẹ.
A. eat food DEF
'ADÉ ate the food.'

The focus marker *rín* is realized in the clause-final position. While the position that the focus marker occupies is uncommon in morphological focus-marking languages, Ìkálẹ̀ is not the only dialect of Yorùbá to have its focus marker in this

⁷I employ the Question-Answer congruence in order to realize focus (Hamblin 1976, Beaver & Clark 2008).

position. Ìlàjẹ is another dialect that morphologically marks focus clause-finally (Akintoye 2020). Similarly, Nupe (Benue-Congo), another Nigerian language, has its focus marker in the clause-final position too (Kandybowicz 2008, Mendes & Kandybowicz 2023).

Similar properties to what we have seen for matrix SF are present in embedded SF. The sentence in (11a) is an embedded subject *wh*-question.⁸ In English, this will not be possible with an overt complementizer *that*. This will result in a *that*-trace effect; except if the complementizer head is null (as in the translation). However, there is no such effect in Ìkálẹ̀. In fact, the complementizer *fi* is obligatory in this case. The answer to (11a) is typically (11b). The embedded SF is realized in the left periphery of the matrix clause. It then leaves an RP behind. What these data show is that resumption in Ìkálẹ̀ is not only present in long-distance dependency but also in matrix sentences (compare 10b and 11b). It is also possible to realize embedded SF in the left periphery of the embedded clause itself. This is the case in (11c). The embedded subject occupies a position in the embedded clausal spine of the sentence and leaves an RP behind. The position of the focus marker *rín* remains the same, i.e. clause finally.

- (11) a. Nẹ̀ẹ́ Adé fọ́ fi *(ó) fẹ̀rà̀n Tolú?
 who A. say COMP RP love Tolú
 ‘Who did Adé say loves Tolú?’
 b. [Bọ́lá]_F Adé fọ́ fi *(ó) fẹ̀rà̀n Tolú *(rín).
 B. A. say COMP RP love T. FOC
 ‘Adé said that BỌLÁ loves Tolú.’
 c. Adé fọ́ fi [Bọ́lá]_F *(ó) fẹ̀rà̀n Tolú *(rín).
 A. say COMP B. RP love T. FOC
 ‘Adé said that BỌLÁ loves Tolú.’

⁸Why there is a difference between the *wh*-phrase in (10a) and (11a) is not entirely clear at this point. The only observation, as far as I know, is that while *nọ̀ọ́* always precedes a verb, *nẹ̀ẹ́* can both precede a verb and a noun. Going by the data, one might want to predict that this has to do with local versus non-local (subject) *wh*-question. However, a closer look shows that *nẹ̀ẹ́* (and not *nọ̀ọ́*) is also used when a local object *wh*-question is formed (i). I will leave the details of their distribution for future research.

- (i) a. Nẹ̀ẹ́ Bọ́lá rí?
 who B. see
 ‘Who did Bọ́lá see?’
 b. *Nọ̀ọ́ Bọ́lá rí?
 who B. see
 ‘Who did Bọ́lá see?’

3.1.1 The High Tone Syllable (HTS) and resumption

In this section, I address an anonymous reviewer's comment on what is classified as an RP in this paper. A reviewer disagrees with calling *ó* an RP, and claims that it should be called a High Tone Syllable (HTS) *ó* instead. While I agree on the existence of a high tonal reflex (floating tone) between a subject and its predicate (as discussed in Section 2 above), I disagree that the pronoun in this context is the HTS.⁹ Firstly, the overt realization of the HTS as *ó* is said to be optional as in (12) below. However, in all the examples where the subject RP is realized, they are obligatory, otherwise the sentence becomes ungrammatical. This is obviously not the case with examples like (12) and (13). If the high tone is identical to the tone on the final syllable of the preceding subject (12), *ó* can be overt, but the high tone docks to the final syllable of the subject when the latter has a different tone (13a). In fact, the overt realization of *ó* can result in a topic interpretation, where the subject is topicalized. For instance, in a contrastive topic context, the sentence in (14a), and not (14b), can be used to answer a question such as *Tell me what the children kill?* The topic *Şògó* is given (i.e. one of the alternatives among the children), but the focus *ekún nẹ* is the new information (cf. Büring 2016). The point here is that (14b) should be perfectly fine if *ó* is only an HTS (as it was when the subject was not contrastively topicalized in (13a)). On a closer look, even the example in (13b) can be analyzed as an aboutness subject topic.

- (12) Bólá (ó) jẹ ejíjẹ nẹ.

B. HTS eat food DEF

'Bólá ate the food.'

- (13) a. Şògó pa ekún nẹ.

S. kill rat DEF

'Şògó killed the rat.'

- b. Şògó ó pa ekún nẹ.

S. HTS kill rat DEF

'Şògó killed the rat.'

⁹There is an ongoing debate, in the Yorùbá literature, as to the status of the HTS *ó*. One group claims that it is a subject marker (cf. Fresco 1970, Oyelaran 1970). Another group argues that it marks perfective aspect (cf. Adewole 1988, Bisang & Sonaiya 1999). A third group sees it as a non-future tense marker (Awobuluyi 1978, 1992, 2006, Ilori 2010). Others call it a predicate junction marker (Bámgbóşé 1967), a subject concord prefix (Stalhke 1974), and so on (see, for some discussions, Ilori 2010).

- (14) a. *Ní ti Sògo, ó pa ekún nẹ.*
 EXPL for S. RP kill rat DEF
 ‘As for Sògo, he killed THE RAT.’
 b. **Ní ti Sògó, pa ekún nẹ.*
 EXPL for S.HTS kill rat DEF
 ‘As for Sògo, he killed THE RAT.’

Secondly, in standard Yorùbá, while linguists like Adesola (2005) analyze *ó* as an RP, others like Ilori (2010) analyze it as HTS when the focus subject is third person and singular, and the RP is assumed to be null (15a). The latter’s analysis is problematic because it leaves us with the question of why a singular subject has a null RP (15a), while a plural subject chooses between a null RP, and an overt one (15b).¹⁰

- (15) Standard Yorùbá (adapted from Ilori 2010: 240)
 a. *Olùkọ_i ni [Ø_i] ó na Akin.*
 Teacher FOC 3SG HTS beat Akin
 ‘It was the teacher that beats Akin.’
 b. *[Olùkọ àti Akòwé]_i ni [Ø/wọ̀n]_i ó na Akin.*
 Teacher and secretary FOC 3SG/3PL HTS beat Akin
 ‘It was the teacher and the secretary that beat Akin.’

The problem with such an analysis is not only with regard to a difference in number features, but also person features. If the focused subject is first person (16a) or second person (16b), the HTS cannot be present, whether as a floating tone or as an overt *ó*. Similarly, in an out-of-focus context, the weak pronouns retain their tones. The confusion seems to concern only the third person singular weak pronoun because it has the same tone as the floating high tone. This is not unexpected.

- (16) Standard Yorùbá
 a. *Èmi ni [mo_i] (*ó) na Akin.*
 1SG FOC 1SG HTS beat Akin
 ‘It was me that beat Akin.’
 b. *Ìwo ni [o_i] (*ó) na Akin.*
 2SG FOC 2SG HTS beat Akin
 ‘It was you that beat Akin.’

¹⁰In these examples, I maintain the gloss for the RP from Ilori (2010), instead of using RP. Both are the same.

Furthermore, Awobuluyi (2006) gives an argument based on the lack of agreement of *ó*. This concerns the grammaticality of constructions such as (17), where *ó* does not agree with the subject in number.¹¹ As a result, Awobuluyi claims that this is not an RP but a preverb. First, these constructions are marked, and are seldom used by speakers. Secondly, cross-linguistic studies have shown that RPs and their antecedents do not always agree in (complete) phi-features (see for example, McCloskey 1990, Aoun et al. 2001, Sichel 2002, Demirdache 2005, Salzmann 2017). Thus, even if the sentences are marginally acceptable, it still falls within the purview of cross-linguistic analysis for resumption. So, this is not a concrete argument against a resumption account.¹²

(17) Standard Yorùbá (adapted from Akintoye & Ariyo 2015: 12962-3)

- a. Àwa ni *ó* lọ.
1PL FOC HTS go
'We were the ones that went.'
- b. Àwọ̀n ni *ó* lọ.
3PL FOC HTS go
'They were the ones that went.'
- c. Òjó àti Olú ni *ó* lọ.
Òjó CONJ Olú FOC HTS go
'Ojo and Olu were the ones that went.'

The tentative conclusion is that the non-future tense is marked with a floating high tone that associates with a preceding subject ending in a low/mid tone syllable (à la Aremu & Weisser 2024). In contrast, the case of a subject ending in a high tone syllable seems to involve the Obligatory Contour Principle (OCP), where one of the adjacent identical high tones gets deleted (cf. Akinlabi & Liberman 2001).¹³ On the other hand, the overt realization of the *ó* syllable is a case of resumption of either a subject topic or subject focus.¹⁴ Since the 3sg RP has a high tone, it also creates the same OCP context with the floating high tone, akin to the canonical subject cases.¹⁵

¹¹See Adesola (2005: ch. 3) for a discussion on agreeing and non-agreeing RPs in Yorùbá.

¹²As a matter of fact, others such as Yusuf (1989), Bámgbòsé (1990), Adesola (2005) analyse *ó* as an RP.

¹³Which one gets deleted is not known at this point.

¹⁴I leave an extensive discussion for future research.

¹⁵For subjects, I will use names that end with a high tone which is the same as the HTS to avoid confusion.

3.1.2 Movement and structural analyses for SF

I propose that A'-movement takes place for both local and non-local SF. The presence of an RP is only an EPP requirement. One way to support my proposal is to apply island tests. Certain syntactic domains obstruct movement out of them. These domains are called *islands* (cf. Ross 1967, Chaves & Putnam 2020). Thus, when an island is sensitive to the (sub)-extraction of an A'-dependency, the result is that the A'-dependency involves movement in a non-island context, like focus. On the other hand, when it is base-generated in the left periphery, we do not expect an island violation; since no movement has taken place. In order to argue for the movement of the subject focus, I employ two island tests: *adjunct island* and *complex noun phrase constraint*.

Adjunct (temporal) clauses ban A'-movement out of them (Cattell 1979, Chaves & Putnam 2020). The sentence in (18a) has a temporal adjunct clause. If the subject of the adjunct clause is sub-extracted, the sentence becomes ungrammatical, as in (18b). This is irrespective of the presence or absence of the RP inside the adjunct clause.

- (18) a. Bólá kú [jí Adé i fẹ́ Tolú].
 B. die before A. PRT marry T.
 ‘Bólá died before Adé married Tolú.’
 b. *[Adé]_F Bólá kú [jí (ó) i fẹ́ Tolú] rín.
 A. B. die before RP PRT marry T. FOC
 Lit. ‘Adé Bólá died before he married Tolú.’

Similarly, the extraction of the subject of a relative clause in a complex noun phrase is prohibited (19b). The presence or absence of the RP does not matter. If the subject dislocation is not as a result of movement, we would expect (19b) to be grammatical since no movement would have occurred from within the island.

- (19) a. Bólá rí [okònrén nẹ́ yí Tolú kí].
 B. see man DET REL T. greet
 ‘Bólá saw the man whom Tolú greeted.’
 b. *[Tolú]_F Bólá rí [okònrén nẹ́ yí (ó) kí] (rín).
 T. B. see man DEF REL RP greet FOC
 Lit. ‘TOLÚ Bólá saw the man who greeted.’

The result above is not only true for SF in root clauses. The fronting of long-distance SF also violate islands. I use adjunct island below (20). Extracting the

subject of the temporal adjunct clause in (20a) is prohibited whether the movement is to the left periphery of the embedded clause (20b), or to the left periphery of the matrix clause (20c). This is regardless of the presence or absence of an RP. In other words, just like the root clause cases, resumption does not repair the island.¹⁶ We now turn to the proposal for the structural analysis of SF in Ìkálẹ̀.

- (20) a. Jídé fò fi Adé ti lọ [jí Tolú ù rí Bólá ní ojà].
 J. say COMP A. PFV go before T. PRT see B. at market
 ‘Jídé said that Adé left before Tolú saw Bólá at the market.’
 b. *Jídé fò fi Tolú Adé ti lọ [jí (ó) ù rí Bólá ní ojà] rín.
 J. say COMP T. A. PFV go before RP PRT see B. at market FOC
 ‘Jídé said that Adé left before Tolú saw Bólá at the market.’
 c. *Tolú Jídé fò fi Adé ti lọ [jí (ó) ù rí Bólá ní ojà] rín.
 T. J. say COMP A. PFV go before RP PRT see B. at market FOC
 ‘Jídé said that Adé left before Tolú saw Bólá at the market.’

Following what has been argued for many languages with morphological focus marking, including Standard Yorùbá and some of its dialects, I propose that the focus marker *rín* occupies the head (Foc⁰) of a Focus Phrase (FocP) (Yusuf 1989, Rizzi 1997, Aboh 1998, Bisang & Sonaiya 2000, Adesola 2005, Ilori 2010, Amaechi & Georgi 2019, Akintoye 2020, Amaechi 2020).¹⁷ Using the SF sentence in (10b), the structure in Figure 1 shows that the subject of the clause *Adé* external merges at the Spec,vP to receive the agent thematic role before moving to Spec,TP. It is at this position that it serves as the goal of a higher probe which is the Foc⁰. Foc⁰ has an interpretable but unvalued [*i*Foc[—]] feature that must be valued. It probes down its c-command domain for an identical valued feature. It finds the uninterpretable but valued subject [*u*Foc[—]], and Agrees with it. The result of the agreement is a feature valuation [*i*Foc[3]] ... [*u*Foc[3]]. In the recent view of agreement, the Spec-head relationship does not necessarily play a role (Chomsky 2000, Baker 2008). In other words, the focused constituent does not need to move to Spec,FocP in order to establish an agreement with Foc⁰, excusing agreement as a trigger for movement. Therefore, I adopt the EPP to account for the presence of the focused constituent in Spec,FocP. The definition that I adopt for

¹⁶I want to thank an anonymous reviewer for pointing out the need to replicate the island test for long-distance SF as well.

¹⁷While some Yorùbá linguists argue that focus construction in the language is a noun phrase (cf. Awobuluyi 1978, 1987, Awoyale 1985), others believe that it is a full sentence whose part (or whole) has undergone information structuring (cf. Bámgbósé 1990, Yusuf 1989, 1990, Owolabi 1981, 1983). I agree with the latter position here.

the EPP requirement is that of Landau (2007: 489) (see also Safir 2019), as given in (21). EPP requires that the Spec of FocP is filled, and the phrase that fills the Spec position must have an overt head. Thus, it is like a feeding relationship, where the focused constituent that moves, is the goal that the probe has agreed with (Chomsky 2001). The presence of an RP in Spec,TP is also due an EPP feature on T. Originally, the EPP requires that every clause must have a subject by obligatorily filling the Spec of TP (Chomsky 1982). Thus, the RP occupies the canonical subject position of the clause.¹⁸

- (21) Landau's (2007) EPP requirement
In $[_{HP} ZP [_{H'} H[_P] \dots]]$, Z must be pronounced.

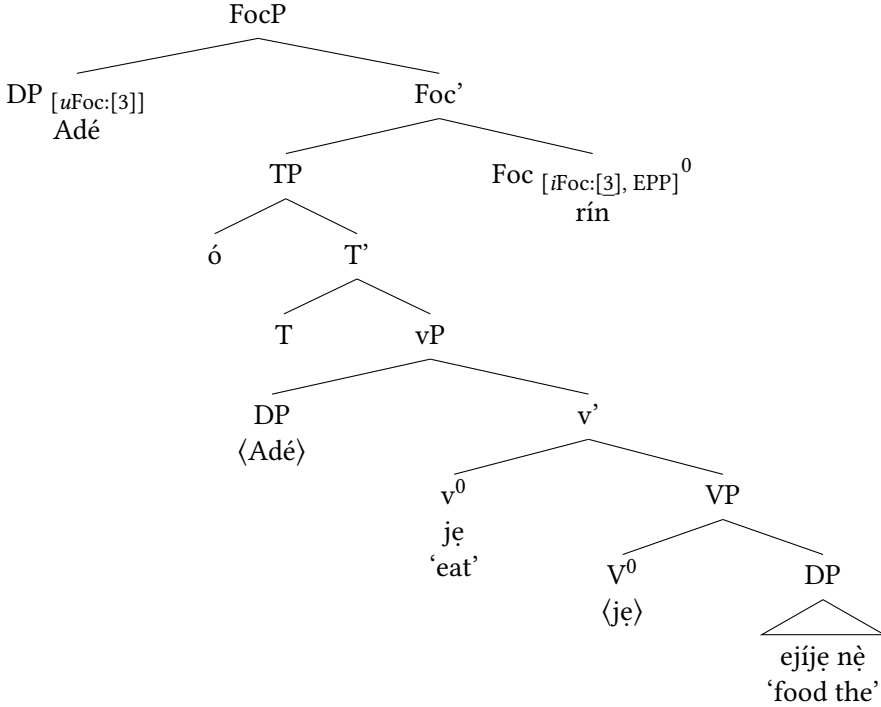


Figure 1: Syntactic tree for (10b)

¹⁸Alternatively, Rizzi's (2006) criterial approach suggests that cross-linguistic studies have shown that resumption can be used to obviate the *subject criterion*. In contrast, there is no object resumption because there is no *object criterion* (see also Rizzi & Shlonsky 2007, Shlonsky 2014, Rizzi 2017).

Having proposed the structure with a head-final focus marker above, we see that this seems to pose a challenge to Kayne's (1994) antisymmetry theory of syntax. Kayne (1994) and the proponents of the antisymmetry theory of syntax argue for a universal asymmetry to c-command, where (underlyingly) the head of a phrase always precedes its complement. Thus, the variation in word order that is present in languages of the world is due to different steps of movements. The implication of this, therefore, is that "languages all have S-H-C (Spec-Head-Complement) order" (Kayne 1994: 45). If we want to be strict with the antisymmetry theory, we would have to assume that the FocP is left-headed. But that would mean that the whole TP has to move (to Spec,FocP, for example) at some point in order to get the surface order, which would result in having the focus marker in a clause-final position. However, this itself is not the best solution because we would be left with the question of where then does the ex-situ SF or OF constituent move to, if the TP is already in Spec,FocP? In fact, such TP to Spec,FocP movement would be unmotivated. There are other potential problems to such a proposal.

An alternative proposal might be to assume a functional projection (FP) above the TP, where any constituent in focus would have to move to first before pied-piping the whole FP to Spec,FocP. Such a structure would look like what is in Figure 2. Assuming that the head of FP has an uninterpretable and unvalued focus feature, plus an EPP [*uFoc*____, EPP], it probes down to Agree with the subject, and then gets valued. Afterward, the subject moves to Spec,FP due to the EPP, and leaves an RP behind. At this point, there are two instances of the focus feature. The *Foc*⁰ head is then merged, having the whole of FP as its complement. *Foc*⁰ has an interpretable but unvalued focus feature that probes downward for a goal. It finds FP which by now is valued as a result of its earlier agreement with the focus DP. The whole of FP is then pied-piped to Spec,FocP. The latter movement allows the focus marker *rín* to appear clause-finally as well as having the focus constituent in the sentence-initial position.

Although this seems to be a plausible analysis, it has some problems. Firstly, going by the principle of economy, the derivation of the structure is uneconomical (cf. Chomsky 1995, 2000). It is built on a number of unmotivated assumptions which is a violation of both the economy of derivation and the economy of representation. For instance, what exactly is the function of FP? Its projection is simply unmotivated. Although it helps in deriving the linear order, based on Kayne's (1994) anti-symmetric syntax, it does not play any role with regard to information structure. Obviously, the sort of motivations that warrant the use of FP elsewhere do not apply here (cf. Cinque 2010, 2023). If we agree that movement is not free, i.e., it is semantically-driven or pragmatically-driven, then the

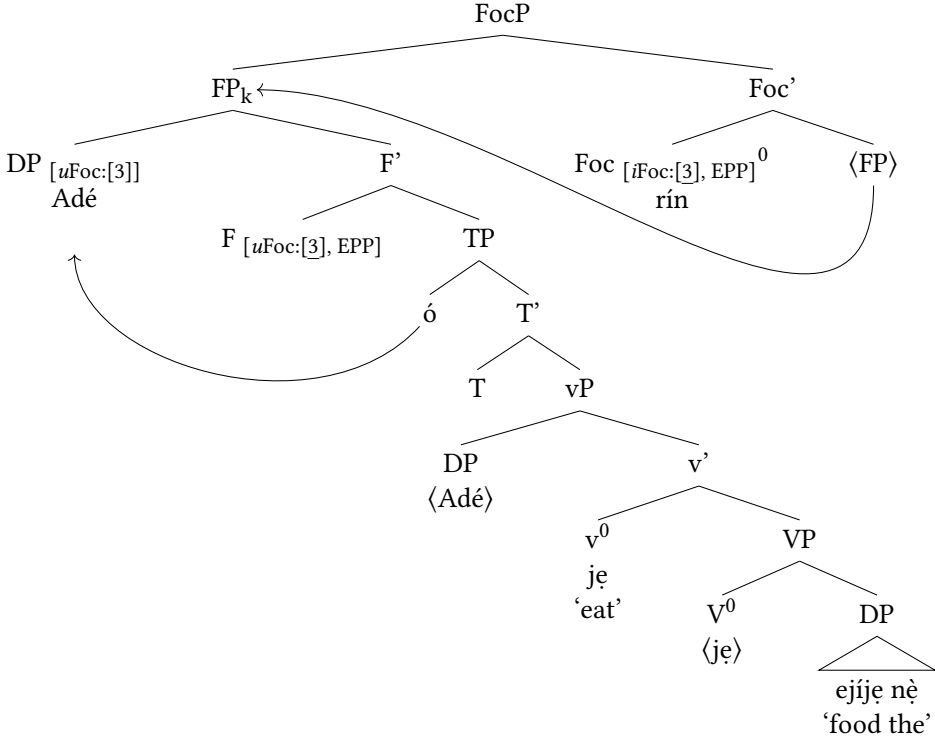


Figure 2: Alternative syntactic tree for (10b)

movement to Spec,FP does not seem to be driven either by semantics or pragmatics.¹⁹ Secondly, the structure in Figure 2 requires the postulation of different operations of agreement which makes the analysis more complex than the structure in Figure 1 above. The latter captures all the necessary steps in the derivation in a simple manner. It is able to capture the semantic property of focus, which is primarily to invoke alternatives. This is realized through F-marking which is overtly expressed via the morphological focus marker *rín*, and/or focus movement in the dialect under investigation. The focus constituent that invokes alternatives enters agreement with the focus head, thereby establishing a syntax-semantic relationship. It only does not fit into the antisymmetry theory of syntax, which is not a problem for the analysis itself (cf. Carstens 2008, 2017, Carstens & Zeller 2020).²⁰ Lastly, the structure in Figure 2 violates Chomsky's (1995) *in-*

¹⁹See, however, Cinque (2010, 2023) and Chomsky (2021) for an unmotivated movement approach.

²⁰See also Occam's razor.

clusiveness condition which primarily restricts the introduction of elements or features which are non-lexical (or lexically related), or the traces which result from the movement of such lexical items. Introducing projections and features that are not lexically-induced, seems to give syntax too much power; a problem Chomsky tries to restrict with the inclusiveness condition.

3.2 Non-subject focus

Non-subject focus (NSF) may involve an object (direct or indirect), an adjunct, a verb, or a verb phrase. All these categories, with exception of the VP, can be focalized both in-situ and ex-situ. NSF differs from SF because while the former can occur in both in-situ & ex-situ positions, the latter only occurs in an ex-situ position (cf. Section 3.1 above).

3.2.1 Object focus (OF)

When an object is in focus, it can be fronted to the left periphery of the clause (22b) or it can stay in-situ – in its external merged position (22c). Unlike SF, OF does not use an RP. It instead leaves a gap in its extraction site (22b). Similarly, the in-situ focused answer cannot be marked with the overt morphological focus marker *rín*. In this context, an ex-situ *wh*-question as in (22a) can be answered with either an ex-situ focus or an in-situ focus, with the former being preferred. However, if the *wh*-phrase is in-situ, the focused answer usually stays in-situ.

- (22) a. Nèé Adé rí-i?
 who A. see-Q
 ‘Who did Adé see?’
 b. [Tolú]_F Adé rí (*ó) **(rín)*.
 T. A. see RP FOC
 ‘Adé saw TOLÚ.’
 c. Adé rí [Tolú]_F **(rín)*.
 A. see T. FOC
 ‘Adé saw TOLÚ.’

Embedded OF can also involve long distance A'-movement, either to the left periphery of the matrix clause (23b) or the left periphery of the embedded clause itself (23c). In these two cases, the focus marker *rín* is obligatorily realized in the clause-final position. Embedded OF can also remain in its base position. It is morphologically unmarked in this case (see ex. 23d). As stated earlier, in cases

where the focus stays in-situ, the context would have to be relied on in order to know which constituent is in focus. This is because focus is marked with a morphosyntactic strategy in *Ìkálẹ̀*, and not intonation. Thus, the absence of this strategy triggers ambiguity.

- (23) a. Nèé Adé fọ fi Bọlá rí ní ọjà?
 who A. say COMP B. see at market
 ‘Who did Adé say that Bọlá saw at the market?’
 b. [Tolú]_F Adé fọ fi Bọlá rí ní ọjà *(rín).
 T. A. say COMP B. see at market FOC
 ‘Adé said that Bọlá saw TOLÚ at the market.’
 c. Adé fọ fi [Tolú]_F Bọlá rí ní ọjà *(rín).
 A. say COMP T. B. see at market FOC
 ‘Adé said that Bọlá saw TOLÚ at the market.’
 d. Adé fọ fi Bọlá rí [Tolú]_F ní ọjà (*rín).
 A. say COMP B. see T. at market FOC
 ‘Adé said that Bọlá saw TOLU at the market.’

So far, I have claimed that ex-situ OF (just like ex-situ SF) undergoes A'-movement to the left periphery. To argue for this claim, I apply both adjunct island (temporal clause) and complex noun phrase constraint (relative clause) tests. The matrix verb in (24a) embeds a temporal clause. The focus extraction of the direct object of the temporal clause *Bọlá* is prohibited (24b). Thus, in focus constructions, the realization of the OF in the left periphery of the clause involves movement and not base-generation. If it were by base-generation, we would expect that the extraction from adjunct islands should be possible, since no movement would be involved.

- (24) a. Adé ti lọ [jí Tolú ù rí Bọlá ní ọjà].
 A. PFV go before T. PRT see B. at market
 ‘Adé left before Tolú saw Bọlá at the market.’
 b. *[Bọlá]_F Adé ti lọ [jí Tolú ù rí — ní ọjà] (rín).
 B. A. PFV go before T. PRT see at market FOC
 ‘Adé left before Tolú saw BỌLÁ at the market.’

Another classic movement test is the Complex Noun Phrase Constraint (CNPC). A complex NP or DP usually involves a noun phrase (or a DP) that takes either a relative clause or a clausal complement. A'-movement is banned from these

domains. Thus, when the object of the relative clause is extracted, the sentence becomes ungrammatical (see 25b).

- (25) a. Adé rí [okònrén nè yí ó jẹ ejíjẹ nè].
 A. see man DET REL RP eat food DEF
 ‘Adé saw the man who ate the food.’
 b. *[Ejíjẹ nè]_F Adé rí [okònrén nè yí ó jẹ —] (rín).
 food DEF A. see man DEF REL RP eat FOC
 Lit. ‘THE FOOD Adé saw the man who ate.’

Not only local object focus, but also non-local object focus involves A'-movement. The adjunct island data below attest this. In (26), the direct object of the adjunct clause cannot move out of the adjunct clause to either the embedded left periphery (26a) or the matrix left periphery (26b). If it does, the sentence becomes ungrammatical (26c).

- (26) a. Jídé fọ fi Adé ti lọ [jí Tolú ù rí Bólá ní ojà].
 J. say COMP A. PFV go before T. PRT see B. at market
 ‘Jídé said that Adé left before Tolú saw Bólá at the market.’
 b. *Jídé fọ fi [Bólá]_F Adé ti lọ [jí Tolú ù rí — ní ojà]
 J. say COMP B. A. PFV go before T. PRT see at market
 (rín).
 FOC
 ‘Jídé said that Adé left before Tolú saw Bólá at the market.’
 c. *[Bólá]_F Jídé fọ fi Adé ti lọ [jí Tolú ù rí — ní ojà]
 B. J. say COMP A. PFV go before T. PRT see at market
 (rín).
 FOC
 ‘Jídé said that Adé left before Tolú saw Bólá at the market.’

The overall consequence of these movement tests is that there is an A'-movement with regards to ex-situ (non)-local OF in Ìkálẹ̀.

Having established the mechanism for realizing ex-situ OF, we can now propose a syntactic analysis for it. The structure proposed for object focus in Figure 3 is similar to what I proposed for subject focus above. The only difference here is that it is the object that possesses the valued focus feature. Thus, it serves as the agreement controller. In addition, unlike SF, which leaves an RP in Spec,TP, OF leaves a movement trace instead in its base position (complement of V). The

alternative analysis that was proposed for the subject focus above faces the same challenge with object focus (and adjunct focus), too. I will not present the structure for want of space.

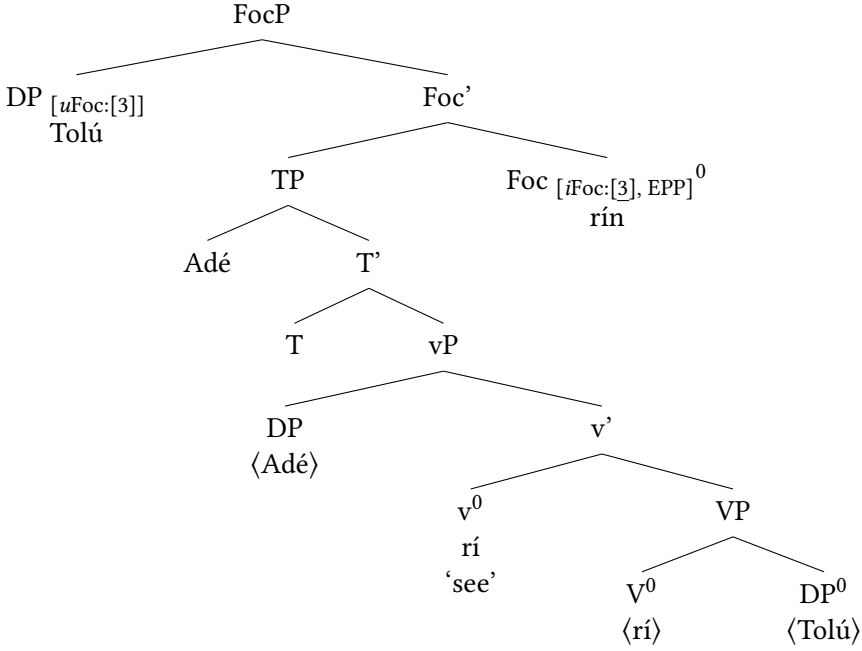


Figure 3: Syntactic tree showing object focus

3.2.2 Other non-subject focus

Adjuncts can also be focused in Ìkálẹ̀. They can either be fronted or realized in-situ. The *wh*-phrase in (27a) inquires the time that Adé saw Tolú. (27b) provides the focused answer by fronting *àná* ('yesterday') to the left periphery, while (27c) has its focus remain in-situ.

- (27) a. Ugbo Ayò rí Tolú?
 when A. see T.
 'When did Adé see Tolú?'
 b. [Àná]_F Ayò rí Tolú *(rín).
 yesterday A. see T FOC
 'Adé saw Tolú YESTERDAY.'

- c. Adé rí Tolú ni [àná]_F (*rín).
 A. see T. at yesterday FOC
 ‘Adé saw Tolú YESTERDAY.’

As expected, the same strategies are present in embedded clauses. The embedded adverbial phrase *ọjá* (‘market’) is fronted to the left periphery of the matrix clause (28b). The adjunct focus can also be realized in-situ and to the left periphery of the embedded CP.

- (28) a. Kúbo Adé fọ fi Bọlá rí Tolú?
 where A. say COMP B. see T.
 ‘Where did Adé say that Bọlá saw Tolú?’
 b. [Ọjà]_F Adé fọ fi Bọlá rí Tolú *(rín).
 market A. say COMP B. see T. FOC
 ‘Adé said that Bọlá saw Tolú AT THE MARKET.’

Another category that can be focused is the verb or verb phrase. The answer to a question like *What did Adé do?* requires a VP focus. In Ìkálẹ̀, VP focus can only be in-situ (29a). However, a verbal focus can be either in-situ or ex-situ. When it is ex-situ, it involves verb doubling (see 29b). The fronted verb is formed by partial reduplication. The onset of the verb *rà* is reduplicated, and then a high toned high front vowel /i/ is inserted, resulting in *rí*. The outcome is a nominalized verb *rírà*. Thus, the moved element has a copy within the clause (see Hein 2020 for an extensive discussion on the typology of V(P) fronting and/or doubling).

- (29) a. Adé [rà ìwé]_F (*rín).
 A. buy book FOC
 ‘Adé BOUGHT A BOOK.’
 b. [Rí-rà]_F Adé rà ìwé *(rín).
 NMZL-buy A. buy book FOC
 ‘Adé BOUGHT a book.’

Before concluding this section, I want to point out that there is a body of research that argues for a semantic and/or pragmatic distinction between in-situ and ex-situ foci (Rizzi 1997, É. Kiss 1998, Drubig 2001, Becker et al. 2019). Aboh et al. (2007: 6) asked the following questions regarding languages with two focus strategies: in-situ and ex-situ (30).

- (30) a. Are there semantic/pragmatic differences between the two focus strategies? And if so, what are they?

- b. Do the ex-situ vs. in-situ strategies comply with the new information versus contrastive focus dichotomy as proposed for certain intonation languages (e.g. É. Kiss 1998, Drubig 2001)?

The effect that a Contrastive Focus (CF) (or *identificational focus* à la É. Kiss 1998) has on the salient set of alternatives is that of the presence of at least one other alternative that is being contrasted in the same context. It, therefore, expresses a quantification-like meaning (É. Kiss 1998). Although the two questions in (30) seem to be the same, the following discussion answers question (30b) explicitly for Ìkálẹ̀.

In Ìkálẹ̀, ex-situ focus, whether subject or non-object, is contrastive in nature, while the in-situ non-object focus (recall that there are no cases of in-situ subject focus) is new informational (non-presupposed information) in nature. Consider the SF data in (31).

- (31) a. [Adé]_F ó jẹ ejíjẹ nẹ *(rín) (kǐi se Bólá). Contrastive SF
 A. RP eat food DEF FOC NEG COP B.
 ‘ADÉ ate the food (and not Bólá).’
 b. * [Adé]_F jẹ ejíjẹ nẹ (rín) (kǐi se Bólá). New information SF
 A. eat food DEF FOC NEG COP B.
 ‘ADÉ ate the food (and not Bólá).’

As shown earlier, SF can only occur in an ex-situ position. Thus, they are always contrastive.²¹ When the proposition is contrasted with the negative phrase in the bracket, the result is that (31a), but not (31b), is grammatical. The implication is that ex-situ SF is contrastive in nature.

Object focus, on the other hand, can be both contrastive and new information. The object focus in (32a) is ex-situ, and has a contrastive reading. However, the in-situ object focus in (32b) has a new information interpretation. The presence of the focus marker *rín* and the negative phrase in the bracket is prohibited. Thus, it cannot be contrastive.

- (32) a. [Ìrẹ̀sì]_F Adé jẹ *(rín) (kǐi se ẹ̀wà). Contrastive SF
 rice A. eat FOC NEG COP beans
 ‘Adé ate RICE (and not beans).’

²¹As to why this is so, is not clear yet. However, this might be due to the fact that subjects are typically topics. Thus, in order to distinguish topics from subject foci in a language that marks focus morphosyntactically (and not prosodically), contrastive focus marking is employed (cf. Rizzi & Shlonsky 2007, Shlonsky 2014).

- b. Adé jẹ́ [ìrẹ̀sì]_F (*rín) (*kíí se ẹ̀wà). New information SF
 A. eat rice FOC NEG COP beans
 ‘Adé ate RICE (*and not beans).’

In this section, I have presented the data on the realization of focus in Ìkálẹ̀, and proposed two structural analyses: one that is free from the antisymmetry theory of syntax, and another that adopts it. In the next section, I will briefly discuss association with focus, with an emphasis on the focus sensitive particle *nùkàn* ‘only’.

4 Association with focus

Certain particles are said to require or interact with the focus in a clause (Rooth 1985, 1992, Krifka 1992, 2006). Such elements are famously called *Focus Sensitive Particles* (FSPs), and the phenomenon is called *Association with Focus* (AwF).²² Although the popular view is that FSPs require overt focus in their scope (or c-command environment), it has been argued that this is not always the case (Beaver & Clark 2008, Grubic 2015). Beaver & Clark (2008) argue that only *Conventional Association Particles* obligatorily require the presence of the focus associate in their scope. Such particles include *only*, *also*, and *even*.²³ This is what Tancredi (1990) calls the *Principle of Lexical Association* (PLA).

(33) Principle of Lexical Association (Tancredi 1990: 30)

An operator like *only* must be associated with a lexical constituent in its c-command domain.

Thus, in (34), (34a) has the reading that the only thing that Adam bought was a car, and nothing else. Here, the focus is on the direct object, and the FSP *only* c-commands its focus associate. Similarly, in (34b), the focus is on the subject and the surface structure shows that *only* c-commands it. (34b) is true if and only if it is only Adam who bought a car, and no one else.

²² Another name for AwF is *Association with Alternative or Alternative Sensitivity*, according to Hartmann & Zimmermann (2008). This makes sense on the ground that FSPs only interact with focus indirectly, and relate directly with the salient alternatives triggered by the focus (cf. Rooth 1992, Beaver & Clark 2008).

²³ There is, however, a crosslinguistic variation regarding the nature of the three particles as conventional operators. For instance, Grubic (2015) argues that in Ngamo (West-Chadic) only *only*, and not *also* and *even*, is a conventional operator (see also Zimmermann 2006, Hartmann & Zimmermann 2008, Grubic & Zimmermann 2011, Duah 2015, Zimmermann 2015).

- (34) a. Adam bought only A CAR.
b. Only ADAM bought a car.

In this section, I will briefly discuss the distribution of the focus sensitive particle *nùkàn* ‘only’ in Ìkálẹ̀. I chose *nùkàn* because, firstly, it has widely been discussed in the literature as a true focus sensitive particle in other languages (see a.o., Lee 2005, Beaver & Clark 2008, Sun 2021, Branen & Erlewine 2023), and secondly, it shows more consistency in its distribution compared to *also* and *even*, for example. In addition, since the association with focus is not the paper’s main focus, it suffices to pick one out of the popular FSPs.

The exclusive focus sensitive particle in Ìkálẹ̀ is *nùkàn*. It has an exclusive interpretation, which presupposes that the only optimal candidate out of a salient set of alternatives is the constituent in focus. The question in (35a) is the context, and (35b) shows an association with SF. Notice that the structure in (35b), without the FSP *nùkàn*, is the same as what we have seen in Section 3.1 where the subject is in an ex-situ position but has an RP in Spec,TP. What is new in (35b) is the adjunction of *nùkàn*. It means that no one else worked apart from Bólá.

- (35) a. Nòó se usẹ: Bólá àbí Adé?
who do work B. or A.
‘Who worked: Bólá or Adé?’
b. [Bólá]_F **nùkàn** *(ó) se usẹ *(rín).
B. only RP do work FOC
‘Only BÓLÁ worked.’

Similarly, in (36) *nùkàn* can associate with the object focus whether in-situ (36a) or ex-situ (36b). While the former lacks the focus marker *rín* because it is in-situ, the latter compulsorily has the focus marker. Both sentences mean that Adé slaughtered only a fowl, and nothing else.

- (36) a. Bólá pa ẹ̀diyẹ **nùkàn**.
B. kill fowl only
‘Bólá slaughtered only A FOWL.’
b. [Ẹ̀diyẹ]_F **nùkàn** Bólá pa *(rín).
fowl only B. kill FOC
‘Bólá slaughtered only A FOWL.’

Structurally speaking, the position of FSPs has been argued to be either *adnominal* or *adverbial* (cf. Jacobs 1983, König 1991, Büring & Hartmann 2001, Mursell

2021). In German, for example, Buring & Hartmann (2001) and Mursell (2021) argue that the FSPs adjoin to *Extended Verbal Projections* (EVPs) such as VP, TP, and CP (37a) (see also Jacobs 1983). English, on the other hand, seems to have a mixed adjunction, i.e., adjunction to both DPs and EVPs, as the case may be (37a & 37b) (König 1991). There are, however, only a few studies that discuss the positions of FSPs in African languages. See, for example, Hartmann & Zimmermann (2008) on Bura, and Carstens & Zeller (2020) on Nguni, respectively.

- (37) a. [_{VP} **only** [_{VP} ...]]
 b. [_{DP} **only** [_{DP} ...]]

Based on the discussion so far, at least two structural questions need to be answered for Ìkálẹ̀:

- (38) a. Does the particle adjoin to the focused DP or to an extended verbal projection? In other words, is *nùkàn* adnominal or adverbial?
 b. How can we account for the rightward position of *nùkàn* in relation to the Principle of Lexical Association (PLA), and considering Kayne's (1994) antisymmetry theory of syntax.

The answer to the first question is that *nùkàn* in Ìkálẹ̀ is both adnominal and adverbial; similar to English. It is adnominal with regards to argument focus, but adverbial with regards to V(P) focus. One good piece of empirical evidence for the adnominal analysis is that *nùkàn* cannot be stranded when its focus associate moves to the left periphery (39a). It must pied-pipe along with its focus associate (39b).

- (39) a. *[Èdiyẹ]_F Bọ́lá pa **nùkàn rín**.
 fowl B. kill only FOC
 'Bọ́lá slaughtered only A FOWL.'
 b. [Èdiyẹ]_F **nùkàn** Bọ́lá pa *(rín).
 fowl only B. kill FOC
 'Bọ́lá slaughtered only A FOWL.'

In verbal or verb phrase focus, however, *nùkàn* is adverbial, and it associates with an extended verbal projection. In an intransitive sentence like (40a), *nùkàn* adjoins to an EVP. However, there is more to the transitive sentence in (40b). Unlike argument focus, *nùkàn* cannot occur immediately after the verb for an

association with the verb. It can only occur after the direct object.²⁴ The structural position of *nùkàn* in (40b) can cause an ambiguous reading, where it could associate with the verb, the direct object, or the entire verb phrase. The three levels of AwF ambiguity are presented in (41).²⁵

- (40) a. Bólá [sùn]_F *nùkàn*.
 B. sleep only
 ‘Adé only SLEPT.’
 b. Bólá pa (**nùkàn*) *ẹ̀diyẹ̀ nùkàn*.
 B. kill only fowl only
 ‘Bólá SLAUGHTERED only a fowl/slaughtered A FOWL/
 SLAUGHTERED A FOWL.’
- (41) a. V: Bólá did nothing else to a fowl, but slaughtered it.
 b. DO: Bólá slaughtered nothing else, but a fowl.
 c. VP: Bólá did nothing else, but slaughtered a fowl.

A proposed structure for adnominal *nùkàn* is presented in Figure 4, where the FSP adjoins to the DP. Figure 5 and Figure 6 represent association with V(P) or DO respectively. It is the context that will determine whether it is an association with V focus or VP focus in (5).²⁶

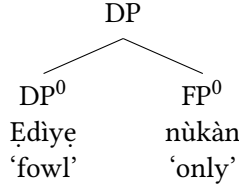


Figure 4: Syntactic structure for adnominal *nùkàn*

²⁴Although *nùkàn* seems to appear after the verb in (40a), it actually adjoins to the entire VP. Example (40b) makes this obvious.

²⁵I do not F-mark any of the constituents because of the ambiguity of the sentence. So, context (e.g., a question) would have to be resorted to in order to know which of the constituents is in focus.

²⁶A reviewer asked why the FSP projection is called FP. I use FP (functional phrase) for the focus sensitive particle projection here because it is a more general name, considering the fact that there are other FSPs such as *also*, *even*, etc. (see Beaver & Clark 2008). It could as well be called Exclusive Phrase (ExclP), OnlyP, Particle Phrase (PrtP), an so on (cf. Hole 2017, Quek & Hirsch 2017, Bayer 2018, Sun 2021). Nothing hinges on this, however.

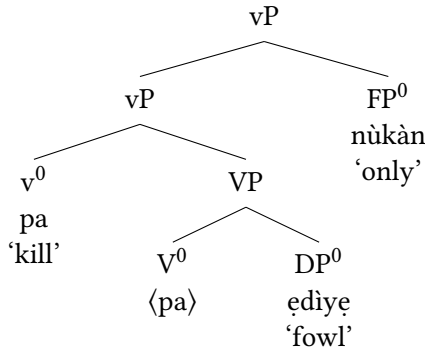


Figure 5: Syntactic structure for adverbial *nùkàn*

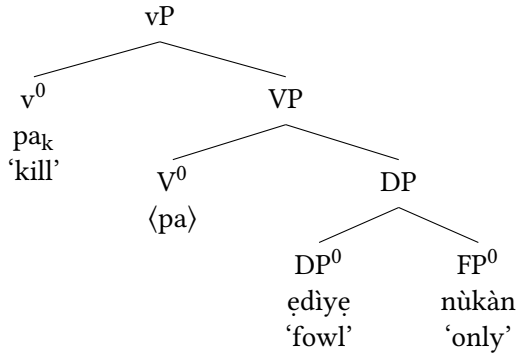


Figure 6: Syntactic structure for *nùkàn* with direct object.

The structures presented lead to the question in (38b). The PLA requires a c-command relation between the FSP and its focus associate (see (33)), and Kayne's (1994) *Linear Correspondence Axiom* (LCA) requires the c-command to be asymmetric. I adopt the formulation in (42) for LCA.

- (42) A lexical item α precedes a lexical item β iff
- a. α asymmetrically c-commands β or
 - b. an XP dominating α asymmetrically c-commands β .
- (Adapted from Carstens & Zeller (2020: 207–208) quoting Hornstein et al. (2005: 277))

However, the data presented show that *nùkàn* occurs in a post-focal position, always to the right of the focus associate. Drawing ideas from Carstens & Zeller's

(2020) analysis for Nguni, there are at least three possibilities that can be considered to answer what we observe in Ìkálẹ̀.

- (43) a. *Nùkàn* does not need to c-command its focus associate.
- b. *Nùkàn* can associate with the trace of the moved focus associate.
- c. *Nùkàn*, as an FP, is an adjunct that can c-command the focus associate to its left.

Similar to what Carstens & Zeller (2020) argue for Nguni, the third possibility (43c) seems to be the best option for Ìkálẹ̀. Adjuncts have been argued to be free from the obligation of the LCA (cf. Takano 2003, Carstens 2008, 2017). In fact, Kayne (1994: 15) says that “specifiers and *adjoined phrases* appear to have no place in the theory (of LCA).” Therefore, the adjoined FSP *nùkàn* can c-command its focus associate to the left.

Option (43a) is out because *nùkàn* needs to c-command its focus associate in order to establish the association, while option (43b) is out because *nùkàn* is adjoined to the focus adnominally (with DPs), or adverbially (with V(P)s). Thus, we are left with the third possibility, which, in fact makes it possible for us to account for the mixed positioning (adnominal and EVP) of the FSP in Ìkálẹ̀. I wish to admit here that there is a need for more arguments for my proposal in this section. However, I will leave this for future research since this section is only meant to give an overview of the syntax of association with focus in Ìkálẹ̀.

5 Conclusion

So far, we have seen that focus realization in Ìkálẹ̀ can be both in-situ and ex-situ. However, only non-subject focus (object, verb, and adverb) can be realized ex-situ or in-situ. Subject focus can only be ex-situ. I argued that both ex-situ subject and non-subject foci are realized via A'-movement (and not via base-generation). The use of resumption by subject focus, however, is due to an EPP requirement on T (in the spirit of Landau 2007). I proposed two syntactic structures for ex-situ focus. The first structure is simpler and devoid of LCA requirements. It merges the focus head to the right clause-final position. The second structure follows the LCA by realizing the focus head to the left. The focused constituent is first moved to the Spec of a higher functional phrase (FP), before pied-piping the whole FP to Spec,FocP. I showed that although this gives us what we want structurally, it is unmotivated and uneconomical, considering the economy of derivation and representation. Thus, I argued that the first structure is better.

Finally, I gave an overview of the structural relationship and distribution between the focus sensitive particle *nùkàn* ‘only’ and its focus associates. The mixed analysis appears to be the best analysis. The FSP can have both adnominal adjunction and EVP adjunction, depending on whether the focus is a DP or a V(P). I showed that following PLA and LCA, the data on AwF seem to pose a challenge at first. However, at a closer look, *nùkàn* happens to be an adjunct that is less strict in compliance to the LCA requirement. It can therefore c-command its associate to the left.

Abbreviations

1	first person	HTS	high tone syllable
3	third person	NEG	negative
COMP	complementizer	PFV	perfective
CONJ	conjunction	PL	plural
COP	copula	PRT	particle
DEF	definite	Q	question particle
EXPL	expletive	REL	relative
FOC	focus	RP	resumptive pronoun
FUT	future	SG	singular

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Chapter 3

Prosody of topic and focus in Tira

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In Tira, word order is determined by information structure. Topic and focus constructions are both characterized by the topic or focused element appearing at the front of the sentence. In this paper, we explore the prosodic properties of these constructions. Cross-word high tone spreading occurs within prosodic constituents, but is blocked between a topic and remainder of the sentence, showing that the topic is prosodically separated. Focus is characterized by a focus particle and by a H tone that appears at the left edge of the auxiliary or verb that directly follows the focused constituent.

1 Introduction

Speakers convey information about the propositional content of utterances, based on shared and presumed knowledge between the speaker and interlocutor. By varying sentence structure, word choice and prosody, speakers package the information in order to guide the interlocutor about new and given information. Prosody is a key component of such information structure, encoding prominence via stress, tone, duration and/or intonation.

In African tone languages, tone encodes lexical contrasts, and can also be employed for grammatical purposes, such as case marking or tense-aspect-mood distinctions (Rolle 2018, Monich 2020). As a result, tone is not often used to mark a topic or focused constituent directly (Aboh et al. 2007, Hartmann 2008, Zerbian et al. 2010, Güldemann et al. 2015), although Akan is reported to use low tone to directly mark focused elements (Kügler 2017). Nevertheless, there may be indirect ways in which tone is employed to express information structure, such as pitch expansion (as in Hausa, Leben et al. 1989), post-focal pitch lowering (as



in Bemba, Kula & Hamann 2017) or blocking of tone spreading (as in Tangale, Hartmann & Zimmermann 2007, Hartmann 2008). Duration is a marker of information structure, and in some Bantu languages, penultimate or phrase-final vowel lengthening occurs to mark prosodic phrase edges. These can correlate with topic and focus information. For example, penultimate lengthening in Sotho-Tswana (Zerbian 2017) indicates phrase edges, and left-dislocated object topics are marked through separate prosodic phrasing, via a boundary pause and penultimate vowel lengthening of the object. Similar patterns are found in XiTsonga (Lee & Riedel 2023), where left dislocated focus constituents exhibit penultimate lengthening.

In this article, we present data on the prosody of topic and focus constructions in Tira, a Niger-Congo language of Sudan. Word order in Tira is determined by information structure. In both topic and focus constructions, the topicalized or focalized constituent occurs first, followed by the verb, or auxiliary if there is one: SVO/SAuxVO or OVS/OAuxSV. Focused adjuncts also have a verb/Aux-second pattern. Word order patterns are reinforced via prosodic phrasing that sets the initial constituent apart, using cross-word high tone spreading as a diagnostic. Duration does not appear to be a factor marking topic or focus constructions. While topic and focus constructions share some syntactic and prosodic properties, they differ in that focus constructions involve a focus marker and High (H) tone at the left edge of the main auxiliary or verb. This H tone is not present in the topic construction. We do not analyze the syntactic structures of topic and focus in detail; more information on a variety of syntactic structures and proposed derivations can be found in Hassen et al. (2022). The present article contributes to the study of how prosody combines with morphosyntax to convey information structure, a pattern that is found in other African tone languages.

This paper is structured as follows. In Section 2, information on the Tira language is provided, including relevant discussion of noun class, case and verb morphology. In Section 3, topic constructions are presented, with detailed comparison of pitch tracks. In Section 4, the properties of focus constructions are discussed, including focus question constructions. Finally, in Section 5, polar questions are addressed.

2 Background

Tira is a West Heiban language of the Kordofanian family (Niger-Congo) (Schadeberg 1981) spoken in the Nuba Mountains of Sudan.¹ Tira was previously de-

¹Recent research (Dimmendaal 2018) has suggested that Kordofanian languages may not be a genetic unity, but are loosely affiliated Niger-Congo languages spoken in the same area. We

scribed in a 1942 grammar edited by Thilo Schadeberg and published as Stevenson (2009), and in an MA thesis (Watters 1993). The data in this paper are provided by Himidan Hassen who speaks the Kadar dialect of Tira, spoken in and around the town of Kauda. He is also fluent in Swahili, English and Sudanese Arabic.

2.1 Tone

Tira has two level tones, H and L (Watters 1993, Kaldhol 2024). There is also a falling tone $\widehat{H}\bar{L}$ which occurs word-finally and contrasts with H and L tones in this position. These are exemplified by the following near minimal pairs.

- (1) Tone patters
- | | | | |
|----|-------|-------------------------|---------------|
| a. | élê | H. $\widehat{H}\bar{L}$ | ‘chief, king’ |
| b. | éré | H.H | ‘fence’ |
| c. | èβè | L.L | ‘bamboo gate’ |
| d. | èràð | L. $\widehat{H}\bar{L}$ | ‘cloth’ |
| e. | ðórà | H.L | ‘fence’ |
| f. | ðólá | H.H | ‘flour’ |
| g. | ðòɽár | L.H | ‘tiger’ |
| h. | ðèrnà | L.L | ‘leather’ |

2.2 Noun class

Tira is a noun class language. Classes are indicated by a word-initial consonant from the set $[\delta \text{ } \text{t} \text{ } \text{s} \text{ } \text{k} \text{ } \text{l} \text{ } \text{r} \text{ } \text{j} \text{ } \text{n} \text{ } \text{ɲ}]$. Nouns are paired in singular/plural combinations that have different noun class consonants (2a) and (2b). Adnominals show concord with the noun class and verbs show noun class agreement. Some nouns are vowel-initial (2d) or have a consonant which is not part of the defined set of noun class consonants (2e). In these cases, noun class is indicated by concord on adnominals and agreement on the verb.

- (2) a. $\eta\acute{e}n$ η -ðlà
 CL η .dog CL η -tall
 ‘The dog is tall.’
- b. $\eta\acute{e}n$ η -ðlà
 CL η .dog CL η -tall
 ‘The dogs are tall.’

do not take a position on this proposal.

- c. ðùṛì ð-ðlà
CLð.tree CLð-tall
‘The tree is tall.’
- d. ùṛì k-ðlà
CLg.tree CLg-tall
‘The trees are tall.’
- e. mùðù k-ðlà
CLg.leopard CLg-tall
‘The leopard is tall.’

2.3 Case marking

Tira shows nominative/accusative case alignment. Accusative case is indicated through a variety of suffixes and tone changes, with little predictability as to which pattern a given noun selects. This is similar to the related language Koalib (or Rere), as described in Quint & Allasonnière-Tang (2023). Some of the case marking patterns are shown in (3).

(3) Case patterns

	NOM	ACC		Pattern
a.	ðàṇàl	ðàṇàl-à	‘sheep’	suffix -à
b.	lèṇár	lèṇár-á	‘field’	suffix -á
c.	iléð	iléð-é	‘hoe, plough’	suffix -é
d.	lídí	lídí-ó	‘clay pot’	suffix -ó
e.	ùjèl	újél-é	‘shadow’	tone change + suffix -é
f.	ṭòlé	ṭòlè-ṗá	‘lion’	tone change + suffix -ṗá
g.	òrì	órí-ṗé	‘water’	tone change + suffix -ṗé
h.	èlàṛà	èlàṛà	‘farmer’	no change
i.	ṇòṛón	ṇòṛón	‘black squirrel’	no change

As can be seen from the examples, several different suffixes are observed. Some suffixes do not engender tone changes on the noun stem (3a-d), while others do (3e-g). Finally, some nouns show no overt marking for case as in (3h-i). If the noun ends in a vowel in the nominative and a vocalic suffix is selected, the suffix replaces the final vowel (3d). See Kaldhol (2024) for more details.

2.4 Verb morphology

Tira has an aspectual distinction between imperfective and perfective, and a verbal deixis distinction between ventive and itive. Depending on the lexical seman-

tics of the verb and its aspect, ventive indicates motion towards or distance away from the speaker, while itive is motion away or proximity to the speaker. The ventive perfective consists of the class agreement marker, the verb root and a final vowel which conveys information on the verb’s inflection class and aspect-mood-deixis distinctions. The itive perfective and the two imperfectives have an auxiliary /a/ which hosts the agreement marker followed by the verb stem, which consists of the verb root and the final vowel. The quality of the final vowel and the tone patterns on the auxiliary and the verb stem serve to distinguish the forms. Tira has a set of seven inflection classes based on the quality of the final vowel (Kaldhol 2024). However, the tone patterns are the same no matter the inflection class. In (4), the conjugation for the verb *vələð* ‘pull’ is shown with the class marker *CLL*.²

(4)	‘pull’	ITIVE		VENTIVE	
		IPFV			
		l-á	vələð-à	l-á	vələð-ó
		CLL-AUX	root-FV	CLL-AUX	root-FV
		H	HL-L	H	LL-H
		PFV			
		l-à	vələð-è		là-vələð-ó
		CLL-AUX	root-FV		CLL-root-FV
		L	HL-L		L-LL-H

3 Topics

Topics are what the discourse is about. At the sentence level, an element is identified as the topic and the rest of the sentence is the comment which provides information about the topic. In Tira, topics occur in the initial position in the sentence. They can only be nominals or pronominals, either subjects or objects. Verbs, adverbs, and adpositional phases cannot occur in initial position of the sentence as topics. Once the topic is introduced, it will typically be referenced further in the discourse by pronominal elements. In Tira, this is accomplished via class agreement or pronominal clitics.

²Since AUX and FV as glosses are uninformative in terms of providing information on tense/aspect/mood/deixis (TAMD), we will adopt the convention of labeling the auxiliary as IPFV if it bears grammatical H tone, and as I.PFV if it bears grammatical L tone. We will label the final vowels as particular TAMD categories depending on what TAMD that particular verb form encodes. However, the reader should bear in mind that the TAMD is expounded via multiple elements in the verbal complex.

3.1 Subject topics

In a subject verb object (SVO) or subject auxiliary verb object (SAuxVO) sentence, the subject is the topic, and the auxiliary, or verb if there is no auxiliary, agrees with it in noun class. In the sentences in (5), the noun class of the subject topic is CLg. The object is marked for accusative case. Prosodic phrasing of the syntactic constituents can be diagnosed with H tone spreading. In (5a) the verb ends in a low tone, and the object is low-toned. In (5b) the verb ends in a low tone, and the object is high-toned. This shows that there is no L-toned spreading from verb to object. However, a single H tone at the end of a word spreads non-iteratively onto a following word and overwrites the low tone of the first tone-bearing unit. In (5c), the H tone at the end of the ventive perfective verb spreads onto the first vowel of the following object *ǎàǎàlà*.³

- (5) a. ùrnò k-á vólèǎ-à ǎàǎàlà-à
 CLg.grandfather CLg-IPFV pull-IT.IPFV CLǎ.sheep-ACC
 ‘Grandfather will pull a sheep (away).’
 b. ùrnò k-á vólèǎ-à ǎén-é
 CLg.grandfather CLg-IPFV pull-IT.IPFV CLǎ.dog-ACC
 ‘Grandfather will pull a dog (away).’
 c. ùrnò kǎ-vólèǎ-ǎ ǎàǎàlà-à
 CLg.grandfather CLg-pull-VEN.PFV CLǎ.sheep-ACC
 ‘Grandfather pulled a sheep (towards).’

Furthermore, the H tone on the auxiliary in (5a) and (5b) spreads onto the verb stem, creating the HL tone pattern on the verb root.⁴

In contrast, H tone does not spread from the subject to the verb, as shown in (6). Both verbs have the same tone pattern in these sentences, even though the subjects that precede them end in different tones. H-tone spreading from word to word is not constrained by the presence of other H tones in the word, as can be seen in (6b), where H spreads from the verb to the object *ǎprí-ǎà*.

- (6) a. ǎprí jǎ-vólèǎ-ǎ ǎàǎàlà-à
 CLǎ.boy CLǎ-pull-VEN.PFV CLǎ.sheep-ACC
 ‘The boy pulled the sheep (towards).’

³The fact that only H tone spreads does not mean that L is unspecified and the language has a H vs. Ø contrast. This is because HL is a contrastive tone and both LH and HL tones can be derived by combining L and H tones. It may be the case that L is the unmarked tone.

⁴H tone spreading does not take place from the auxiliary to the ventive imperfective verb root due to independent reasons; see Kaldhol (2024) for discussion.

- b. ðàŋàl ðà-vəlèð-ó áprí-ŋá
 CLj.boy CLj-pull-VEN.PFV CLj.boy-ACC
 ‘The sheep pulled the boy (towards).’

If there is an auxiliary following the subject, H tone also does not spread from the subject to the auxiliary. This can be seen with the itive perfective form which has a L auxiliary.

- (7) àprí k-à vólèð-è ðàŋàl-à
 CLj.boy CLg-IT.IPV pull-IT.IPV CLð.sheep-ACC
 ‘The boy pulled a sheep (away).’

The lack of H tone spreading is evidence of a prosodic break between the subject and the verb or auxiliary, and prosodic phrasing of the auxiliary, verb and object together, as shown in (8), where PPh refers to phonological phrase.⁵

- (8) Phrasing of SVO sentences
 [Subject]_{PPh} [(Aux) Verb Object]_{PPh}

3.2 Object topics

Object noun phrases (NPs) can also appear in initial position as topics, and this requires OVS order, as shown in (9b). The fronted noun phrase is the topic, and the rest of the sentence provides information about the topic, or the comment. The first sentence (9a) shows the SVO order in which the subject is the topic, and the second (9b) the OVS order. In this position, the nominal cannot be marked for accusative case; it appears in the nominative, or unmarked form.⁶

- (9) a. tólé tð-nòŋ-ó ŋén-é únéré
 CLt.lion CLt-watch-VEN.PFV CLŋ.dog-ACC yesterday
 ‘The lion watched the dog yesterday.’

⁵We are agnostic on precisely what type of prosodic phrase this is and choose phonological phrase for simplicity.

⁶H tones are lowered in phrase-final position in Tira, so many elicited sentences include a time adverb in final position to ensure the tone patterns of the final nominal are indicated. Note that the adverb *únéré* is usually pronounced with lowered tones in final position. We abstract away from utterance final lowering in the transcriptions, but we do note cross-word H tone spreading. In the examples in (9a) and (9b), the nominal after the verb spreads H tone to the adverb *únéré* to produce [únéré].

- b. *ɲén ɲə-nòn-ó tólé únéré*
 CL_ɲ.dog CL_ɲ-watch-*VEN*.PFV CL_ɲ.lion yesterday
 ‘As for the dog, the lion watched it yesterday.’

If the initial NP ends in a H tone, this H tone does not spread to the verb. However, there is H tone spreading from the verb to the post-verbal subject, where *tólé* (LH tone) is pronounced [tólé] (HH tone). This indicates that topic objects are phrased independently from the rest of the sentence in the same manner as topic subjects.

Verbs with auxiliaries have OAuxSV order when the object is the topic. The itive perfective has L HL(L) tone (10a). However, when the subject is positioned after the auxiliary, the H tone appears at the left edge of the following subject *ðàɲàl* rather than on the verb (10b), which is low-toned. This is evidence that the H tone of the auxiliary is floating and appears on whichever element occurs to the right of the auxiliary, rather than being assigned specifically to the verb root. It further suggests that i) the auxiliary and the subject are prosodically phrased together, and ii) when auxiliary and verb stem are adjacent, they are also prosodically phrased together.

- (10) a. *ðàɲàl ð-à vólèð-è àprí-ɲá únéré*
 CL_ð.sheep CL_ð-PFV pull-*IT*.PFV CL_j.boy-ACC yesterday
 ‘The sheep pulled the boy (away) yesterday.’
 b. *àprí j-à ðàɲàl vèlèð-è ùnééré*
 CL_j.boy CL_j-PFV CL_ð.sheep pull-*IT*.PFV yesterday
 ‘As for the boy, the sheep pulled him (away) yesterday.’

Finally, H-tone spreading can also take place from a post-auxiliary subject onto the following verb stem, as shown by the itive imperfective in (11b). The verb stem is L-toned when following a L-toned subject, but has an initial H tone when following a H-toned subject.

- (11) a. *àprí j-á ðàɲàl vólèð-à*
 CL_j.boy CL_j-IPFV CL_ð.sheep pull-*IT*.IPFV yesterday
 ‘As for the boy, the sheep will pull him (away).’
 b. *ðàɲàl ð-á àprí vèlèð-à*
 CL_ð.sheep CL_ð-IPFV CL_j.boy pull-*IT*.IPFV yesterday
 ‘As for the sheep, the boy will pull it (away).’

Based on the diagnostic of H-tone spreading, the phrasing is schematized in (12). For SVO and OVS, the same phrasing is found regardless of the grammatical role of the NPs. In the sentences with auxiliaries, the topic is phrased separately; the position of the Subject and Object differentiates the sentences. H-tone spreading can occur within the second Phonological phrase from any element to the next: Verb to Object, Aux to Verb, Aux to Subject and post-Aux Subject to Verb.⁷ No H-tone spreading occurs from the Topic to the rest of the sentence.

(12) Phrasing of Topics

	Topic	Comment
SVO/OVS	[NP] _{PPh}	[Verb NP] _{PPh}
SAuxVO	[NP] _{PPh}	[Aux Verb NP] _{PPh}
OAuxSV	[NP] _{PPh}	[Aux NP Verb] _{PPh}

3.3 Prosodic phrasing and pitch tracks

Given the NP Verb NP word order for SVO/OVS, case marking can be used to distinguish the sentences. Consider the sentences in (13). Both nominals belong to the same noun class, CLg. The only distinction between them is the accusative case marker on ‘leopard’ in (13a).

- (13) a. èlàɾà kə-ðɪŋ-ó múð-ò
 CLg.farmer CLg-fear-VEN.PFV CLg.leopard-ACC
 ‘The farmer fears the leopard.’
 b. èlàɾà kə-ðɪŋ-ó múðù
 CLg.farmer CLg-fear-VEN.PFV CLg.leopard
 ‘As for the farmer, the leopard fears him.’

We have determined that lack of H tone spreading does mark topics as prosodically separate. However, it is sometimes the case that pitch raising or lowering could also mark topics and distinguish them from neutral declarative sentences. In Tira, though, as information structure determines word order, neither sentence type is neutral with respect to information structure. We have no evidence to suggest that SVO is the “basic” order and OVS a more marked order. There are no clear pitch raising or lowering effects pertaining to the topic, and no discernible difference between SVO and OVS sentences in terms of pitch. This can be seen in the following pitch tracks.

⁷H tone spreading can also occur in double object constructions from one object to the next.

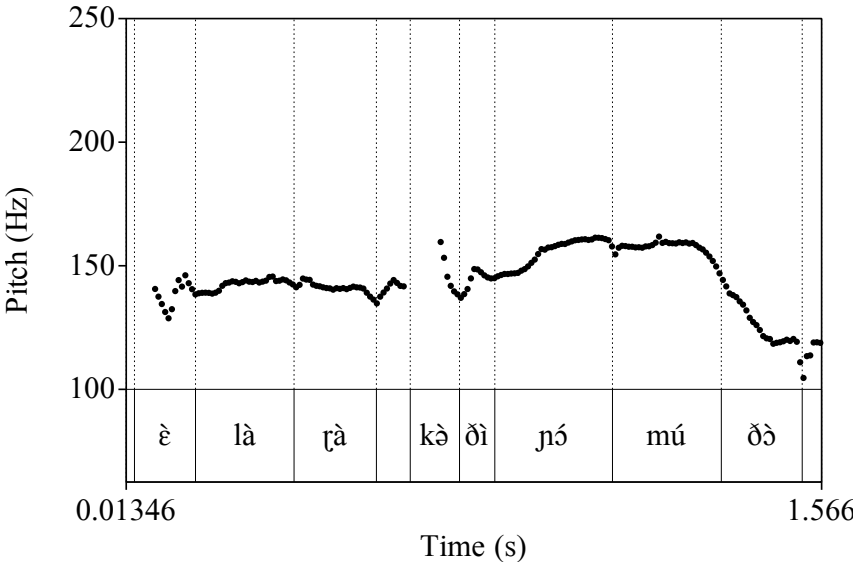


Figure 1: Intonation of (13a)
'The farmer fears the leopard'

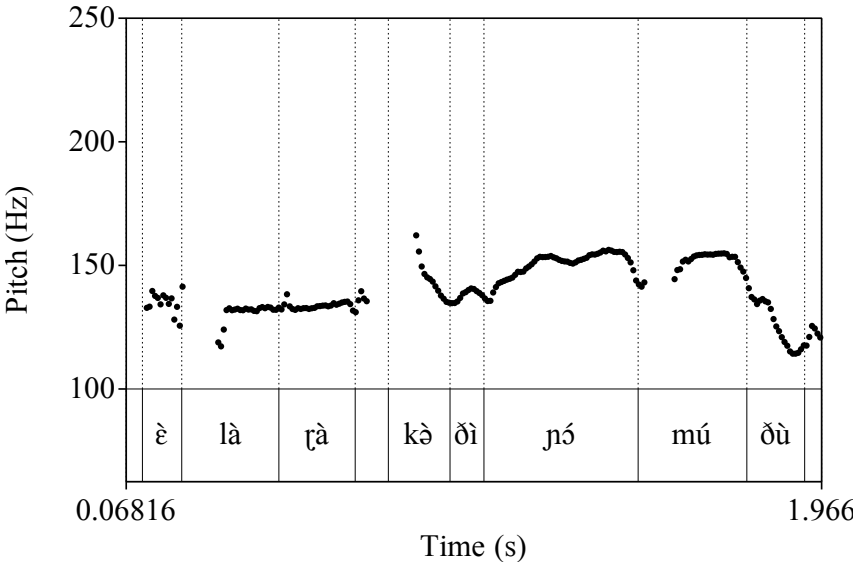


Figure 2: Intonation of (13b)
'The farmer, the leopard fears him'

We examined other SVO/OVS sentences where the topic has H tone to see if there were prosodic differences with respect to pitch levels of H tones with objects versus subject topics.

- (14) a. ðùɽú ð-àmòrð-ó áprí-já
 CLð.ostrich CLð-bite-VEN.PFV CLj.boy-ACC
 ‘The ostrich bit the boy.’
 b. ðùɽú ð-àmòrð-ó áprí
 CLð.ostrich CLð-bite-VEN.PFV CLj.boy
 ‘As for the ostrich, the boy bit it.’

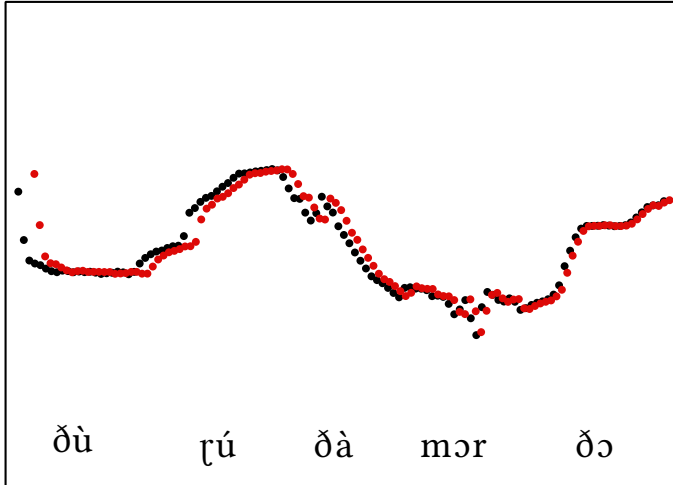


Figure 3: Intonation of topic ðùɽú and verb ðàmòrð of (14a) and (14b)

There were no differences in the pitch of the topic or its juncture with respect to the verb. This can be seen in the superimposed pitch tracks of the topic and verb portion of the sentences in (14); the pitch tracks are identical for subject topics (red) and object topics (black). Note that the ending of each sentence is different because of case-marking on the object, so we only show the first part of the sentence.

Duration could also be employed as an indicator of prosodic phrasing, but we have not noted any consistent use of vowel lengthening in these constructions.

Languages can sometimes use a pause to mark initial topics, so called “comma intonation”. For example, in Akan (Kügler 2017), there is often a pause and pitch reset following the topic. Such pauses are possible in Tira, but are optional. In

Tira texts, this is indicated with a comma. For example, in the Tira alphabet book in which the property of beds is discussed, the following sentence (15) is written with a comma.

- (15) Ahangarr ye, yerrdo ramanya mahu
àṛàṅgár j-é j-èṛḍ-ó rómân-pà màṛù
CLj.bed CLj-DEM CLj-have-VEN.PFV CLr.leg-ACC four
‘This bed, it has four legs.’

When read aloud by Himidan, there is a pause between the topic NP and the verb, and the demonstrative has a H tone that is pronounced higher than other H tones in the sentence (indicated by a super-high tone). A raised pitch in this manner functions to indicate a continuation in the sentence.

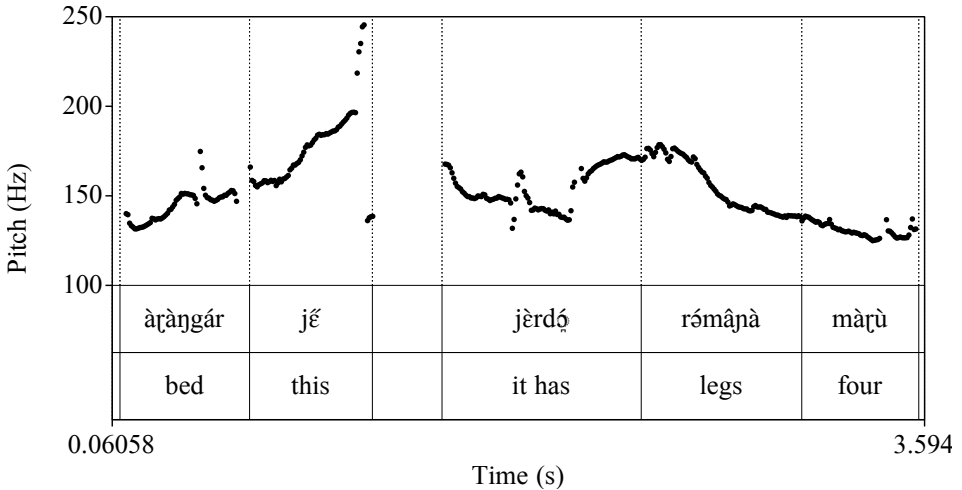


Figure 4: Intonation of (15) ‘This bed, it has four legs.’

However, Himidan notes that such pauses are not obligatory, and in the case of the read text, might be due to the formality of the speech and the presence of the comma. In the example pitch tracks in Figure 1, Figure 2 and Figure 3, there is no pause or pitch raising.

If the initial object is plural, a 3PL object marker *-l* appears at the end of the verb or auxiliary (16a). This is another means to distinguish SVO from OVS word order.

- (16) a. lárò l-àmòṛḍ-ó áprí-pà
CLl.chicken CLl-bite-VEN.PFV CLj.boy-ACC
‘The chickens bit the boy.’

- b. lárò l-àmòrð-ó-l áprí
 CLg.chicken CLL-bite-VEN.PFV-3PL.OBJ CLj.boy
 ‘As for the chickens, the boy bit them.’

However, sometimes, there is no morphology or prosody distinguishing SVO and OVS sentences. The word for ‘farmer’ has no distinction for nominative and accusative case. Therefore, if it occurs in post-verbal position, the sentence has two possible interpretations (17a). Himidan commented, “they are the same – this can mean either.”

- (17) a. mùðù kə-ðìj-ó élàrà
 CLg.leopard CLg-fear-VEN.PFV CLg.farmer
 ‘The leopard fears the farmer.’ or ‘As for the leopard, the farmer fears him.’

Topics can only be NPs. It is not possible to topicalize adpositional phrases or adverbial phrases (18c), with either CLK or CLL agreement. However, objects of adpositional phrases can appear as topics at the front of the sentence. When this occurs, there is a locative marker *-l* that appears on the verb and the postposition appears following the subject (18b).

- (18) a. ðòndò ð-ìd-òt-ó lóbú kárí únéré
 CLð.gourd CLg-fall-APPL-VEN.PFV CLL.well inside yesterday
 ‘The gourd fell inside the well yesterday.’
 b. ləbú l-ìd-òt-ó-l ðòndò kárí únéré
 CLL.well CLL-fall-APPL-VEN.PFV-LOC CLð.gourd inside yesterday
 ‘As for the well, the gourd fell inside it yesterday.’
 c. *ləbú kárí k/l-ìd-òt-ó-l ðòndò únéré
 CLL.well inside CLK/l-fall-APPL-VEN.PFV-LOC CLð.gourd yesterday
 Intended: ‘As for inside the well, the gourd fell in it yesterday.’

3.4 Content questions and topichood

Tira has two kinds of content questions. In focus questions, the interrogative phrase appears in initial position with a focus marker (so called “ex-situ” content questions). These are discussed in Section 4. It also has content questions in which the interrogative word occurs post-verbally (so called “in-situ” content questions). In the latter type of construction, the interrogative word cannot appear in the topic position because, by definition, a question asks for new information, and topics are given or old information. As far as we can ascertain, there

is no meaning distinction between these two types of content questions. We first explore the properties, syntactic and prosodic, of object questions and then turn to subject questions. The following pair in (19) shows simple substitution of the object with the interrogative word.

- (19) a. Subject topic sentence
 ʔòlé ʔ-á nón-à ɲén-é
 CLʔ.lion CLʔ-IPFV watch-IT.IPFV CLɲ.dog-ACC
 ‘The lion will watch the dog.’
 b. Object content question
 ʔòlé ʔ-á nón-à ɲndì
 CLʔ.lion CLʔ-IPFV watch-IT.IPFV CLg.what
 ‘The lion will watch what?’

Content questions do not show intonational differences from their declarative counterparts. Both sentences have a phrase-final fall, attributable to a low boundary tone. This lowers the word ‘dog’ from *ɲén-é* to [ɲénɛ̃], and there is a slight dip for the low-toned *ɲndì*. This is similar to what has been reported for the related language, Moro, wherein wh-in-situ questions have the same basic intonational contour as declaratives, although the H tones in Moro can be pronounced slightly higher (Piccinini & Rose 2017). We do not observe that pattern in Tira. The only difference in the two pitch tracks is the H tone on the object *ɲén-é* (Figure 5), compared to L tone on *ɲndì* (Figure 6).

Both constructions also show H-tone spreading from the verb onto the following object as in (20). While *ɲén-é* already begins with H tone and there is no change, *ɲndì* does not, but gets its H tone from the verb. Due to final lowering, however, the two objects are realized in a similar fashion.⁸

- (20) a. Declarative
 ʔòlé ʔ-à-nón-ó ɲén-é
 CLʔ.lion CLʔ-watch-VEN.PFV CLʔ.lion-ACC
 ‘The lion watched the dog.’
 b. Object content question
 ʔòlé ʔ-à-nón-ó ɲndì
 CLʔ.lion CLʔ-watch-VEN.PFV CLg.what
 ‘What did the lion watch?’

⁸Because *ɲndì* has no case marking, the sentence in (20b) can also be interpreted as ‘As for the lion, what watched it?’

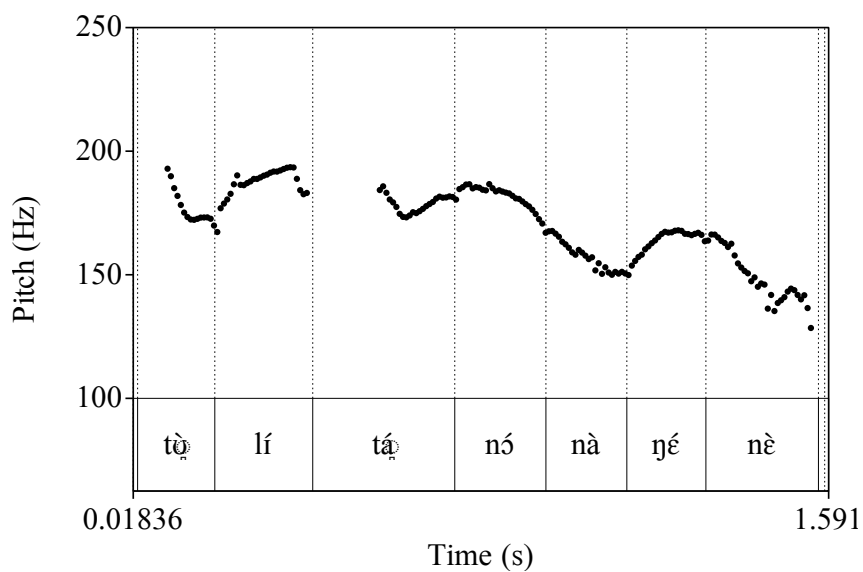


Figure 5: Intonation of (19a)
'The lion will watch the dog.'

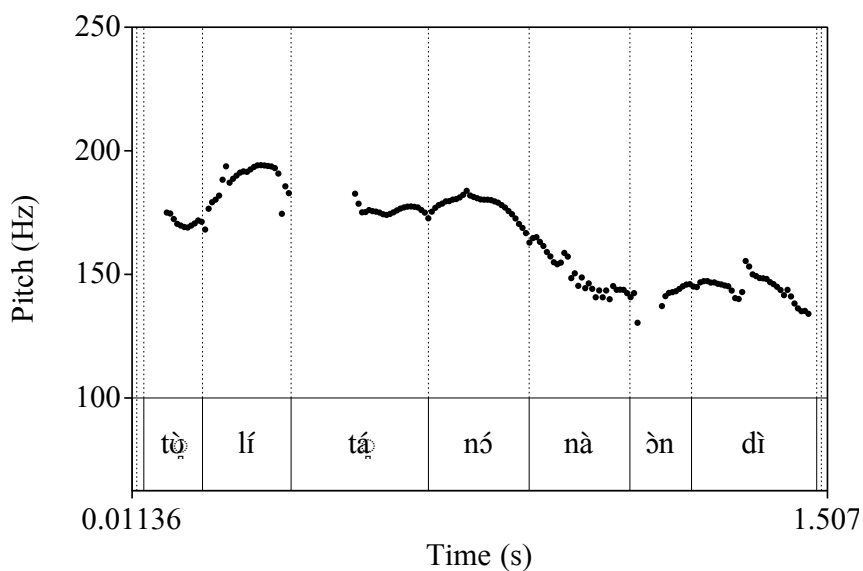


Figure 6: Intonation of (19b)
'What will the lion watch?'

A superimposition of the two pitch contours in (20) shows similar pitch contours; if anything, the question shows slightly lower pitch overall.

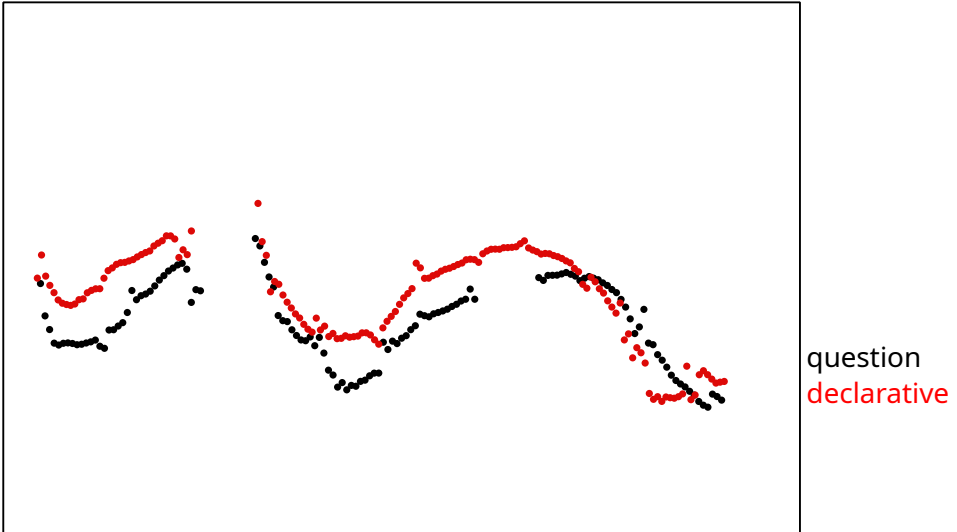


Figure 7: Comparison of intonation pitch contours of (20a) and (20b)

The answers to object questions are varied, as shown in (21). One can answer simply ‘the dog’, but it must have accusative case marking (21a), reflecting its position in the declarative sentence. One can answer with a focus marker *àn^H*, but there can be no accusative case marking (21b), or one can answer with the verb and object, where the verb’s class agreement denotes the subject (21c).

(21) Answers to content object question

- a. *ɲén-é*
CL_ɲ.dog-ACC
‘The dog.’
- b. *àn ɲén*
FOC CL_ɲ.dog
‘The DOG.’
- c. *ɬə-nòn-ó ɲén-é*
CL_ɬ-watch-VEN.PFV CL_ɲ.dog-ACC
‘It (the lion) watched the dog.’

Subject content questions with topics cannot have SVO word order. The object appears in the topic position, and the interrogative word must appear in the post-verbal position, so OVS (or with an auxiliary, OAuxSV) (22). This indicates that the topic in content questions is the other argument, in this case the ‘lion’.

(22) Subject content question

tòlé t̀à-rl̀èní ́j́j́
CL₁.lion CL₁-chase-*VEN*.PFV CL_g.who
‘Who chased the lion (here)?’

This sentence cannot be misconstrued as ‘Who did the lion chase (here)?’ because ́j́j́ is marked with a case suffix when it occurs as an object following the verb: ́j́j́-*ḡù* (who-ACC).

The structure of content questions that question an argument is as follows (transitive sentences). The subscript Q indicates the interrogative word.

(23) Phrasing of Content questions

SVO/OVS	[NP] _{PPh}	[Verb NP _Q] _{PPh}
SAuxVO	[NP] _{PPh}	[Aux Verb NP _Q] _{PPh}
OAuxSV	[NP] _{PPh}	[Aux NP _Q Verb] _{PPh}

In conclusion, sentences with SVO or SAuxVO order have subjects as topics. Prosodic phrasing shows no H tone spreading from subject to the following verb. With object topics, the word order is OVS or OAuxSV, and the same prosodic phrasing applies, again with a separation between the topic and the verb, as diagnosed via a lack of H tone spreading. H-tone spreading occurs between elements in the comment portion of the sentence. The verb or auxiliary spreads H onto the following post-verbal subject or object. The auxiliary spreads H tone onto the following Verb, and the subject spreads H tone onto the following verb. Apart from these diagnostics, there is no special prosody associated with topic sentences. Content questions show the same pattern, and the content question word appears in the post-Aux or post-verbal position, never in the topic position.

4 Focus

Content questions are a useful way to determine the expression of focus as they often presuppose an event has occurred, and invite the interlocutor to provide new information. The interrogative word narrows on the unknown portion of the event, what Aissen (2023) refers to as the “value” of the word in the proposition.

Answers can involve narrow focus, zeroing in on a particular constituent, or broad focus, such as the answer to a general question such as ‘What happened?’. Focus constructions in Tira involve an initial focused element and a focus particle, as well as H tone at the left edge of the verb or auxiliary. In this section, we present focus questions and focus answers.

4.1 Focus subject questions and answers

Focus content questions have an initial interrogative word, which occurs to the left of the verb and must be preceded by the focus marker *àn*^H.⁹ This is shown for a subject focus question in comparison to a subject topic sentence in (24). The interrogative word ‘what’ references a non-human. In focus questions, the verb agrees with the interrogative that precedes it, as indicated by CLg agreement.¹⁰ The question also shows a tone change on the verb. The ventive perfective normally has the tone pattern LLH, as seen in (24a), but in the question, it is realized as HLH (24b). This cannot be attributed to tone spreading from the previous word, as that word ends in a L tone. This H tone marks the syntactic construction. We will show in Section 4.4 that it also occurs in relative clauses, so it is not unique to focus constructions. It therefore appears to mark filler-gap dependencies. Hassen et al. (2022) analyze it in a minimalist framework as marking $\bar{A}$ -movement.

(24) Subject focus question

- a. *tòlé t̃à-nòn-ó ɲén-é únéré*
 CL_T.lion CL_T-watch-VEN.PFV CL_g.dog-ACC yesterday
 ‘The lion watched the dog yesterday.’
- b. *àn óndì k̃à-nòn-ó ɲén-é únéré*
 FOC CL_g.what CL_g-watch-VEN.PFV CL_g.dog-ACC yesterday
 ‘What watched the dog yesterday?’

There are two possible answers to the question in (24b), either a fragment (25a) or a full sentence (25b). Both answers must contain the *àn*^H focus marker. In addition, a H tone appears at the left edge of the verb, just as it did in the question. The focus answer matches the structure of the focus question. In these sentences, the basic word order has not changed, so the presence of *àn* and the H tone on the verb serves to mark the sentence as a focus construction, with

⁹See Hassen et al. (2022) who analyze it as a focus copula.

¹⁰The class marker will be CLl if the interrogative word is plural.

focus on the subject. Moreover, the word for ‘lion’ *tólé* is LH when in the subject position of a subject topic SVO sentence, as in (24a), but is realized as HH in both the focus question and the answers. Similarly, the word for ‘what’ *óndì* is LL when in a non-case marked position following a low-toned word (19b), but is HL in focus position (24b). We analyze this H tone as a floating H tone located at the right edge of *àn* that is realized on the following word rather than as distinct tonal marker of focus. This is a different H tone from the one that appears at the left edge of the verb. There are two reasons for this analysis. First, there is an alternate form of the focus marker *áné*, so it stands to reason that if the final [ɛ] were deleted, the H tone would remain and be realized on the following word, an instance of tonal stability. Second, the H tone is always at the left edge of the word immediately following the *àn*, suggesting its source is to its left.

(25) Subject focus answers

- a. *àn tólé*
 FOC CL_T.lion
 ‘The LION.’
- b. *àn tólé tǎ-nòn-ó ɲén-é únéré*
 FOC CL_T.lion CL_T-watch-VEN.PFV CL_ɲ.dog-ACC yesterday
 ‘The LION watched the dog yesterday.’

4.2 Focus object questions and answers

When objects are focused, the word order is OVS. The object appears to the left of the verb in the focus position, and the subject appears post-verbally. An object focus question is shown in (26). The verb agrees with the focused object rather than the post-verbal subject. This is similar to the word order seen with object topic constructions.

(26) Object focus question

- àn óndì ká-ɾlɛɲ-í tólé únéré*
 FOC CL_g.what CL_g-chase-VEN.PFV CL_T.lion yesterday
 ‘What did the lion chase yesterday?’

The fragment answer to the question contains the focus marker and the object in nominative (or unmarked) form (27a). The full sentence answer shows the same structure as the focus question (27b). It has OVS word order, the focus marker, and H tone on the verb. One cannot answer simply *ɲén* ‘the dog,’ or even *ɲén-é* ‘the dog’ with accusative case marking, the latter of which is possible in topic constructions.

(27) Object focus answers

a. àn ɲén
FOC CLɲ.dog
'The DOG.'

b. àn ɲén ɲǎ-ɪlɛɲ-í tólé únéré
FOC CLɲ.dog CLɲ-chase-VEN.PFV CLɪ.lion yesterday
'It's the DOG that the lion chased yesterday.'

The post-verbal subject phrases with the verb, as is evident by the H-tone spreading. The subject 'lion' has LH tone (*tólé* in initial position of a subject topic sentence (24a), but has HH tone following the perfective verb in (27b), which ends in a H tone. It is not possible to tell whether there is a prosodic break between the focused constituent and the verb based on tone spreading since the verb must begin with a H tone in focused constructions.

Content questions with multiple interrogative words must employ the focus construction. It is not permissible to have an interrogative word in the initial position unless it is focused, and only one focused element can appear in that position. The other interrogative word occurs post-verbally. It is not possible to have VSO word order with multiple interrogatives.

(28) án ɔ́jó k-íj-ɔ́ ɔ́ndì únéré
FOC CLg.who CLk-eat-VEN.PFV CLg.what yesterday
'Who ate what yesterday?'

In non-subject focus questions with auxiliaries, such as the itive perfective, the subject appears following the auxiliary and before the verb stem. While imperfective auxiliaries have H tone, the itive perfective auxiliary has a L tone, with H tone at the left edge of the verb root. In focus content questions, the auxiliary is realized with H tone (29b). This demonstrates that the H tone marking focus constructions appears on the auxiliary if there is one, otherwise on the verb. Note that the verb stem has an initial H tone in (29b) following the subject due to H tone spreading. If it were to follow a subject ending in a low tone, the verb stem would be low-toned in this position. See Kaldhol (2024) for more details.

(29) a. ɲén ɲ-à ɪlɛɲ-ì tólè-ɲá
CLɲ.dog CLɲ-IT.PFV chase-IT.PFV CLɪ.lion-ACC
'The dog chased (away) the lion.'

- b. àn óndì k-á ɲén flèɲ-i?
 FOC CLg.what CLg-IT.PFV CLɲ.dog chase-IT.PFV
 ‘What did the dog chase (away)?’

Focused elements can be either arguments or adjuncts. The following examples illustrate a non-focused (30a) and focused prepositional phrase (30b). In the latter, instead of agreement with the object of the preposition, there is default CLg agreement on the verb. In addition, there is a suffix *-l* that appears at the end of the verb, as was also seen with a topicalized locative NP. In this case, there is no H tone on the element following *àn*, but instead the H tone appears on the focus marker itself, an instance of “self-docking” of tone (McPherson & Heath 2016). It appears that the H tone can only dock onto a following nominal. This may indicate that the prepositional phrase forms a prosodic constituent apart from the focus marker.

(30) Adjunct (PP) focus

- a. ùrá ɲ-gàr-àṭ-ó ðów-é nè lèvǎr
 CLg.child CLg-break-APPL-VEN.PFV CLð.stick-ACC in CLl.field
 ‘The child broke the stick in the field.’
- b. àn nè lèvǎr ɲ-gàr-àṭ-ó-l úrá ðów-é
 FOC in CLl.field CLg-break-APPL-VEN.PFV-LOC CLg.child CLð.stick-ACC
 ‘It’s in the field that the child broke the stick.’

In conclusion, the assumed prosodic phrasing of the focus constructions with transitive verbs are shown in (31).

- (31) S VO/O VS [àn^HNP]_{PPH} [^HVerb NP]_{PPH}
 S AuxVO [àn^HNP]_{PPH} [^HAux Verb NP]_{PPH}
 O AuxSV [àn^HNP]_{PPH} [^HAux NP Verb]_{PPH}
- PP VSO [àn^HPP]_{PPH} [^HVerb NP NP]_{PPH}
 PP AuxSVO [àn^HPP]_{PPH} [^HAux NP Verb NP]_{PPH}

4.3 Long-distance content questions and H tone marking

Long-distance content questions are those in which the “gap” of the filler-gap relationship is in the dependent clause. In these constructions, both the main and dependent clauses show the word order patterns associated with focus. The subject appears post-verbally in the main clause as well as the dependent clause

if a non-subject is focused, and both verbs show noun class agreement with the focused element. In addition, there is a marker *-Cé* which appears on the main verb. This marker agrees in noun class with the focused interrogative word and appears when the focused element is something other than the argument of the main verb (certain kinds of adjuncts and non-main clause arguments). However, the H tone that marks focus constructions only appears on the main verb, not the dependent verb (32).

- (32) àn óndì k-ár-ó-gé lurnò kè-rlèŋ-í
 FOC CLg.what CLg-say-VEN.PFV-CLg CLL.grandchild CLg-chase-VEN.PFV
 ɲén
 CLŋ.dog
 ‘What did the grandchildren say that the dog chased?’

There are reports of H tone on verbs in similar long-distance constructions in other Niger-Congo languages, namely Asante Twi (Korsah & Murphy 2020) and Bamileke Medumba (Keupdjio 2020). Related Heiban languages also show similar patterns. In Rere (or Koalib), a Central Heiban language, the same type of focus constructions with H tone at the left edge of the main verb (Rodríguez 2019) is found. In Moro, a West Heiban language, the verb is instead marked with a H-toned prefix *é-* (subject focus) or *ó-* (non-subject focus) (Rose et al. 2014). This is a possible source of Tira’s focus-marking H tone. In Moro, the H-toned prefix also only appears on the main verb. For more on the syntactic structure of long-distance questions and focus constructions in Tira, see Hassen et al. (2022).

4.4 Relative clauses

Relative clauses resemble focus constructions in Tira. While they do not use the focus marker, they do show the H tone at the left edge of the verb. This indicates that the H tone on the verb/auxiliary and the presence of *àn^H* are independent. It provides an argument that the H tone on the verb marks a type of filler-gap construction rather than focus directly (Hassen et al. 2022), as it appears in relative clauses as well, which do not have a focus marker, but do have a filler-gap dependency.

- (33) Subject relative clause
 a. ɲén ɲè-rlèŋ-í tólè-ná
 CLŋ.dog CLŋ-chase-VEN.PFV CLŋ.lion-ACC
 ‘The dog chased the lion.’

- b. t̥òlé [t̥ó-ɾlèŋ-í ɲén-é] t̥-ìtʃò
 CL_L.lion [CL_L-chase-VEN.PFV CL_ŋ.dog-ACC] CL_L-bad
 ‘The lion who chased the dog is bad.’

NPs can be complex and can themselves contain a relative clause. In the following sentence, the sheep is the focused object. It is modified by an object relative clause, which has VS word order and H tone at the left edge of the verb. The main clause also shows VS word order and H tone at the left edge of the verb. In terms of intonation, there is a pitch rise at the end of the relative clause, which signals a continuation.

- (34) àn ðáŋàl [ðó-ɾlèŋ-í élàɾà ùnééré]
 FOC CL_ð.sheep [CL_ð-chase-VEN.PFV CL_g.farmer yesterday]
 ðó-ɾŋ-ó kámàɾà-ŋà
 CL_ð-kill-VEN.PFV CL_g.friend-ASSOC.PL
 ‘It’s the sheep that the farmer chased yesterday that my friends killed.’

In conclusion, focus constructions can focus an NP, PP or AdvP. The focus marker *àn^H* precedes the focused element, and the H tone is realized at the left edge of a following nominal; otherwise, on the focus marker itself. In addition, a H tone appears at the left edge of the auxiliary if one is present, or at the left edge of the verb, if no auxiliary. This H tone does not appear in dependent clauses, even if the gap is in the dependent clause. It also appears in relative clauses, an indication that focus constructions and relative clauses both have filler-gap dependencies.

5 Polar questions

Polar questions are those that seek information about the truth value of the whole proposition, answerable by ‘yes’ or ‘no’. In many sub-Saharan African languages they are indicated by so-called “lax prosody” (Rialland 2007, 2009), whereby the final portion of the question may have a low-toned question particle, low intonational tone, and/or breathy voice. Research on the related Heiban languages, Moro (Piccinini & Rose 2017) and Rere (or Koalib) (Chai et al. 2022) has shown that they fit this profile. Both have final lowering (a L% boundary tone), and a low-toned question particle -à. Moro shows tone compression leading to neutralization of tone distinctions in final position. However, both languages show raising of H tones earlier in the question. In this section, we present data on polar questions in Tira and analyze how topic and focus are realized within them.

The main distinction between a declarative SVO sentence and a corresponding polar question in Tira is raising of H tones. Tira has an optional final low-toned question particle -à, but Himidan does not always use it. Intonationally, L tones remain at the baseline pitch, but H tones at the end of the utterance are raised. Consider the following utterances. The H tone on ‘Himidan’ is pronounced lower than preceding H tones in the answer (35b), showing evidence of automatic downstep (gradually lowering of H tones across the utterance). However, in the question (35a), the H tone on ‘Himidan’ is pronounced at the same level as the preceding H tone, showing raising of H tone, or, alternatively, suppression of downstep.

- (35) a. á-g-á ðúr-ò ká-nón-à ìmídán-ɲw-à
 2SG-CLg-IPFV want-IT.IPFV CLg-see-IT.IPFV Himidan-ACC-Q
 ‘Do you want to see Himidan?’
 b. ái íɲ-g-á ðúr-ò ká-nón-à ìmídán-ɲù
 yes 1SG-CLg-IPFV want-IT.IPFV CLg-see-IT.IPFV Himidan-ACC
 ‘Yes, I want to see Himidan.’

Let us now consider SVO and OVS sentences to see how the intonation may differ in polar questions. In a sequence of H tones, the final one is at the highest pitch peak. In the following examples, the H tone of the verb spreads to the following object. Compared to the declarative sentence illustrated in Figure 8, both of these H tones are raised in the polar question, Figure 9, with the second of the two Hs forming the peak of the upsweep. This is not suppression of downstep, as these are the only H tones in the utterance. Instead they show pitch raising.

- (36) a. èlàɾà kè-ðìɲ-ó múð-ò
 CLg.farmer CLg-fear-VEN.PFV CLg.leopard-ACC
 ‘The farmer fears the leopard.’
 b. èlàɾà kè-ðìɲ-ó múð-ò
 CLg.farmer CLg-fear-VEN.PFV CLg.leopard-ACC
 ‘Does the farmer fear the leopard?’

In the corresponding OVS sentences, the same pattern is observed (Figure 10 and Figure 11). The only distinction between the pairs of sentences is the case marking on the post-verbal nominal in (36) and the lack of case marking in (37), which manifests as a different final vowel on ‘leopard’, but the same tone pattern.

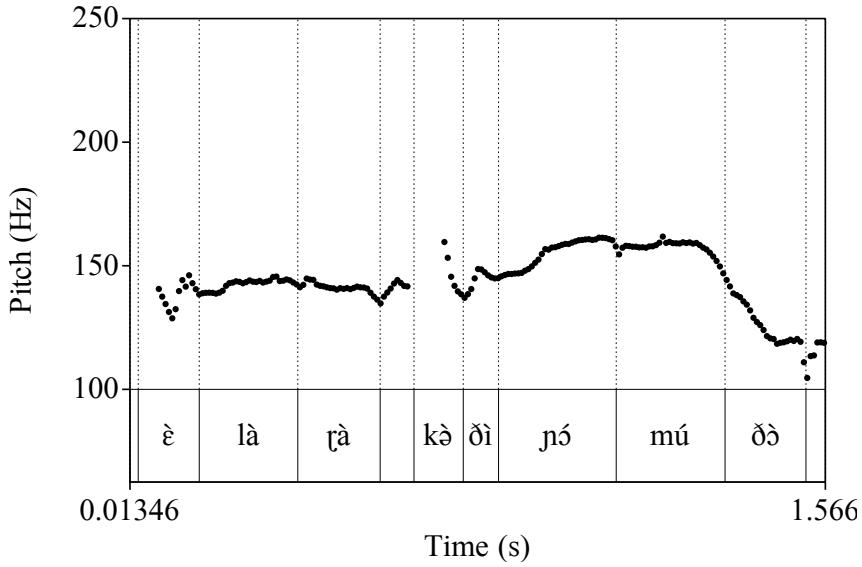


Figure 8: Intonation of (36a)
'The farmer fears the leopard.'

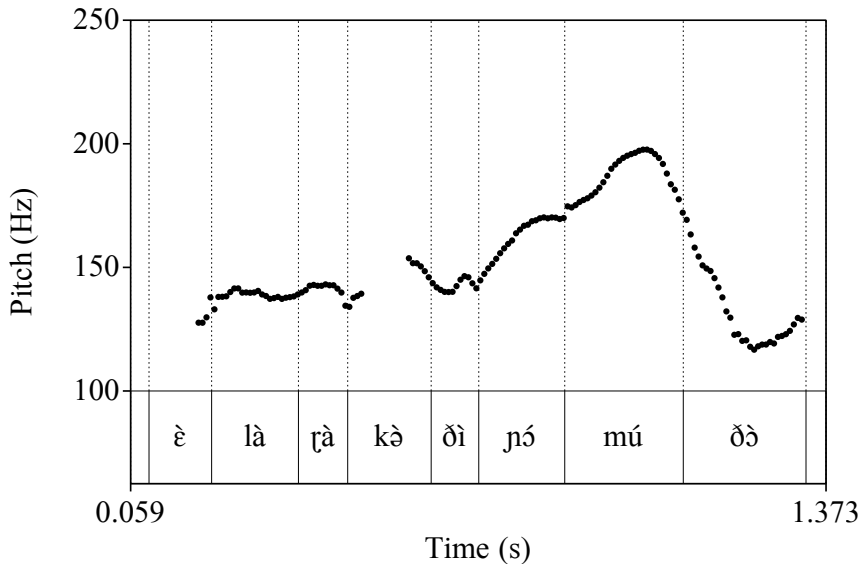


Figure 9: Intonation (36b)
'Does the farmer fear the leopard?'

- (37) a. èlàɾà kə-ðìɲ-ɔ múðù
 CLg.farmer CLg-fear-VEN.PFV CLg.leopard
 ‘The farmer, the leopard fears him.’
 b. èlàɾà kə-ðìɲ-ɔ múðù
 CLg.farmer CLg-fear-VEN.PFV CLg.leopard
 ‘The farmer, does the leopard fear him?’

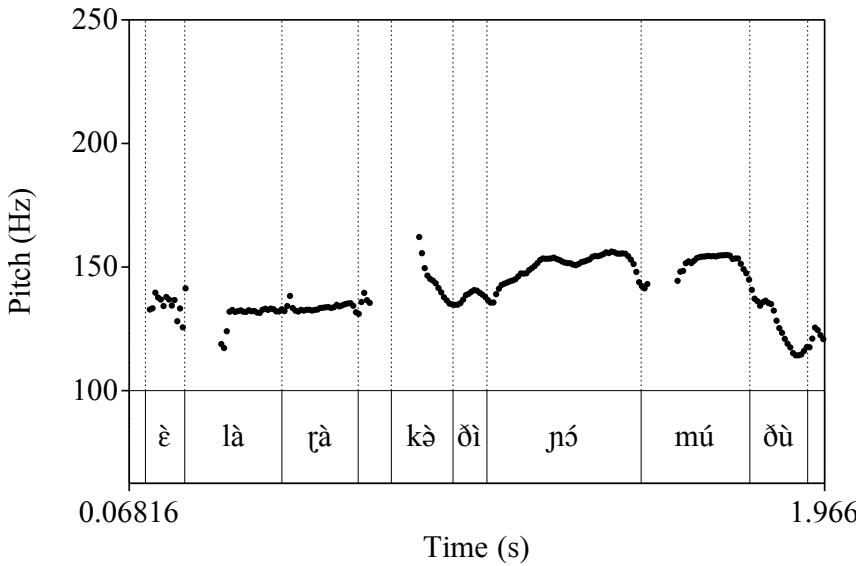


Figure 10: Intonation of (37a)
 ‘The farmer, the leopard fears him.’

As with declarative topic sentences, there is no distinct intonational prosody that would signal the grammatical role of the NPs in the sentence. There is no special intonation that is placed on the topic itself in either declarative sentences or questions. But polar questions are distinguished from declarative sentences with raised H tones.

Focus polar questions are similar to topic polar questions. The following are examples of a subject focus declarative sentence (38a) compared to its polar question (38b).

- (38) a. àn ðàɲàl ð-ámòrð-ɔ áprí-ɲá únéré
 FOC CLð.sheep CLð-bite-VEN.PFV CLj.boy-ACC yesterday
 ‘It was the SHEEP who bit the boy yesterday.’

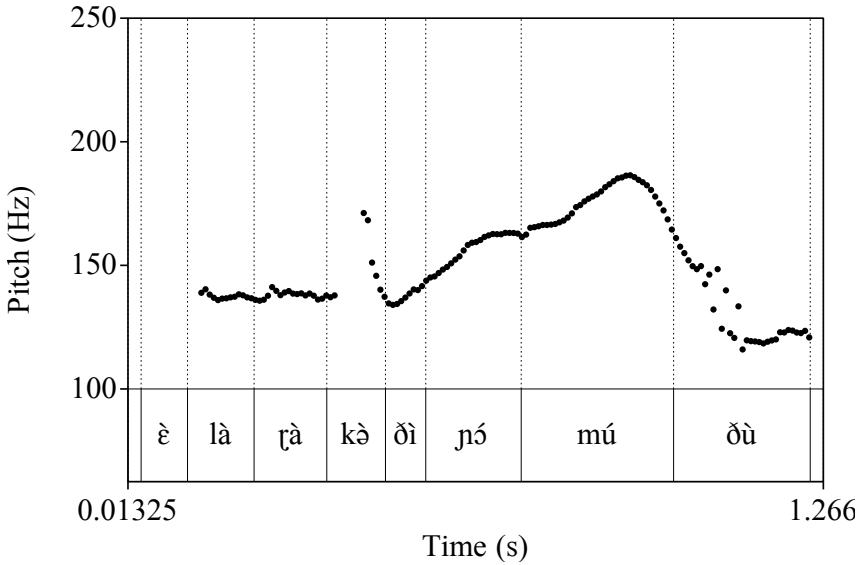


Figure 11: Intonation (37b)
‘The farmer, does the leopard fear him?’

- b. àn ðàŋàl ð-ámòð-ó áprí-ɲá únéré
FOC CLð.sheep CLð-bite-VEN.PFV CLj.boy-ACC yesterday
‘Was it the SHEEP who bit the boy yesterday?’

They show the same intonational patterns as topic sentences with pitch raising, but the pitch of the entire utterance is raised for both H and L tones in the question (Figure 13) relative to the declarative sentence (Figure 12), except for the beginning and end of the utterance. This entails a sharp pitch drop at the end of the question, which we attribute to a L boundary tone. There is less steep declination for the polar question than the declarative, presumably due to the lack of pitch raising. The following pitch contours show focused subjects.

The pitch contours of the polar question and declarative can be superimposed as in Figure 14 to illustrate the raising effect.¹¹

Focus polar questions show pitch raising in earlier parts of the utterance, but show a fall phrase finally, a pattern that is consistent with other African languages. This pattern is observed with both topic and focus constructions alike.

¹¹These examples were recorded on the same day.

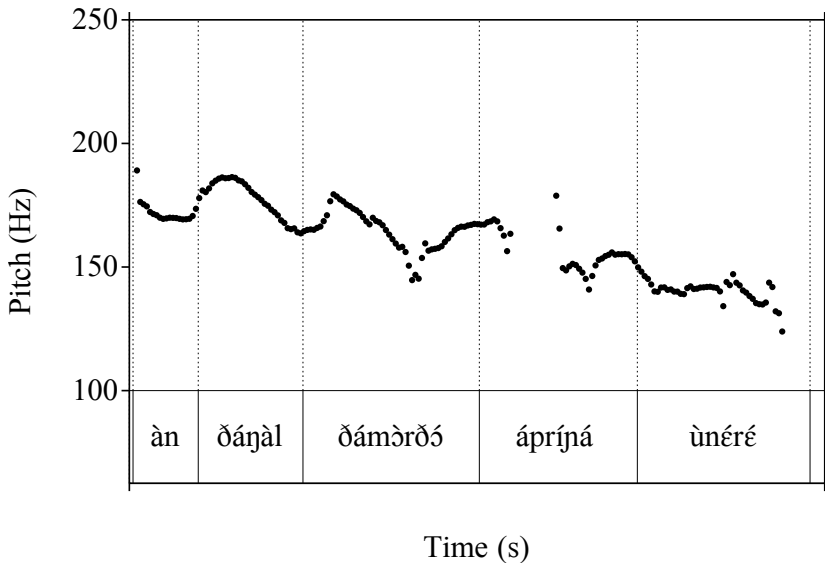


Figure 12: Intonation of (38a)

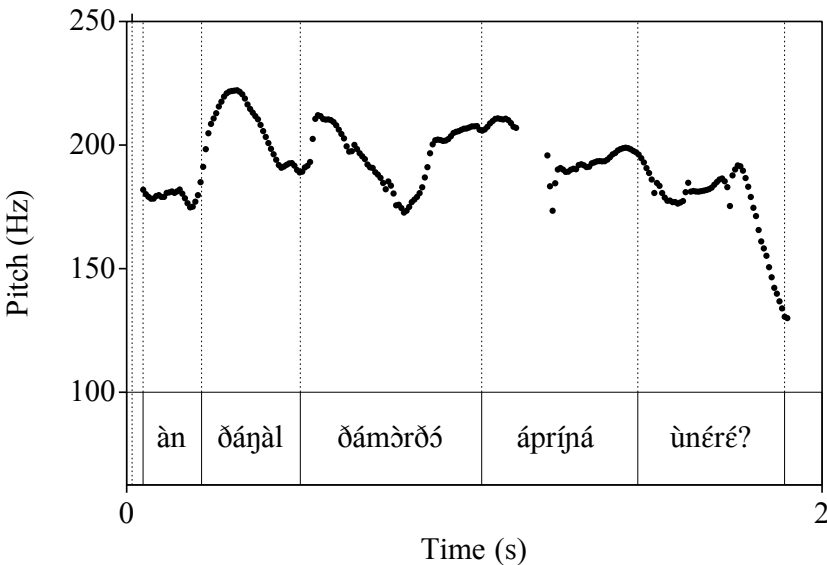


Figure 13: Intonation of (38b)

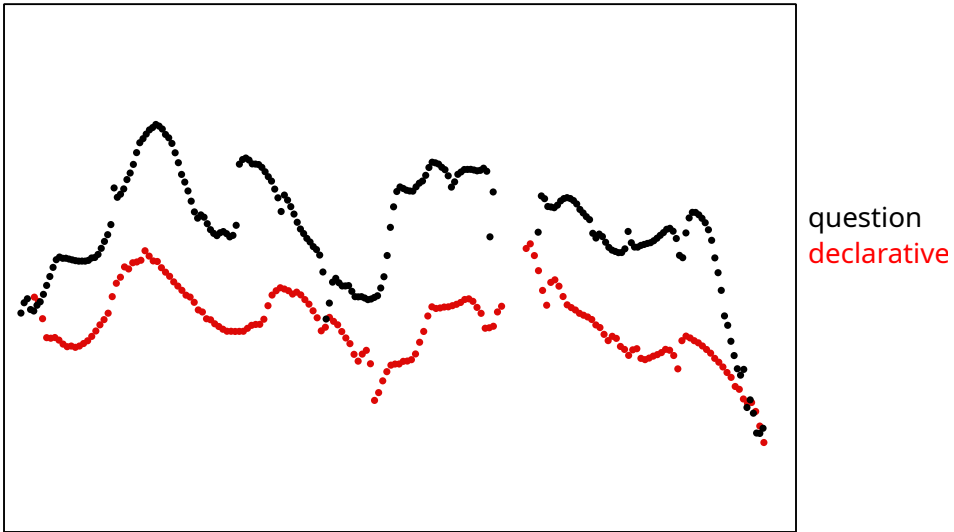


Figure 14: Intonation of subject focus statement and question compared

6 Conclusion

This paper presents novel data on the structure and prosody of focus and topic constructions in Tira, a Heiban language of Sudan. We show that topics can be either a subject or object noun phrase at the beginning of the sentence with which the verb agrees. With topic objects, the subject appears after the auxiliary or the verb. Topic constructions exhibit no special prosodic marking such as H tone. However, the topic is prosodically separated from the rest of the sentence, as there is a lack of H tone spreading from the topic to the following auxiliary or verb. Focus constructions involve an element at the front of the sentence preceded by a focus particle. In addition, there is H tone marking at the left edge of the verb or auxiliary. With non-subjects in focus, the subject must appear post-verbally, and it phrases with the verb, or the auxiliary, if one is present. Because of the H tone at the left edge of the verb/auxiliary, H-tone spreading cannot be used as a diagnostic of prosodic phrasing. Finally, polar questions are primarily indicated by raising of pitch across the utterance in both topic and focus constructions, ending with a fall due to a low boundary tone.

Abbreviations

1	first person	IT	itive
2	second person	LOC	locative
ACC	accusative	OBJ	object
APPL	applicative	PFV	perfective
ASSOC.PL	associative plural	PPh	phonological phrase
AUX	auxiliary	PL	plural
CL	noun class	Q	question particle
FOC	focus	SG	singular
FV	final vowel	VEN	ventive
IPFV	imperfective		

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Chapter 4

Tone-intonation interactions in Ede Chaabe

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This paper presents an analysis of the intonation of yes/no questions, statements and exclamations in Ede Chaabe (cbj, Benin). The paper shows that yes/no questions in Ede Chaabe not only have a low boundary tone (i.e. L%), but also have an extended pitch lowering effect that spans over the last few syllables of the utterance, unlike in statements. The presence of the extended lowering in yes/no questions is what differentiates them from exclamations with which they share the low boundary tone. The L% is shown to be *locally accommodated* on the utterance-final syllable, whereas the extended pitch lowering effect is *superposed* on the last few syllables of the utterance. The extent of the extended lowering is experimentally shown to vary as a function of utterance length, with potential prosodic phrasing effects. The paper further argues that superposition in Ede Chaabe is phonological, and best analyzed independently of the boundary tone it co-occurs with. The implications of the results for existing models of intonation are discussed.

1 Introduction

Intonation plays an important role in the ways information is structured and conveyed in most languages, including signaling different types of focus, marking sentence type (e.g. questionhood), among other functions. Tone languages are especially interesting in this regard due to the added complexity stemming from tone and intonation interactions at different levels of the grammar. As Hyman & Monaka (2011) put it, lexical tone “can be subject to considerable manipulation by the postlexical phonology, where juxtaposition, syntactic conditioning,



or phonological phrasing can modify the word-level input and introduce additional pitch features” (Hyman & Monaka 2011: 267). In African tone languages, intonation is often approached as (i) emergent from the cumulative effects of local interactions between lexical tones as seen in processes such as downdrift (Welmers 1959, Clements 1979, Ladd 1990, Genzel & Kügler 2011, Genzel 2013); as (ii) the manifestations of prosodic domain boundary tones (Rialland 2007, 2009, Cahill 2012, Kügler 2017), and/or (iii) some form of restricted register shifts, i.e. register raising or lowering affecting a stretch of lexical tones near the end of a given utterance (Inkelas et al. 1986, Downing & Rialland 2017). These analytical approaches are collectively known as downtrends.

The study of downtrends as a result of interactions between tone and intonation has given rise to a four-way typology of such interactions, namely avoidance, submission, accommodation (Hyman & Monaka 2011), and superposition (sometimes called “superimposition,” Pierrehumbert & Beckman 1988), based on the degree of hospitality (for accommodation, submission, and superposition) or hostility (avoidance) of lexical tones toward intonation. Avoidance constitutes a case when intonation is minimized or non-existent. Submission is when intonational tones override or neutralize lexical tones. Accommodation is when lexical tones and intonation “peacefully” co-exist, “perhaps on different syllables or with intonation superimposed on lexical tone (...)” (Downing & Rialland 2017: 5). Superposition can in fact be considered a sub-type of the accommodation strategy, i.e. a case when the intonational tone affects lexical tones without overriding them, but where its effects span over many such tones (or corresponding Tone Bearing Units (TBUs)). When accommodation is achieved at the very edge of some domain or on a specific tone/TBU, as when a single TBU is shared by both tone and intonation, it will be referred to as “local accommodation”, which we consider to be a second sub-type of accommodation.

In Ede Chaabe, yes/no (or polar) questions seem to differ from other sentence types in having what we can descriptively call an extended lowering effect (extended lowering hereafter) of the intonational boundary tone on a stretch of lexical tones near the end of the utterance (Mamadou 2023). This is so termed due to its co-occurrence with the low boundary tone in yes/no questions. The purpose of this paper is to determine whether or not the extended lowering is phonological, on the one hand and to experimentally study its extent on the other hand. More specifically, we investigate the question: is the observed extended lowering in yes/no questions an instance of phonological superposition or is it a case of a low level phonetic co-articulatory effect of the boundary tone on adjacent lexical tones? This phonological status will be investigated using two approaches:

the first consists in identifying a prosodic minimal pair, where the extended lowering is the only prosodic difference between two different sentence types. The second approach experimentally studies the f_0 of syllables at and near the end of the utterance in an attempt to determine how many syllables are affected by the lowering.

2 Extended lowering across languages

The status of extended lowering has been shown to greatly vary from language to language. In Chichewa for instance, Downing (2017) showed that there is a boundary LHL% melody in yes/no questions that is superposed on the last two Low-toned syllables of the intonational phrase;¹ in Hausa, the tones of the last phrase of yes/no questions are phonologically raised,² with the insertion of a High register tone (Inkelas et al. 1986, Inkelas 1989); in Mandarin, the globally high f_0 cues questionhood regardless of whether or not there is a boundary tone, which is why it is treated by Shen (1992) as being phonological. Akan questions have a high or expanded global register in addition to their low boundary tone (L%), but a perception study in Genzel & Kügler (2020) concluded that the global raising has no effect in question identification – which we interpret here as having a phonetic status. That is, in Akan, the low boundary tone is the most important question cue, rather than the global raising. This brief overview of the different manifestations of yes/no question intonation suggests that in addition to intonational boundary tones, some languages can have superposed phonological register shifts that affect the entire utterance (e.g. Mandarin) or a few lexical tones within a well-defined prosodic domain near the end of the utterance (e.g. Hausa), while others can have a similar overall pitch expansion, but with a phonetic status (e.g. Akan). The register expansion noted in Akan is attested in a number of African languages, even in those known to have the low boundary tone (the so called “lax prosody” languages, Rialland 2007, 2009). In fact, all but one³ of the 12 languages studied in Downing & Rialland (2017) have register expansion/raising along with some boundary tone in their yes/no questions. As we study the status of the extended lowering in Ede Chaabe below, we will show

¹The LHL melody is simplified when these last two syllables are High-toned.

²In addition, downdrift is suspended within such phrases, and a boundary L% is optionally added.

³The only language that lacked register expansion/raising in Downing & Rialland (2017) is Mambila. Connell (2017) attributed this to the fact that Mambila has four lexical tones. See Mamadou (2023) for further discussion of this.

that it crucially differs from many of the languages in that collection in the sense that it has register compression/lowering toward the end of the utterance.

The rest of the paper is organized as follows: §3 presents background information on Ede Chaabe; §4 gives an overview of sentence prosody in Ede Chaabe and further motivates the research question; §5 describes the experiment with §5.1 presenting the stimuli, the participant pool and the analysis method and §5.2 presenting the results, followed by a discussion in §6. §7 concludes.

3 Background on Ede Chaabe

Ede Chaabe [èdèè tʃáàbɛ́] is an Eastern Ede language of the Benue-Congo family spoken in the Central and South-Central part of Benin, with Savè representing the district with the largest number of Ede Chaabe speakers (87,177 people, according to the 2013 RGPH4 census). It is a language with three lexical level tones, L, M, and H, as shown in the minimal triplet in (1). Different aspects of Ede Chaabe have previously been studied, including its interrogative pronouns (Idohou 2009), the sociolinguistics of its personal nouns (Moufoutaou 2013), the morphophonology of its numerals (Mamadou 2013), and its phonetics and phonology (Alo 1991). In describing the phonetics and phonology of the language, Alo (1991) shows pitch tracks, spectrograms and waveforms of Ede Chaabe words in isolation. Alo (1991) was the only instrumental study of Ede Chaabe until Mamadou (2023) conducted a quantitative study of its intonation.⁴

- (1) Three tone levels
- a. kí ‘greet’
 - b. kī ‘set fire/burn’
 - c. kì ‘praise/recite one’s panegyric’

Just like its better studied sister language Yoruba, Ede Chaabe contrasts three lexical tones and shares some of the word-level tonal constraints and processes found in Yoruba, including a tonotactic constraint against H tones on the initial vowel of vowel-initial words (e.g. VCV, VCVCV). Other phonological processes that are relevant for the discussion below include vowel and tone deletion processes, as well as [l]/[n] alternation. The vowel deletion process occurs in V1V2 contexts, where V1 usually deletes, though there are cases where it is V2 that deletes. If the tone of the deleted vowel is M, then the tone deletes as well, but

⁴The results presented here are a subset of the data from Mamadou (2023).

not if its tone is H or L. In Ede Chaabe, [l] and [n] are in complementary distribution.

As far as its syntax is concerned, Ede Chaabe has an SVO word order.⁵ Moreover, non-pronominal subject NPs are followed by a resumptive pronoun [ó/ǒ],⁶ which harmonizes with the final vowel of the preceding word. Its H(igh) tone then creates a contour with the tone of the preceding syllable when that tone is non-high.⁷ The examples below follow the Africanist tradition of treating contour tones as sequences of level tones (Yip 2002, Hyman 2011), which is why each tone is on a separate vowel in the examples.⁸

4 Sentence prosody in Ede Chaabe

In Ede Chaabe, prosody is sometimes the main, if not the only way, of conveying sentence type. Yes/no questions represent a prime example of this observation. They are expressed in two different ways: the morpho-syntactic expression, whereby the particle *tʃé* or *nʃé* is placed at the beginning of the utterance as in (3a) and the prosodic expression, consisting of appending a boundary L% at the end of yes/no questions as in (3b) (Mamadou 2023). These two strategies are in complementary distribution, so the morpho-syntactic strategy is not compatible with a boundary L%. The morpho-syntactic strategy in (3a) is often used in formal speech, whereas the prosodic strategy is used in casual speech, and is by far the most widely used question form by Ede Chaabe speakers, as is the case in Yoruba (Laniran & Clements 2003).

(2) Statement⁹

[bíjò ló gbówó fólówó]
bíjò nī ó gbé-ówó fú-óní-ówó.

Biyo TOP 3SG give-money to-AGEN-money

‘It was Biyo who gave some money to the rich man.’

⁵NPs in Ede Chaabe impose a strict order on post-nominal modifiers: head nouns are followed by adjectives, numerals, determiners, and quantifiers, in that order.

⁶The vowel quality depends on ATR harmony, like the harmony found in Yoruba.

⁷This H tone is sometimes loosely described as marking the right edge of non-pronominal subjects in Yoruba.

⁸Another reason for making this assumption is that vowels with contour tones are robustly longer than those with a single tone.

⁹The first lines in the examples represent the surface forms of the sentences and the second lines, the underlying forms. Two key phonological processes are of note in these sentences: (i) the deletion of V1 in V1V2 sequences as a hiatus resolution strategy, and (ii) n/l alternation before [-high, -low, +ATR] vowels (i.e. /n/ → [l]/_ [o, e]).

(3) Yes/no questions

- a. [tʃé/ɲɛ́ bíjọ́ ló gbówó fólówó]
 tʃé/ɲɛ́ bíjọ́ nī ó gbé-òwó fú-òní-òwó?
 Q Biyo TOP 3SG give-money to-AGEN-money
 ‘Was it Biyo who gave some money to the rich man?’
- b. [bíjọ́ ló gbówó fólówó]
 bíjọ́ nī ó gbé-òwó fú-òní-òwó? L%
 Biyo TOP 3SG give-money to-AGEN-money Q
 ‘Was it Biyo who gave some money to the rich man?’

Yes/no questions with the question particles *ɲɛ́* or *tʃé* as in (3a) have the same overall prosody as statements as in (2).¹⁰ This can be seen with the following sentence in (4a) with all-M tones and its question counterpart (4b). The pitch tracks are illustrated in Figure 1.

(4) Statement and yes/no questions with all-M tones

- a. [ī kē mōmī ējā ōkō]
 ī kē mō-ōmī ējā ōkō
 you FUT drink-water fish farm
 ‘You will drink the farm-fish broth.’
- b. [tʃé ī kē mōmī ējā ōkō]
 tʃé ī kē mō-ōmī ējā ōkō
 Q you FUT drink-water fish farm
 ‘Will you drink the farm-fish broth?’

The use of prosody to distinguish between sentence types goes beyond the question-statement distinction in Ede Chaabe. An example of another use of prosody comes from a form of exclamatory sentence with the same syntactic make-up as statements, but with a unique prosodic signature as in (5).

(5) Exclamatory Sentence

- [bíjọ́ ló gbówó fólówó]
 bíjọ́ nī ó gbé-òwó fú-òní-òwó L%
 Biyo TOP 3SG give-money to-AGEN-money EXCL
 ‘Of course it was Biyo who gave some money to the rich man!’

¹⁰Yes/no questions with *tʃé* are biased towards a ‘yes’ response, whereas those with *ɲɛ́* are neutral.

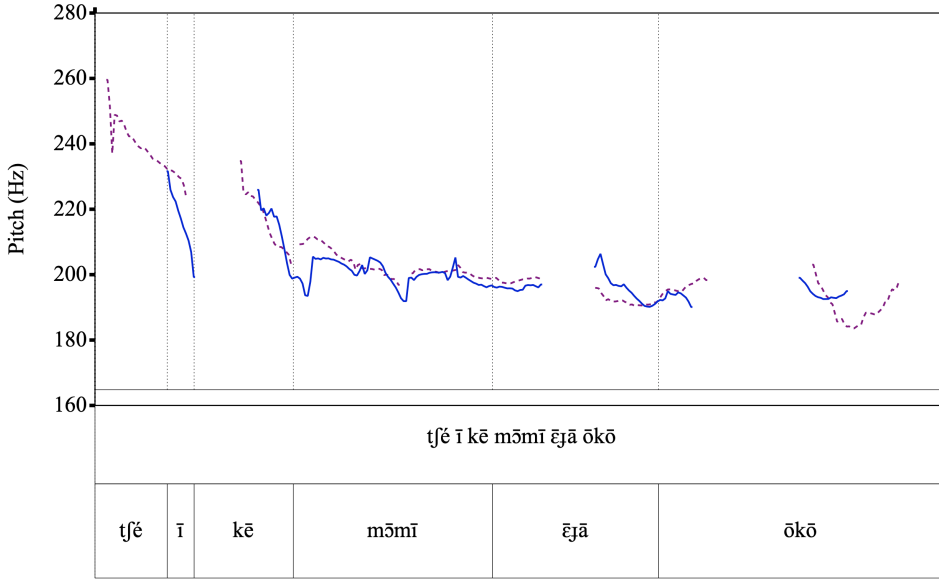


Figure 1: Overlaid pitch tracks of the *statement* in (4a) (plain blue line) and the morpho-syntactic *yes/no question* in (4b) (dashed purple line), showing prosodic similarity between both sentence types, produced by a female (RI).

Along with statements and yes/no questions, this yields a 3-way contrast in sentence modality based solely on prosodic differences in Ede Chaabe, as in Figure 2. Yes/no questions and exclamatory sentences differ from statements in that they both have a boundary L%, and differ from each other in the sense that yes/no questions have extended lowering of many lexical tones near the end of the utterance, an effect that is not observed in exclamatory sentences. The experimental study presented below addresses the extent and status of this extended lowering in the Ede Chaabe grammar.

It is worth mentioning that exclamatory sentences can optionally take the morpheme *kóò/kǎò* at the beginning of the sentence, though crucially it is in addition to having the low boundary tone (not in lieu of it) as shown in (6).

(6) Exclamatory sentences with non-high tones

- a. [(kǎò) māmī ējā ōkō]
 (kǎò) māmī ējā ōkō L%
 2SG drink-water fish farm EXCL
 ‘Go ahead, drink the farm fish’s soup!’



Figure 2: Pitch-based 3-way contrast, showing the *statement* in (2) in (a), the *exclamation* in (5) in (b), and the *prosody-only yes/no question* in (3a) in (c).

- b. [(kóṣò) fòṣò gbì ìbàbà ònè]
 (kóṣò) fī-òṣò gbì ìbàbà ònè L%
 2SG with-rain sow millet road EXCL
 ‘Go ahead, sow the roadside millet!’

So far, we have attributed the falling pitch at the end of both exclamatory sentences and yes/no questions to a L%. Let us establish that the exclamatory sentences do in fact end in a L% and that they form a minimal pair with yes/no questions. One possible analysis of the exclamatory sentences’ lack of the extended lowering could be that they end in a boundary HL% instead of the assumed L%. While such an analysis will effectively account for the difference between the two sentence types, especially since the H in the HL% melody would effectively block the L% from having any effect beyond the very last syllable, it is not supported by data. This is because the HL% analysis predicts that when these sentences have all-Mid and all-Low tones, they will show a slight rise in pitch right before the final pitch fall, similar to Chichewa (Downing 2017). However, that is not what we observe in the data, as shown in Figure 3 and 4.

4 Tone-intonation interactions in Ede Chaabe

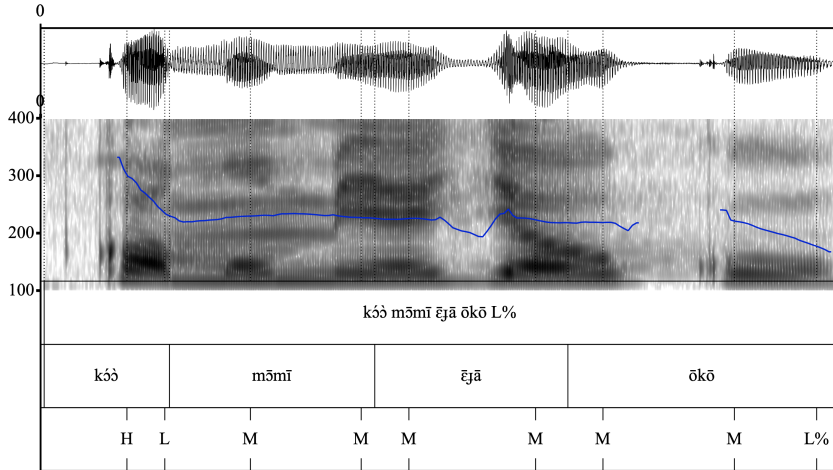


Figure 3: Pitch track of the exclamatory sentence in (6a) with all-Mid tones, produced by a male participant (AY).

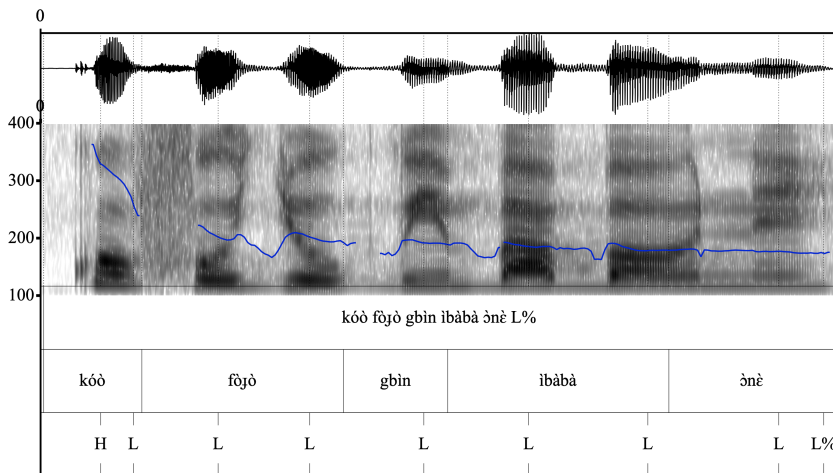


Figure 4: Pitch track of exclamatory sentence in (6b) with all-Low tones, produced by the male participant (AY).

What is observed instead, is that regardless of the tonal make-up of exclamatory sentences, pitch does not rise before falling at the end of the sentence. In fact, exclamatory sentences with the all-Low tones have a flattened pitch throughout the utterance, including utterance-finally as shown in Figure 4. So, since the last Low lexical tone in that utterance is indistinguishable from the boundary L%, we can conclude that exclamatory sentences do in fact have a boundary L% in Ede Chaabe but its presence is only observable when the utterance-final lexical tone is H or M. Now, as to whether yes/no questions also have a boundary L%, we do not need to look any further than Figure 2, which shows a clear abrupt pitch fall on the very last syllable of the yes/no question along with the extended lowering effect noted much earlier on the preceding H or M tones, but not on L tones.¹¹ Note that while this effect is clearly noticeable on the lexical tones, the intonational tone does not override these lexical tones as would be the case if the latter were submitting to the former. The interim conclusion here is that the extended lowering constitutes the sole difference between the minimal pair of exclamatory sentence (in 5) and its yes/no question counterpart (in 3b). Based on this determination, the extended lowering is phonological and is accomplished by way of lowering H and M lexical tones near the right edge of the utterance. Table 1 summarizes the key characteristics of the different utterance types in Ede Chaabe.

Table 1: Summary table of the prosodic characteristics of different sentence types in Ede Chaabe.

Sentence type	Boundary tone	Extended effect	Tones affected
Statement	None	None	None
Prosodic yes/no question	L%	Yes	H & M, not L
tʃé/ɲé yes/no question	None	None	None
Exclamation	L%	None	None

We now turn to the experimental approach, which aims to further support the phonological status of the extended lowering and determine its extent.

¹¹This is true in utterances with like tones. It is to be seen whether or not those with mixed tones will also display the extended effect and if they do, to what extent.

5 Extended lowering: An experiment

The goals of the experimental study reported below are three-fold: to support the phonological status of the extended lowering in Ede Chaabe, to determine the number of affected tones and how utterance length plays into the extent of the effect. A phonological phenomenon will be expected to be systematic and to supersede any micro-prosodic effects, i.e. effects of surrounding segments on f0 in this case (see De Moraes & Rilliard 2023 for a recent account of micro-prosodic effects). Hausa exemplifies a case where an extended raising effect, or to use Inkelas et al.'s (1986) terminology – a phonological raising – affects the last phonological phrase of a sentence. Inkelas et al. (1986) show the pitch tracks of several yes/no questions where the sentence-final phrase appears to be raised and/or in which downdrift is suspended, regardless of the syllable count of that sentence-final phrase. Another example is Moro, where pitch compression on the object in SVO sentences is “the strongest indication of polar question” (Piccinini & Rose 2017: 28), i.e. pitch compression is the phonological signature of polar questions. While the phonological status of the extended lowering in Ede Chaabe yes/no questions is no longer at issue and despite the determination that the extended effect spans over several lexical tones, the following questions remain: (i) does the extended lowering affect a fixed number of lexical tones regardless of the utterance length or does the extent of its effect vary according to the length of the utterance? If the latter, (ii) are its effects strictly constrained within some phonological (i.e. prosodic) domain? The current experiment investigates the question in (i), leaving the question in (ii) for future work.

- **Hypothesis 1: Superposition vs. Local Accommodation**

In yes/no questions, the extended lowering affects a stretch of several lexical tones (or TBUs) without neutralizing them (superposition), and not just the very last tone in the utterance (local accommodation). Schematically, this hypothesis predicts the following different surface patterns for superposition (1a) and for local accommodation (1b):

- 1a. Superposition: (- - - - ... - - - -)_i
- 1b. Local Accommodation: (- - - - ... - - - -)_i

- **Hypothesis 2: Varying vs. Fixed Extent of Effect**

The extended lowering varies according to utterance length, such that it affects more tones (or TBUs) in long utterances as compared to its effect on short utterances (2a). That is, its effect does not remain invariant across different utterance lengths (2b).

- 2a. Varying Extent
Long: (- - - - -)_i
Short: (- - -)_i
- 2b. Fixed Extent
Long: (- - - - - - -)_i
Short: (- - - -)_i

Hypothesis 1 predicts that the onset of the extended lowering will occur much earlier than the ultimate syllable of the sentence in yes/no questions. Under this hypothesis, not only will the utterance-final tone have significantly lower f0 values in yes/no questions, but also multiple lexical tones before the ultima will be lowered, which will support the superposition strategy over the local accommodation one. As for Hypothesis 2, it predicts that more tones will be lowered near the end of long yes/no questions than in the short ones.

5.1 Methods

5.1.1 Speakers and materials

A production study was conducted with 24 native speakers of Ede Chaabe, aged between 18 to 34 years (13 male and 11 female). The participants were recorded in a quiet room at the University of Abomey-Calavi (UAC) in Benin. Each participant produced 36 utterances (3 tones * 2 lengths * 2 modality * 3 repetitions) for a total of 864 utterances.¹² Each sentence used in this experiment is made of like-tones (i.e. all-H, all-M, and all-L). The two modalities here are yes/no questions and statements, each in two lengths, short and long.

The author, who is also a native speaker of Ede Chaabe, produced and recorded the twelve stimuli utterances, which were then used to cue the production of statements and yes/no questions from the participants. The participants were instructed to produce a statement when they heard a question and a question when they heard a statement, keeping everything else in their production the same as the stimulus sentence.¹³ The audio stimuli were reinforced with a visual image on

¹²The participants produced an additional 10 sentences each, with varying sentence structure and mixed tones, representing the set of filler sentences. The data from two of the participants was discarded due to one participant's failure to follow instructions (repeating exactly the stimuli instead of changing its modality), and to the presence of dysfluencies in the other participant's productions, which resulted in longer than normal pauses between words.

¹³It is important to note that the participants did not repeat verbatim the stimulus; instead, the sentence they produced was of the opposite sentence-type (either statement or question) than the stimulus.

a laptop screen, showing a question mark (“?”) when the participant is expected to produce a question and a red dot (“.”) when they should produce a statement. The participants would hear a new stimulus sentence after pressing the space-key on the laptop, they would then produce the appropriate sentence, and repeat the same process until the end of the experiment. This design is primarily due to Ede Chaabe not having a written tradition and lacking a widely-adopted standardized writing system.¹⁴ The all-H long utterances (in 7a and 7b) have seven (7) syllables, whereas the all-H short utterances (in 8a and 8b) have four (4) syllables. Their syntactic structures are shown in (7c) and (8c), respectively, and the corresponding prosodic structures are shown in (7d) and (8d), respectively, assuming isomorphism between syntactic and prosodic phrases, following Selkirk (2011). As such, lexical words, NPs/DPs/PPs/ and Clauses/TPs/CPs respectively correspond to the prosodic word (ω), phonological phrase (ϕP), and intonational phrase (iP) (see also Kügler 2017). Only these three levels of the prosodic hierarchy will be assumed, unless the data analysis to follow suggests otherwise. ϕP and iP have been assumed for many African languages in Downing & Rialland (2017) on the basis of Selkirk’s (2011) syntax-prosody mapping, but only a few of the languages in that collection have actual diagnostic phonological processes of phonological phrasing — Bàsàà (Makasso et al. 2016) being one of the examples. We also assume recursive prosodic phrasing.

(7) Long sentences (7-syllables)¹⁵

- a. [bǐjò gbówó fólówó]
 bǐjò H- gbé-ōwó fù-ōní-ōwó
 Biyo SBJ give-money to-AGEN-money
 “Biyo gave some money to the richman.”
- b. [bǐjò gbówó fólówô]
 bǐjò H- gbé-ōwó fù-ōní-ōwó L%
 Biyo SBJ give-money to-AGEN-money Q
 “Did Biyo give some money to the richman?”
- c. [_{CP}[_{TP}[_{NP} bǐjò] [_{VP} gbé- [_{NP} ōwó] [_{PP} fù-ōní-[_{NP} ōwó]]]]]
- d. ((bǐjò)_φ (gb- (ówó)_φ (f-ól-ówó)_φ)_φ)_ι

¹⁴The Centre National de Linguistique Appliquée (CENALA) (2008), has an alphabet for writing Beninese languages. While the alphabet is used in literacy efforts, its usage is not widespread among speakers of Ede Chaabe.

¹⁵The examples are presented in their fluent speech forms; e.g. *gbówó* is made up of the morphemes *gbé+ōwó*; either V1 or V2 deletes as a hiatus resolution strategy depending on the input vowels. This creates the new phonological word *gbówó*.

(8) Short sentences (4-syllables)¹⁶

- a. [bíjǒ néwó]
 bíjǒ H- né-ōwó
 Biyo SBJ spend-money
 “Biyo spent some money.”
- b. [bíjǒ néwô]
 bíjǒ H- né-ōwó L%
 Biyo SBJ spend-money Q
 “Did Biyo spend some money?”
- c. [CP[TP[NP bíjǒ] [VP né- [NP ōwó]]]]
- d. ((bíjǒ)_φ (né (-wó)_ω)_φ)_t

The experiment was presented to the participants on a laptop in three blocks, using Psychopy (Peirce 2007). The data was collected using a Marantz PMD660 Solid State Recorder with the Shure SM10A Professional Unidirectional Head-worn Dynamic microphone.

5.1.2 Data processing and statistical analysis

The collected data was saved as .wav files, and pre-processed in Praat (Boersma & Weenink 2017) before being automatically segmented with the help of the Montreal Forced Aligner (McAuliffe et al. 2017). The aligned data were then inspected and hand adjusted to make sure boundaries were placed appropriately. A Praat script was then used to extract duration, f0, F1, and F2 data (f0, F1 and F2 values were taken from the midpoint of each vowel, which is a standard practice when all tones in the utterance are identical). Only the f0 values are reported below.

The statistical analysis was conducted in R. Based on our research questions, several Bayesian models were fitted using the R package *bmrs* (Badr & Bürkner 2021). f0 was the dependent variable, sentence *modality* and syllable *position* were the independent variables, and *participant* was used as the random effect. Because we were specifically interested in how f0 values changed as the sentence unfolded through the different syllables, the *position* variable was contrast coded, using the Reverse Helmert Coding (a sliding difference technique), which consists of comparing f0 values at each level of the *position* variable to the mean

¹⁶The H- tone in the examples marks the right edge of non-pronominal subjects by docking on the last syllable of the said-subject. It is particularly noticeable when the lexical tone on the docking site is non-high. When that tone is high, however, the two tones merge into one.

f0 values at the previous levels. A similar coding was done for the *modality* variable.

For the purposes of the statistical analysis, the data was normalized and subsetted into two: the all-H-long dataset (7-syllable utterances with all High tones) and the all-H-short dataset (4-syllable utterances with all High tones). Two models were fitted to each normalized data subset, the first model uses *non-informative priors* with a mean of 0 and a standard deviation of 1 for all parameters (i.e. *Intercept*: normal(0,1); β : normal(0,1); etc.). Priors are information we already know about the question under investigation from previous studies, like the average measurement of a High tone in the language under investigation or a language with a similar tonal system. Bayesian models allow for these kinds of information to be made explicit in the model, rather than simply assuming it. The second model uses *weakly informative priors* (i.e. *Intercept*: normal(100,10); β : normal(160,10); etc.). The weak priors (namely that of the β parameter) were based on the mean f0 value for High tones in different consonantal environments reported for Ede Chaabe in Alo (1991) and from the initial H in Yoruba reported in Laniran & Clements (2003), which is relevant here due to Ede Chaabe and Yoruba being closely related languages. These are weak priors because the same value of 160Hz, which is an average (across gender) from initial High tones in Yoruba and from a variety of consonantal environments in Ede Chaabe, is used as an approximation for all syllable positions in the fitted Bayesian models, although we know from the exploratory data analysis and our discussion above that sentence-final lexical High tones are greatly lowered due to the intonational boundary tone.

To check for the predictive power of the two fitted models, the Leave-one-out (loo) validation method which holds out data for model testing from the initial dataset, was used. Both model types converged (*Rhat* = 1), but the models with weakly informative priors proved to be better (*looic* = 8183.8) than the ones with non-informative priors (*looic* = 8202.0) since they had a lower *loo* value. The model with weakly informative priors also performed best on the all-H-short data (*looic* = 4670.7 for the weakly informative priors compared to 4703.6, for the uninformative ones).

5.2 Results

This section only reports the results of the all-H utterances due to the fact that the effects of the extended lowering and that of the final L% are most noticeable on sentences with all-H tones, and because the all-H sentences are sufficient in answering our research question in this paper.¹⁷

¹⁷For the complete results, see Mamadou (2023).

Table 2 shows the descriptive statistics (Mean f0 and Standard Error) of the results plotted in Figure 5.

Table 2: Mean f0 and Standard Error (SE) for H-toned long and short utterances averaged across all 3 repetitions from the 22 participants. Each syllable position ranges from “1” to “7” for long utterances and from “1” to “4” for short utterances where position “1” is the utterance-final position. All values are in Hz.

Modality	Utt.Length	Stats	7	6	5	4	3	2	1
Question	Long	MeanF0	207.1	203.1	196.5	194.9	189.6	179.6	157.9
		SE	3.2	2.8	2.4	2.4	1.9	1.8	2.5
	Short	MeanF0	-	-	-	199.0	193.0	183.1	159.5
		SE	-	-	-	3.3	2.6	2.5	2.2
Statement	Long	MeanF0	204.6	202.6	199.8	202.0	201.8	197.7	197.5
		SE	2.7	3.2	3.5	2.8	2.5	2.4	3.2
	Short	MeanF0	-	-	-	202.7	204.6	203.3	202.0
		SE	-	-	-	3.9	2.9	2.9	2.5

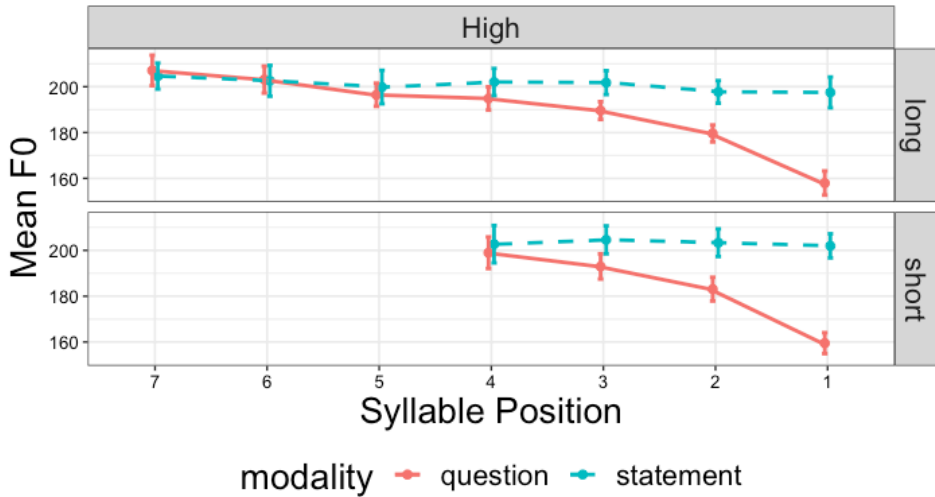
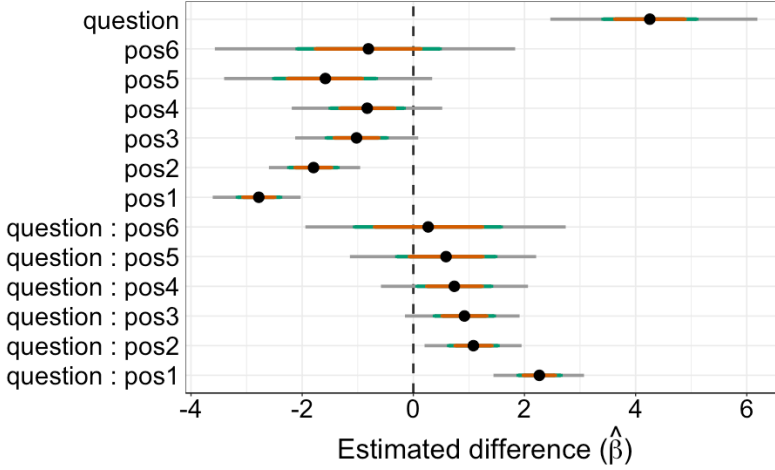


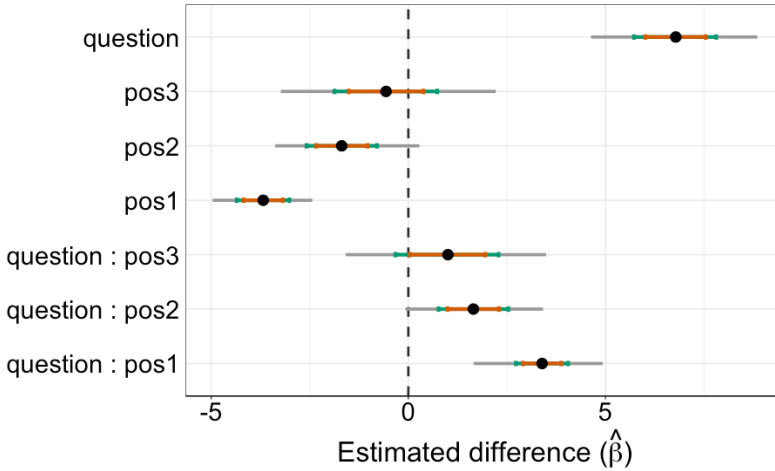
Figure 5: Plot of the mean f0 values per syllable across all participants, faceted by utterance length. Error bars indicate Standard Error (SE).

The plots of the posteriors for the two models with weakly informative priors are shown in Figure 6. In that figure, the effect of syllable positions is determined

by whether or not the error lines (the horizontal lines with different colors) overlap with β line (the dashed vertical lines), such that the further they are from the line, the stronger the effect of modality on the f0 of that specific syllable position.



(a) All_H Long



(b) All_H Short

Figure 6: Posterior plots of the all-H-long model (a) and all-H-short model (b) both with the weakly informative priors. Position 1 indicates final syllable and position 6 (in the long model) and 3 (in the short model) are the peninitial syllables. The horizontal grey lines indicates the range of the distribution, the orange regions constitute 85% Credible Intervals, and the green the 95% Credible Interval.

That being said, in the question modality for long utterances, we see an effect of modality as early as 3 syllables (position 3) from the end (see “question:pos3” in Figure 6a), whereas the same effect starts 2 syllables (position 2) from the end in short utterances (see “question:pos2” in Figure 6b). In order to further confirm these results, the *hypothesis()* function from the *brms* library was used for hypothesis testing in each syllable position. This method showed that the posterior probability exceeded 95% in positions 3 through 1 for the all-H-long model, whereas the posterior probability only exceeded 95% in positions 2 and 1 – crucially not in position 3 – for the all-H-short model.

More specifically, given the (weak) priors and corresponding all-H-long model, there is a 95% probability that the true mean f_0 amount by which questions are lower than statements are as follows: in utterance-final position (position 1), questions are lower than statements by 37.9Hz; in the penult (position 2), it is by 19.3Hz; in the antepenult (position 3), it is by 21.8Hz; in the fourth syllable from last, it is by 6Hz, etc. As for the all-H-short model, there is 95% probability that the true mean f_0 amount by which questions are lower than statements is 38Hz in utterance final position, 23.1Hz in the penult, 14.4Hz in the antepenult and 7.1 in initial position. Table 3 shows all the Estimated Marginal Means (the means of a given variable when the other is held constant) of the difference between questions and statements by syllable position.

6 Discussion

Hypothesis 1, which holds that the extended lowering observed in yes/no questions is a case of superposition and not one of local accommodation is supported in the results presented above. This finding is somewhat expected and supports the argument in Section 4 on how the sole difference between yes/no questions and the exclamatory sentences comes from the presence on the former, of an extended lowering, while the latter has a local lowering. The results also address the question of the extent of the extended lowering in the sense that the lowering was most noticeable on both the final syllable (the location of the boundary L%) and the penultimate one in short utterances, whereas in long utterances the lowering affected the ante-penultimate, the penultimate and the final syllables, which is also what Hypothesis 2 predicted.

Hypothesis 2 predicted that the effects of the extended lowering will vary as a function of utterance length and that is indeed what the results show: three tones/TBUs were significantly lowered in the long utterance against only two in the short utterance. Of course, the effect of the extended lowering does not seem

Table 3: Estimated Marginal Means of the question (Q) minus statement (S) contrast from the Bayesian model output for the all-H-long and all-H-short datasets. The lower 95% CI and upper 95 % delineate the range of the 95% Credible Interval. Position ‘1’ is the final syllable position. All values are in Hz.

Model	Position	Contrast	Estimate	l.95% CI	u.95% CI
Long	1	Q-S	-37.89	-41.86	-33.76
	2	Q-S	-19.28	-23.32	-15.33
	3	Q-S	-21.84	-25.8	-17.92
	4	Q-S	-6.05	-10.19	-2.15
	5	Q-S	-4.72	-10.27	0.88
	6	Q-S	-1.03	-6.71	4.49
	7	Q-S	0.74	-4.8	6.43
Short	1	Q-S	-38.01	-44.14	-32.13
	2	Q-S	-23.12	-28.94	-17.35
	3	Q-S	-14.35	-20.17	-8.39
	4	Q-S	-7.10	-12.94	-1.17

to vary by the same syllable count by which the long utterance is longer than the short one. That is, the long utterance (with 7 syllables) is three syllables longer than the short one (with 4 syllables), but the extended effect only significantly affects one more syllable in the former than in the latter. This suggests that while the extent of the extended lowering clearly varies according to utterance length, there may be additional factors at play in Ede Chaabe. Although it is beyond the scope of the present paper to determine exactly what such factors are, prosodic phrasing seems to be a prime candidate due to the fact that the long utterance in the experiment has a 3-syllable-long prosodic phrase sentence-finally, while the short utterance has a 2-syllable-long prosodic phrase in that same position. Note that these are exactly the number of syllables affected by the extended lowering at the end of each sentence type in the experiment. It is possible that a correlation between prosodic phrasing and extended lowering is purely accidental. However, consider the example sentences in (9) with 10 syllables.

(9) Long sentences (9-syllables)

- a. [bíjò ló gbówó kówó fólówó]
bíjò nī ó gbé-ówó kí-ówó fú-ōní-ōwó
Biyo TOP 3SG give-money bad-money to-AGEN-money
‘It was Biyo who gave some bad money to the rich man.’

- b. [bǐjǒ ló gbówó kówó fólówô]
 bǐjǒ nī ó gbé-ówó kí-ówó fú-óní-ówó L%
 Biyo TOP 3SG give-money bad-money to-AGEN-money Q
 ‘Was it Biyo who gave some bad money to the rich man?’
- c. [_{CP}[_{TP}[_{NP} bǐjǒ nī ó] [_{VP} gbé- [_{NP} ówó kí ówó] [_{PP} fú-óní-[_{NP} ówó]]]]]
- d. ((bǐjǒ ló)_φ (gb- (ówó k-ówó)_φ (f-ól-ówó)_φ)_φ)_l

Figure 7 shows that having a higher syllable count does not translate into a longer extent of the extended lowering. Instead, it shows that only the 3-syllable-long final prosodic word in the yes/no question is affected by the extended lowering, as was the case with the 7-syllable-long utterance in the experiment. Ultimately, whether or not the prosodic phrase does, in fact, affect the extent of the extended lowering would need to be systematically investigated with an appropriate experimental setup. All we have evidence for in the results above is the hypothesized idea that utterance length affects the extent of the extended lowering with potentially some effects of prosodic phrasing. We now turn to the significance of the results and how to model them.

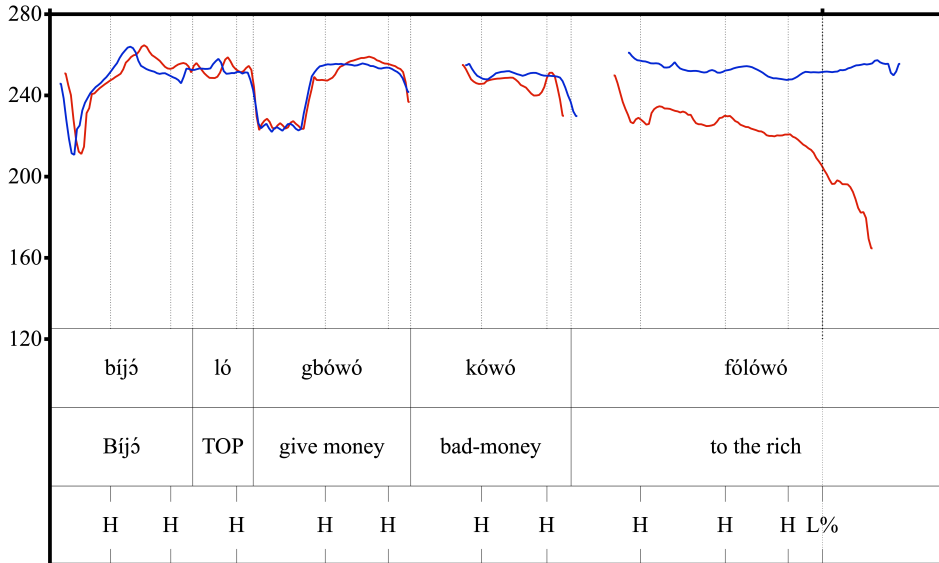


Figure 7: Comparative pitch track of a statement and yes/no question version of the sentence in (9). This pair of utterances was produced by the female participant (RI).

Based on Gussenhoven's (2007) definition of intonation as "the structured variation in pitch which is not determined by lexical distinctions as in tone languages" (Gussenhoven 2007: 253), and in light of the results presented in §5.2, both the boundary L% and the extended lowering collectively contribute to the structured variation in pitch that is characteristic of yes/no questions in Ede Chaabe. Despite using Pierrehumbert & Beckman's (1988) representation of the boundary tone as one that attaches at the end of intonational phrases for the purposes of hypothesis testing, there are aspects of Ede Chaabe that suggest that the boundary L% is realized as a lexical tone would be, i.e. with a very localized effect. This is because the effects are perceivable only on the accommodating utterance-final syllable. More specifically, the fact that the L% in exclamatory sentences does not have an extended lowering in pitch (as does the L% in yes/no questions) seems to suggest that it attaches at a different level of the prosodic hierarchy in Ede Chaabe. However, such an analysis would force one to assume that there are two different intonational boundary L% tones, one that attaches at phrase level (yes/no question) and another that attaches like a lexical tone (exclamation).¹⁸ Adopting such analysis for Ede Chaabe will inevitably require an increase in the number of intonational tones. We thus reject it here because it unnecessarily complicates the grammar; this leaves us with the need for another way of representing the extended lowering.

Previous work has treated this type of phonological shift in pitch, whether global or local, as a shift in the "register" feature (*h* and *l*), which, along with the "primary" feature (*H* and *L*), is used to represent lexical tones (Yip 1980, Snider 1999), and to capture global intonational shifts (Ladd 2008, D'Imperio & Michelas 2014). For instance, the series of papers on the intonation of Hausa by Inkelas and colleagues (namely Inkelas et al. 1986, Inkelas 1989, and Inkelas & Leben 1990) consistently analyzed the "extra High" and "raised Low" at the end of Hausa yes/no questions as a result of a high register that originates from the lexical *H* tones, and by extension any lowering effect can be attributed to low register features from lexical tones. However, such accounts make the extended lowering dependent on the L%,¹⁹ which we do not want because a move like that will predict extended lowering for exclamatory sentences as well. Instead, we propose a slight variation for the analysis of Ede Chaabe, one where yes/no questions have

¹⁸Downing (2017: 388) had similar accounts of intonational tones in Tumbuka and Chichewa: in the former, *H*% attaches at phrase level (with more global effects), whereas in the latter *H*% attaches in a lexical tone fashion (with more restricted and local effect). A similar analysis was proposed for Hausa as well (Inkelas et al. 1986).

¹⁹Admittedly, the terminology "extended lowering" also suggests that the lowering effect depends on the L%. Remember though that the term is simply used descriptively here.

two different intonational features: a *l*-register feature that attaches to the maximum projection of the last prosodic phrase, and a boundary L% that attaches at the end of the utterance. Under this account, the *l*-register feature is responsible for the lowering in pitch of the lexical tones near the end of the utterance and the L% is responsible for the more dramatic pitch fall at the very end of the utterance. This proposal maintains that the boundary L% is locally accommodated in both exclamations and yes/no questions, and that the extended lowering in the latter emanates from the *l*-register feature, which crucially does not come from lexical tones or intonational tones. Under our current formulation, the register feature is allowed to directly enter the representation of global or prosodic phrase-bound intonational shifts.

Table 4: A 4-way sentence typology with a binary register feature and two boundary tones H% and L%. The grayed out row indicates that Ede Chaabe has no sentence type with a register *l* and a boundary H%.

Ede Chaabe	Register	Boundary tone
Question	<i>l</i>	L%
Exclamation	<i>h</i>	L%
Statement	<i>h</i>	–
–	<i>l</i>	H%

It is not uncommon to talk about pitch range compression (or expansion) to capture the extended lowering observed in Ede Chaabe (e.g. Moro by Piccinini & Rose 2017, and many contributions to Downing & Rialland 2017). However, pitch range compression/expansion still needs to be represented in the grammar if used contrastively as it is in Ede Chaabe. The proposal of an independent register feature satisfies the need for a representational device for pitch range compression/expansion. More importantly though, it correctly accounts for the 3-way pitch-based contrasts observed among sentence types in Ede Chaabe. If as proposed, the register feature – which is binary (i.e. can be *h* or *l*), is represented independently from lexical and intonational tones and is able to combine freely with intonational boundary tones to induce contrast between different sentence types, we can get up to four sentence types based on independent combinations of register features and intonational boundary tones utterance-finally²⁰ as shown in Table 4. Here, the register features are being allowed to directly and

²⁰Obviously, this 4-way typology will be greatly expanded if, as suspected, we have grounds to factor in prosodic domains and their different levels.

independently enter the representation of intonation without being first passed through the representation of lexical tones (see Lionnet To appear). This account crucially assumes a model of tonal representation devoid of the register features in order to avoid any duplication of register effects²¹; such a register-free model of tonal representation can be found in Mamadou (2023) and allows register to be exclusively an intonational feature where it can have effects spanning over entire prosodic units.

7 Conclusion

This paper established that yes/no questions in Ede Chaabe have a characteristic boundary L%, which is locally accommodated on the utterance-final syllable as well as an extended lowering of pitch over the last couple of syllables. The extent of the extended lowering varied according to utterance length. In light of the results, the paper argued for an account of superposition as a phonological tone-intonation strategy in Ede Chaabe that warrants a representational status that is at least semi-independent of the boundary tone. The proposed representational model predicts a 4-way pitch-based contrast from a combination of a binary register feature, *h* and *l*, and two boundary tones, H% and L% (with local effects). Three out of the 4-way contrast are employed to distinguish between statements, yes/no questions and exclamations in Ede Chaabe. Future work is needed to determine if the extent of the lowering is further conditioned by prosodic phrasing, and whether or not the predicted 4-way typology in pitch-based intonation is cross-linguistically adequate.

Abbreviations

2	second person	SG	singular
3	third person	SBJ	subject
AGEN	agent	TOP	topic
EXCL	exclamation	ω	prosodic word
FUT	future	φ P	phonological phrase
PL	plural	<i>l</i> P	intonational phrase
Q	question particle		

²¹Cf. Connell (2017: 162) for a similar idea.

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Chapter 5

DP-internal focus affects agree: Insights from Bamileke Ngemba

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This paper presents and analyses a cross-linguistically rare effect of focus-induced DP-internal reordering on agreement, using data from Ngemba. In Ngemba, adnominal possessive pronouns in information-structure-neutral order (N > POSS) agree in noun class with the head noun. When these modifiers are contrastively focused, in which case they precede the noun (POSS > N), we get animacy-only exponence with singular head nouns, but full noun class exponence with plurals. I argue that the spell out of animacy in singular contexts tracks semantic agreement, syntactic agreement having failed. That syntactic agreement fails in singular POSS > N order, but not in the plural, I show, is due to three fairly standard assumptions, namely (a) DP is a phase (Svenonius 2004, Heck et al. 2008, Kramer 2010, Bošković 2012), (b) phase edges are non-recursive (Müller 2010, 2011), and (c) number is privative (Nevins 2011). These assumptions combined derive the observed asymmetry. Overall, the paper provides empirical evidence for privative number features in Ngemba, and strengthens the theoretical contribution that phase edges need not be recursive.

1 Introduction

Languages use different strategies to mark a constituent as focused, topicalized, new and/or given. These notions are generally studied under information structure (see Krifka 2008, among others) and have been a central topic of research in linguistics over the last few decades. Research on this topic has, for example, established that focus marking can be achieved by (a) prosody (English, French, Hungarian, Italian etc.; Samek-Lodovici 2005, Szendrői 2017), (b) dedicated focus



Leonel Tadjó Fongang. 2025. DP-internal focus affects agree: Insights from Bamileke Ngemba. In Akinbiyi Akinlabi, Sampson Korsah, Sharon Rose & Abdul-Razak Sulemana (eds.), *Cross-disciplinary approaches to Information Structure in Niger-Congo languages*, 111–134. Berlin: Language Science Press. DOI: 10.5281/zenodo.16875458 

particles (Gungbe Aboh 2004, Igbo Georgi & Amaechi 2023, Medumba Keupdjio 2020, etc.) and (c) reordering by movement or base-generation (Igbo; Georgi & Amaechi 2023). Most, if not all, the languages I cited above combine two (and perhaps all three) of these strategies to mark certain types of focus. In Medumba, for example, the combination of (b) and (c) can result in exhaustive focus (Keupdjio 2020). In Hungarian, (a) and (c) interact in fronted foci in the sense that focus movement (reordering) is triggered by prosodic constraints on stress assignment (Szendrői 2017).

Reordering of constituents in many languages (with or without information-structure (IS) effects) can affect agreement in different ways. This can lead to situations where an intervener now stands in-between a probe and a goal (intervention effects), or a scenario where one is no more local enough to the other (or is too local; Anti-locality in agreement) for agreement to take place. In Ngemba (Cameroon, Grassfields Bantu), DP-internal modifier focus affects word order which, in turn, affects agreement in interesting ways.¹ Possessive pronouns agree in noun class with the head noun, as (1) shows.²

- | | | |
|-----|--|--|
| (1) | <p>a. Ø-kwòʔ Ø-á
 1-chair 1-POSS
 ‘my chair’</p> | <p>b. mò-kwòʔ m-á
 6-chair 6-POSS
 ‘my chairs’</p> |
|-----|--|--|

In (1a), the possessive pronoun root á ‘my’ agrees in noun class with the head noun kwòʔ ‘chair’, and this is reflected by the insertion of the exponent Ø- that attaches to both the head noun and the possessive pronoun. When the head noun is plural, as in (1b), the possessive pronoun also agrees with it, and this is manifested by the insertion of Class 6 morphology, namely mò-. Interestingly, if the possessive pronoun is focused, in which case it precedes the head noun (2), the exponents for noun class are replaced by one of two glides, if the head noun is singular. Specifically, the glides w- or j- are prefixed to the possessive pronoun

¹Ngemba is a Central Bamiléké language that belongs to the Ghómáláʔ sub-group. It has five dialects which are spoken in Baméka, Bamougoum, Bafounda, Bansoa and Bamendjou (Mba 1997). Data that I present and analyse in this paper are from Baméka-Ngemba, and were collected during a fieldtrip to Cameroon (April 2023). I thank Glwadys Matsinkou, Enoch Tadjó, Claudine Ngamgueu and Mysteline Makougoum for sharing their knowledge of Ngemba with me. I also happen to be a native speaker of the language and as such, some data patterns were cross-checked with my own native speaker judgments.

²Throughout this paper, I take nominal prefixes to be the spell out of both gender and number information (Carstens 1991, Fuchs & van der Wal 2021, among others). How exactly the two combine will be relevant in deriving the puzzle this paper sets out to investigate.

depending on whether the head noun is animate (*w-*) or inanimate (*j-*). If the head noun is plural, nothing changes. This is shown in (2) below.

CONTEXT: *There are (MUST BE) many chairs in the context of the conversation and, an event E has happened that affected a chair or many of them (a falling event, for example). (2a) and (2b) restrict the understanding of “falling” to X’s chair (2a) or chairs (2b), to the detriment of all the other chairs that are present in the context.*

- | | |
|---|--|
| (2) a. <i>j-ǎ</i> <i>Ø-kwò?</i>
INANIM-POSS 1-chair
‘MY chair, as compared to
X’s’ | b. <i>m-ǎ</i> <i>mà-kwò?</i>
6-POSS 6-chair
‘MY chairs, as compared to
X’s’ |
|---|--|

If the head noun were animate, *w-* (4a) would be inserted in place of *j-*. The examples in (3) show possessive pronoun agreement with an animate head noun which, as in (1), does not pose any problem.³ As soon as the possessive pronoun is focus-marked, the glide needs to be inserted, if the head noun is singular (see 4a). If it is plural, nothing about the agreement exponent changes (4b).⁴

- | | |
|--|--|
| (3) a. <i>Ø-pǔsì Ø-à</i>
1-cat 1-POSS
‘my cat’ | b. <i>mà-pǔsì m-á</i>
6-cat 6-POSS
‘my cats’ |
| (4) a. <i>w-ǎ</i> <i>Ø-pǔsì</i>
ANIM-POSS 1-cat
‘MY cat, as compared to X’s’ | b. <i>m-ǎ</i> <i>mà-pǔsì</i>
6-POSS 6-cat
‘MY cats, as compared to
X’s’ |

One might think, at first glance, that the insertion of the glide is sensitive to the absence of a concrete noun class exponent in (1a) and (3a). This is clearly not the case, as possessive pronouns associated with Class 5 nouns (marked *Ø-N* and *n-POSS*) also undergo the same process with singular head nouns. This is shown in (5) and (6).

³Class 1 in Ngemba features both animate and inanimate nouns, based, mostly, on morpho-phonological factors.

⁴Also note the tone changes which, for reasons of space, I will not discuss in detail.

- | | |
|---|--|
| <p>(5) a. Ø-kəsálà n-á
5-cassava 5-POSS
'my cassava'</p> | <p>b. mǝ-kəsálà m-á
6-cassava 6-POSS
'my cassavas'</p> |
| <p>(6) a. j-ǎ Ø-kəsálà
INANIM-POSS 5-cassava
'MY cassava, as compared to
X's'</p> | <p>b. m-ǎ mǝ-kəsálà
6-POSS 6-cassava
'MY cassavas, as compared
to X's'</p> |

Comparing (2a) to (4a) shows that the choice between the glides is sensitive to the animacy of the head noun. *j-* is compatible with inanimate head nouns, and *w-* with animate head nouns. The ungrammaticality of (7a) and (7b) is therefore tied to this distinction.

- | | |
|--|--|
| <p>(7) a. *j-ǎ Ø-pǔsì
INANIM-POSS 1-cat
'MY cat, as compared to X's'</p> | <p>b. *w-ǎ Ø-kwǝʔ
ANIM-POSS 1-chair
'MY chair, as compared to
X's'</p> |
|--|--|

To summarise, adnominal possessive pronouns can undergo contrastive focus in Ngemba. This affects canonical word order which, in turn, affects agreement in the following ways: (a) plural nouns keep their noun class prefix unchanged, and (b) singular nouns see their noun class prefix replaced by one of two glides (*w-* or *j-*), each of which are sensitive to the animacy of the head noun.

In this paper, I propose an account of the puzzle just described. I argue that the spell out of animacy in singular contexts tracks semantic agreement, syntactic agreement having failed. That syntactic agreement fails in singular POSS > N order, but not in the plural, I show, is due to three fairly standard assumptions, namely (a) DP is a phase (Svenonius 2004, Heck et al. 2008, Kramer 2010, Bošković 2012), (b) phase edges are non-recursive (Müller 2010, 2011), and (c) number is privative (Nevins 2011). These assumptions combined derive the observed asymmetry.

The paper is structured as follows. Section 2 gives an overview of what noun classification may look like in Ngemba. Because the system has, to my knowledge, not been described before, I will also rely on existing literature on related languages (Voorhoeve 1968, Hyman et al. 1970, Nkemnji 1995, Maho 1999, Nchare 2011, Déchaine & Keupdjio 2016) to propose a classification. In Section 3, I provide details of DP-internal word order in Ngemba, and propose how this order is derived. Section 4 sketches the proposal, and Section 5 concludes the paper.

2 Towards a noun class system for Ngemba

Previous studies of Grassfields Bantu noun classes (see, among others, Voorhoeve 1968, Hyman et al. 1970, Nkemnji 1995, Maho 1999, Nchare 2011, Déchaine & Keupdjio 2016) agree on the fact that these classes have considerably reduced nowadays, compared to Proto-Bantu that had about twenty-three (23) noun classes (Maho 1999). For instance, while Nchufie has only about four classes (Nkemnji 1994), Medumba has five to six classes today (Kouankem 2013, Keupdjio 2020). Nweh has eight classes (Nkemnji 1995), Shupamem: nine (Nchare 2011), Aghem: thirteen (Hyman 2010), Babanki: twelve (Akumbu & Kießling 2022), and Lamnso': ten (McGarrity & Botne 2001).

The literature also agrees on the fact that in most of these languages, noun class identification mostly takes into account noun prefixes and concord morphology on nominal modifiers (possessive and demonstrative pronouns, adjectives, quantifiers, numerals etc.). The number and type of nominal modifiers that show agreement vary from language to language. In Aghem, Babanki, Bafut and Lamnso', for example, almost all nominal modifiers agree with the head noun (McGarrity & Botne 2001, Tamanji 2006, Hyman 2010, Akumbu & Kießling 2022). In Ngemba, only possessive and demonstrative pronouns tend to show clear signs of noun class agreement.⁵

Besides, unlike 'proper' Bantu languages (Chichewa, Swahili, Bemba etc. cf. Fuchs & van der Wal 2021) in which class prefixes can spread to other constituents of the clause (verbs, for example), noun class agreement tends to be strictly DP-internal in most Grassfields Bantu, especially in the languages spoken in West Cameroon. Other Grassfields Bantu languages have subject agreement markers (Aghem, Babanki, Awing, cf. Fominyam & Georgi 2021), but their status as being DP-internal or external (on verbs) is a topic of debate (see, for example, McGarrity & Botne 2001 for arguments that subject markers are clitics to DPs and not to verbs in Lamnso').

Although most, if not all the literature I cite above, acknowledge and use the classificational trend proposed for Proto-Bantu (see Maho 1999 for an overview), they are unanimous in claiming that it is very unclear today if semantics has a key role to play in noun classification in Grassfields Bantu. For instance, there appears to be no clear correlation between the semantics of a noun (human, for example) and morphological marking in most Grassfields Bantu, as is the case with Proto-Bantu. Class 1 grouping in Proto-Bantu is mostly made up of humans, and is morphologically marked by the prefix *mu-*. Class 2 is made up of plurals

⁵Solange Mekamgoum (p.c.) thinks that numerals from 1 to 5 in Ngemba also show traces of noun class agreement.

of Class 1 nouns and is marked by the prefix *ba-* (see Maho 1999 for a complete picture). The picture in Grassfields Bantu is somewhat blurred today in the sense that, as Keupdjio (2020) observes for Medumba, nouns that hitherto belonged to different classes in Proto-Bantu are now organised into one single class according to concord morphology with possessives. The same situation is observed in Fe'Fe, Ghomálá?, Nweh and Shupamem (Voorhoeve 1968, Nkemnji 1995, Nchare 2011). This implies that in some of these languages, Class 1 morphology, for example, is not as semantically restricted (to humans) as it is in Proto-Bantu.

Ngemba manifests the same types of complications, as one can find all sorts of nouns in hitherto Proto-Bantu Class 1 grouping (at least semantically). As a matter of fact, most trees, plants and inanimates (Class 3 in Proto-Bantu), liquids (Class 6 in Proto-Bantu), animals (Class 9 in Proto-Bantu) etc. form their singulars with Class 1 morphology, and their plurals with Class 6 morphology. This, in my view, indicates that the best and most productive way to classify them today is morpho-phonological (see Msaka 2019, for arguments that phonology also plays a key role in noun classification in the Bantu language Chichewa). This does not mean that we should abandon semantic-oriented classifications altogether, but more emphasis, I think, should be laid on morpho-phonology, rather than semantics.

With what precedes in mind, Ngemba, has about six noun classes, with singular-plural pairings.⁶ I summarize them in Table 1 below.⁷

As Table 1 shows, nominal prefixes in singular classes are always segmentally null.⁸ In the plural, class prefixes alternate between *p(ə)-* and *mə-*. Class 2 only contains humans.⁹ Class 1 contains all sorts of nouns (humans, animals, body parts and miscellaneous). Non-humans in Class 1 form their plural in Class 6. Classes 3 and 4 share the same agreement prefix, namely *ts-*. It is unclear to me if there is a Class 4 in Ngemba because Class 3 nouns tend to be inherently plural. *nəvəm* 'stomach', for example, can be understood as having internal parts. The same holds for *nələ?* 'yellow yam', which always has a many-reading. The singulative is marked by means of the numeral classifier *ndəŋ*, that makes reference

⁶There might be a dedicated class for infinitives. I do not discuss them here because their behaviour is unclear to me. They are marked by the prefix *nə-*, and tend to have agreeing prefixes reflecting all the singular classes in Table 1. Examples include: *nə-rà ts-á* 'my speaking' (Class 3 agreement prefix), *nə-rə n-à* 'my going' (Class 5) and *nə-k^hù ə-à* 'my running' (Class 1). They do not have plural counterparts.

⁷I only use possessive pronouns here because only they allow to straightforwardly see the behaviour of tones. That only they show some consistency in this regard is not new in Grassfields Bantu (see, for example, Nweh; Nkemnji 1995).

⁸In Section 4.1, I use this as evidence for adopting a privative number system for Ngemba.

⁹Ngemba also has an associative plural marker (*pə*) that, unlike in Medumba (see Keupdjio 2016 for details), can only be used with proper names referring to human entities. There probably is a link between the *p(ə)-* of Class 2 and this associative plural marker.

Table 1: Summary of Ngemba noun classes

Class	Nominal prefix	Examples	Agreeing prefix	Poss(1SG/PL)+ tone
1	Ø-	Ø- <i>mé(n)</i> ‘child’ Ø- <i>ndá</i> ‘house’ Ø- <i>kwò?</i> ‘chair’ Ø- <i>pǔsì</i> ‘cat’ Ø- <i>tòŋ</i> ‘ear’	Ø-	a +Low
2	p(à)-	<i>p-áé</i> ‘children’	p-	a +High
3	Ø-	Ø- <i>nàlǎ?</i> ‘yellow yam’ Ø- <i>nàvám</i> ‘stomach’	ts-	a +High
4 (?)	mà-	<i>mà-nàlǎ?</i> ‘yellow yam’ (types) <i>mà-nàvám</i> ‘stomach’ (types)	ts-	a +High
5	Ø-	Ø- <i>kàsàlà</i> ‘cassava’ Ø- <i>pùmá</i> ‘orange’	n-	a +High
6	mà	<i>mà-ndá</i> ‘houses’ <i>mà-kwò?</i> ‘chairs’ <i>mà-tòŋ</i> ‘ears’ <i>mà-pǔsì</i> ‘cat’ <i>mà-kàsàlà</i> ‘cassava’ <i>mà-pùmá</i> ‘orange’	m-	a +High

to the shape of the yam. Class 3 nouns can be pluralised. When this happens, they have a type-reading.¹⁰ I assume that the tones on possessive pronouns are underlying. They can, of course, differ on the surface.¹¹

¹⁰This can be taken as evidence that *mà* also functions as a plural marker. This would make sense, considering the fact that Class 6 is generally taken to be the default plural class in Grassfields Bantu (see, for example, Keupdjio 2016 on Medumba). Ngemba confirms this in the sense that non-humans in Class 1, as well as Class 5 nouns, form their plural in Class 6. The fact that associative *pà* exists still suggests that some notion of gender is still in play here.

¹¹I do not discuss tone changes on possessive pronouns because they can be impacted by the tones of the nominal root. For related discussion in Grassfields Bantu, I refer the reader to Akumbu & Hyman (2017).

3 DP-internal word order in Ngemba

In this section, I look into the order of nominal modifiers in information-structure-neutral contexts, which is relevant for the asymmetry in agreement related to word order changes. For this reason, it is important for the reader to know what basic order is attested within Ngemba DPs and how such is derived.

The basic word order internal to Ngemba DPs is given in (8). This order is exemplified with the root *pùmá* ‘orange’ in the singular (9) and in the plural (10).¹²

(8) ADJ > N > POSS > DEM > NUMERAL.

- (9) a. Ø-pùmá n-á
1-orange 1-POSS
‘my orange’
b. mmá Ø-pùmá n-á
big 1-orange 1-POSS
‘my big orange’
c. mmá Ø-pùmá n-á n-è
big 1-orange 1-POSS 1-DEM
lit.: ‘that my big orange’
d. mmá Ø-pùmá n-á n-è zhímó?
big 1-orange 1-POSS 1-DEM one
lit.: ‘that my one big orange’
- (10) a. m-è-pùmá m-á
6-orange 6-POSS
‘my oranges’
b. m-è mmá pùmá m-á
CL6 big orange 6-POSS
‘my big oranges’
c. m-è mmá pùmá m-á m-è
CL6 big orange 6-POSS 6-DEM
lit.: ‘those my big oranges’

¹²I leave quantifiers out because their position depends on their scope properties. Besides, they do not show agreement and, as such, are not directly relevant for the topic under discussion.

- d. mǎ mmá pùmá m-á m-è sǎmbà
 CL6 big orange 6-POSS 6-DEM seven
 lit.: ‘those my seven big oranges’

Adjectives precede the head noun and can be as many as need be. Possessive pronouns follow the head noun. They are in turn followed by demonstrative pronouns. Numerals come last. There can only be one POSS, DEM or NUMERAL per DP. What is striking about (10) is that class prefixes have to precede the adjective.¹³ This is evidence that they have to be as close to the head noun as possible.¹⁴ In the next section, I propose that they are adjoined to *nP*. Having established the order of DP-internal modifiers, we can now turn to how such an order is derived. I do this in the next section.

3.1 Deriving DP-internal word order in Ngemba

In the previous section, I showed that DP-internal modifiers are ordered as in (11), in an IS-neutral context. This section proposes how the order in (11) is derived.

- (11) ADJ > N > POSS > DEM > NUMERAL.

I assume that all DP-internal modifiers project. Possessive pronouns head a PossP, demonstrative pronouns a DemP, and numerals a QP.¹⁵ The head D of DP takes PossP as its direct complement. Poss takes DemP, and Dem takes QP. QP is in turn followed by a NumP which takes *nP* as its complement. Adjectives are in Spec*nP*.¹⁶ This is schematized in Figure 1.

The surface word order in (12) is derived by moving NumP, and everything it dominates, to SpecDP as in Figure 2.

- (12) mǎ mmá pùmá m-á m-è sǎmbà
 CL6 big orange 6-POSS 6-DEM seven
 lit.: ‘those my seven big oranges’

¹³Class prefixes can also simultaneously appear before the adjective and on the head noun, with the reading ‘the big oranges amongst the oranges’. I do not discuss such cases here because the contrastive reading brings in issues related to information structure. Besides, I think such cases involve complex DP structures with potentially two nouns. One is deleted for economy. For these reasons, I will not consider such examples in this paper.

¹⁴In Section 3.1, I will use it as evidence that adjectives are adjoined to Spec*nP*, and that, although gender features are located on *n*, noun class is expounded on Num, via a local step of agreement.

¹⁵I assume QP for numerals because they quantify.

¹⁶Adjectives cannot be in the specifier of a potential $\sqrt{\text{root}}P$ because this would make the wrong prediction, from a compositional semantics point of view, that adjectives are to be interpreted as part of the nominal root.

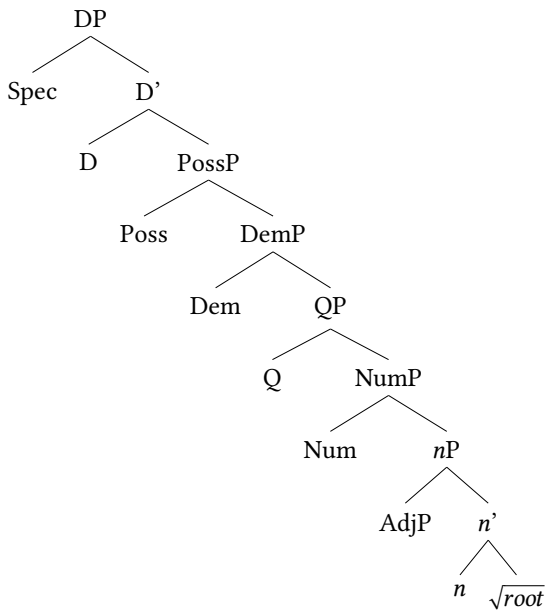


Figure 1: Basic DP structure

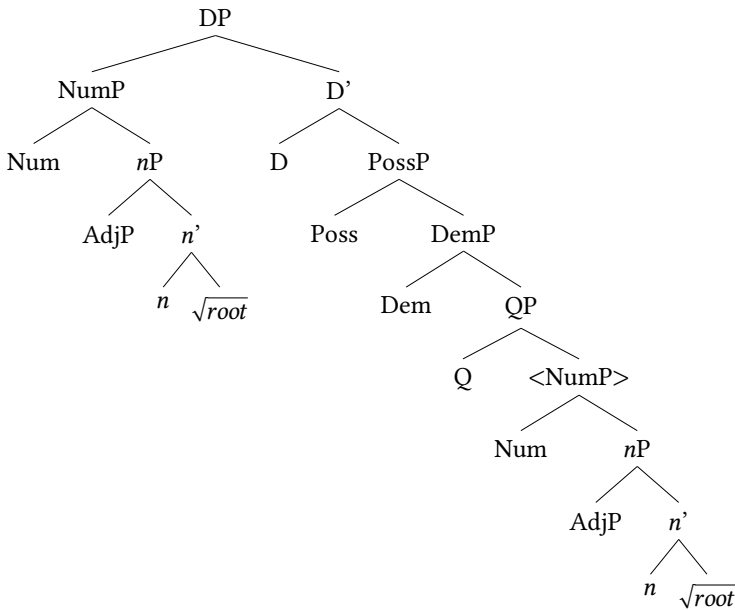


Figure 2: Deriving surface word order

The next step is to account for the position of the class prefix in (12). It precedes the adjective which, in turn, is followed by the head noun. I adopt the fairly standard assumptions that (a) number features are on Num (Carstens 1991, Ritter 1991, Nkemnji 1995, Preminger 2020, Fuchs & van der Wal 2021), and (b) gender features are on *n* (Kramer 2015, 2016, Fuchs & van der Wal 2021). I follow Carstens (1991) and Fuchs & van der Wal (2021) in assuming that noun class exponents on nouns in Bantu are the spell out of gender and number features. What this means is that, at some point, both features need to be on the same terminal node to spell out nominal prefixes. The fact that adjectives stand in-between the class prefix and the head noun suggests that noun class is exponed on the head Num, considering the structure in Figure 1. I propose that Num has a φ -probe that is active whenever related features are indeed present on the number head. This φ -probe copies gender features from *n* to ensure that noun class be exponed on the head noun as a combination of number and gender information. In the absence of a number feature, this probe is silent, and no agreement takes place. I will propose that this is what happens in singular contexts because number is privative in Ngemba. With this background in mind, we can turn to deriving the asymmetry. I do this in the next section.

4 The proposal

In this section, I derive the asymmetry between singular and plural head nouns in modifier focus contexts. As a reminder, adnominal possessive pronouns can undergo contrastive focus in Ngemba. This affects canonical word order which, in turn, affects agreement in the following ways: (a) plural nouns keep their noun class prefix unchanged, and (b) singular nouns see their noun class prefix replaced by one of two glides (*w-* or *j-*), each of which are sensitive to the animacy of the head noun. I present the proposal in two steps. In the first, I lay out the theoretical assumptions I adopt for the analysis. In the second, I describe the workings of Agree.

4.1 Theoretical assumptions

The analysis I adopt in this paper is rooted in Distributed Morphology (Halle & Marantz 1993). Specifically, I will adopt, among other things, the idea that syntax feeds morphology, as morphological exponents spell out terminal nodes that host features from the syntax (inherent features as well as those from syntactic operations such as Agree). The choice of the exponent that is the best match for a terminal node is subject to the Subset Principle (13)

(13) *Subset Principle*

A vocabulary item V is inserted into a functional morpheme M iff (a) and (b) hold:

- a. The morpho-syntactic features of V are a subset of the morphosyntactic features of M .
- b. V is the most specific vocabulary item that satisfies (a).

I further follow DM in assuming that nouns are formed by merging a category-neutral root with the categorizing head n (Marantz 1997, Arad 2003, Harley 2014, among others). This head hosts gender features in Ngemba. In other words, when n nominalizes the root in this language, it gives it gender information (see also Kramer 2015, 2016, Fuchs & van der Wal 2021, for similar proposals in Bantu and beyond). Num hosts number features (Carstens 1991, Ritter 1991, Nkemnji 1995, Preminger 2020). Nominal prefixes are the spell-out of number and gender features (Carstens 1991, Fuchs & van der Wal 2021). The head Num has a φ -probe that is active whenever related features are indeed present on the number head. This φ -probe copies gender features from n to ensure that noun class be the exponence of both number and gender information.¹⁷ In the absence of a number feature, this probe is silent, and no agreement takes place. I will propose that this is what happens in singular contexts because number is privative in Ngemba. These assumptions are exemplified in the simplified structure in Figure 3, for plural DPs.

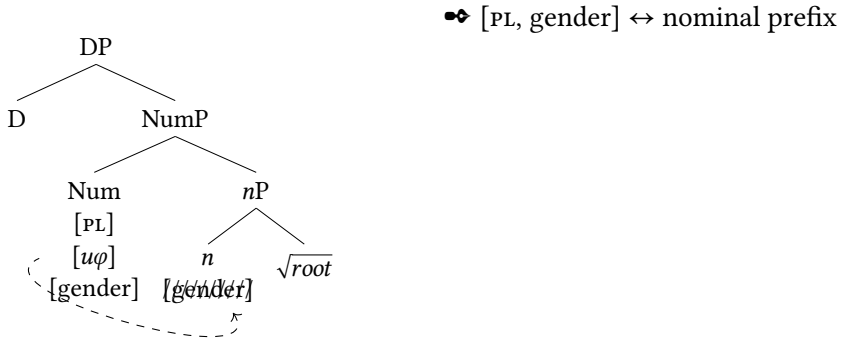


Figure 3: The locus of φ -features in Ngemba and spell-out of nominal prefixes

¹⁷Having both number and gender features on n from the start makes the wrong theoretical prediction that Ngemba can have a number-based noun classification system akin to Kipsigis (Kouneli 2021).

I assume that number is privative in Ngemba. Partial evidence for this comes from the way the language marks type-plurals, as shown in Class 4 (?) in Table 1. If *mə* can function as a plural marker, then the fact that it exists points towards the presence of a plural feature within Ngemba DPs. There is, to my knowledge no singulative exponent in Ngemba. This, to me, suggests the absence of a singular feature. I interpret this as evidence for a privative number system in the language. In other words, singular is the absence of number features (Nevins 2011).¹⁸

Looking closely at singular class exponents in Table 1 shows that this assumption might be on the right track. In fact, nominal prefixes are always segmentally null in the singular. If φ -agreement with *n* does not take place in singular contexts because number features are absent, and nominal prefixes need to expone both features, we would expect class prefixes to be null on nouns in the singular because both features never end up on the same terminal node. The gender features on *n* can then be interpreted at LF to group nouns into classes.¹⁹ This is exactly the reasoning I adopt here. Specifically, I propose that nominal prefixes are null in singular contexts because the φ -probe on Num is inactive. This is due to it lacking number features. Since this probe is silent, number and gender never get together to spell out nominal prefixes. As a result, no class prefix appears on nouns on the surface. These nouns' gender features are interpreted at LF.²⁰

4.2 The workings of Agree

This section derives the Asymmetry by relying on three key assumptions. First, Agree is downwards. Second, DP is a phase (Svenonius 2004, Heck et al. 2008, Kramer 2010, Bošković 2012), and third, phase edges are non-recursive (Müller 2010, 2011). I adopt the version of the PIC in (14).

(14) *Phase Impenetrability Condition (PIC)*

In phase α with head H, the domain of H is not accessible to operations outside α ; only H and its edge are accessible to such operations

¹⁸The Number projection is not totally absent in this context because, as Ritter (1991) and Preminger (2020) argue, there are independent reasons for postulating a functional projection in-between D and *n*P. Ritter (1991) further proposes that this projection is NumP. It cannot be GenderP because gender information is already encoded by *n*.

¹⁹The empirical data shows that one has to rely on agreeing modifiers to classify singular nouns into classes. This can also be taken as evidence that class is not expounded on nouns in the singular.

²⁰Agreeing modifiers can, of course, have access to the gender features on *n*. Different flavours of *n* will produce the different singular agreeing prefixes.

That phase edges are non-recursive simply means that if a phase edge has more than one projection, only the highest (together with its head) is accessible to operations outside the said phase domain. In other words, if DP is a phase, and its edge is similar to that in Figure 4, only the phase head D and the highest projection in the edge of DP and its head (NumP and Num) would be accessible to operations above DP.

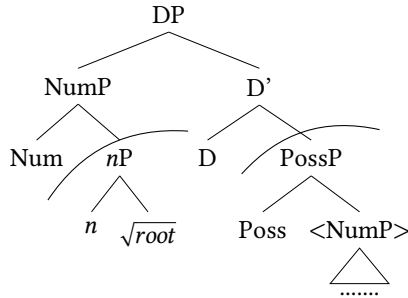


Figure 4: Phase edges are non recursive

4.2.1 Deriving agreement in N > POSS order

In IS-neutral order, regular class prefixes are used in both singular and plural contexts. Example (1), repeated as (15), illustrates this.

- (15) a. Ø-kwò? Ø-á
 1-chair 1-POSS
 ‘my chair’
 b. mǝ-kwò? m-á
 6-chair 6-POSS
 ‘my chairs’

Agreement works as in Figure 5, in singular contexts, and Figure 6 in plurals.²¹

In singular contexts as in Figure 5, the φ -probe responsible for the spell out of nominal prefixes as a combination of number and gender features is silent because number features are absent within the DP. For this reason, gender features are not copied onto the number head. The φ -probe on Poss scans its c-command domain and copies gender features from *n*. Singular Poss will be sent to the morphology with the feature specification [gender, poss]. This spells out Ø-á in (15a).

²¹Since agreement is downwards, it takes place before Num-to-SpecDP movement, that, I assume, derives the correct linear order. Ordering movement before agreement makes the wrong predication that agreement would fail.

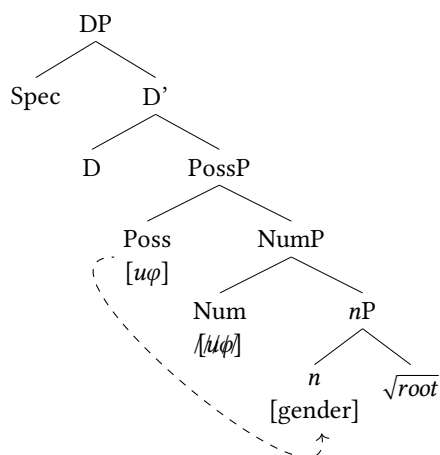


Figure 5: Singular context

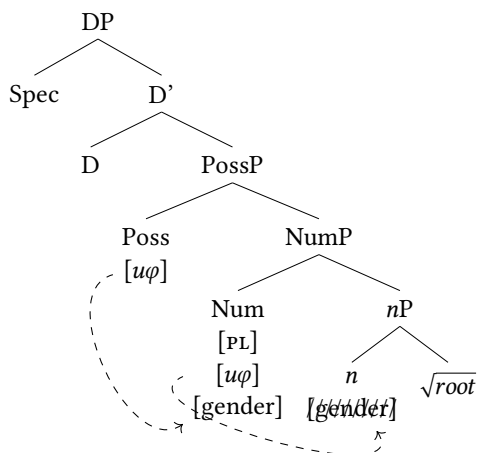


Figure 6: Plural context

In the plural Figure 6, the φ -probe on Num is active because it has number features. It copies the gender features from *n* to Num. This will spell out the nominal prefix *m̩* (15b) in the morphology. The probe on Poss then copies all the features on Num. It will be sent to the morphology with the specification [PL, gender, poss]. This spells out *m-á* in (15b). NumP, in both singular and plural contexts, then moves to SpecDP to derive the surface word order (c.f. Figure 2).

4.2.2 Deriving agreement in POSS > N order

I adopt, from van de Velde (2022), the idea that in some Bantu languages, the possessive pronoun in POSS > N order is appositional. What this means is that POSS in POSS > N order is different from the one in N > POSS order (see van de Velde 2019 for related details). The former is adnominal, and the latter appositional. This view excludes the movement approach to POSS > N, wherein this order would be derived by movement of POSS to a position where it precedes N. The tone difference could be used to back up this line of analysis. Recall that all possessive pronouns in POSS-N order surface with a rising tone (e.g. *j-ǎ Ø-kwǎ?* INAN-POSS 1-chair ‘MY chair, as compared to X’s’ (2)). Unless we think of the tone change as the by-product of the interaction between the default tone on possessive pronouns and a floating tone in the new position, I would tend to think that the tone change suggests that these possessive pronouns are different. The floating low tone could well be a tone augment of the type we see in Eton (van de Velde 2017). Partial evidence that these possessive pronouns might indeed be different comes from the reading associated with POSS > N order. In fact, this reading is of the type ‘My X(s) among the Xs or among your Xs,’ which suggests

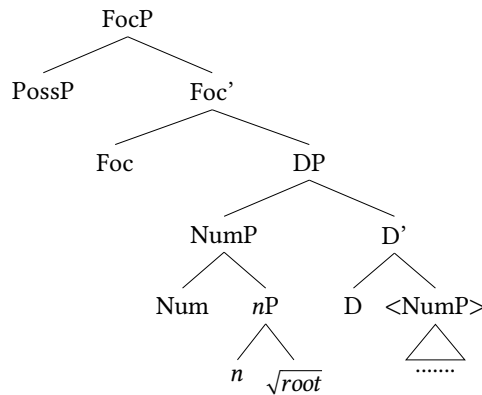


Figure 7: Appositional POSS is base-generated in SpecFocP

two determiners associated with a noun; the higher one having a close apposition reading which, as the literature suggests, is restrictive (Asiimwe et al. 2023). I assume that this type of apposition takes place after fully-formed adnominal possession i.e., after, in addition to adnominal possessive pronoun agreement, NumP has moved to SpecDP.²²

With this in mind, the structure of POSS > N looks like Figure 7, where Appositional POSS is base-generated in SpecFocP.²³

Agreement works as in Figure 8 in the singular and as in Figure 9 in the plural. Agreement with the gender features on *n* fails because *n* is too deeply embedded in the phase edge to be visible. Num lacks features, and is invisible to the φ -probe on PossP. In Figure 9, however, the φ -probe on Num is active, given the fact that Num has plural features. This probe copies gender features onto Num to ensure that nominal prefixes be the spell out of both gender and number information. PossP probes and can access the phase edge. Given the fact that edges are non-recursive, only NumP and Num are visible. Num has all the relevant features. PossP copies them. They will spell out the class prefix on the appositional possessive pronoun.

If agreement fails in Figure 8, why is it that the derivation does not crash? I follow, among others, Preminger (2014) in assuming that Agree failures do not

²²The behaviour of demonstrative pronouns provides empirical evidence for the existence of both adnominal and appositional modifiers. Both adnominal and appositional demonstrative pronouns can co-occur, giving rise to DEM > N > DEM. The first DEM only has animacy distinctions in the singular, but full noun class in the plural, as it is the case with possessive pronouns. This is exemplified in (iii). Both (ii) and (iii) have the same interpretation, but, as (iii) shows, both adnominal and appositional DEMs can co-occur.

- | | | | | | | | | |
|-------|----|--|-----------|-------|----|--|-----------|-------|
| (i) | a. | Ø-kəsálà | n-è | | b. | mə-kəsálà | m-è | |
| | | 5-cassava | 5-DEM | | | 6-cassava | 6-DEM | |
| | | 'that cassava' | | | | 'those cassavas' | | |
| (ii) | a. | y-ě | Ø-kəsálà | | b. | m-ě | mə-kəsálà | |
| | | INANIM-DEM | 5-cassava | | | 6-DEM | 6-cassava | |
| | | 'THAT cassava, as opposed to this one' | | | | 'THOSE cassavas, as opposed to these ones' | | |
| (iii) | a. | y-ě | Ø-kəsálà | n-è | b. | m-ě | mə-kəsálà | m-è |
| | | INANIM-DEM | 5-cassava | 5-DEM | | 6-DEM | 6-cassava | 6-DEM |
| | | 'THAT cassava, as opposed to this one' | | | | 'THOSE cassavas, as opposed to these ones' | | |

²³I assume that projections, as well as heads, can serve as probes (Clem 2022). I thank one of the anonymous reviewers for suggesting this reference to me.

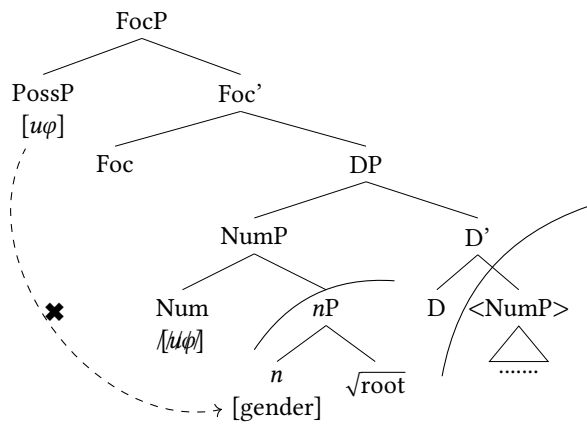


Figure 8: Singular context

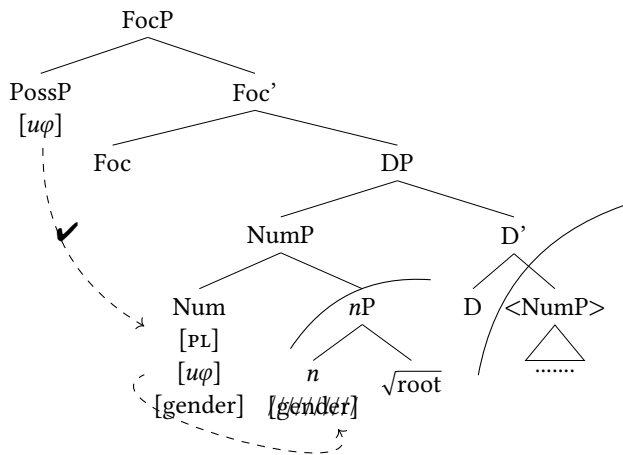


Figure 9: Plural context

necessarily cause the derivation to crash. This can sometimes be repaired by inserting a default exponent. Crucially, I assume for Ngemba, that when syntactic agreement fails, semantic agreement takes over.

In discussing semantic agreement in Bantu, van de Velde (2022: 917) cites Corbett (1979) who claims that “we speak of semantic agreement, as opposed to syntactic agreement, when the choice of agreement pattern depends on the meaning of the controller noun, rather than on its morphological class as defined by an overt class marker.” In the system I adopt in this paper, morphological class markers on agreeing modifiers track features from an Agree operation in the syntax. My understanding of semantic agreement, based on this, is therefore that it does not track features from the syntax, but rather is controlled by the meaning of the noun it associates with.

Interestingly, semantic agreement in Bantu is generally construed as animacy agreement (Garbo & Verkerk 2022, van de Velde 2022, to cite only these few).²⁴ Garbo & Verkerk (2022) argue, in their typological study of Northwestern Bantu gender systems, that verbs and pronouns are most likely to show animacy agreement. van de Velde (2022) claims that possessive pronouns are less likely to show semantic agreement because they tend to be closer to nouns than other agreeing elements. This in turn suggests an understanding of semantic agreement based on locality and the possibility for semantic agreement to be the repair for syntactic Agree failures. We saw that the probes that fail to agree are phase-external, hence non-local to features that would ensure that syntactic agreement obtains, hence spell out of noun class. With all this in mind, I propose that when syntactic agreement fails in Ngemba, semantic agreement takes over.²⁵

5 Conclusion

In this paper, I looked into DP-internal focus-induced reordering in Ngemba and the way it affects agreement. The data show that in IS-neutral word order (N > POSS), we always find regular noun class agreement morphology. In the focused order (POSS > N), we only find regular noun class agreement morphology if the head noun is plural. If it is singular, regular noun class morphology is replaced by one of two possible glides. Data suggest that the choice between these is not

²⁴See van de Velde (2019) for other types of semantic agreement in Bantu languages.

²⁵It is unclear to me how to implement this in the architecture of the grammar I assume in this paper. For space reasons, I will not develop this any further. I think, however, that it is a theoretical challenge for all those who assume semantic agreement in Bantu. Since I do not have a clear-cut answer to this question, I leave it out for future research.

phonologically conditioned. Rather, it reflects the animacy of the head noun. I argued that the spell out of animacy in singular contexts tracks semantic agreement, syntactic agreement having failed. That syntactic agreement fails in singular POSS > N order, but not in the plural, I show, is due to three fairly standard assumptions, namely (a) DP is a phase (Svenonius 2004, Heck et al. 2008, Kramer 2010, Bošković 2012), (b) phase edges are non-recursive (Müller 2010, 2011), and (c) number is privative (Nevins 2011). These assumptions combined derive the observed asymmetry.

Overall, the paper provides empirical evidence for privative number features in Ngemba, and strengthens the theoretical contribution that phase edges need not be recursive, such that only the highest projection (and its head) in a phase edge are accessible to operations outside a given phase domain.

Abbreviations

ANIM	animate	NUM	number
CL	noun class	PL	plural
DEM	demonstrative	POSS	possessive
FOC	focus	SG	singular
INANIM	inanimate		

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Chapter 6

Definitely a question: Definiteness markers in Akan ex-situ questions

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This paper contributes to the emerging literature on cross-categorial definite determiners — determiners that appear in non-DP contexts—by investigating the semantic analysis of the clause-final marker *nó* in Akan ex-situ wh-questions. I argue that clause-final *nó* in Akan ex-situ wh-questions presupposes familiarity of the existential closure of the question nucleus. I demonstrate that questions with and without the clause-final *nó* are both information-seeking questions but differ in the kind of possible answers they allow. Clause-final *nó* wh-questions are analogous to cleft questions in terms of semantics. They encode a hard existence presupposition that cannot be suspended nor canceled across speakers. The paper, therefore, also adds to the discussion on whether definiteness determiners encode uniqueness or familiarity. In the nominal domain, uniqueness and weak familiarity overlap in many contexts, but weak familiarity provides a more accurate explanation for the distribution of these markers in the clausal contexts.

1 Introduction

In Akan (a Niger-Congo language spoken in Ghana and Côte d'Ivoire), the definite determiner *nó* occurs in a typical DP as in (1), where it takes an NP complement. The whole DP then functions as the object of the verb *eat*. Notably, this morpheme also occurs at the end of fronted wh-questions such as (2), where it cannot be construed as taking a nominal complement since Akan pronouns do not take determiners.¹

¹This paper presents data from the Asante-Twi dialect of Akan; examples from other dialects will be cited explicitly. The data comes from two consultants, who are trained linguists, and the author's native-language judgment. The data are written in Akan orthography with tone added. Data from other sources will be explicitly specified.



- (1) Kòfí dì-ì àdùàné (nó). nominal definite determiner
Kofi eat-PST food DEF
'Kofi ate the food.'
- (2) **Context:** Kofi informed us this morning that he asked someone about an issue we had. But I have forgotten who he claimed to have spoken to.
Héná nà Kòfí bísá-à nò (nó)? clause-final determiner
who FOC Kofi ask-PST 3SG.OBJ DEF
'Who did Kofi ask?'

This paper examines the semantic contributions of clause-final *nó* in constructions such as (2).

The structure of the remaining sections of the paper is as follows: In Section 2, I offer a brief overview of *nó* in the nominal domain by summarizing the three primary current analyses of the determiner. The clause-final *nó* in declarative clauses is the topic of Section 3, which introduces the non-DP uses of *nó*. Section 4 delves into the central matter of the paper. I examine the meaning of ex-situ wh-questions with clause-final *nó* by comparing them to ordinary ex-situ wh-questions. Section 5 is the conclusion.

2 The nominal definite marker

Since Christaller (1875), the morpheme *nó* in Akan noun phrases, such as (3a) and (3b), has been translated in English as the definite determiner *the*. The determiner generally occurs in contexts where English will use *the*. The relationship between *nó* and *the* is explored by Amfo (2006, 2007), Fretheim & Amfo (2008), and Amfo (2010) who argue that *nó* has the same cognitive status as *the*, i.e., uniquely identifiable. However, the equivalence between English and Akan is not perfect, as noted by Arkoh (2011) and Arkoh & Matthewson (2013). Consider the following sentence in example (4). The definite determiner is required in the English translation, but it is missing in the Akan sentence.

- (3) a. Pàpá **nó** bà-à hà.
man DEF come-PST here
'The man came here.'
- b. Kòfì tò-ò pén. Ná pén **nó** yè fitáá.
Kofi buy-PST pen PST pen DEF COP white
'Kofi bought a pen. The pen was white.'

- (4) **Context:** The beginning of a documentary on the solar system
 Àwíá (*nó) yè ñsórómá.
 sun DEF COP star
 ‘The sun is a star.’
- (5) Àwíá (nó) rè-bó ènné.
 sun DEF PROG-hit today
 ‘The sun is shining today.’ (Bombi 2018: 150)

In contexts such as (5) on the other hand, the English determiner is required, but *nó* is optional.

There are at least three formal semantics analyses of the nominal definite determiner: *weak familiarity* (Arkoh & Matthewson 2013), *uniqueness* (Bombi 2018), and *weak familiarity + anti-uniqueness* (Owusu 2022). Given that Arkoh & Matthewson (2013) and Owusu (2022) both assume weak familiarity, I will group them under *familiarity* analyses and refer to them as such. I will discuss all three analyses briefly.

Arkoh & Matthewson (2013) has received a great deal of attention in the literature as the first formal semantics analysis of the nominal definite determiner *nó*. They make two very important contributions. First, they argue that *nó* encodes *weak familiarity*. Weak familiarity, as defined by Roberts (2003), is analogous to Prince’s (1992) concept of “hearer-old” entities; it can be satisfied by non-linguistic antecedents. Secondly, they claim that Akan, like Schwarz’s (2009) analysis of German, distinguishes between uniqueness and familiarity definites. *Nó* encodes familiarity, whereas uniqueness denoting nouns are bare as in (4). Arkoh & Matthewson (2013) predict that *nó* is felicitous in (6), where there is a linguistic antecedent for the definite description, as well as in (7), where the antecedent of the definite description is part of the discourse context.

- (6) Ám’á hù-ù òkyèrèkyèrèní bí né sógyàní bí. ó-kyèà-à
 Ama see-PST teacher INDF CONJ soldier INDF 3SG.SBJ-greet-PST
 sógyàní nó.
 soldier DEF
 ‘Ama saw a teacher and a soldier. She greeted the soldier.’
- (7) **Context:** A man and a woman are arguing in the street. Ama and Kwame are sitting in front of their house, where they can see but cannot hear them. Kofi walks in and sees them staring. He says oh...
 Pàpá nó dè mààmé nó ká.
 man DEF owe woman DEF debt
 ‘The man owes the woman money.’

However, as pointed out by Bombi (2018), the analysis does not correctly predict that (4) is infelicitous since world knowledge makes the referent of *sun* familiar in all contexts.

To account for the infelicity of *nó* in (4) and other contexts such as (8), Owusu (2022) proposes that in addition to weak familiarity, *nó* also carries an *anti-uniqueness* presupposition which was proposed by Robinson (2005) and Dayal & Jiang (2021) for demonstratives.

- (8) **Context:** A newscaster in Ghana is talking about the Ghanaian president
àmànpànín (***nó**) bé-blémí òbí.
president DEF FUT-blame someone
'The president will blame someone.'

According to Owusu (2022), *nó* takes a property and returns the unique familiar entity that satisfies that property on the condition that there is more than one satisfier of that property in the world. As such, (4) is infelicitous because there is only one sun in the solar system, (8) is infelicitous because *president* is interpreted as 'president of Ghana' and there is only one such individual. On the other hand, examples (6) and (7) are felicitous because they satisfy the weak familiarity presupposition and do not violate the anti-uniqueness presupposition; multiple entities possess the property of being a soldier in (6) or a man or woman in (7).

Bombi (2018) takes issue with Arkoh & Matthewson's (2013) assertion that *nó* presupposes weak familiarity while bare nouns encode uniqueness. She proposes that the core meaning of *nó* is uniqueness; it denotes the unique referent with the description content of the NP in a particular situation. "Informational uniqueness"—the notion of uniqueness relevant here—evaluates uniqueness relative to the informational state of the discourse participants, not the world (Roberts 2003). A uniqueness analysis, therefore, also correctly predicts that *nó* is felicitous in (6) and (7); in both cases, there is a unique entity with the NP property relative to the informational state of the discourse participants. Nevertheless, the analysis of uniqueness fails to account for the infelicity of *nó* in example (8), as there is indeed a single entity that possesses the characteristic of being a president. About the status of bare nouns, Bombi (2018) contends that given that the bare noun is always in object position in the examples provided by Arkoh & Matthewson (2013) to support their bare weak definite analysis, "it is unclear whether we are dealing with a generalized definite import of the bare noun or whether the alleged uniqueness examples are not instead pseudo-incorporating or weak definite constructions."

Although the core of their analyses differs, Arkoh & Matthewson (2013) and Bombi (2018) make similar predictions regarding the distribution of *nó*, and both

analyses explain all the contexts in which *nó* is present. Still, they cannot exclude the contexts in which *nó* is absent. The anti-uniqueness presupposition proposed by Owusu (2022) explains the contexts in which *nó* is considered infelicitous. Despite the inherent differences among the three theories, a shared consensus is that *nó* presupposes existence. In other words, when a speaker utters a sentence containing “DP *nó*,” they are committed to the existence of an entity that possesses the descriptive property of the NP.

3 The declarative clause-final *nó*

A variant of the nominal definite marker occurs in non-DP contexts such as clause-final in a relative clause (9).² In this context, Saah (2004) refers to it as a *clausal determiner* and glosses it as CD.³

- (9) Abofrá nó [_{RC} áà Kofi hú-u nó **no**] á-!ba.
 child DEF REL Kofi go-PST 3SG.OBJ CD PRF-come
 ‘The child whom Kofi saw has come.’ (Saah 2010: 94)

Previous literature, such as Amfo (2006), has regarded clause-final *nó* as a marker for subordinate clauses. However, Owusu (2022) demonstrates that this variant can also appear at the end of simple declarative sentences, as shown in example (10), where it cannot be analyzed as marking a subordinate clause.

- (10) **Context:** Kofi is having problems with linguistics. His parents have been encouraging him to ask their neighbor, Dr. Abrefa, for help, but Kofi has been too shy to ask him. This morning Kofi informs his mother that he has finally asked their neighbor for help. His mom reports to his father...
 [[_{TP} Kòfí nyá kò-hú-ù Dr. Àbréfá] énorà **nó**].
 ‘Kofi get MOT.-see-PST Dr. Abrefa yesterday DEF
 Kofi went to to see Dr. Abrefa yesterday.’ (Owusu 2022)

In (10), *nó* cannot be construed as a nominal definite determiner since proper names in Akan do not typically co-occur with definite determiners; it rather attaches to the whole clause.

The rest of this section explores the semantic contribution of the clause-final *nó* in the declarative sentence and its relation to the nominal determiner.

²See Bombi et al. (2019) for discussions suggesting that the clause-final *nó* in relative clauses functions at the NP level, indicating that the modification pertains to the NP rather than the clause. Nonetheless, refer to Owusu (2022) for counterarguments to this position.

³This should be interpreted as a categorization rather than an analysis.

3.1 A familiar clause

Like the nominal *nó*, clause-final *nó* is not felicitous in out-of-the-blue contexts. For instance, while both sentences in (11) have the same truth condition - being true if “Kofi has gone to see Dr. Abrefa” and false otherwise - only (11a) is felicitous at the beginning of a discourse.

- (11) a. Kòfi kó kò-hú-ù Dr. Àbréfà énorà.
Kofi MOT go-see-PST Dr. Abrefa yesterday
‘Kofi went to see Dr. Abrefa yesterday.’
b. Kòfi kó kò-hú-ù Dr. Àbréfà énorà **nó**.
Kof MOT go-see-PST Dr. Abrefa yesterday DEF
‘Kofi went to see Dr. Abrefa yesterday.’

A context, such as (12), must license the clause with *nó* (11b).

- (12) **Familiar context:** Kofi is having problems with linguistics. His parents have encouraged him to ask their neighbor, Dr. Abrefa, for help, but Kofi has been too shy to ask him. This morning, Kofi informs his mother that he has finally asked their neighbor for help. His mom *can* report the news to Kofi’s father by uttering (11b).

Owusu (2022) argues that clause-final *nó* encodes the information that the propositional content of the clause is familiar. She contends that it takes a TP complement, in contrast with previous literature such as Saah (1995), Boadi (2005), Ofori (2011), Bombi et al. (2019) who treat *nó* as an event determiner, i.e, it takes a VP complement. As an event determiner, the morpheme is informally described as marking an event as familiar or presupposed. While the VP analysis retains some validity, the declarative contexts under consideration are more amenable to a propositional analysis. I present one piece of evidence in favor of the propositional analysis.

Negation affects the accessibility of discourse referents for anaphoric reference, (Karttunen 1976). Individual discourse referents introduced below negation do not survive beyond the scope of negation. For instance, referring back to the car in (13) with either a definite description or pronoun is not possible.

- (13) a. Bill does not have a car_i.
b. [#The car]_i/#it_i is black. (Karttunen 1976: 4)

In addition to individual discourse referents, a sentence (14a) may introduce event and propositional discourse referents, as illustrated by (14a).

- (14) a. Ede didn't steal the cookies.
 b. [_{NegP} Ede didn't [_{TP} t_{Ede} t_{did} [_{vP} t_{Ede} t_{steal} steal the cookie
 ↪ d'_{prop} ↪ d''_{prop} ↪ d'''_{event} (Krifka 2013)
]]]

With event anaphora, reference to an event under negation is possible (15a). However, you cannot make an anaphoric reference to the negated event itself (15b).

- (15) a. Ede didn't [steal the cookies.]_i John did it_i.
 b. Ede [didn't steal the cookies.]_i #It_i was easy.

Propositional anaphora allow anaphoric reference to the positive clause and the negated clause. Both the NegP (16a) and TP(16b) introduce propositional discourse referents which are available for anaphoric reference.

- (16) Ede didn't steal the cookie. (Krifka 2013: 5)
 a. and he can actually prove *it*. → that he didn't steal the cookie.
 b. even though people believed *it*. → that he stole the cookie.

Hence, negation can serve as a criterion for distinguishing an anaphoric reference to a proposition from an anaphoric reference to an event. Clause-final *nó* takes negated clauses as complements (17). As the context shows, the only possible antecedent for *nó* clause is the negated sentence.

- (17) Context: Ama told Kofi not to sleep, and he didn't sleep.
 Kòfí á-ń-ná éńórà nó.
 Kofi PRF-NEG-sleep yesterday DEF
 'Kofi didn't sleep yesterday.'

Propositional analysis readily explains negative clause-final *nó* sentences by positing an anaphoric reference to the NegP propositional discourse referent. A VP analysis would require additional mechanics, particularly to define what it means for a negated event to be familiar, to account for this data.

3.2 Familiar but not in the common ground

This section focuses on the kind of contexts that can introduce antecedents for the clause-final *nó*. I begin with the kind of contexts where *nó* is infelicitous. Clause-final *nó* is incompatible with facts and universal truths (18).

- (18) a. Àwíá pùé wò àpùèí débíáá (#nó).
 sun come.HAB be.located east everyday DEF
 ‘The sun rises in the east everyday.’
 b. Accra wò Ghana (#nó).
 Accra be.located Ghana DEF
 ‘Accra is located in Ghana.’

Clause-final *nó* is also not infelicitous in contexts such as (19), where the sentence functions as a reminder.

- (19) **Context:** The department sent out a list of first years to everyone, and five people are on the list. There were no prior expectations about the number of people. Ama and Yaa both received this email, so they have this information. Ama cannot remember the number of first years and needs to be reminded. Yaa says...
 First yearsóó nó yè five àmpá (#nó).
 first years.PL DET COP five truly DEF
 ‘There are five first years, truly (as we know).’

What (18) and (19) have in common is that they are both contexts where the truth of the proposition is established prior to the utterance of the utterance. In other words, in both cases, the propositional content of the *nó*-clause is already in the *common ground*. The common ground (CG) includes all propositions that participants have publicly and mutually committed to, and whose truth participants are assumed to have taken for granted (Stalnaker 1978).

A typical context in which *nó*-declarative clauses are licensed is one where a previous issue has been raised and remains unresolved. The resolution of the issue is typically achieved through the use of the *nó*-clause. For instance, prior to Abena’s statement in (20), the question of whether or not Kofi had gone to the market remained unresolved.

- (20) **Context:** Kofi informed Ama and Abena, who are his roommates, that he would go to the market later in the day. Later in the evening, Abena is looking for him. Ama tells Abena...
 [[Kòfí nyá kò-ò dwám énorà] nó.]
 Kofi get go-PST market yesterday DEF
 ‘Kofi went to the market yesterday’ (as we said he will).

By uttering (20), Abena resolves this issue. Since the context before the *nó*-clause is such that the issue is unresolved, let us paraphrase it as (21). Modal

sentences make two propositions salient; the modal proposition itself (21a) and the prejacent of the modal (21b) (von Fintel & Gillies 2007).

- (21) Kofi will necessarily go to the market.
 a. ϕ = ‘that it is necessary that Kofi go to the market,’
 b. ψ = ‘Kofi goes to the market.’

Both propositions introduce propositional discourse referents, which are available for anaphoric reference. These propositions, however, have different effects on the context. According to von Fintel & Gillies (2007), the prejacent proposition is asserted content (added to the common ground), while the modalized proposition is “proffered” content (not added to the common ground directly). As demonstrated earlier, declarative *nó* cannot refer back to common ground propositions. Thus, only the prejacent proposition remains accessible to the determiner. As expected, Ama’s response to Abena in (20) is interpreted as referring to the prejacent proposition rather than the modal proposition.

What does this indicate about the status of the propositions that could potentially be antecedents for *nó*? Snider (2017) presents a list of propositions that introduce discourse referents and can be referred to later in the text. More significant for our purposes, he highlights that these propositions may hold varying degrees of speaker commitment. Discourse participants are committed to the truth of the propositional content of the discourse referent a declarative sentence introduces. However, propositional contents of discourse referents introduced by the prejacent of negation are asserted to be false, and speakers do not make any commitment to the propositional content of discourse referents introduced by the prejacent of epistemic modals. Of these three propositions discussed, only the ones introduced by declarative sentences are in the common ground. In other words, discourse participants can be mutually aware of a proposition without committing to it. Portner (2008) models the space where information that participants in a conversation are mutually aware of but not committed to as the *common propositional space* (CPS). Owusu (2022) concludes that in declarative sentence contexts, *nó* makes anaphoric reference to CPS propositions. Clause-final *nó* declarative sentences make two semantic contributions (22):

- (22) a. *Assertion*: ϕ -*nó* asserts ϕ .
 b. *Presupposition*: the propositional content of their complement has an antecedent in the discourse (familiarity).

Owusu (2022) argues that the restriction that common ground propositions cannot be antecedents of *nó*-clauses arises from the fact that clause-final *nó*

declarative sentences, like ordinary declarative sentences, have the illocutionary force of an assertion. Assertions are “proposals to change the common ground in a certain way” (Stalnaker 2014: 39) and such one cannot reassert information already in the common ground. Owusu’s (2022) propositional analysis is thus different from Wespel (2008) who argues that the Haitian Creole clausal determiner has a meaning akin to Himmelfmann’s (1996) recognitional use of nominal definite description; their role is to highlight the information that is already common knowledge and has some significance to the present conversation.

4 The wh-question clause-final *nó*

Ex-situ wh-questions in Akan may also license clause-final *nó*. Akan employs two main strategies for constructing wh-questions: the ex-situ strategy, which involves fronting with morphological marking (23a), and the in-situ strategy, which does not use fronting and lacks morphological marking (23b). Ex-situ wh-questions have five features: (i) the wh-word is located at the left periphery of the clause, (ii) the focus particle *na* immediately follows the wh-word, (iii) a resumptive pronoun occurs at the canonical position of the wh-word (Marfo & Bodomo 2005), (iv) high tone overwrites low tone on the root (Korsah & Murphy 2020), and (v) the copular phrase *è-ýé* ‘it is’ may optionally precede the question word.

- (23) a. (è-yè) héná nà Kòfí hyiá-à nó (nó)? EX-SITU QUESTION
3SG.SBJ-COP who FOC Kofi meet-PST 3SG.OBJ DEF
'Who did Kofi meet?'
- b. Kofi hyiá-à héná (*nó)? IN-SITU QUESTION
Kofi meet-PST who DEF
'Who did Kofi meet?'

The clause-final *nó* only occurs in ex-situ questions.

Whether the two strategies have distinct underlying syntax is a subject of ongoing discussion. Saah (1995) argues there is no structural complexity difference between in-situ and ex-situ questions in Akan, while Boadi (2005), Kobele & Torrence (2006), Larrey (2020) contend that ex-situ wh-questions are structurally complex because they involve movement. For instance, Kobele & Torrence (2006) demonstrates that ex-situ questions show intervention effects with operators such as *negation*, *only* and *even*, but in-situ do not. The syntactic distinction between ex-situ and in-situ strategies and how this distinction contributes to the

(in)ability to license *nó* is an important question; however, I will leave that for future research. This section focuses on the semantic contribution of *nó* in ex-situ questions.

4.1 Interpreting a wh-question without *nó*

Regarding the meaning, Marfo & Bodomo (2005), building upon previous assertions by Boadi (1990), contends that there is no discernible distinction between these ex-situ (24a) and in-situ (24b) wh-questions. They argue that the focus marker *na* does not add additional meaning in ex-situ questions, as Q-words are already inherently focus-marked.

- (24) a. Héná nà Kòfí bísá-à nó?
 who FOC Kofi ask-PST 3SG.OBJ
 ‘Who did Kofi ask?’
 b. Kòfí bísá-à héná?
 Kofi ask-PST who
 ‘Who did Kofi ask?’

If we adopt a Hamblin (1976)-Karttunen (1977) semantics of questions, according to which the meaning of questions is the set of possible answers, then the fact that the questions in (24) have the same meaning means that they elicit the same possible answers. Suppose we have two individuals in our context named *Kwame* and *Ama*, then the set of possible answers for (24a) and (24b) includes all of the propositions in (25):

- (25) { Kofi asked Ama, Kofi asked Kwame, Kwame FOC Kofi asked, Ama FOC Kofi asked, Kofi didn’t invite anyone }

Both the ex-situ and in-situ questions allow for focus and declarative answers. Further, they both permit a negative answer *nobody*.

In general wh-questions are assumed to presuppose that at least one of its Hamblin answers is true (Katz & Postal 1964). Therefore, wh-questions express an existential presupposition, as illustrated in the examples in (26).

- (26) a. Who smokes? $\Rightarrow$ Someone smokes. ($\Rightarrow$: ‘presupposes’)
 b. Which semanticists danced? $\Rightarrow$ Some semanticist danced. (Uegaki 2021: 2)

Abusch (2010) argues that the existential presupposition associated with wh-questions is defeasible and thus treats it as a *soft presupposition*.⁴ Dayal (2016) points out that the existence presupposition is easily cancellable, particularly across multiple speakers. By responding with “nobody” in (27), speaker B cancels the presupposition that an answer to the question exists.

- (27) Speaker A: Who did Kofi ask?
Speaker B: Nobody.

The existence presupposition may be suspended, even with the speech of a single speaker. For example, a speaker might start a question by explicitly stating that an answer may not exist (28).

- (28) I don’t know if Kofi asked anyone but if he did, who did he ask?

Considering that both in-situ and ex-situ wh-questions in Akan allow for negative answers, we can conclude that they both trigger a soft existence presupposition. They are comparable to canonical questions in English. Moreover, given that they allow the same possible answers, Marfo & Bodomo’s (2005) assertion that they are semantically equivalent is right. However, it must be emphasized that this only applies to ex-situ wh-questions without the copular *ε-yε* ‘it is’ phrase. When the copular phrase is present, the wh-question is interpreted as a cleft question with hard existence presupposition. Like canonical wh-questions, cleft wh-questions convey an existence presupposition, but in this case, the presupposition cannot be canceled even across multiple speakers (Dayal 2016).

- (29) Speaker A: Who is it that Kofi asked?
Speaker B: #Nobody.

Speaker B cannot respond to the question in (29) with *nobody*.

4.2 Interpreting clause-final *nó* in wh-questions

As mentioned in the previous section, ex-situ wh-questions without the copular phrase *ε-yε* ‘it is’ trigger a soft existence presuppose, and therefore allow negative answers. In contrast, clause-final *nó* ex-situ wh-questions do not allow negative answers across speakers (30) or the existential presupposition to be suspended (31).

⁴Due to the defeasible nature of these presuppositions, Brandtler (2019), for instance, treats them as conversational implicatures.

- (30) a. Speaker A:
Hénfòò ná Kòfí bísá-à ómmó *nó*?
who.PL FOC Kofi ask-PST 3PL DEF
‘Who all did Kofi ask?’
- b. Speaker B:
*ò-è-m-mìsá òbíárá.
3SG.SBJ-PRF-NEG-ask anyone
‘He did not ask anyone.’
- (31) I don’t know if Kofi asked anyone but ...
Hénfòò ná Kòfí bísá-à ómmó (**nó*)?
who.PL FOC Kofi ask-PST 3PL DEF
‘Who all did Kofi ask?’

Clause-final *nó* wh-questions, therefore, pattern with cleft questions in English and wh-questions with overt cleft markers in Akan.

Based on the sameness in meaning, Kobele & Torrence (2006), for instance, have proposed that *nó* wh-questions are covert cleft questions. In other words, every *nó* wh-question is preceded by *è-yé* and *nó* is a cleft definite determiner. Their analysis builds on analyses of cleft constructions that relate them to definite descriptions (Hedberg 1990, 2000, Daniel & Križ 2013, Hedberg 2013, Križ 2017, Tieu & Križ 2017). While this analysis is reasonable, it fails to account for the declarative *nó* and how it relates to the wh-question *nó*.⁵

Wh-question *nó*, I argue, takes the same semantic object as complement in both declarative sentences and wh-questions. *Nó* in a question presupposes that some proposition is familiar, that is, an antecedent for the proposition exists. In a question such as (30a) what is presupposed to be familiar is the existential closure (32a) of the open proposition in (32b).

- (32) a. Existential closure: $\exists x$.Kofi asked x
b. Open proposition: Kofi asked $x : x \in person$

I propose that (30a) has the syntax in Figure 1, not Figure 2. In other words, I assume *nó* attachment occurs prior to the formation of questions.

Since the clause-final *nó*-question presupposes the familiarity of the existential closure of the TP proposition, it is not felicitous in out-of-the-blue contexts such as (33) and although not obligatory in familiarity contexts such as (34), it is preferred.

⁵Also see Grubic et al. (2019) for an analysis of ex-situ focus constructions with and without *nó* in Akan and their presuppositions.

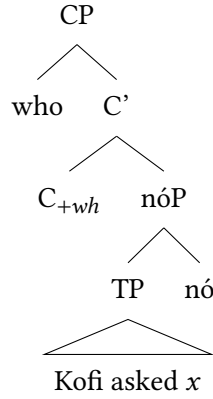


Figure 1: ✓ *Nó* attaches to the TP before the C_{+wh}

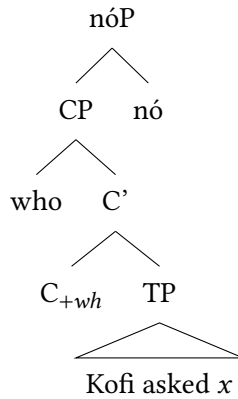


Figure 2: # *Nó* attaches to the question CP

- (33) **Context:** We are looking for volunteers to represent the department at the dean's annual party.

Héná nà í-bé-kó (*nó)?

who FOC 3SG.SBJ-FUT-go DEF

'Who will go?'

- (34) **Context:** Kofi informed us this morning that he asked someone about an issue we had. But I have forgotten who he claimed to have spoken to.

Héná nà Kòfí bísá-à nò nó?

who FOC Kofi ask-PST 3SG.OBJ DEF

'Who did Kofi ask?'

Owusu (2022) argues that in declarative sentences, the proposition discourse referent of *nó*-clauses although familiar, must not be in the common ground. As a result, *nó*-declarative sentences cannot be used to remind someone of a proposition they have forgotten. Clause-final *wh nó*-questions, however, as illustrated in (34), are typically used when speakers want to be reminded of information they have forgotten. The status of the propositional content of the antecedent of clause-final *nó*, i.e., whether it is in the common ground or not, is not a property of the morpheme itself. Rather, it arises through interaction with the speech act it is embedded within. Since one cannot assert a common ground proposition (Stalnaker 1976), that is ruled out in *nó* assertions, but one can question a proposition in the common ground.

5 Conclusion

I have argued that clause-final *nó* in ex-situ *wh*-questions in Akan presupposes familiarity with the existential closure of the question nucleus. Even with a familiarity presupposition, these questions satisfy the felicity conditions of the speech act of questioning. The speaker seeks information they do not know and believes the addressee does know. They are comparable to cleft questions in terms of semantics. *Wh*-questions with clause-final *nó* do not permit the cancellation or suspension of the existence presupposition, unlike *wh*-questions without clause-final *nó*.

Furthermore, I have also demonstrated that *nó* in all three contexts shares a core meaning: a weak familiarity and existence presupposition. *Nó* functions as an anaphoric marker and thus requires an antecedent with the same descriptive content as its complement. In the nominal domain, this discourse referent is an individual, whereas in the clausal domain, it is a proposition.

One question that remains open is why in-situ *wh*-questions are incompatible with clause-final *nó*. This distribution of *nó* is especially puzzling for analyses that contend that in-situ and ex-situ *wh*-questions have the same structural complexity and that in-situ strategies involve covert movement.

Abbreviations

3	third person	DEF	definite
COP	copular	FOC	focus
CONJ	conjunction	FUT	future
CONS	consecutive marker	INDF	indefinite

MOT	motional prefix	PRF	perfect
NEG	negation	PROG	progressive
OBJ	object	SBJ	subject
PL	plural	SG	singular
PST	past		

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Chapter 7

The descriptive and syntactic properties of the *ànáá* particle(s)

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This paper gives an overview of the distribution(s) and interpretation(s) of the particle *ànáá* in Akan. This particle has previously been discussed as a question particle, distinguishing a polar question from a declarative statement. In this paper, I identify three additional syntactic and semantic core functions of the particle, namely (a) tag function, (b) disjunctive marker, and (c) discourse particle, and provide the semantic and syntactic properties for each of them. By syntactic analysis, I propose that each function can be situated on a different syntactic slot along the left periphery based on previous work by Dayal (2023), thus highlighting the relevance of this latter paper to the study of information structure. Finally, the paper also provides a cross-linguistic perspective by showing how the patterns of *ànáá* in certain respects are comparable to particles from other languages such as the German *wohl* and the Hindi *kya*.

1 Introduction

The particle *ànáá* has been discussed extensively as a question particle in Akan (see Dolphyne 1988, Boadi 1990, Marfo 2005, Kügler 2017, etc.). However, apart from functioning as a question particle, *ànáá* can occur in other instances and have different functions, such as indicating a tag, indicating the speaker's attitude, and disjoining nouns and sentences. For example, consider (1)

- (1) a. *Kòfí kɔ-ɔ sùkúù mú hó ànáá.*
Kofi go-PST school LOC LOC QPRT
'Did Kofi go to the school?'



- b. Kòfí ànáá Ám¹á nà è kó-ò sùkúù mú h́.
Kofi or Ama FOC 3SG go-PST school LOC LOC
'It was either Kofi or Ama who went to the school.'
- c. ànáá, Kòfí kò-ò sùkúù mú h́.
PRT Kofi go-PST school LOC LOC
'Probably Kofi went to the school.'
- d. Kòfí kò-ò sùkúù mú h́, ànáá?
Kofi go-PST school LOC LOC, TPRT
'Kofi went to the school, didn't he?'

In (1), apart from the question marking function (1a), there are three other instances of *ànáá*, one as a disjunctive marker (1b), as a discourse particle (1c) and as a tag (1d). These other instances have received little or no discussion in the literature (to my knowledge). Are all these instances in (1) evidence of the same particle or distinct particles? This study discusses four prominent *ànáá* cases and argues that these four particles have distinct semantics and occupy different syntactic slots within the clausal spine. The following sections are organized as follows: Section 2 discusses *ànáá* as a question particle. Section 3 discusses *ànáá* as a disjunctive marker. Section 4 examines *ànáá* as a tag question. Finally, Section 5 will focus on *ànáá* as a discourse particle, and Section 6 concludes.¹

2 Question marking function

The *ànáá* particle in Akan can function as a question marking particle. The particle marks polar questions and subordinate interrogative questions. In this section, I focus on the question marking function of the particle and propose its syntactic position in light of Dayal (2023)'s left periphery cartography.

2.1 Polar questions

Polar questions in Akan are either prosodically or morphologically marked (cf. Dolphyne 1988, Boadi 1990, 2005, Marfo 2005, Genzel 2013, Kügler 2017). Morphologically, according to Marfo (2005), polar questions in Akan are encoded using

¹For data and judgments, the study relied on four native Akan (Akwapim, Kwahu, and Asante dialects) speakers between the ages of 20–35 years. The elicitation was twofold: (1) speakers were given a context and asked if the target *ànáá* construction is compatible with the said context; (2) speakers were given the context and were asked whether they would use *ànáá* in any of the contexts. In addition to the consultants, the author's self-introspection was also relied upon. Examples are written in Akan orthography, but tone has been added.

two question particles (QPrt) namely *ànáá* and *àsò*. These particles differ in their clause position and are in complementary distribution. For example, *ànáá* occurs in the final position (2a), while *àsò* occurs in sentence-initial position (2b).²

(2) Akan (Marfo 2005:32)

a. Yàw à-dá *ànáá*?

Yaw PRF-sleep QPRT

‘Has Yaw slept?’

b. Àsò Yàw à-dá?

QPRT Yaw PRF-sleep

‘Has Yaw slept?’

In terms of prosody, polar questions are characterized by a raised pitch register and a sentence-final downward glide which Kügler (2017) and Marfo (2005) call a Low Intonation Phrase Boundary Tone (L%). Like the morphological particles, the L% distinguishes between interrogative and declarative statements. See Kügler (2017) and Genzel & Kügler (2020) for further exploration of the prosodic distinction between declarative and polar questions.

2.2 Subordinate interrogative clause

Aside from marking polar questions, *ànáá* also marks subordinate interrogative clauses. The complementizer *sê* in Akan does not distinguish between declarative and interrogative complements. Having *ànáá* in an embedded clause is sufficient to achieve an interrogative interpretation. For instance, let us consider the complements of a responsive predicate ‘know,’ which can take both declarative [C_{-WH}] and interrogative complements [C_{+WH}] (as shown in 3 and 4). In (3) and (4), the complementizer remains the same, however, the presence of the *ànáá* particle in (3) triggers an interrogative reading which is absent in (4).

(3) Kòfí nìm sê Ám¹á á-dà *ànáá*.

Kofi know COMP Ama PRF-sleep QPRT

‘Kofi knows whether Ama is asleep.’

(4) Kòfí nìm sê Ám¹á á-dà.

Kofi know COMP Ama PRF-sleep

‘Kofi knows that Ama is asleep.’

²It should be noted that *àsò* is not available in many of the Akan dialects and among many younger speakers. Note that the original transcription of *ànáá* in Marfo & Bodomo (2005) was *àná* with a short final vowel.

The presence of the question particle in the complements also helps distinguish between the equivalent of the rogative predicate ‘wonder’ and the anti-rogative predicate ‘think’ in Akan. These predicates have the same morphological form *dwéné*. Their difference depends on the presence or absence of a question particle in their complements, as shown in (5) and (6). In (5), the predicate *dwéné* takes *ànáá* interrogative complement, which is more like a rogative predicate ‘wonder,’ while (6) looks more like an anti-rogative predicate ‘think.’

- (5) Kòfí dwénè-è sê Ám¹á á-dà ànáá.
 Kofi think-PST COMP Ama PRF-sleep QPRT
 ‘Kofi wondered if Ama was asleep.’
- (6) Kòfí dwénè-è sê Ám¹á á-dà.
 Kofi think-PST COMP Ama PRF-sleep
 ‘Kofi thought that Ama was asleep.’

The ability of the *ànáá* particle to mark both polar question and subordinated interrogatives, therefore, indicate that the particle is indeed a clause-typing particle in keeping with the analysis in Marfo (2005) and Kügler (2017), and not just a polar question particle, as Bhatt & Dayal (2020) and Dayal (2023) proposed for the Hindi *kya*.

Let us dedicate a few lines to distinguish between a clause-typing particle and a polar question particle using Akan *ànáá* and Hindi’s *kya*. A clause-typing particle is a particle that distinguishes between clause types.³ I propose that *ànáá* is a clause-typing particle that distinguishes between declarative and interrogative clauses. A polar question particle, on the other hand, marks only polar questions and does not have the property to distinguish between clause types. Bhatt & Dayal (2020) propose Hindi *kya* as a polar question particle. This particle can only appear with polar questions and can also be optional. Bhatt & Dayal (2020) also observe that *kya* is ungrammatical with responsive predicates such as ‘to know’ which can take both declarative (C_{-WH}) and interrogative complements (C_{+WH}) as shown in (7).

³See Cheng 1997 for discussions on clause-typing hypothesis.

(7) Hindi (Dayal 2023: 10)

- a. ravi ja:nta: hai [ki anu ja:egi:.]
Ravi knows [SUB Anu will.go]
'Ravi knows that Anu will go.'
- b. *ravi ja:nta: hai [ki kya: anu ja:egi:.]
Ravi knows [SUB PQP Anu will.go]
Intended: 'Ravi knows that Anu will go.'

The particle *kya:* is only compatible with predicates that take only C_{+WH} such as *to ask* as in (8a) and *the question is* (8b).

(8) Hindi (Dayal 2023: 11)

- a. Ti:char-ne anu-se pu:cha: [ki kya: vo ca:i piyegi:.]
teacher-ERG Anu-INS asked [SUB PQP she tea will.drink]
'The teacher asked Anu if she will drink tea.'
- b. sava:l yeh hai [ki kya: nayi: vyavastha: ka:gar sa:bit
question this is [COMP PQP new arrangement effective prove
hogi:.]
will.be]
'The question is whether the new arrangement will prove effective.'

Bhatt & Dayal (2020), thus, dubbed *kya:* as a polar question particle that is only compatible with predicates that take only $[C_{+WH}]$. Akan *ánáá* differs from Hindi *kya:*, in that *ánáá* is compatible with both responsive (in which *ánáá* does clause type-distinction between declarative and interrogative) and rogative predicates. Table 1 summarizes the difference between the two particles regarding the predicate they can occur with.

Table 1: The differences between Hindi *kya:* and Akan *ánáá*

Particle	Responsive predicate	Rogative predicate
<i>kya:</i>	✗	✓
<i>ánáá</i>	✓	✓

2.3 Neutral versus biased questions

Apart from polar and subordinated interrogative, there are other question categorizations (precisely, neutral and bias questions) that are worth mentioning in discussing question particles in Akan.

The difference between neutral and biased questions is observed in English between rising declarative and subject inversion questions. According to Buring & Gunlogson (2000) and Gunlogson (2008), in English, a rising declarative, as in (10) (a biased question), cannot be uttered out of the blue. That is, it requires a well-defined context for it to be uttered. However, a neutral question such as the subject inversion question in (9) can be uttered out of the blue.

(9) Is it raining? Neutral Q

(10) It is raining? Biased Q

Using the idea of commitment, Gunlogson (2008) proposes that with a rising declarative in English, the speaker has a tentative commitment to the proposition and is only seeking ratification from the addressee who has authority or expertise in a said proposition. With the subject inversion question, on the other hand, the questioner has no such commitment to questions and only poses the question out of ignorance.

In Akan, both the morphologically *ànáá*-marked and marked prosodic questions have neutral readings. These questions can be uttered out of the blue and embedded under rogative predicates such as ‘ask’ as in (11) and (12).

(11) Kòfí bisá-à sê Ám'á á-dà ànáá.
Kofi ask-PST COMP Ama PRF-sleep QPRT
‘Kofi asked if Ama was asleep.’

(12) Kòfí bisá-à sê Ám'á á-dà. (L%)
Kofi ask-PST COMP Ama PRF-sleep
‘Kofi asked if Ama was asleep.’

We will defer the discussion on biased questions in Akan to Section 5.1. In sum, *ànáá*-marked polar questions have neutral readings similar to the prosodically marked polar questions.

2.4 Possible syntactic position

To indicate the syntactic position of the *ànáá*, we subscribe to Dayal (2023)’s cartography of the left periphery. Dayal (2023) proposes splitting the interrogative

left periphery into three “layers”: Speech Act Phrase (SAP), Perspective Phrase (PerspectiveP), and the Complementizer Phrase (CP), as in (13).

- (13) [SAP [SA_{ASK} [PerspectiveP Persp_{CQ} [CP C_{O+WH} [TP...]]]]]

Let us discuss each layer further. Dayal (2023) defines C_{O+WH} as the position where declaratives and interrogatives are distinguished. For example, in English, an embedded interrogative is distinguished from an embedded declarative at C_{O+WH} position by the complementizers *whether* or *if* as in (14).

- (14) John asked [whether it will rain] Embedded interrogative

- (15) John knows [that it will rain] Embedded Declarative

The perspective phrase (PerspectiveP)/layer introduces an individual for whom the embedded CP is active. For instance, in (16) the “seeker” of the information conveyed by the embedded question is the matrix subject *every physicist*. Thus, the matrix subject is said to be the perspectival center or the judge (cf. Pearson 2013) and the embedded CP (or the centered question (CQ)) is potentially active for the subject as in (17). Such embedded questions are said to be quasi-subordinated (cf. Dayal 2023).

- (16) Every physicist_i wonders [will he_i be awarded the Nobel prize].

- (17) a. $\llbracket \text{PerspCQ} \rrbracket = \lambda Q \lambda x: Q \text{ is P(potentially)-ACTIVE for } x. Q$
 b. $\forall x \forall Q [P\text{-ACTIVE-for}(Q, x) \rightarrow \neg \text{know}(x, \text{Ans}(Q))]$

At the Speech Act level (SAP), however, the question denoted by the interrogative becomes a requested information from the speaker to an addressee. In other words, the speaker requires an answer from an addressee. An example of a Speech Act Phrase structure is a matrix question such as (18).

- (18) Will he be awarded the Nobel prize?

In (18), the speaker puts the addressee under an obligation to provide an answer to the question provided. Therefore, the semantics of the Speech Act Layer is defined as in (19).

- (19) a. $\llbracket \text{SA}_{\text{ASK}} \rrbracket = \lambda Q \langle \langle s, t \rangle, t \rangle \lambda x \lambda y: y \text{ puts } x \text{ under obligation to } \text{ASSERT}(\text{Ans}(Q)). Q$
 b. $\llbracket \text{SAP} \rrbracket = \text{Speaker}_C \text{ puts addressee }_C \text{ under obligation to } \text{ASSERT}(\text{Ans}(Q)). Q$

In summary, with Dayal's (2023) cartography, the CP layer houses questions that are fully embedded, the perspective layer houses quasi-subordinated questions, and the Speech Act layer houses matrix questions.

I propose that the *ànáá* question particle occurs as the head of a CP as in (20) in keeping with the property of being a clause typing particle, that is, distinguishing between declaratives and interrogatives.

(20) [SAP [SA_{ASK} [PerspectiveP Persp_{CQ} [CP C_{O+WH} *ànáá*[TP...]]]]]

In semantics, following Bolinger (1978) and more recently Dayal (2023), we assume that the basic meaning of a polar question is a singleton propositional set as in (21).

(21) $\llbracket C_{+WH} \rrbracket = \lambda q \lambda p [q=p]$

This singleton set, however, gets coerced into plural propositional sets, and the coercion in English is triggered by the presence of *whether* or *if* or *se* in Italian (cf. Dayal 2023).

In Akan, we assume that polar questions (cf. 22) begin as single propositions—formalized in (23a) and represented in (23b) and are then shifted to a set of propositions by the presence of *ànáá* as shown in (23c).

(22) Yàw à-dá *ànáá*
 Yaw PRF-sleep QPRT
 'Has Yaw slept?'

(23) a. $\llbracket C_{+WH} \rrbracket = \lambda q \lambda p [q=p]$
 b. $\llbracket TP \text{ Yàw à-dá} \rrbracket = \{\wedge \text{Yaw has slept}\}$
 c. $\llbracket CP \text{ Yàw à-dá } \textit{ànáá} \rrbracket = \{\wedge \text{Yaw has slept}\} \{\wedge \neg \text{Yaw has slept}\}$

In effect, *ànáá* is the overt realization of the clause-typing C_{+WH} head of CP.

Another function or instance of *ànáá* is the disjunctive function, which is discussed below.

3 Disjunctive function

3.1 *ànáá* as a disjunctive marker

ànáá in Akan also appears as a disjunctive marker. It combines both word level and phrasal level categories. For instance, it can combine simple nominals as in (24), TPs/CPs as in (25) and PPs as in (26).

- (24) Kòfí tò-ò { m̀pá } *ànáá* { m̀pá dùá. } [NPs/DPs]
 Kofi buy-PST { bed } or { bed wood }
 ‘Kofi either bought a bed or a bed frame.’
- (25) { Wó pè s̀ká } *ànáá* { wó pè èdán. } [TPs/CPs]
 { 2sg want money } or { 2sg want house }
 ‘Either you want money, or you want a house.’
- (26) Kòfí hyè-è { àkònwá nó ásé } *ànáá* { àkònwá nó àk̀yí } [PPs]
 Kofi hide-PST { chair DET under } or { chair DET behind }
 ‘Kofi hid under the chair or behind the chair.’

It should be mentioned that speakers consider combining verbs or verb phrases with the disjunctive marker ungrammatical or downgraded as shown in (27).⁴

- (27) *Kòfí { k̀ò-ò s̀úkúù } *ànáá* { tèná-à fié } [VPs]
 Kofi { go-PST school } or { stay-PST home }
 ‘Kofi went to school or stayed at home.’

The disjunctive marker and the question *ànáá* particle can co-occur and this results in disjunctive questions discussed below.

3.2 Disjunctive questions

In English, non-wh disjunctive questions have been considered ambiguous between polar questions (POLQ) and alternative questions (ALTQ). The ALTQs, according to Biezma & Rawlins (2015), come with a strong presupposition for an exhaustive alternative. That is, the possible answers to the ALTQs can only correspond to the propositional disjuncts. For example, in (28b), the possible answers can only be *coke* and *iced tea*. With the polar questions (POLQ), such presupposition is absent. The possible answers to these questions are non-exhaustive. For instance, the answer alternatives in (28a) can be *coke*, *iced tea*, *coffee*, or *water*.

- (28) a. Did Sandy want Coke L *H– or iced tea L *H–H%? [Polar question]
 b. Did Sandy want Coke L *H or iced tea H *L–L%? [Alternative question]

According to Bartels (1997), these two questions also can be differentiated by pitch accent, prosodic break, and final intonation, as shown in (28a) and (28b).

⁴We will leave the discussion on why this is the case to future studies.

Whereas (28b) is characterized by a pitch accent on the first disjunct and a final falling contour, (28a) lacks a pitch on the first disjunct and has a final rising contour.

In Akan, the disjunctive marker *ànáá* can occur with the question marker *ànáá* to derive disjunctive questions. However, unlike English, the final intonation of *ànáá*-marked disjunctive question cannot distinguish between polar disjunctive and alternative questions. For instance, according to Kügler (2017) the shape of the final contour for *ànáá*-marked questions is determined by the particle *ànáá*. This, therefore, implies that a question with a final question particle will always have its final contour determined by the contour associated with the particle. Given such an observation, final intonation may not distinguish between polar and alternative questions in Akan.

The difference between polar disjunctive and alternative questions can, however, be observed in the differences between an *ànáá*-marked disjunctive question and a prosodically marked disjunctive question. Whereas the disjunctive marker and a question particle (*ànáá*) can only result in non-exhaustive lists or open alternatives (polar question (POLQ)), as shown in (29), the prosodically marked disjunctive questions result in closed alternatives or exhaustive lists (alternative question (ALTQ)) as shown in (30).

- (29) a. Mike:
 wó nà wó bé nóá pástà nó *ànáá* ènám nó *ànáá*?
 2SG FOC 2SG FUT cook pasta DET or meat DET QPRT
 ‘Are you the one going to prepare the pasta or the meat?’ (POLQ)
- b. John:
 dààbí, mé yé stèw nó
 no, 1SG.FUT do stew DET
 ‘No, I will be preparing the stew.’
- (30) a. Mike:
 wó nà wó bé nóá pástà nó *ànáá* ènám nó? (L%)
 2SG FOC 2SG FUT cook pasta DET or meat DET
 ‘Are you the one going to prepare the pasta or the meat?’ (ALTQ)
- b. John:
 #(dààbí,) mé yé stèw nó
 no, 1SG.FUT do stew DET
 ‘No, I will be preparing the stew.’

In (29), the alternatives are not limited to just the mentioned alternates but the possible alternates are endless, and thus, John’s response is appropriate. In (30b),

however, all my consultants consider John's response as inappropriate, which thus, indicates that speakers consistently rate the disjunctive question marked by *ànáá* as polar disjunctive question and the disjunctive question marked by prosody as an alternative question that elicits exhaustive alternatives or closed alternatives.

Another reading which results from the disjunctive marker is the difference between choice and cancellation reading, and these are discussed in the following section.

3.3 Choice and cancellation reading

Szabolcsi (2015), Hirsch (2018), and more recently, Dayal (2023) observe that a question with disjunction at the clausal level allows for two readings: choice reading and cancellation reading. With the choice reading, the options are available for the addressee to select from the listed options. With the cancellation reading, on the other hand, the speaker cancels his first assertion and then offers the second assertion as the most preferred choice.

In Akan, the *ànáá* disjunct allows only for the choice reading in which the speaker expects the addressee to select from the available options. The cancellation reading is absent when the disjunct is *ànáá*. The cancellation reading is only available when the disjunct occurs with *sê* as in *ànáá sê(è)* as shown in (31) and (32).⁵

- (31) Yè frě wò sén ànáá wò frì hén?
 1PL call 2SG what or 2SG come where
 'What is your name or where are you from?' [choice reading]
- (32) yè frě wò sén ànáá sê wò frì hén
 1PL call 2SG what or sê 2SG come where
 'What is your name, or rather, where are you from?' [cancellation reading]

Let us consider (31) and (32) in the following context adapted from Dayal (2023): A secretary wants to retrieve data on somebody and asks the person to either provide his name or where he comes from. Here, (31) is used. The speaker, in this context, leaves the decision to the addressee. In (32), the speaker cancels the first statement and is now interested in the second statement (the reading is more pronounced when there is a pause after the first option). The cancellation reading is unavailable in (31) even if a pause follows the disjunct. According

⁵The *sê* in *ànáá sê* differs from the complementizer *sê*. I gloss it as *se*, differentiating it from the complementizer, which I gloss as *COMP*.

to Dayal (2023) the cancellation reading is only available for Speech Act Phrase (SAP) disjunctions. For example,

(33) Mary is asking what is your name? or what is your SSN?

(34) [SAP Mary is asking what is your name] or rather [SAP Mary is asking what is your SSN.]

Here, if we interpret (33) as (34), then here the speaker cancels his first assertion, ‘Mary is asking what is your name?’ and opts for the second assertion of the disjunct Mary is asking, ‘what is your SSN?’.

Since the cancellation reading is unavailable for the *ànáá* disjunct, I, therefore, propose that *ànáá sè* is an SAP disjunct in Akan following Dayal (2023) and *ànáá* a disjunct for combining structures lower than the Speech Act Phrase, as shown in (35).

(35) [SAP [[CP *ànáá* CP][[TP *ànáá* TP]...]]]

It should be noted that although the question particle and the disjunctive marker appear homophonous, they can co-occur as mentioned earlier. The co-occurrence of *ànáá* as the question particle and the disjunctive marker as in (29), repeated here as (36), indicates that these particles are independent.

(36) Mike:

wó nà wó bé nóá pástà nó *ànáá* ènám nó *ànáá*?

2SG FOC 2SG FUT cook pasta DET or meat DET QPRT

‘Are you the one preparing the pasta or the meat?’

According to Bailey (2013), in languages whose question particles tend to be homophonous to their disjunctive marker, the question particles are primarily heads of the Disjunct Phrase. As such, these particles cannot introduce an embedded clause. The question particle, however, can be embedded, as shown in (3) repeated as here as (37).

(37) Kòfí nìm sê Ám¹á á-dà *ànáá*.

Kofi know COMP Ama PRF-sleep QPRT

‘Kofi knows whether Ama is asleep.’

The embeddability property of the question particle provides evidence to consider the *ànáá* question particle distinct from the *ànáá* disjunct.

Finally, cross-linguistically, Akan *ànáá* patterns with other languages such as Japanese *ka* (Jayaseelan 2008) and Hausa *kō* in having a question particle and

disjunctive markers which are homophonous. These are exemplified in Hausa as follows:

- (38) Hausa (Cowan & Schuh 1976: 216 cited in Heine & Kuteva 2002: 226 and Bailey 2013: 39)

- a. *kō nī kī kai*
 either 2:SG or I
 ‘either you or I’
- b. *kō kā sàmi gyàdà màiyawà?*
 Q you get peanuts many
 ‘Did you get a lot of peanuts?’

4 Tag function

4.1 Tag questions

Another instance of *ànáá* in Akan is used to request confirmation. This function of *ànáá* appears in the form of a tag. Tag questions in Akan have two parts: the declarative (anchor) and the reduced interrogative question or the tag, just like in English (cf. Tottie & Hoffmann 2006 and Kim & Ann 2008). The tag in Akan can be a tag phrase or only *ànáá* (which I will gloss from hereon as tag particle (TPRT)). Let us first discuss the tag phrase in detail.

4.1.1 Tag phrase

The tag phrase in Akan consists of a third-person inanimate pronoun as the subject, the verb ‘be like’, the demonstrative pronoun ‘that,’ and the question particle ‘*ànáá*’. The subject does not agree in phi features with the subject in the anchor. It is a form of dummy ‘it.’ Only the demonstrative pronoun *sàá* has a referential use, and here it refers to the whole proposition. (39) and (40) exemplify the tag phrases in their negative and positive forms.

- (39) *Kòfí kò-ɔ̀ sùkúù énorà, è-n-té sàá ànáá?*
 Kofi go-PST school yesterday, 3SG.INANIM-NEG-be.like DEM QPRT
 ‘Kofi went to school yesterday. Is it not like that?/is it not the case?’
- (40) *Kòfí á-n-kó sùkúù énorà, é-té sàá ànáá?*
 Kofi PRF-NEG-go school yesterday, 3SG.INANIM-be.like DEM QPRT
 ‘Kofi didn’t go to school yesterday. Is it like that?/is it the case?’

The tag phrases in Akan are mainly used when the speaker requires the addressee to acknowledge the truth of the anchor. They, therefore, resemble a lawyer's cross-examination questions in court. The speaker only requires the addressee to confirm the anchor as presented. Thus, the speaker has a "strong" tentative commitment towards the anchor.

4.1.2 Tag particle *ànáá*

Aside from the tag phrase, Akan also uses the tag particle (TPRT) *ànáá*. With the tag particle, the anchor is followed by a pause before the tag particle is uttered. The tag particle can also occur with both negative and positive anchors, with its primary function being requesting confirmation. For example,

- (41) Àbàn nó á-tò pétrol nó mú bióm, ànáá?
government DET PRF-increase petrol DET LOC again, TPRT
'The government has increased the petrol prices, haven't they?'
- (42) Àbàn nó í-tò-ò pétrol mú bióm, ànáá?
government DET NEG-increase-PST petrol LOC again, TPRT
'The government has not increased the petrol prices, have they?'

In both (41) and (42), the tag particle is used to request confirmation from an addressee whom the speaker considers an expert in petroleum and governance. Here, the speaker has evidence to support his proposition but only requests confirmation.

In addition to requesting confirmation, the tag particle can sometimes be used for a ‘genuine’ information-seeking reading. For instance, consider the following context: *Kofi’s mother has been out of town. She recently returned and decided to pay a surprise visit to her son at his school, and she learned her son is not in school.*

- (43) Kòfí kò-ò sukúù énorà, ànáá?
Kofi go-PST school yesterday, TPRT
'Kofi went to school yesterday, didn't he?' [Information seeking]
[Confirmation]
- (44) Kòfí á-ò-kó sùkúù énorà, ànáá?
Kofi PRF-NEG-go school yesterday, TPRT
'Kofi did not go to school yesterday, did he?' [Information seeking]
[Confirmation]

The mother's utterance in (43) or (44) has two readings. The first is the confirmation reading, where she asks the addressee to confirm whether the child was in school. In addition, it is also seeking information. The mother has no tentative commitment to the proposition with the information-seeking reading and is asking out of ignorance. She only finds the addressee as the authority to provide the information needed. This information-seeking reading is unavailable when the tag phrase (*é-tí sàà ànáá* 'is it the case?') is used. However, in a context where the mother drove her son to school yesterday and saw him entering the school premises and was later told that her son had not been coming to school upon her visit, her utterances in (43) and (44) only seek confirmation.

4.2 The differences between the *ànáá* tag and tag phrase

Although the speaker expresses tentative commitment towards the proposition with both the tag phrase and the tag particle, the "strength" of the tentative commitment differs among the two tag forms. With the tag phrase, the speaker has a "strong" tentative commitment; hence, it is compatible with overtly expressed direct and indirect evidential, as shown in (45) and (46), respectively.

- (45) Direct evidential
 Mè hù-ù sê Kòfí á-tò kàrà, è-n-té sàá ànáá?
 1SG see-PST COMP Kofi PRF-buy car, 3SG-NEG-be.like DEM QPRT
 'I saw that Kofi has bought a car. Is it not the case?' [Confirmation]

- (46) Indirect evidential
 Yè sé Kòfí á-tò kàrà, è-n-té sàá ànáá?
 3PL say Kofi PRF-buy car, 3SG-NEG-be.like DEM QPRT
 'They say Kofi has bought a car. Is it not the case?' [Confirmation]

The *ànáá* tag, on the other hand, is compatible with an overtly expressed indirect evidential. With a direct evidential expressed as 'I saw,' the structure becomes a self-directed question, as shown in (47).

- (47) Direct evidential
 Mè hù-ù sê Kòfí á-tò kàrà, ànáá?
 1SG see-PST COMP Kofi PRF-buy car, TPRT
 'I saw that Kofi has bought a car, didn't I?' [Confirmation from self/speaker]

For indirect evidential (hearsay or reportative), tag particle constructions require ratification from an addressee (external addressee), as shown in (48).

- (48) Yè sé Kòfí á-tò kàrà, ànáá?
 3PL say Kofi PRF-buy car, TPRT
 ‘They said Kofi had bought a car, didn’t they?’ [Confirmation from addressee]

According to Bhadra (2020), a speaker can be an involved or uninvolved source, depending on their commitment to the evidence’s validity. That is, a speaker is an involved source if he is committed to the validity of the evidence and uninvolved if otherwise. In Akan, with the *ànáá* tag, the speaker must be an uninvolved source; however, with a tag phrase, the speaker can be both an involved and uninvolved source.

Table 2 summarizes the differences between the tag phrase and the tag particle based on the evidential type.

Table 2: The differences between tag phrase and tag particle

Type of evidence	Tag phrase	Tag particle
Direct (overt)	Confirmation (addressee)	Self-directed question
Indirect	Confirmation (addressee)	Confirmation (addressee)

4.2.1 Selectional restriction of *ànáá* TPRT

The *ànáá* tag particle has a selection restriction on its anchor. It selects a declarative anchor (CP [C_{-WH}]) but not an interrogative anchor, as shown in (49) and (50) respectively.

- (49) Ám’á á-dà, ànáá?
 Ama PRF-sleep TPRT
 ‘Ama slept, didn’t she?’
- (50) ?ànáá Ám’á á-dà, ànáá
 QPRT Ama PRF-sleep, TPRT
 *‘Is Ama asleep? Isn’t she?’⁶

The oddity of (50) results from the meaning of the two particles. Questions marked by *ànáá* are neutral questions. With neutral questions, according to Gunlogson (2008), the speaker has no bias or tentative commitment towards the question. That is, the speaker is unaware of whether P or ¬P holds and therefore

⁶Note as pointed out by an anonymous reviewer, (50) can be grammatical when one of the *ànáá* is interpreted as ‘probably’ as in ‘Is Ama probably asleep?’. See Section 5 for further discussion of *ànáá* as a discourse marker.

depends on the addressee for information. With the tag question, however, the speaker is biased toward the proposition and only seeks confirmation from the addressee. Given these differences, the property of the polar questions and the function introduced by the tag contradict, hence the oddity in (50).

The two particles, however, can co-occur when the question particle is embedded (51). This is because the question particle only marks the interrogative complement as shown in (51b) and not the matrix statement.

- (51) a. Kòfí bisá-à sê ànáá Ám'á á-dà, ànáá?
 Kofi ask-PST COMP QPRT Ama PRF-sleep, TPRT
 'Kofi asked if Ama was asleep, didn't he ask?'
 b. [ànáá] [CP Kofi asked [CP_{+WH} [ànáá] [TP Ama slept]]]

4.2.2 A possible syntactic position

What is the syntactic position of the *ànáá* tag particle? As aforementioned, the speaker of the tag questions must have a tentative commitment to the proposition, making the *ànáá* tag particle speaker-oriented and, thus, it cannot be embedded. For instance, in (52), the tag particle questions the matrix clause, not the embedded clause. (52) can roughly be translated as: 'Didn't Yaw say that Kofi asked if Ama is asleep?' The tentative commitment is also not expressed by the matrix or the embedded subject but by the speaker, indicating that the tag structure cannot be embedded.

- (52) Yàw kà-à sê Kòfí bisá-à sê ànáá Ám'á á-dà, ànáá?
 Yaw say-PST COMP Kofi ask-PST COMP QPRT Ama PRF-sleep, TPRT
 'Yaw_i said that Kofi asked if Ama was asleep, didn't he_i? *didn't he (Kofi)?'

Given the unembeddable property of the tag particle and following Dayal's (2023) cartography, we propose that the *ànáá* tag occurs as the head of a Speech Act Phrase (SAP), a projection that cannot be embedded.

- (53) [SAP_{ASK} [ànáá_{TPRT}] [PersP Perspective [CP C [ànáá_{QPrt}] [TP...]]]]

4.3 Is the question marker the same particle as the *ànáá* TPRT?

As shown in Section 4.2, the tag and the question particle can co-occur if the interrogative is an embedded complement. The co-occurrence, therefore, suggests that the two particles are distinct.

We have also shown in Section 2.1. that *àsò* and *ànáá* are in complementary distribution. To decide whether the *ànáá* TPRT particle is the same as the question particle, we will substitute all instances of *ànáá* QPrt with *aso* QPrt in tag constructions. Consider (54) and (55).

In the tag phrase (54), *àsò* can replace the *ànáá* particle with no changes in meaning, indicating that *ànáá* in (54) is a question particle.

- (54) Kòfí kò-ò sùkúù énorà, àsò è-n-té sàá?
 Kofi go-PST school yesterday, QPRT 3SG.INANIM-NEG-be.like DEM
 ‘Kofi went to school yesterday. Is it not like that?/is it not the case?’
 [Confirmation (available for 54)]
 [Information seeking (unavailable for 54)]⁷

However, when we substitute *aso* with *ànáá* TPRT in as in (55), we only have the strict information-seeking polar questions and not biased reading as expected from the *ànáá* TPRT; this change reading, therefore, indicates that *ànáá* TPRT is worth considering as a distinct particle from the question marker.

- (55) Àsò ná Kòfí wó sùkúù hó énorà?
 QPRT PST Kofi LOC school LOC yesterday
 ‘Was Kofi at school yesterday?’
 [Information seeking (available for 55)]
 [Confirmation (unavailable for 55)]

Other differences between *ànáá* and the tag particle *ànáá* are summarized in Table (3) below.

⁷One reviewer pointed out that both question particles may occur in the same question in Akan. *ànáá* may occur as a final particle in the tag phrase, which has the initial particle *aso*, as in the following examples:

- (i) Kòfí kò-ò sùkúù énorà, àsò è-n-té sàá ànáá?
 Kofi go-PST school yesterday, PRT 3SG-NEG-be.like DEM PRT
 ‘Kofi went to school yesterday, isn’t it?’
 (ii) Kòfí kò-ò sùkúù énorà, àsò ànáá è-n-té sàá?
 Kofi go-PST school yesterday, PRT PRT 3SG-NEG-be.like DEM.
 ‘Kofi went to school yesterday, isn’t it?’

One possible explanation is to assume that *aso* and *ànáá* in (i) and (ii) are performing different functions. In this case, *ànáá* indicates the speaker’s tentative commitment (its discourse function, see Section 5), and *aso* is the question particle. Hence, the co-occurrence of the two does not show that the two are not complementary, but rather, the two are different morphemes performing different functions.

Table 3: *ànáá* QPRT versus *ànáá* TPRT

Properties	<i>ànáá</i> QPRT	<i>ànáá</i> TPRT
Use for neutral questions	✓	✗
Use for biased questions	✗	✓
Can be embedded	✓	✗

4.4 Taking stock

We have discussed three distributions of the *ànáá* particle. When *ànáá* is a question marker, it can be substituted with the *aso* question marker. The disjunctive marker is distinct from the question marker as both can co-occur in the same structure. The *ànáá* tag marker is distinct from the question marker as its substitution for *aso* only results in the formation of polar questions. Let us consider the final instance of *ànáá*.

5 Discourse function

5.1 Discourse particle *ànáá*

The final instance of *ànáá* here is referred to as the discourse particle. This particle expresses possibility, or it expresses the speaker's uncertainty. The *ànáá* discourse particle does not affect the truth condition of its proposition. It only indicates the speaker's degree of commitment toward the proposition, which must be weaker.

Let us consider this context: *Kofi's mum realizes that her son did not attend school today; she utters (56a) to one of Kofi's friends (Mike), but Mike has no idea where Kofi is, but his best guess is that Kofi may be at home playing video games; therefore, he will utter (56b).*

- (56) a. Kòfí wò hén?
 Kofi LOC.be where
 'Where is Kofi?'
 b. Kòfí wò fié ànáá
 Kofi LOC.be house PRT
 'Kofi may be at home.' [Guessing]

The use of *ànáá* by Mike in (56b) indicates the speaker's weak commitment to the proposition. He is uncertain of his answer, and Kofi being at home is his

best guess. Therefore, *ànáá* is infelicitous if speaker B is firmly committed to the proposition.

The behavior of the *ànáá* particle aligns with the German discourse particle *wohl*, as Zimmermann & von Fintel (2011) discussed. *Wohl*, like *ànáá*, expresses the participant's weak commitment to the proposition's truth and is also infelicitous if the participant is strongly committed to the proposition. For instance, in (57), B's utterance is infelicitous if he has strong evidence that Hein is at sea.

(57) German (Zimmermann & von Fintel 2011)

A: I can't see Hein.

B: Er ist wohl auf See.

'He may be at sea.'

[Guessing]

Table 4 summarizes the similarities between the Akan *ànáá* discourse particle and the German *wohl* particle (cf. Zimmermann & von Fintel 2011).

Table 4: *ànáá* versus *wohl* discourse particles

Properties	<i>ànáá</i>	<i>wohl</i>
Expresses the speaker's uncertainty	✓	✓
Can be embedded	✓	✓
Invisible to the interrogative operator	✓	✓

5.2 *ànáá* versus modal auxiliaries

Although *ànáá* expresses possibility, it is not a modal or an auxiliary adverb. Zimmermann & von Fintel (2011), following Hacquard (2006), indicates that modals and auxiliary adverbs contribute to the descriptive content of their proposition. For instance, the modal auxiliaries and adverbs are visible to the interrogative's operators and, thus, become part of the set of possible interrogative alternatives. However, discourse particles such as *ànáá* do not form part of the response alternatives to a question. In a question, let us consider *ànáá* particle and the modal auxiliary *ébíà* 'presumably.'

(58) a. Ébíà Kòfí kò-ò fíé *ànáá*?

presumably, Kofi go-PST home QPRT

'Has Kofi presumably gone home?'

b. {presumably Kofi is at home, presumably ¬Kofi is at home} [response alternatives]

- (59) a. *ànáá*, Kòfí kò-ò fié *ànáá*?
 PRT Kofi go-PST home QPRT
 ‘Has Kofi presumably gone home?’
 b. [Kofi is at home, ¬Kofi is at home] [response alternatives]

In (58a), the modal *ebia* forms part of the question alternatives. However, in (59a), the *ànáá* particle does not form part of the alternatives. The function of the particle is thus to introduce a tentative commitment to the proposition. That is, the *ànáá* presents bias reading to the question. Consider (60):

- (60) a. *ànáá*, òsúó rè-tò *ànáá*?
 PRT, water PROG-fall QPRT
 ‘It is raining?’ [If a speaker has evidence]
 b. òsúó rè-tò *ànáá*?
 water PROG-fall QPRT
 ‘Is it raining?’ [If no such evidence exists]

The question in (60a) can be uttered in a windowless basement context, where someone walks in with a wet umbrella. Here speaker A has some form of evidence to support his proposition and only needs ratification from the addressee. (60b), however, is used if no such evidence exists and the speaker is asking out of ignorance.

5.3 Syntactic position of discourse particle

Although the discourse particle *ànáá* appears to be addressee/speaker oriented, it can allow for embedding. For example,

- (61) Ám'á dwéné sê *ànáá* Kòfí á-dídí.
 Ama think COMP PRT Kofi PRF-eat
 ‘Ama presumably thinks that Kofi has eaten.’

In (61), the weak commitment is expressed by the matrix subject (Ama) and not the speaker. That is, in (61), it is Ama who presumably thinks that Kofi has eaten and not the speaker. Thus, although the discourse particle *ànáá* is addressee/speaker oriented, it can allow for some form of embedding, suggesting that it must occur lower than the SAP phrase within Dayal's cartography. Therefore, we propose the Perspective phrase as the syntactic slot for the *ànáá* discourse particle, as shown in (62).

(62) [SAP_{ASK} [ànáá] [PersP [ànáá] [CP C [ànáá] [TP...]]]]⁸

6 Conclusion

This paper provided an overview of the properties of the Akan particle *ànáá*. Four different functions are presented, and syntactic, semantic and syntactic properties are provided for each of them. The paper also provides a cross-linguistic perspective in showing how *ànáá* patterns in certain respects with comparable particles from other languages such as German and Hindi. Finally, the paper contains a syntactic analysis of the various functions of *ànáá* which is based on previous work by Dayal (2023). The paper makes a contribution to the area of information structure and semantics, as well as to African linguistics.

⁸Aside from situating the different functions of *ànáá* into Dayal's cartography, there are other alternative analyses that can be proposed for the particle(s). For instance, one reviewer suggested an analysis along the lines of vP ellipsis. According to this analysis, the various interpretations of *ànáá* can be because of ellipsis of the preceding or following vP constituent. Consider example (i).

- (i) wó nà wó bé nóá pàstà nó ànáá ènám nó ànáá?
 2SG FOC 2SG FUT cook pasta DET or meat DET QPRT
 'Are you the one to prepare the pasta or the meat?'

The initial *ànáá* may involve vP disjunction with ellipsis in the second VP or DP disjunction as schematized as (ii) and (iii) below:

- (ii) [[[vP [VP wó bé nóá pàstà nó] ànáá [vP [VP ... ènám nó.]]]
 (iii) [[DP pàstà nó] ànáá [DP ènám nó]].

The second *ànáá* similarly has a vP complement that is deleted:

- (iv) [[[vP [VP wó bé nóà pàstà nó] ànáá [[vP [VP ... ènám nó]]] ànáá [vP [VP ...]]].

The co-occurrence of the *ànáá* particle, therefore, appears to involve recursive application/copying of a disjunctive rule rather than projection of several distinct syntactic heads. In this alternative analysis, the polar question function arises from ellipses of the vP preceding or following *ànáá*.

Although vP ellipsis can account for some of the distributions of the *ànáá* particle, it cannot account for the interpretation differences between the particles. For instance, *ànáá* as a tag is always associated with biased questions, while the question particle is always a neutral information-seeking question. Assuming a recursive vP ellipsis may seem syntactically economical, but the meanings associated with the instances of *ànáá* do not support such an approach.

Abbreviations

1	first person	NEG	negation
2	second person	PRT	particle
COMP	complementizer	PST	past tense
DET	determiner	PRF	perfective
ERG	ergative	PQP	polar question particle
FOC	focus	SG	singular
INANIM	inanimate	QPRT	question particle
LOC	locative marker	TPRT	tag particle

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Chapter 8

Verb phrase ellipsis in Likpakpaanl

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This paper discusses verb phrase ellipsis in Likpakpaanl, a Mabia (Gur) language belonging to the Niger-Congo family and predominantly spoken in the Eastern part of the Northern Region and the Oti Region of Ghana. I argue that VP ellipsis is licensed by the modals *ɲmàn* and *bàn*, which project separate modal phrases (ModP) in the clausal structure. The data drawn from the Licahbol dialect show that only the higher ModP position occupied by *ɲmàn* and *bàn* hosts the ellipsis [E]-feature which licenses VP ellipsis by deleting the phonological content of their complements. I show that even though both modals license VP ellipsis, *bàn* targets an entire CP for deletion, while *ɲmàn* deletes the VP and its internal arguments. Finally, I adduce evidence from Antecedent Contained Deletion (ACD) and Reconstruction effects and propose a base-generation analysis to account for the presence of the resumptive pronoun in the distribution of *bàn*, and also assume a PF-deletion approach for Likpakpaanl VP ellipsis.

1 Introduction

In natural language, we communicate through the use of words, which are strung together to convey our intended meaning. This meaning comes from the interlocutor's ability to create a meaningful link between sound and meaning. However, under ellipsis, the observed mapping of form-to-meaning is disrupted, resulting in a mismatch between form and meaning, although the meaning is not lost due to the omission of background information. This paper investigates the syntax of verb phrase ellipsis in Likpakpaanl and contributes to the ongoing effort to examine the nature of VP ellipsis licensing from a cross-linguistic perspective. There are few works on ellipsis in the Mabia languages, or African languages in general. The few works on ellipsis have come from Basàá (Bassong



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2014), Gungbe (Lipták & Aboh 2013) and Nupe (Mendes & Kandybowicz 2023). Data from a less researched language like Likpakpaanl advances our understanding of the typological variation of VP ellipsis phenomena. The goal of this paper is to discuss VP ellipsis in Likpakpaanl and try to answer the following questions:

- i. What are the strategies for licensing VP ellipsis in Likpakpaanl?
- ii. Which functional element hosts the ellipsis feature that triggers VP ellipsis?
- iii. What evidence is there to support a PF-deletion account of Likpakpaanl VP ellipsis?

There are two main approaches in the study of ellipsis, viz the PF-deletion and null-Pro-form. This study adopts the former approach. The PF-Deletion approach, first proposed by Merchant (2001), argues that ellipsis involves an underlying syntactic structure that is present in Logical Form (LF) but deleted in Phonetic Form (PF). Under this analysis, the elided constituent in (1) contains the full syntactic structure, allowing it to be interpreted properly at LF, but the phonological content ‘eat pizza’ is deleted and, therefore, not pronounced (Merchant 2001, Frazier & Clifton 2005). PF-Deletion, as noted by Merchant (2001), applies post-syntactically at the interface between syntax and phonology and is driven by an ellipsis [E]-feature that triggers deletion as shown in the *strikeout* text in (1).

- (1) John ate pizza last night, and Mary did *eat pizza* too.

One advantage of the PF-Deletion approach is that it accounts for both the interpretability and silence inherent in ellipsis constructions while maintaining the presence of syntactic structure (Merchant 2004). Subsequent research has supported the PF-Deletion account through evidence from binding connectivity effects in ellipsis sites (Johnson 2001, Frazier & Clifton 2005, Aelbrecht & Harwood 2019).

Beyond this introduction, the rest of the paper is structured as follows: Section 2 explains the notion of ellipsis in general and verb phrase ellipsis in particular, while Section 3 provides a background to the language and the causal structure. Section 4 introduces the various modal auxiliaries and their interaction with the tense and aspectual systems of the language and lays the basis for discussing VP ellipsis in the language. Section 5 examines the various licensing conditions for VP ellipsis and shows that not all modals can license VP ellipsis except *ɲmàn* and *bàn*, while Section 6 presents the evidence in support of a

movement plus PF deletion. Finally, Section 7 summarises the paper's findings and conclusions.

2 The notion of ellipsis

Ellipsis refers to a linguistic phenomenon where some elements of a sentence are omitted, but we are still able to interpret the meaning based on the context of the sentence (Ross 1967, Merchant 2019). Ellipsis has been identified as a cross-linguistic phenomenon common in most languages. However, the language-internal properties of each language play a crucial role in determining the licensing conditions of ellipsis. For instance, Dagnac (2010) observed that in French and Spanish, it is modal verbs that license a type of VP ellipsis, which they term “Modal complement ellipsis” (MCE) (Aelbrecht 2010). In the examples in (2), the crossed-out materials represent unpronounced syntactic constituents (the ellipsis site), which, though present in the syntax, are not overt at LF (Johnson 2001, Merchant 2001, Aelbrecht 2010, Dagnac 2010). From the literature, different ellipsis phenomena, such as nominal ellipsis, gapping, sluicing, and verb phrase ellipsis, are derived from syntactic movement followed by the deletion of the phonological form (PF) part of the clause (Merchant 2001, Aelbrecht 2010, van Craenenbroeck 2017). Verb Phrase ellipsis (VPE), which is the focus of this paper, describes a linguistic phenomenon where the verb phrase, together with its internal arguments, are elided if they meet the licensing and identification conditions proposed by Merchant (2001) and Lobeck (1995). Merchant (2001) explains licensing as a locality constraint on syntactic heads that determine the structure in a particular language that can be omitted under ellipsis, while identification (syntactic and semantic) refers to the process of recovering missing information that is expected to be present in overt syntax. Under the PF-deletion approach to VP ellipsis, the “deleted” predicate in (2) gets a full semantic interpretation because it has an appropriate antecedent in the first conjunct clause. The predicates in the second conjunct of (2a) and (2b) are not overtly expressed, yet the sentences are interpreted as ‘Anthony will eat the plantain’ and ‘I could buy Susan the oranges’, respectively.

- (2) a. The children will eat the plantain, and Anthony will [~~eat the plantain~~] too.
 b. Adelina couldn't buy Susan the oranges, but I could [~~buy Susan the oranges~~].

Johnson (2001), Merchant (2001) and Aelbrecht & Harwood (2015) note that research into VP ellipsis has revolved around trying to examine the syntactic environments in which it is licensed, investigating the structural relationship between the elided verb phrase and its antecedent, and exploring how the meaning at the ellipsis site is recovered. Johnson (2001) and Aelbrecht & Harwood (2015) point out that finite auxiliaries license VP ellipsis in English. Thus, the example in (3) shows that the finite auxiliaries *could* and *had* cannot be “deleted” because they license VP ellipsis in English. However, lexical verbs, whether finite or non-finite, need to be deleted in VP ellipsis as illustrated in (3a). If a sentence does not have a finite auxiliary to license the ellipsis in English, a dummy auxiliary ‘do’ is inserted in the structure, as in (3d); otherwise, the resulting sentence will be ungrammatical. Mutual entailment between an antecedent clause and an elided clause is necessary for the felicitous interpretation of verb phrase ellipsis (VPE) (Dalrymple et al. 1991). Dalrymple et al. (1991) note that in VPE like (3a) and (3b), the interpretation is possible because the antecedent clause *I thought the auxiliary hadn’t disappeared* and the elided clause *it had* stand in a mutual entailment relationship, such that each clause entails the other. This creates the parallelism required for resolving the missing verb ‘disappeared.’ Consequently, in (3c), the VPE interpretation fails, indicating that mutual entailment between the antecedent and elided clauses is necessary for licensing VPEs and arriving at the intended interpretation (Dalrymple et al. 1991, Hardt 1999).

- (3) a. An elephant can’t fly, but maybe a rhino *(could) [~~fly~~].
b. I thought the auxiliary hadn’t disappeared, but it *(had) [~~disappeared~~].
c. *The chicken didn’t put the tuna on the table, but the penguin put [~~the tuna on the table~~].
d. The chicken didn’t put the tuna on the table, but the penguin did [~~put the tuna on the table~~].

(Aelbrecht & Harwood 2015: 63).

As a general principle, elements that undergo ellipsis must be subject to the recoverability condition, which requires that whatever is elided should be able to establish a mutually entailed interpretation between the antecedent and the ellipsis site. Thus, elements at the ellipsis site must be retrievable via a salient linguistic antecedent (Johnson 2001, Merchant 2001, van Craenenbroeck 2017). Fiengo & May (1994) view the identity relation between the ellipsis site and the antecedent as a pure syntactic identity and propose a “Strict Isomorphism” condition, which requires elided elements to have a syntactically identical antecedent.

3 Background to Likpakpaanl and the clausal structure

This sub-section briefly explains the properties of Likpakpaanl to provide a background for understanding the discussion of VP ellipsis licensing. Likpakpaanl (also known in the literature as Konkomba) belongs to the Gurma subgroup of the Oti-Volta branch of North Central Mabia¹ (Manessy 1971, Naden 1989). The people refer to themselves as *Bikpakpaam*, their language as *Likpakpaanl* and their land as *Kikpakpaanl*. The language is predominantly spoken in the Eastern Corridor of the Northern region of Ghana as well as in the North-East, Oti and Ahafo regions. Likpakpaanl is also spoken in the Western part of the Republic of Togo, with a population of 112,000 and over 1,187,000 speakers in Ghana (Joshua Project 2022, World Population Review 2023).

Likpakpaanl has a system of 15 noun classes based predominantly on class affixes, which also bear number semantics of number agreement (Tait & Strevens 1954, Winkelmann 2012, Bisilki & Akpanglo-Nartey 2017). Noun class affixes are used to distinguish singular and plural nouns. The affixes also play a central role in marking agreement patterns in the internal structure of the DP, such as demonstratives and possessives. Structurally, Likpakpaanl, like other Mabia languages, is an SVO language such that a canonical simple sentence can have the pattern SVO, SV or SVA, depending on whether the verb is transitive or intransitive, as illustrated in (4a), (4b) and (4c), respectively.²

- (4) a. Tàmànjà nà̀n tər̀ ù-bò gbà̀n. (SVO)
 T. PST help 1-child DEF
 ‘Tamanja helped the child.’
- b. Wumbei là̀. (SV)
 W. laugh.PFV
 ‘Wumbei has laughed.’
- c. Naja nà̀n chù̀n sù̀-sù̀. (SVA)
 N. PST walk slow-slow
 ‘Naja walked slowly.’

In ditransitive constructions, the verb is immediately followed by the indirect object before the direct object and adverbials, as illustrated in (5a). Sentence (5b)

¹I use the term ‘Mabia’ in the sense of Bodomo (2020) to refer to over 80 languages spoken across the West African sub-region. The Mabia languages, numbering about 70 languages, are also spoken in south-eastern Mali, Burkina Faso, northern Benin, northwestern Nigeria and northern Ivory Coast.

²Odd numbers represent singular noun affixes, while even numbers encode plural noun classes

illustrates a serial verb construction, where the introduction of the verb *dì* selects the direct object, which precedes the indirect object, as its internal argument.

- (5) a. Adelínà gà tì Ama lí-nùù-l gbààn dín.
 A. FUT give A. 7-yam-7 DEF today
 ‘Adelina will give Ama the yam today.’
 b. Kànjo gà dì lí-nùù-l gbààn tì Ama.
 K. FUT take 7-yam-7 DEF give A.
 ‘Kànjo will give the yam to Ama.’

Despite the SVO clause structure, information-structural-related constructions can trigger the extraction of these elements out of their canonical positions and moved to the sentence-initial position, resulting in some cases to focus (6b) and topic (6c) interpretations.

- (6) a. Ponaa dàà ì-ṇùò. (Canonical)
 P. buy.PFV 6-goat
 ‘Ponaa has bought goats.’
 b. Ì-ṇùò_i lé Ponaa dàà t_i. (Focus)
 6-goat FOC P. buy.PFV
 ‘Ponaa has bought GOATS.’
 c. Ì-ṇùò gbààn_i mà-kàn Ponaa dàà t_i. (Topic)
 6-goat DEF TOP P. buy.PFV
 ‘As for the GOATS, Ponaa bought them.’

It is also worth noting that Likpakpaanl verbs do not inflect for tense, aspect, person, or number. Therefore, tense, aspect, and mood are marked by independent preverbal particles in the language. In Likpakpaanl, the present tense is not morphologically marked; the past tense is marked by preverbal particles that categorise into immediate, external, and remote past. The immediate past tense or hodiernal, marked by the particle *ba*, describes events or actions that occurred the same day or shortly before the current moment of speaking (7a). There is the hesternal past tense or “yesterday past” (7b), which refers to events that took place exactly a day before the speech time. Events that happened more than three days, a week, to years before the present moment are marked by the remote past particle *nàn* as in (7c) (glossed as PST), while the future tense is illustrated in (7d).

- (7) a. Mbeijom bà jìn sàkolà.
 M. HOD.PST eat 3.fufu
 ‘Mbeijom ate fufu (within the day).’

- b. Mbeijom fè jìn sàkòlà.
 M. HEST.PST eat 3.fufu
 ‘Mbeijom ate fufu (yesterday).’
- c. Mbeijom nàn jìn sàkòlà.
 M. PST eat 3.fufu
 ‘Mbeijom ate fufu (some time ago).’
- d. Mbeijom gà jìn sàkòlà.
 M. FUT eat 3.fufu
 ‘Mbeijom will eat fufu.’

The perfective aspect in Likpakpaanl, like many other Mabilia languages, is unmarked and a bare verb such as *màà* in (8a) receives a perfective interpretation, signalling that an event or action has been completed. The imperfective aspect is morphologically expressed by a preverbal particle *bì* as in (8b) and demonstrates that the act of “building” is still ongoing at the time of the utterance (see also, Bodomo 1997, Issah 2013, Abubakari 2019).

- (8) a. Kando màà kì-dìi-k.
 K. build.PFV 11-room-11
 ‘Kando has built a room.’
- b. Kando bì màà kì-dìi-k.
 K. IPFV build 11-room-11
 ‘Kando is building a room.’

The existence of such preverbal particles is a key property of the clause structure of the Mabilia languages like Dagaare (Bodomo 1997, 2020), Dagbani (Issah 2013, 2020), Kusaal (Abubakari 2018, 2019), and Gurene (Kropp Dakubu 2003, Atintono 2013) among others.

4 Interaction between modals, tense and aspect in Likpakpaanl

This sub-section provides a background to modal construction in Likpakpaanl, the basis upon which the discussion of VP ellipsis will be centred. The modal system of verbs in Likpakpaanl is denoted by particles that express root and epistemic modality. Kratzer (1981) and Nuyts (2001), among others, have argued that modals constitute quantifiers that create possible “worlds” whose restriction is contextually provided. She identifies two broad classes of modals: epistemic

modals and root modals. Kratzer (1981) notes that while the former express a speaker's assessment of the probability of a proposition being true, focusing on the speaker's knowledge and opinions about the world, the latter, conversely, convey notions of obligation, permission, or necessity and thus indicate whether an action is required, allowed, or forbidden (Hacquard 2006, 2010). According to Kratzer (1981), there are two broad classes of modals: epistemic and root modals; the former express a speaker's assessment of the probability of a proposition being true, focusing on the speaker's knowledge, confidence or lack of it in the truth of a proposition. Examples include *might*, *could*, *may*, *must*, *probably*, and *likely*. Root modals, conversely, convey notions of obligation, permission, or necessity and thus indicate whether an action is required, allowed, or forbidden (Nuyts 2001, Hacquard 2006, 2009).

4.1 The syntax of Likpakpaanl modal verbs

Likpakpaanl modal elements like *ɲmàn* 'can', *bàn* 'want' and *nibààkàn*³ 'maybe' are used to encode ability, possibility, and probability, respectively. A root modal like *ɲmàn* in the examples in (9a) and (9b) expresses ability, indicating that Takai and Mpopiin' are able to pluck the mango and weed the farm, respectively.

- (9) a. Takal nàn ɲmàn kìr mǎngù gbààn.
 T. PST can pluck mango DEF
 'Takal was able to pluck the mango.'

³Adopting Hoyer (1997) and Nuyts (2001)'s criteria for modal adverbs, I analyse *nibààkàn* as a modal adverb based on its syntactic patterning. Examples (i.a)-(i.c) demonstrate *nibààkàn* can appear in a clause's initial, medial, or final position without any change in its meaning.

- (i) a. Nibààkàn Jàbàb gà dàn dìn.
 maybe J. FUT come today
 'Maybe/Probably, Jabab will come today.'
 b. Jàbàb gà dàn dìn, nibààkàn.
 J. FUT come today maybe
 'Jabab will come today, maybe/probably.'
 c. Jàbàb gà dàn, nibààkàn dìn.
 J. FUT come maybe today
 'Jabab will maybe/probably come today.'

This positional flexibility indicates that it syntactically takes sentential scope and modifies the truth value of the entire proposition rather than a single constituent. Semantically, it expresses epistemic modality by qualifying the likelihood of the proposition 'Jabab will come' being true.

- b. Mpopiin nàn ɲmàn kùùr kí-sàà-k gbààn.
 M. PST can weed 11-farm-11 DEF
 'Mpopiin was able to weed the farm.'

In its distribution, another modal *bàn* can optionally occur with the complementiser *kè* to express desires, preferences, intentions, or requests. For instance, the use of *bàn* in (10a) demonstrates the intentional willingness of *Powon* to go to school while (10b) expresses *Kunji*'s desire to eat groundnuts. Example (10c) also illustrates that the same modal can be used to make a polite request.

- (10) a. Powon_i bàn (kè) ù_i bùén sikùù.
 P. want that 3SG go 3.school
 'Powon wants to attend school.'
- b. Kunji_i bàn (kè) ù_i ɲmɔ simà.
 K. want that 3SG eat 3.groundnuts
 'Kunji wants to eat groundnuts.'
- c. M bàn (kè) à tér mi.
 1SG want that 2SG help me
 'I want you to help me.'

A major observation that can be made from the distribution of *bàn* is that it selects a CP complement introduced by the optional complementiser *that*. Thus, the sentence in (10a) translates literally as 'Powon_i wants that he_i goes to school.' Such sentences are ambiguous since the pronoun could also refer to someone other than 'Powon', and the only way to disambiguate it is by using the appropriate co-indexation or situating it within the right context.

In Likpakpaanl, interpretations of obligation and necessity arise when the copular *yè* 'be' co-occurs with the complementiser *kè* 'that'. Thus, the bi-partite modal *yè* *kè* denotes an objective necessity to indicate that something has to happen. The interpretation of *yè* *kè* in (11) means that a 'child' needs to obligatorily submit to the authority of elders as a socio-cultural requirement, as far as the African society is concerned.

- (11) Û-bo yè kè ù pàk bí-nínkpíí-b.
 1-child COP that 3SG respect 2-elder-2
 'A child must respect elders.'

Given different contexts, the expression *yè* *kè* can elicit different interpretations. For example, in a context where there are dark clouds, a sentence like (12a)

is uttered as an inference of a high probability, based on observable evidence, that rain is likely to fall. On the contrary, the *yè kè* in (12b) points to the fact that Binlù has to fulfil an obligation and ensure that the meat is fried.

- (12) a. Nì yè kè ù-tààl nù.
it COP that 1-rain fall
'It must rain.'
b. Nì yè kè Bìnlù kìn tì-nàn gbààn.
it COP that B. fry.PFV 14-meat DEF
'Binlu must fry the meat.'

It is also important to note that a construction with only the copular verb, as presented in (13a), yields a cleft-focus construction and requires the focus particle for the sentence to be grammatical. Example (13b) is, therefore, ungrammatical because of the absence of the complementiser *kè* to convey the desired meaning.

- (13) a. Nì yè Bìnlù lé kìn tì-nàn gbààn.
it COP B. FOC fry.PFV 14-meat DEF
'BINLU fried the meat/ It is BINLU who fried the meat.'
b. *Nì yè Bìnlù kìn tì-nàn gbààn.
it COP B. fry.PFV 14-meat DEF
Intended: 'Binlu must fry the meat.'

The use of bi-partite markers to express modality is not new in the literature of African languages. Many Niger-Congo languages, including Lingala (Heine 1991), Ewe (Ameka 2008), and Igbo (Uchekchukwu 2008), among others, use bipartite markers comprising two distinct elements to convey modal interpretations. According to Ameka (2008), in Ewe, an obligation is expressed when the third person singular pronoun *é-* is combined with the expletive verb and followed by the complement clause. In (14), the 3SG-pronoun, which is the subject, is followed by a verb and the complementiser *bé*, which introduces the complement clause to express the obligation.

- (14) Ewe (Ameka 2008: 147)
É-dze bé nà-vá kpɔ-m.
3SG-fit that 2SG:SUBJV-VENT see-1SG
'You must come to see me.'

Having looked at root modals, I also briefly examine the distribution and use of some epistemic modals. These modals indicate a speaker's degree of certainty, familiarity with a proposition or judgment about the likely occurrence of a proposition rather than its truth value. In (15), the epistemic modal expresses the general

possibility that Pando will buy a canoe. *Nibààkàn*, meaning ‘probably/maybe,’ belongs to a group of modals which have been referred to in the literature as modal adverbs (Nuyts 1993, Hoyer 1997, Cinque 1999, Nuyts 2001). Example (15a) demonstrates that the epistemic adverb can also appear in the verbal domain of the clause without significant differences in the semantic interpretation. It does not immediately come before the verb as a modal adverb but can occur in clause-initial (15a), medial (15b) or final position (15c). In (15c), the modal adverb occurs between the raised subject in the specifier of a topic phrase (Spec-Top) and the tense marker. The base position of the moved subject is replaced with a resumptive pronoun (see also the Italian example in (16b)).

- (15) a. *Nibààkàn*, Pando gà dàà bù-ɲɔb.
 maybe P. FUT buy 9-canoe
 ‘Maybe/probably, Pando will buy a canoe.’
 b. Pando gà dàà bù-ɲɔb, *nibààkàn*.
 P. FUT buy 9-canoe maybe
 ‘Pando will probably buy a canoe.’
 c. Pando_i mà-kán, *nibààkàn*, ú_i gà dàà bù-ɲɔb.
 P. TOP maybe RP FUT buy 9-canoe
 ‘As for Pando, he will buy a canoe.’

The Likpakpaanl modal adverb shares some similar syntactic behaviour with its Italian counterpart *probabilmente* ‘probably’. According to Giorgi (2013) *probabilmente* is able to semantically scope over an entire sentence whether it precedes or follows the subject without resulting in a change of meaning as in (16a). Giorgi (2013) predicts that modal adverbs can either occur above the tense phrase (TP) in the clause-initial position or in the base-generated position as the complement of the verb.

- (16) Italian (Giorgi 2013: 99)
 a. *Probabilmente* Gianni vincerà la gara.
 probably G. win the race
 ‘Probably Gianni will win the race.’
 b. Gianni *probabilmente* vincerà la gara.
 G. probably win the race
 ‘Probably Gianni will win the race.’

Furthermore, the modal *li* translated literary as ‘should’ can also be used to express either epistemic or root modality. In (17a), *li* conveys necessity or obligation by indicating that *fathers* are required, according to rules, expectations, or

moral standards, to take good care of their children. In example (17a), however, in a context where someone sees Wongol entering a compound and people follow up and do not find him in the yard, then the sentence in (17b) is uttered to express a degree of possibility that once Wongol was in the yard, then the belief is that he is in one of the rooms. The examples above show that the context determines whether *lì* functions as a root or epistemic modal.

- (17) a. Tè-tiib lì lìk bì àà-bim pù.
 father-2 should look 3SG POSS-child LOC
 ‘Fathers should look after their children.’
 b. Wongol gà lì kòò kì-dù-k nì.
 w. FUT should enter 11-room-11 in
 ‘Wongol should be in the room.’

The data presented so far have demonstrated that some modal verbs can convey both root and epistemic modality, thus expressing both the underlying necessity of actions or events and the speaker’s degree of certainty. The fact that the modals can be used to express root and epistemic modality in (17a) and (17b) is not novel since the same observation has been made in many different languages. For instance, Hacquard (2016) contends that the modal ‘must’ can express either root or epistemic modality in English. The sentence in (18a) expresses epistemic modality because the speaker believes that if John is not in his office, then he will certainly be at home. The same modal ‘must’ in the (18b) denotes an obligation for Dickson to pay the fine for wrong parking. He has no choice.

- (18) English (Hacquard 2016: 45)
 a. John is not in his office. He must be home.
 b. Dickson parked illegally. He must pay a fine.

Additionally, Butler (2003), highlighting the difference between root and epistemic modals in Yoruba, notes that the modal *gbòdó* in (19) could have a dual interpretation suggesting a strong conviction for a young person to marry or an obligation for them to do so.

- (19) Yoruba (Butler 2003: 972)
 Ó gbòdó ní iyàwò.
 he must have bride
 ‘He must be married.’ (root/epistemic)

As a result of these subtle interpretational differences in modal verbs, many authors have claimed that the distinction can be accounted for syntactically or lexically even though other scholars like Kratzer (1981), Cinque (1999), Butler (2003), Hacquard (2016) among others, also assert it is semantically or pragmatically determined.

4.2 Interaction between modal, tense and aspect

From this modal construction overview, I will examine the interaction between modals and the aspectual system in Likpakpaanl. The interaction between tense, aspect and modal in Likpakpaanl elicits different interpretations. In (20a), we have a default perfective verb co-occurring with the modal *ɲmàn* to express the past perfective aspect of Bilikuni possessing the ability to fix the bicycle. In (20b), the auxiliary verb ‘will’ is used as a root modal to demonstrate the capability of Bilikuni to repair the bicycle in the future. In (20c), tense and aspect interact with two modals to suggest a moderate degree of certainty that Bilikuni could be riding a bicycle, but it does not imply necessity or obligation.

- (20) a. Bilikuni ɲmàn tòòr chééchè gbààn.
 B. can repair.PFV 3.bicycle DEF
 ‘Bilikuni has been able to repair the bicycle.’
- b. Bilikuni gà ɲmàn tòòr chééchè gbààn.
 B. FUT can repair.PFV 3.bicycle DEF
 ‘Bilikuni will be able to repair the bicycle.’
- c. Bilikuni gà ɲmàn lì bì tòòr chééchè gbààn.
 B. FUT can should IPFV repair.PFV 3.bicycle DEF
 ‘Bilikuni could be repairing the bicycle.’
- d. Bilikuni ààn ɲmàn lì bì tòòr chééchè gbààn.
 B. NEG.FUT can should IPFV repair.PFV 3.bicycle DEF
 ‘Bilikuni could not be repairing the bicycle.’

Moreover, in the sentences in (21), using different tense and aspectual markers in the presence of the same modal verb serves different modal interpretations related to time and certainty. The use of *lì* in (21a) expresses a sense of past necessity where Noah was expected to buy a canoe. The use of *lì* in (21b) indicates a future possibility or likelihood of Noah acquiring a canoe. Example (21b), therefore, suggests that there is a good chance that Noah will buy a canoe in the future, but does not necessarily imply a necessity as in (21a).

- (21) a. Noah bà lì dàà bù-ḡɔb gbààn.
 N. HOD.PST should buy 9-canoe DEF
 ‘Noah should have bought the canoe.’
 b. Noah gà lì bì dàà bù-ḡɔb gbààn.
 N. FUT should IPFV buy 9-canoe DEF
 ‘Noah would be buying the canoe.’

Descriptively, the examples from (20) and (21) show that in the interaction between tense, aspect and modals in Likpakpaanl, the modal phrase occurs below the TP but above aspect and the VP. Thus, in (22a), the modal *ḡmàn* and *lì* project below TP but above the aspectual phrase and the vP. The ungrammaticality of (22c) occurs because the aspectual phrase precedes the modal. The data further indicates root modals can co-occur, and when they do, *ḡmàn* referring to ‘ability’ must precede, *lì* as captured in the ungrammaticality of (22c).

- (22) a. [TP Binlu gà [ModP ḡmàn [ModP lì [IPFV bì [vP_i kìr [t_i
 B. FUT can should IPFV pluck
 màngù.]
 2.mango
 ‘Binlu could be plucking a mango.’
 b. [TP Binlu ààn [ModP ḡmàn [ModP lì [IPFV bì [vP_i kìr [t_i
 B. NEG.FUT can should IPFV pluck
 màngù.]
 2.mango
 ‘Binlu would not be plucking a mango.’
 c. *[TP Binlu gà [ModP lì [IPFV bì [ModP ḡmàn [vP_i kìr [t_i
 B. FUT should IPFV can pluck
 màngù.]
 2.mango
 Intended: ‘Binlu could be plucking a mango.’

The data on the interaction between the various modals in Likpakpaanl brings into question the hierarchy of appearance. The relative ordering of modals as functional categories within the causal architecture has been the subject of considerable research, and the general consensus is that epistemic modals are higher than roots. Following Cinque (1999), I assume that Likpakpaanl modal verbs are heads of functional projections of a Modal Phrase (ModP), which project above aspect and below the tense Phrase (TP). The various modals such as *ḡmàn*, *lì*, *bàn*, etc., constitute different sub-categories of modals in the hierarchy of root modals in Likpakpaanl as proposed by Cinque (1999).

5 Derivation of Likpakpaanl VP ellipsis

In the previous section, we examined the distribution of modal auxiliaries, and the discussion in this section proceeds to discuss how modals license the derivation of VPE in Likpakpaanl. Even though Section 4.1 showed the presence of different modals, I show that among the various modals, it is only *ɲmàn* and *bàn* that are able to license VP ellipsis in the language. I also argue that when these two modals co-occur, the functionally higher modal *ɲmàn*, which bears the [E]-feature, licenses the deletion. Likpakpaanl illustrates a type of VP ellipsis where the verb phrase and its internal argument in the complement of a modal auxiliary are deleted. The examples will show that the modal can elide its complements and still have the same interpretation in relation to an antecedent clause. Such phenomena have been found in Dutch (Aelbrecht 2010), Italian (Lobeck 1995) and French Dagnac (2010) and they refer to this type of ellipsis as *Modal Complement Ellipsis* (MCE).⁴

5.1 The syntax of *ɲmàn*

The first modal auxiliary I consider is the root modal *ɲmàn*. In (23a), it elides its VP complement, and this is possible because the elided verb phrase *read the book* has an appropriate antecedent in the first conjunct. The elided constituent is semantically understood to mean *Kunji can't read the book*. As predicted by the identify and e-GIVENness condition, (23b) is disallowed under ellipsis since the deleted VP (the struck-out element in the ellipsis site) is new in the discourse and therefore fails to satisfy the e-GIVENness condition, which stipulates that an element, a VP, can be elided if it has an appropriate antecedent and has a mutually entailing relationship with the antecedent. Thus, only previously mentioned elements in a discourse can undergo ellipsis (see Schwarzschild 1999, Merchant 2001).

- (23) a. Ama bà ɲmàn kàrn kì-gbàŋ gbààn àmàà Kùnjí àà bà
 A. HOD.PST can read 11-book DEF but K. NEG HOD.PST
 ɲmàn [kàrn kì-gbàŋ gbààn.] .
 can [read 11-book DEF]
 ‘Ama can read the book, but Kunji can’t.’
 b. *Ama bà ɲmàn kàrn kì-gbàŋ gbààn àmàà Kùnjí àà bà
 A. HOD.PST can read 11-book DEF but K. NEG HOD.PST
 ɲmàn [kìrì ì-jàn gbààn.]
 can [fry 6-fish DEF]

⁴I still use the conventional term “Verb phrase ellipsis” to refer to the phenomenon in Likpakpaanl

Apart from occurring in local clauses, *ɲmàn* is also able to license VP ellipsis in non-local clauses such as (24a) where it elides the entire VP complement inside the embedded clause. Thus, the unacceptability of (24b) can be explained in terms of its violation of the e-GIVENness and identity condition on ellipsis.

- (24) a. Nèinà bì sà̀n sà̀kòlà lè Olàmbà; mù dàk kè ù; gà *ɲmàn*
 N. IPFV pound fufu and o. too think that 3SG FUT can
 [sà̀n sà̀kòlà.]
 [pound fufu]
 ‘Neina is pounding fufu, and Olamba thinks he will be able to ____.’
- b. *Nèinà bì sà̀n sà̀kòlà lè Olàmbà; mù dàk kè ù; gà *ɲmàn*
 N. IPFV pound fufu and o. too think that 3SG FUT can
 [yèí lì-nùù-l.]
 [peel 7-yam-7]
 Intended: ‘Neina is pounding fufu, and Olamba thinks he can peel
 yam-too.’

Ungrammatical sentences like (24b) demonstrate that syntactic identity is essential in licensing VP-ellipsis because only constituents that satisfy the syntactic identity condition can be elided. Merchant (2013), however, notes that there are instances where VP ellipsis permits voice mismatches between elements at the ellipsis site and the antecedent. For example, in (25), the verb phrase in the antecedent clause is in the active voice, but the elided VP occurs in the passive structure. Despite the fact that (25) violates the syntactic identity condition on ellipsis since the passive phrase is elided based on the active antecedent, the sentence is nevertheless grammatical.

- (25) The janitor must remove the trash whenever it is apparent that it should
 be removed. (Merchant 2013: 78).

The e-GIVENness condition requires a mutual entailment relationship between the ellipsis site and the antecedent as illustrated in the example in (26), adapted from Merchant (2001) where the most appropriate and accepted interpretation is (26a), and not (26b). The difference in interpretation between the two examples is because the content of the antecedent verb phrase in (26b) ‘...insult Francis’ does not express a logical relationship or establish the true condition presented in (26a), ‘...call Francis an idiot’ has a semantically entailed interpretation with the elided VP.

- (26) a. Adelina called Francis an idiot after Bianca did [~~call Francis an idiot~~].
 b. ≠Abby called Chunk an idiot after Ben did [~~insult Francis~~].

Thus, it is possible to elide the VP in (26a) because it is e-GIVEN, and its interpretation is entailed in the antecedent. The sentence in (26b) is infelicitous because it lacks an appropriate antecedent. Thus, it cannot be elided because it violates the syntactic and semantic identity condition of ellipsis because Bianca can insult Francis without necessarily ‘calling him an idiot.’ Therefore, it falsifies the statement in the antecedent VP. It is, however, important to state that the antecedent is not always a linguistic element because an elided constituent can also find an antecedent in the non-linguistic context (Merchant 2004, Biloa & Bassong 2020). There is also evidence that other syntactic elements, such as temporal adverbs and adjectives, can also be contained in the E-site in VP ellipsis in Likpakpaanl. For instance, in (27a), the VP and its internal arguments, the DP *woman*, and the temporal adverb are all elided, while in (27b), the modal deletes the verb *dàà* ‘buy’ together with the DP as well as the numeral and qualitative adjective modifiers.

- (27) a. Bileṇàn lèn kè Ntaadiù; gà yòòr ù-píí lí-bín-l lí-mìnà, àmàà tí
 B. say that N. FUT take 1-wife 7-year-7 7.DEM but 1PL
 mómòk nyì kè ù; ààn ṇmàn [yòòr ù-píí lí-bín-l lí-mìnà.]
 all know that 3SG NEG.FUT can [take 1-wife 7-year-7 7.DEM]
 ‘Bileṇàn said that Ntaadi would marry this year, but we all know that
 he can’t [~~marry this year~~].’
 b. Takai dàà ṇì-nààtāk pùàn ṇì-lèè tìi ù-nà lè Nàntòb; mù
 T. buy 8-sandal new 8-two give 3SG-mother and N. too
 dàk kè ù; gà ṇmàn [dàà ṇì-nààtāk pùàn ṇì-lèè tìi
 think that 3SG FUT can [buy 8-sandal new 8-two give
 ù-nà.]
 3SG-mother]
 ‘Takai bought two new sandals for her mother, and Nantob thinks
 (that) she can [~~buy two new sandals for her mother~~], too.’

5.2 The syntax of *bàn*

The data examined so far has looked at *ṇmàn* as a licenser for VP ellipsis in Likpakpaanl. This sub-section discusses how *bàn* licenses the deletion of the PF-content of its complement. The semantic interpretation of the elided constituent is due to the fact that its meaning can be inferred and “recovered” from the antecedent clause in the first conjunct because of the mutual entailment between the unpronounced constituent in the ellipsis site and the antecedent.

The data in (28a) illustrate the fact that ellipsis is licensed by only *bàn* where the CP complement *that he slaughter a pig* in the second conjunct undergoes dele-

tion. The ungrammaticality of (28b) and (28c), on the other hand, demonstrates that VP ellipsis in Likpakpaanl is blocked if the licensing head is a complementiser or a resumptive pronoun supporting Merchant's (2001) assertion about the selectional restrictions on lexical items in a particular language that possess the required [E]-feature to license VP ellipsis and ellipsis in general.

- (28) a. Dawòn bàn (ké) ù kər í-gbèèr lè Naadi mù bàn [(ké) ù
D. want that RP slaughter 6-pig and N. too want [that RP
kər í-gbèèr.]
slaughter 6-pig]
'Dawon wants to slaughter a pig, and Naadi also wants to.'
- b. *Dawòn bàn (ké) ù kər í-gbèèr lè Naadi mù bàn (ké) [ù
D. want that RP slaughter 6-pig and N. too want that [RP
kər í-gbèèr.]
slaughter 6-pig]
Intended: 'Dawon wants to slaughter a pig, and Naadi also wants to.'
- c. *Waatan bàn (ké) ù sànbànché béé ù àà bàn (ké) ù
W. want that RP pound 2.cassava or 3SG NEG want that RP
[sànbànché.]
[pound 2.cassava]
Intended: 'Does Waatan want to pound cassava or she doesn't want to.'

The distribution of *bàn* in the examples above shows the presence of an optional complementiser *ké* 'that' and an obligatory resumptive pronoun. Thus, unlike *ɲmàn*, which targets a VP for deletion, *bàn* selects a CP as its complement. I argue that even though both *bàn* and *ɲmàn* target their complements for deletion, they vary in the size of the ellipsis site they elide. Thus, while the former targets an entire complementiser phrase (CP) for deletion, the latter elides just its VP complement and its internal arguments.

The critical question is how to account for the occurrence of the resumptive pronoun in the syntax of *bàn*. The data on the distribution of *bàn* show a resumptive pronoun (RP) in the embedded clause, and I contend that the presence of the RP suggests that the subject in the matrix clause is base-generated in Spec-TP in the left periphery of the clause and binds the RP. The motivation for such an assumption comes from island and reconstruction effects in Likpakpaanl. The specifier of the CP has been argued to constitute an "escape hatch" for elements that are undergoing movement out of embedded clauses (Chomsky 2000). Following that assertion, one will expect (29b) to be ungrammatical if the RP is a

remnant of the moved subject. This is because the intermediate landing site of the moved DP is already occupied by another DP island and, therefore, blocking any movement out of it. Thus, (29b) is grammatical because the subject, Talido, as the antecedent c-commands and therefore binds the resumptive pronoun in its base-generated position (Chomsky 1980, 1981, Reinhart & Reuland 1993).

- (29) a. Talido nàn bàn ké ù dì ì-líl gbààn dàà nì-tùùn.
 T. PST want that RP take 6-money DEF buy 8-beans
 ‘Talido wanted the money to buy beans.’
- b. Talido nàn bàn [DP ì-líl gbààn_t [CP ké [TP ù dì t_i dàà
 T. PST want 6-money DEF that RP take buy
 nì-tùùn.]]]
 8-beans
 ‘Talido wanted the money to buy beans.’

Using reconstruction as a diagnostic for movement, I propose a base-generation account in examples such as (30b) where the RP is unable to reconstruct because it is not c-commanded at LF by the antecedent that binds it and, therefore, cannot receive a bound reading. Thus, if reconstruction is understood as a property of chains created by movement as argued by Chomsky (1980), then the data suggest there is no movement involved in (30b) but that the RP is bound by Kotin.

- (30) a. Kotin_i gà bàn ké ù_i tèr ù-bà_i.
 K. FUT want that RP help 3SG-self
 ‘Kotin will want to help himself.’
- b. *Ù-bà_i lé Kotin_i gà bàn ké ù_i tèr t_i.
 3SG-self FOC K. FUT want that RP help
 Intended: ‘Kotin will want to help himself.’

5.3 VP ellipsis licensing restrictions in Likpakpaanl

The literature on ellipsis has shown that cross-linguistically, the [E]-feature is syntactically restricted, and as a result, not every lexical item can be a licenser for ellipsis. Having focused on the modals *ɲmàn* and *bàn*, the rest of this section looks at other modal verbs like *yè kè* ‘must’, *nibààkàn* ‘maybe’ and *lì* ‘should’ as well as tense, aspect and negation particles and show that they cannot license VP ellipsis in the language. The ungrammatical sentences in (31a) and (31b) involving the modal *yè kè* lends credence to that conclusion.

- (31) a. *Sukpen yè kè ù pén kí-dìi-k gbààn àmàà Mpopiin àà yè
 s. COP that RP roof 11-room-11 DEF but M. NEG COP
 kè [ù pén kí-dìi-k gbààn.]
 that [RP roof 11-room-11 DEF]
 Intended: ‘Sukpen must roof the room, but Mpopinn mustn’t.’
- b. *Sukpen yè kè ù pén kí-dìi-k gbààn àmàà Mpopiin àà yè
 s. COP that RP roof 11-room-11 DEF but M. NEG COP
 [kè ù pén kí-dìi-k gbààn.]
 [that RP roof 11-room-11 DEF]
 Intended: ‘Sukpen must roof the room, but Mpopinn mustn’t.’

The examples in (32) further demonstrate that the modal adverb *nibààkàn* ‘maybe’ and the deontic modal *lì* are unable to license VP ellipsis. It does not matter if *nibààkàn* occurs clause-initially, medially or finally, the resulting sentence is ungrammatical (32b).

- (32) a. *Kojo lì dàà chééchè àmàà Fred àà lì [dàà chééchè.]
 K. should buy.PFV 3.bicycle but F. NEG should [buy bicycle]
 Intended: ‘Kojo should buy a bicycle, but Fred shouldn’t.’
- b. *Yelson gà nibààkàn gbìi bànehè àmàà Philip ààn nibààkàn
 Y. FUT maybe dig 2.cassava and P. NEG.FUT maybe
 [gbìi bànehè.]
 [dig 2.cassava]

This section explored VP ellipsis in Likpakpaanl and showed that there is a restricted set of modal auxiliaries which can license VPE, viz *bàn* and *ɣmàn*. The data further showed that both modals target their complements for deletion, although they elide different constituents. I also presented evidence that demonstrates the inability of the modals *yè kè*, *nibààkàn* and *lì* to license VP ellipsis in Likpakpaanl.

6 On the derivation of Likpakpaanl VP ellipsis

The empirical data on Likpakpaanl VP ellipsis have shown that it is licensed by a modal verb, which licenses the deletion of its complement. The data also confirm the assertion that ellipsis is a cross-linguistics phenomenon even though the licensing heads vary from one language to another. I account for the Likpakpaanl data by adopting Merchant’s (2001) ellipsis [E]-feature analysis and postulate that VP ellipsis in Likpakpaanl is triggered by an E-feature, which is

merged on the head of the modal phrase (Mod^0). According to Merchant (2001), the [E]-feature embodies semantic, syntactic, and phonological feature bundles that interact with the licensing head to trigger ellipsis. Semantically, only constituents with a salient antecedent, i.e. already mentioned in a previous discourse, e-GIVEN, in the sense of (Schwarzschild 1999, Merchant 2001), can undergo deletion. In terms of syntax, each language has certain syntactic categories that are eligible for the E-feature which licenses ellipsis. For example, only finite auxiliary verbs may bear the feature in English, while in French, it is the modal verb (Johnson 2001, Aelbrecht 2010, Dagnac 2010). Merchant (2001) further argues that the phonological component of the [E]-feature is responsible for removing the PF content of the complement of the head bearing the feature.

From the data presented in this paper, I contend that Likpakpaanl modals possess the [E]-feature for licensing VP ellipsis. Syntactically, the [E]-feature in Likpakpaanl is located on the head of the modal phrase (Mod^0) and targets the complement to the modal for deletion. Once the [E]-feature is activated on (Mod^0), it triggers the “deletion” of the PF content of its complement if it has an appropriate salient antecedent. After the ellipsis, the E-site becomes inaccessible for any further syntactic operations and Merchant (2004) uses that as a basis to explain why some island violations are obviated under ellipsis. Thus, I assume that the sentence in (33) has the underlying syntactic structure represented in Figure 1.⁵

- (33) Nantob àà nànn ɲmàn dàà lòòr pùàn gbààn àmàà Mpopiin gà ɲmàn
 N. NEG PST can buy 3.car new DEF but M. FUT can
 [dàà lòòr pùàn gbààn.]
 [buy 3.car new DEF]
 ‘Nantob was not able to buy the new car, but Mpopiin will be able to [buy
 the new car].’

I assume, following Munn (1987), that the conjunction *àmàà* is a functional head projecting a Boolean Phrase (BP). The first conjunct in the example in (33) occurs in the specifier to the Boolean head while the complement of the B-head, in which VP ellipsis takes place, is the second conjunct of the coordinated constituent (Munn 1992). To derive the structure in Figure 1, the VP is merged with *little v* followed by a head movement of the *dàà* from V to *little v*. The vP external argument *Mpòpíín* then moves from its base-generated position in Spec-vP to Spec-TP to satisfy the Extended Projection Principle (EPP) requirement on T. The [E]-feature on the lexical head *ɲmàn* licenses the deletion of the PF-content

⁵The marked rectangular area on the tree structure represents the ellipsis site.

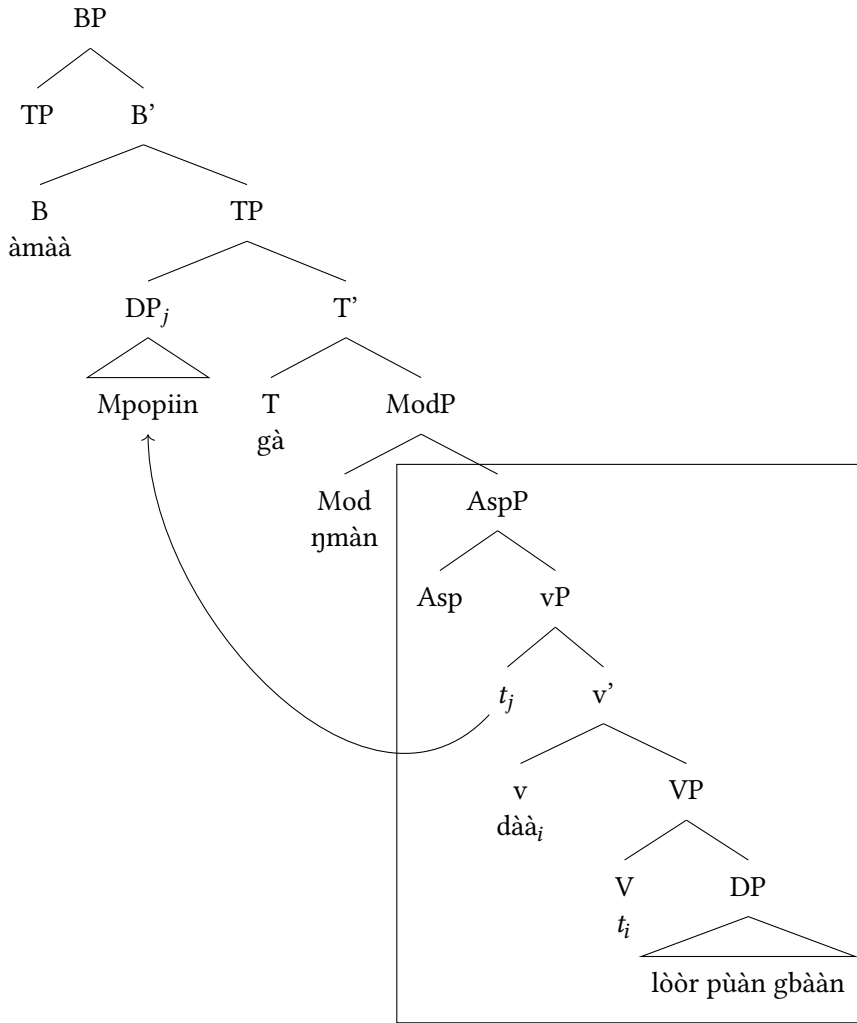


Figure 1: Syntactic tree for (33)

of vP complement and its internal arguments since they satisfy the e-GIVENness condition.

On the basis of the distribution of *bàn*, I assume that the VP ellipsis in Figure 2 is licensed by the modal head *bàn* bearing the [E]-feature that deletes the CP-complement to the (Mod⁰). The major difference between *ɲmàn* and *bàn* as shown in (34) and illustrated in Figure 2 is that the latter targets an entire complementiser phrase rather than the VP as in the case of Figure 2. It has been

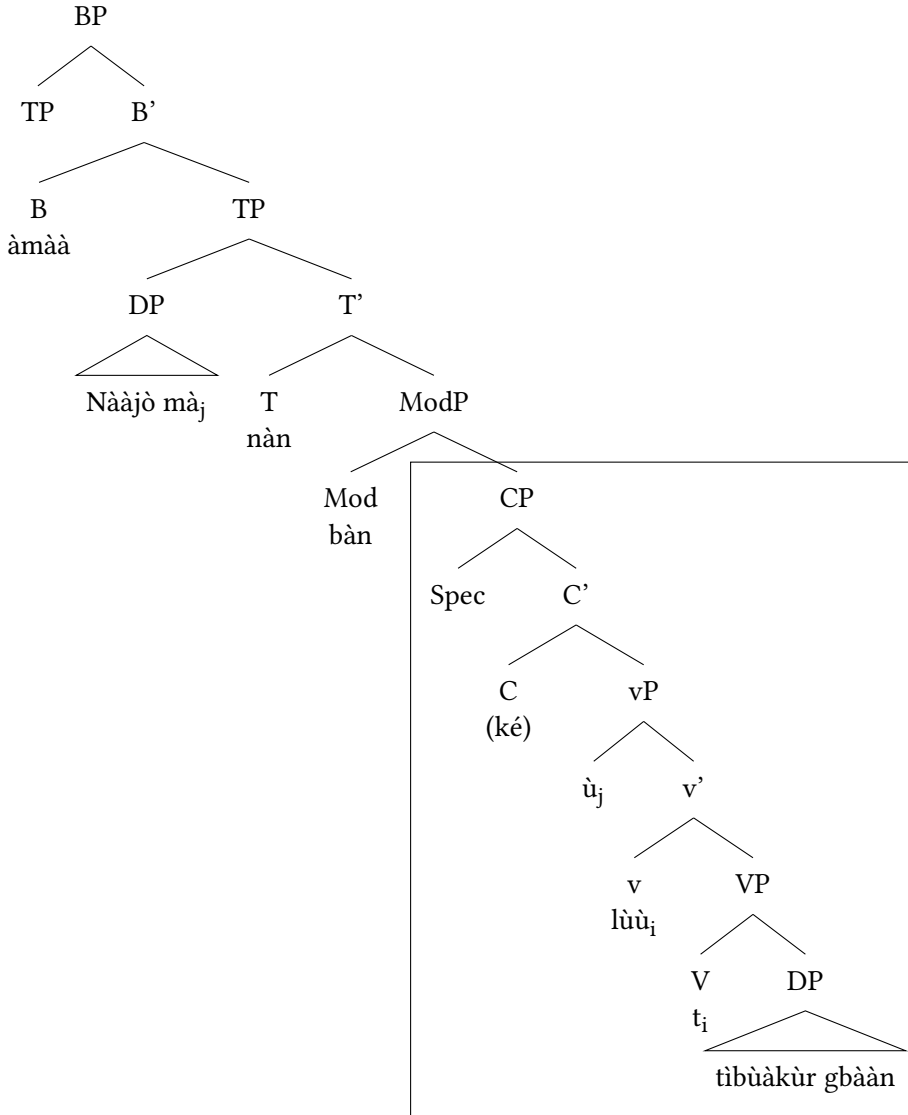


Figure 2: Syntactic tree for (34)

established already in (30) and (31) that the resumptive pronoun inside the CP is base-generated in Spec-TP and gets its interpretation from the c-commanded antecedent in Spec-TP of the matrix clause. Thus, even though the VP *...weave the two baskets* is present throughout the derivation, it is subject to deletion at PF, and the semantics provide the interpretation at LF.

- (34) Kulma àà nànn bàn (ké) ù lùù tì-bùàkùr tì-lèè gbàànn àmàà Nààjò
 K. NEG PST want that RP weave 14-basket 14-two DEF but N.
 mà nànn bàn [(ké) ù lùù tìbùàkùr gbàànn.]
 mà PST want [that RP weave 14-basket 14-two DEF]
 ‘Kulma didn’t want to weave the two baskets, but as for Naajo, she
 wanted to [weave the baskets.]’

Merchant (2019, 2001) notes that cross-linguistically, there is a restriction on syntactic elements and categories that can license ellipsis. Likpakpaanl VP ellipsis data show that the [E]-feature is syntactically restricted and optionally occurs on only modal heads that select the site. The ungrammatical sentences in (35), thus, demonstrate that lexical elements marking tense (35a), aspect (35b) and even negation (35c) are unable to license VP ellipsis in Likpakpaanl.

- (35) a. *Bààyi àànn kpiìn ì-nyòò gbàànn, àmàà Nímbé gà [kpiìn ì-nyòò.]
 B. NEG.FUT feed 6-goat DEF but N. FUT [feed 6-goat]
 Intended: ‘Baayi will not feed the goats, but Nimbe will.’
 b. *Pònjà bì kòr ì-gbéér àmàà Wùmbèi àà bì [kòr
 P. IPFV slaughter 6-pig but W. NEG IPFV [slaughter
 ì-gbéér.]
 6-pig]
 Intended: ‘Pònjà is slaughtering pigs, but Wùmbèi isn’t.’
 c. *Nààjòì bàn ké ùì chààr ì-mùùl gbàànn, bèè ù àà [bàn ké
 N. want that RP winnow 10-rice DEF or 3SG NEG [want that
 ùì chààr ì-mùùl gbàànn.]
 3SG winnow 10-rice DEF]
 Intended: ‘Does Nààjò want to winnow rice or she doesn’t?’

The discussion so far has shown that both *nyàn* and *bàn* can license VP ellipsis, but one theoretical question that needs to be answered is whether each of these modals possesses the [E]-feature and if so, how is VP ellipsis licensed when both modals co-occur in a sentence as (36a). (36a) shows that when the two modals co-occur, it is the hierarchically higher root modal *nyàn* that can license the ellipsis of its complement and not the lower modal *bàn* as reflected in the ungrammatical sentence in (36b).

- (36) a. Kùnjì gà ñmàn bàn_i ù_i dàà lòòr àmàà ù àà kpà í-lík
 K. FUT can want 3SG_i buy 2.car but 3SG_i NEG have 6-money
 nìmàpùnà, ù àà ñmàn [bàn ù dàà lòòr.]
 so, 3SG_i NEG can [want RP buy 2.car.]
 ‘Kùnjì would want to buy a car, but he has no money, so he can’t.’
 b. *Kùnjì gà ñmàn bàn ù dàà lòòr àmàà ù àà kpà
 K. FUT can.MOD want 3SG_i buy 2.car but 3SG_i NEG have
 í-lík nìmàpùnà, ù àà ñmàn bàn [ù dàà lòòr.]
 6-money so, 3SG_i NEG can want [RP buy 2.car.]
 Intended ‘Kùnjì would want to buy a car, but he has no money, so he
 can’t.’

The type of VP ellipsis manifested in Likpakpaanl is similar to what obtains in French, Italian, and Dutch. The examples in (37a) and (37b) show that VP ellipsis in French and Spanish, respectively, are licensed by modals.

- (37) a. French (Dagnac 2010: 158)
 Tom a pu voir Lee, mais Marie n’a pas pu.
 T. can.PST see L. but M. not can.PST
 ‘Tom could see Lee, but Mary couldn’t.’
 b. Spanish (Dagnac 2010: 158)
 Tom pudo ver a Lee, pero María no pudo.
 T. can.PST see (to) L. but M. not can.PST
 ‘Tom could see Lee, but Mary couldn’t.’

A similar observation has been made by Aelbrecht (2010) showing that in Dutch and Italian, too, only root modal verbs can license VP ellipsis as illustrated in (38a) and (38b) for Dutch and Italian, respectively.

- (38) a. Dutch (Aelbrecht 2010: 14)
 Jelle zal wel betalen, maar Johan kan niet.
 J. will PRT pay but J. can not
 ‘Jelle will pay, but Johan can’t.’
 b. Italian (Aelbrecht 2010: 14)
 Antonio pagherà, ma Stefano non può.
 A. will.pay but S. not can
 ‘Antonio will pay, but Stefano can’t.’

The argument advanced in this paper is that the syntactic position where the [E]-feature can be hosted to license VP ellipsis is in the position occupied by *ñmàn*.

As a result, *bàn* may only license ellipsis in the absence of *ɲmàn*. The syntactic distribution of *bàn* is similar to Dutch root modal verbs, which, according to Aelbrecht (2010), select a TP complement and not a VP. He also notes that when the two modals co-occur as in (39a), it is the hierarchically high modal *moet* and not *kunnen* that licenses ellipsis. Unlike in Likpakpaanl where the modal *ɲmàn* occurs higher and licenses ellipsis when it co-occurs with *bàn*, in Dutch, it is the opposite; the root modal *kunnen* which expresses “ability” does not license ellipsis as shown in the ungrammatical sentence in (39b).

(39) Dutch (Aelbrecht 2010: 151)

- a. Hij zal geen Frans kunnen spreken, maar ik vind dat hij eigenlijk
he will no French be.able speak but I find that he actually
wel moet [~~? Frans spreken~~ / ? Frans kunnen spreken.]
PRT must [French speak French be.able speak]
‘He won’t be able to speak French, but I think that he actually should
(speak French/be able to speak French).’
- b. *Hij zal geen Frans kunnen spreken, maar ik vind dat hij eigenlijk
he will no French be.able speak but I find that he actually
wel moet kunnen [~~Frans spreken.~~]
PRT must be.able [French speak]

Thus, while VPE in English can be licensed by a finite auxiliary (40a), a modal (40b), or the infinitival marker *to* (40c) as a licenser for VP ellipsis, it is not the same in Likpakpaanl and Dutch.

- (40) a. Mr Honeypie was happily eating pumpkins, but his wife certainly
wasn’t.
- b. Laura was hit by the elephant trunk, and Lola was [~~hit by the
elephant trunk~~], too.
- c. Bettina wants to hear Susanna’s story, and I also want to.

(Aelbrecht & Harwood 2019: 506, 512)

Johnson (2001) and van Craenenbroeck (2017) note that in English, only finite auxiliaries can license VP ellipsis even though there are instances such as (41) in English where the sentential negator, *not*, can sometimes license VP ellipsis.

- (41) John is leaving but Mary’s not [~~leaving~~]. (Lobeck 1995: 143)

The behaviour observed with the modal auxiliary *bàn* is not unique to Likpakpaanl because previous research on VP ellipsis has shown that remnant elements,

including aspectual and voice heads, can intervene between the modal and the elided verb. As Merchant (2013) explains, in (42), the finite auxiliary *should* elides the vP, but the non-finite auxiliary intervenes between the licensing head and the ellipsis site.

- (42) The janitor must remove the trash whenever it is apparent that it should be removed. (Merchant 2013: 78)

We have illustrated VP ellipsis in Likpakpaanl and demonstrated that such ellipsis is licensed by root modal auxiliaries, which elide the phonological content of their complements. Furthermore, I have argued that the ellipsis site possesses a complex syntactic structure that is absent from the LF. In Section 6.1, I adduce syntactic evidence for a PF-deletion account for Likpakpaanl VP ellipsis.

6.1 Motivating a PF-deletion approach to Likpakpaanl VP ellipsis

This section presents evidence supporting a PF-deletion analysis for VP ellipsis in Likpakpaanl. Under the PF-deletion approach, the ellipsis site is argued to have a full-fledged structure. Therefore, there are no semantic or syntactic differences between elliptical and non-elliptical sentences, except that the [E]-feature deletes the PF structure in the former. This approach to ellipsis postulates that the ellipsis site has a fully-fledged syntactic structure and therefore has the same underlying syntax as their non-ellipsis elided counterparts (Ross 1969, Johnson 2001, Merchant 2001, 2019).

Drawing on examples from Antecedent-Contained Deletion (ACD) and island constraints, I argue for the presence of structure at the ellipsis site in Likpakpaanl VP ellipsis. Antecedent-contained ellipsis describes a phenomenon where an elided verb phrase is contained within its own antecedent. For instance, in (43a), the verb phrase in the main clause appears to license ellipsis inside the relative clause, which modifies its object. The VP *buy all the clothes* has a salient antecedent (e-GIVEN) in the first clause and is elided inside the relative clause. The deleted VP at the ellipsis site can only mean *buy all the clothes*, indicating that the object, though raised out of the VP, is still interpreted in a VP-internal position. (43b) still exemplifies the same ACD except that in this case, the ellipsis site targeted by the modal *bàn* is the entire CP and not just the verb phrase and its arguments. The presence of such ACD constructions, which require the DP to be moved from a base position prior to ellipsis, supports the assertion that the ellipsis site has a fully-fledged syntactic structure that is unpronounced at LF.

- (43) a. Kunjam nànn dàà ò-kèkèn ò-mòmòk_i ù gà òmàn [dàà t_i] nà.
 K. PST buy 8-cloth 8-all 3SG FUT can [dàà] REL.DET
 ‘Kunjam bought all the clothes that he could.’
 b. Obed nànn bàn ké ù kàrn tì-gbàn tì-mòmòk_i Alex nànn bàn [ké
 O. PST want that RP read 14-book 14-all A. PST want [that
 ù kàrn t_i] nà.
 RP read] REL.DET
 ‘Obed wanted to read every book that Alex wanted.’

Additional A-bar movement examples (44a) demonstrate that it is possible to extract an object out of the verb phrase prior to the ellipsis and leave a trace in the base generated position. I assume that A-bar moved constituents would not have a position (trace) in their base-generated position if the ellipsis site did not contain any underlying syntactic structure (cf: Merchant 2001, 2004, van Craenenbroeck 2017, Lipták & Aboh 2013). As predicted, when the DP is inside an island as in (44b), it is impossible for extraction to take place because it violates the complex NP island constraint, which prohibits movement out of the island.

- (44) a. Philemon nyì ò-kòlà òj Líyàb_i nànn òmàn kòr nà,
 P. know 6-fowl 6.REL L. PST can slaughter REL.DET
 àmàà ù àà nyì ò-kòlà òj ù àà nànn òmàn [kòr t_j]
 but 3SG NEG know 6-fowl 6.REL 3SG_i NEG PST can [slaughter]
 nà.
 REL.DET
 ‘Philemon knows the fowls that Liyab was able to slaughter, but he doesn’t know which ones he couldn’t.’
 b. *Nyoja nànn òmàn tér bì-kpààb bì kpà ò-nà-jà_i nà, àmàà
 N. PST can help 2-farmer 6.REL have 6-cow-man REL.DET but
 m àà nyì [DP ò-nà-jà òjà] ù nànn òmàn [tér
 1SG. NEG know 6-cow-man how.many 3SG PST can [help
 bì-kpààb bì kpà t_{DP}] nà.
 2-farmer 6.rel have] REL.DET
 Intended: ‘Nyoja was able to help the farmers who had bulls, but I don’t know how many bulls he could.’

In this sub-section, I have attempted to provide some formal evidence to support the argument that there is an unpronounced underlying syntactic structure at the ellipsis site in Likpakpaanl VP ellipsis.

7 Conclusion

This paper provided an account of the derivation of verb phrase ellipsis in Likpakpaanl and showed that the language exhibits a type of VP ellipsis where the complement to the modal auxiliaries is targeted for deletion. Drawing on empirical data, I showed that VP ellipsis is licensed by a restricted class of modals, namely, *ɣmàn* and *bàn*, which functionally project a Modal phrase (ModP) below T and above AspP and vP. In light of the current ellipsis theory of Merchant (2001) and Aelbrecht (2010), I assume that it is the [E]-feature on *ɣmàn* and *bàn* that licenses the deletion of the PF content of the complement to the licensing head (Mod^o). I further adduced evidence from ACD and complex noun phrase islands to support the assumption that there is an underlying syntactic structure at the ellipsis derivation of VP ellipsis in Likpakpaanl. The significant contribution of this paper to the understanding of ellipsis is that it has provided novel data as further evidence in support of Modal Complement Ellipsis (MCE), which has been shown to exist in French, Italian and Spanish. Aside from the empirical material that contributes significantly to the literature on VP ellipsis in Mabia languages, formal accounts have been provided for the descriptive facts on Likpakpaanl VP ellipsis.

Abbreviations

1	first person	NEG	negative
2	second person	NEG.FUT	negative future
3	third person	PFV	perfective
COP	copular	PL	plural
DEF	definite	PRT	particle
FUT	future	PST	past tense
FOC	focus	REL.DET	relative determiner
HEST.PST	hesternal past	REL	relativiser
HOD.PST	hodiernal past	RP	resumptive pronoun
IPFV	imperfective	SG	singular
LOC	locative	VENT	ventive
MOD	modal		

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Chapter 9

On the clausal syntax of Dschang predicate focus doubling

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This paper investigates the syntax of Predicate Focus Doubling (PFD) in Dschang, a Grassfields Bantu language of Cameroon. PFD is similar to the well-known predicate cleft construction found in a number of other West African languages, because it encodes verbal focus and involves two instances of the verb. However, unlike the predicate cleft construction, PFD involves the right clausal edge, not the left. Through analyzing PFD, we propose that Dschang provides strong crosslinguistic support for the existence of a low focus position near the vP edge. We further argue that the surface word order in the PFD construction is derived by head movement of the verb (understood as copying) to the low focus position. Finally, we provide evidence that there is movement of a constituent, containing minimally the verb and direct object, to a position higher than that of the copy of the verb in low focus.

1 Introduction

This paper describes and analyzes PREDICATE FOCUS DOUBLING (PFD), found in Dschang (and related Grassfields Bantu languages):

(1) Dschang

- a. mən̩ lè lì
1SG DIST.PST sleep
'I slept.'
- b. mən̩ lè lì lí
1SG DIST.PST sleep sleep
'I SLEPT.'

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It is important to note that we distinguish PFD from PREDICATE CLEFT constructions (found in Niger-Congo, West African languages, Atlantic creoles (and beyond)) in which a verbal copy appears in a left-peripheral position:

- (2) Krachi (Kwa, North Guang; Ghana) (Kandybowicz & Torrence 2021)

- a. okyɪ wɔ ɛ-dɪkɛ i-gyo
 woman the PST-cook PL-yam.
 ‘The woman cooked yams.’
- b. Kɛ-dɪkɛ yɪ Okyɪ wɔ ɛ-dɪkɛ i-gyo.
 NMLZ-cook FOC woman the PST-cook PL-yam
 ‘It was COOKING that the woman did to yams (not, say, eating).’

- (3) Vata (Kru) (Koopman 1984)

- pā n kǎ mǎ pá.
 throw I will it throw
 ‘I will THROW it.’

- (4) Haitian Creole (Torchon 2012)

- a. Jan te ekri poэм lan.
 Jan PST write poem the.
 ‘Jan wrote the poem.’
- b. Se ekri Jan te ekri poэм lan.
 SE write Jan PST write poem the.
 ‘Jan WROTE the poem.’

- (5) Yiddish (Bleaman 2022)

- red-n mame-loshn red ikh (*mame-loshn).
 speak-INF mama-language speak.1SG I mama-language
 ‘As for speaking Yiddish, I speak it.’

In this paper, we argue that PFD in Dschang provides evidence for the existence of a low focus position near the vP edge (as proposed in, for example, Jayaseelan 2001 and Belletti 2004). As Figure 1a shows, the head of the focus phrase, represented as μ (canonically realized as a mora), takes vP as its complement. Specifically, we argue that the surface word order in PFD is derived in two steps: (i) a verb marked with a focus feature undergoes copy movement to this low focus position, and then (ii) a phrase consisting minimally of the verb and its internal arguments moves to a position higher than the copied verb. This is schematized in Figure 1:

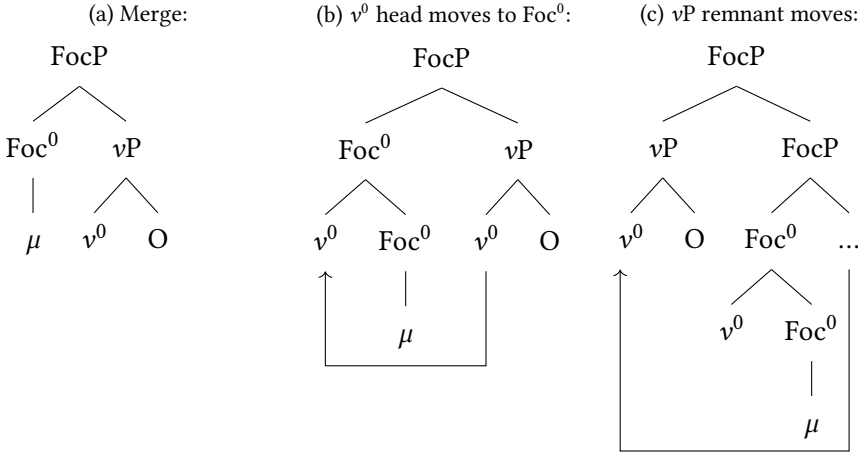


Figure 1: Derivation of predicate focus doubling

The rest of this paper is organized as follows: in Section 2 we provide a language background, in Section 3 we discuss the interpretative properties of PFD (where and how it is used), in Section 4 we discuss “high” and “low” focus in Dschang, and provide a derivation for PFD, and in Section 5 we conclude.

We note that we take PFD to be distinct from the predicate cleft construction, which canonically involves verb copying on the *left* edge of the clause. On the other hand, the PFD construction found in Grassfields Bantu languages, seems to involve verb copying on the *right* edge of the clause. The PFD construction is also of interest because it leads to insights about the general syntax of Dschang and other Grassfields languages. Work on PFD connects to efforts to understand how OV structures arise in VO languages under certain conditions (Kayne 1994, Aboh 2004). In the Grassfields literature, this construction has been analyzed, but has overall received surprisingly little attention (Nkemnji 1995 for Nweh, Nchare 2012 for Shupamem, and Zimmermann & Kouankem 2024 for Medumba).

2 Language background

Dschang (ISO 639-3 ybb; also called *Yemba* and *Bamileke-Dschang*) is a Grassfields (Bantu) language of Western Cameroon, in the Bamileke language cluster (with Fe’fe’, Ghomálá’, Kwa’, Mengaka, Nda’nda’, Ngyemboon, Ngomba, Ngombale, and Ngwe), with approximately 300,000 speakers.

Dschang has two tones, High and Low, and four surface level tones as both tones can be downstepped (Hyman 1985). It also has both lexical and grammatical

tone. The unmarked word order of Dschang is Subject > Tense > Verb > Object, as shown in (6), (7), and (8):

- (6) Shùfò lè pwàp-á màngò.
 shufo DIST.PST catch-O.CONC mango
 ‘Shufo caught the mango.’
- (7) Shùfò lè pwáp mè-màngò.
 shufo DIST.PST catch.O.CONC CL.PL-mango
 ‘Shufo caught the mangos.’
- (8) Shùfo lè lá-á fùfù.
 Shufo DIST.PST cook-O.CONC fufu
 ‘Shufo cooked fufu.’

The transitive verbs in (6), (7), and (8) display “object concord”. Object concord in Dschang and other Bamileke languages has not been extensively studied, and there is no comprehensive description or analysis (see Anderson 1983 for discussion of closely related Ngiemboong). In Dschang, the realization of object concord may be segmental, tonal, or both, depending on the noun class of the object and the shape of the verb stem (Harro & Haynes 1991). Typically concord surfaces on the verb. However, in some cases where the concord is tonal, it may surface on the object itself (Harro & Haynes 1991). In (6) and (8), the concord is a clear suffix. In contrast, in (7), it manifests as a high tone on the verb, which has overwritten the underlying verbal low tone. To our knowledge, object concord only occurs when the direct object immediately follows the verb and is not related to definiteness/specificity. We do not explore the complexities of object concord here.

Like other Bamileke languages, Dschang possesses multiple past and future tenses, approximately five pasts and five futures (Hyman 1980). We mostly restrict ourselves to the “distant past” in this paper (but there are a couple of examples in the hodiernal past, to show that PFD is not restricted to this tense.) We refer to the “zone” between the subject and lexical verb as the “preverbal field”.

The template of a simple sentence given above is augmented by a number of “particles”, auxiliaries and adverbs that can surface in the preverbal field. In the distant past, when one (or more) of these occurs in the preverbal field, the verb occurs in the “consecutive” form, characterized by the presence of a nasal prefix, which may trigger a change to the initial consonant of a verb. For detailed description and analysis of consecutive markers in other Grassfields languages,

see Tamanji (2014) for Bafut and Fominyam (2021) for Awing. The consecutive marker has a schwa allomorph that occurs in connected fast speech:

- (9) a. mènɛ lè lì sàà
 1SG DIST.PST sleep at.market
 ‘I slept at the market.’
 b. mènɛ lè sàà n-diì
 1SG DIST.PST at.market CONS-sleep
 ‘I slept at the market.’

While it is true that the consecutive form is triggered by the presence of preverbal auxiliaries and adverbs, for example, it does not occur *only* in those circumstances. For example, in the hodiernal past tense, the verb is in the consecutive form, and a floating high tone mora occurs between the subject and verb. This floating tone/mora docks to the right edge of the subject and may be realized as vowel lengthening, as in (10). In (11) the subject pronoun has low tone. However, (12) shows that the consecutive high tone may overwrite the low tone on a subject pronoun:

- (10) Shùfò-ó ɲ-káɲ mbàp.
 shufo-HOD.PST CONS-fry meat
 ‘Shufo fried meat earlier today.’
 (11) mènɛ lè zùk-ù váp
 1SG DIST.PST sweep-O.CONC dust
 ‘I swept the dust.’
 (12) mènɛ n-zùk-ù váp
 1SG CONS-sweep-O.CONC dust
 ‘I swept the dust earlier today.’

Negation in the distant past is bipartite, with *tè* occurring in the preverbal field and a high/rising tone mora on the right edge of the clause, which is realized in different ways depending on the rightmost segmental material:

- (13) Méri lè tè zá`à kéndòɲ-ó
 Mary DIST.PST NEG cut plantain-NEG
 ‘Mary did not cut plantains.’

We note that the non-future tenses precede the *tè* negation, while the future tenses follow this marker. Templatically, Dschang clauses are as follows:

(14) S T/PST Neg1 T/FUTURE V O Neg2

In the remainder of this section, we provide a brief overview of focus constructions in Dshang. As these constructions have not been studied in detail, there are necessarily a number of open empirical and analytical questions. There are basically three focus constructions that we have identified in Dshang: PFD, left peripheral focus, and post-verbal focus.

The PFD construction involves two instances of the verb. One is in “ordinary” position, following the preverbal field, while the second occurs to the right of direct objects and other clausal material (as we discuss in 4.3-4.5). For the present paper, one important point is that in PFD, the first instance of the verb has the same form (including object concord) as in its correlate non-PFD clause. The second instance of the verb in PFD, the one canonically in clause-final position, does not display object concord and may be tonally and/or segmentally distinct.

Analytically, we take this to indicate that the first instance of the verb is in the same position as in a non-PFD clause. Whatever mechanism accounts for object concord in non-PFD clauses (Spec-Head, AGREE, clitics, etc.) behaves the same way in PFD clauses for the first instance of the verb. In a PFD clause, the first instance of the verb is indistinguishable from that in a non-PFD clause. In contrast, in PFD, the second instance of the verb, in the clear cases, has a suffixal focus marker, so in most cases is distinct, depending on the shape of the verb root or stem. A precise characterization of these changes is difficult at this point in our investigation, as the conditioning factors are poorly understood. For instance, for CVC_{+STOP} verbs, there is typically an overt high toned vocalic suffix, which may be a copy of a preceding vowel (e.g., 15). For some CV verbs, there may be only a difference of tone in the second verb (e.g. 1b) or there may be changes of vowel quality (e.g. 16) or truncation of the stem (e.g. 17).

- (15) mènɿ lè v'ét v'ét-ú
 1SG DIST.PST tremble tremble-FOC
 ‘I TREMBLED instead.’
- (16) mènɿ lè kó'ə y'im kó'-ó
 1SG DIST.PST climb baobab climb-FOC
 ‘I CLIMBED the baobab instead.’
- (17) mènɿ lè là-à nnà là-á
 1SG PST cook-O.CONC soup cook-FOC
 ‘I COOKED soup instead.’

The second construction is a left peripheral (cleft-like) focus that involves an *ààà* (18b-18c) or *iiî* (18e) that immediately precedes the focused item, which may be followed by a noun-class agreeing marker that is also found in relative clauses. For focused subjects, this marker is obligatory (18b):

- (18) a. *mbhù lè lùŋ-è kùnà*
dog DIST.PST bite-O.CONC pig
‘The dog bit the pig.’
- b. *ààà mbhù *(i-y’i) lè lùŋ-è kùnà*
FOC.ANIM dog CL-REL DIST.PST bite-O.CONC pig
‘It’s the dog that bit the pig.’
- c. *ààà kùnà (i-y’i) mbhù lè lùŋ*
FOC.ANIM pig CL-CL.REL dog DIST.PST bite
‘It’s the pig that the dog bit.’
- d. *mèn lè tyúè è-gèm zò*
1SG DIST.PST cut.down CL-baobab.tree yesterday
‘I cut down the baobab tree yesterday.’
- e. *iiî è-gèm (è-sè) mèn lè tyúè zò*
FOC.INAN CL-baobab.tree CL-CL.REL 1SG DIST.PST cut.down yesterday
‘It’s the baobab tree that I cut down.’
- f. *ààà zò mèn lè tyúè è-gèm*
FOC.ANIM yesterday 1SG DIST.PST cut.down CL-baobab.tree
‘It’s yesterday that I cut down the baobab tree.’
- g. **ààà/*iiî tyúè mèn lè tyúè è-gèm*
FOC.ANIM/FOC.INAN cut.down 1SG DIST.PST cut.down CL-baobab.tree
Intended: ‘It’s cut down the baobab tree that I did.’

Example (18g) shows that verbs cannot be focused using the left peripheral strategy. We have given the relative markers as optional. But we do not know whether the forms with and without the relative markers represent a single construction. As indicated, the difference between the use of *ààà* or *iiî* may have to do with animacy. But, this is not clearcut since it is possible to use the *ààà* with some inanimates, as in (18f). Generally the use of *ààà* seems to be much freer than *iiî*, but we leave this as an open question here.

The third focus construction involves the post-verbal domain. Postverbal constituents are focused when immediately preceded by an *àà*. This can be seen for a direct object (19b) and adverb (19c) below:

- (19) a. mènɿ lè zhùè mbàp zò
 1SG DIST.PST buy meat yesterday
 'I bought meat yesterday.'
- b. mènɿ lè zhùè áà mbàp zò.
 1SG DIST.PST buy FOC meat yesterday
 'It was meat that I bought yesterday.'
- c. mènɿ lè zhùè mbàp áà zò
 1SG DIST.PST buy meat FOC yesterday
 'It was yesterday that I bought meat.'
- d. *mènɿ lè zhùè ìì mbàp zò
 1SG DIST.PST buy FOC meat yesterday
 'It was yesterday that I bought meat.'

Example (19d) shows that the inanimate focus marker does not occur in the postverbal focus construction. From this, we may tentatively conclude that the postverbal focus markers are distinct from the left peripheral focus markers or at least have a different distribution from them.

It is also possible to focus a verb by adding a V̂V̂ (HL) sequence at the right edge of the verb. Although the tonal contour seems to be the same as with the áà postverbal focus marker, the vowel quality varies, depending on the shape of the verb. At this point, it is unclear whether this focus marker is the same as the other postverbal focus marker áà. Note, too, that this focus marker is phonologically distinct from the PFD suffixal focus marker. This *predicate focus suffixation* is shown below. At this stage, too, it is unclear how this focus marker interacts with object concord on the verb. Thus in the examples below, the final vowel(s) and tones of the verb differ in neutral (20a), PFD (20b) and PFS (20c) clauses:

- (20) a. mènɿ lè zòòtì mè-là
 1SG DIST.PST insult CL-vampire
 'I insulted the vampires.'
- b. mènɿ lè zòòtì mé-là zòòtí-í *Predicate focus doubling*
 1SG DIST.PST insult CL-vampires insult-FOC
 'I INSULTED the vampires instead.'
- c. mènɿ lè zòòté-è mé-là. *Predicate focus suffixation*
 1SG DIST.PST insult-FOC CL-vampires
 'I INSULTED the vampires instead.'

Similarly, (21b) and (21c) show the difference in form between a PFD verb and its PFS counterpart:

- (21) a. mènɲ lè tyùé kéndónɲ
 1SG DIST.PST harvest plantain
 ‘I harvested the plantain.’
- b. mènɲ lè **tyùé** kéndónɲ **tyùé-é** *Predicate focus doubling*
 1SG DIST.PST harvest plantain harvest-FOC
 ‘I HARVESTED the plantain instead.’
- c. mènɲ lè **tyú-ù** kéndónɲ *Predicate focus suffixation*
 1SG DIST.PST harvest-FOC plantain
 ‘I HARVESTED the plantain instead.’

We now turn to the analytical background.

2.1 Analytical background

The analysis that we argue for in this paper, predicated on the existence of a low focus position in the clause for verb focus, belongs to a class of analyses posited for multiple Niger-Congo languages. Collins & Essizewa (2007) report on Kabiye, a Gur/Mabia language of Togo, which has a verb doubling focus construction that looks similar to that found in Dschang:

- (22) Kabiye (Collins & Essizewa 2007)
 ma-ní-ʊ kabiye kí ní-ʊ ma-a yɔɔd-ʊ kó.
 1SG-understand-IPFV Kabiye KI understand-INF 1SG-NEG speak-IPFV it
 ‘I only understand Kabiye. I don’t speak it.’

They argue that verb doubling focus like (22) above involves a low focus phrase (FocP) that is immediately above VP, in addition to a left peripheral focus position. Verb focus only involves the low focus position. The phrase immediately above the FocP is *KIP*, which is headed by the element *kí*. In a verb focus clause like (22) above, an infinitival copy of the verb is made and moves to Spec,FocP. The entire VP, with the object and the first merged instance of the verb, then raises to Spec,*KIP* (and from this position, the verb then head raises to *INFL*).

Within Grassfields, Becker & Nformi (2016) analyze verb focus in Limbum, in the Nkambe branch within Eastern Grassfields. They argue that Limbum has both high (left peripheral) and low focus positions, that low focus sits just above *vP* (and is headed by *bá*) and is associated with both verb focus and *wh*-questions

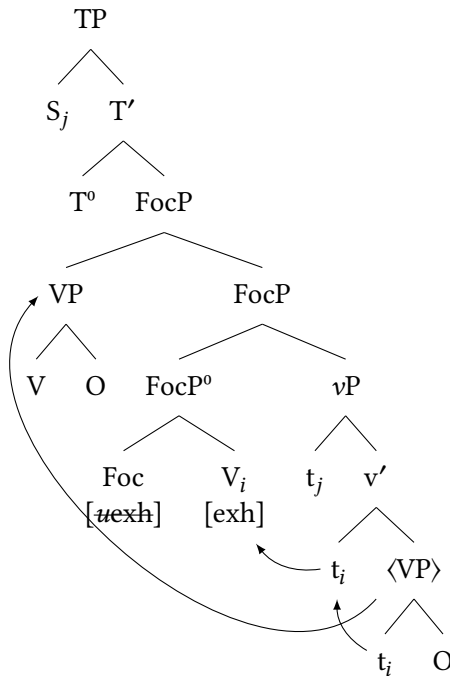
(23b), for example. As in Dschang, the focused verb is doubled and sandwiches a direct object, as in (23a):

(23) Limbum (Becker & Nformi 2016)

- a. njínwè fō bí yū msān yú
 woman DET FUT1 buy rice buy
 ‘It is buying that the woman will do to the rice.’
- b. à bā zhē bá ndà bāā?
 EXPL PST1 eat FOC who fufu
 ‘Who (if not X/of them) ate fufu?’

They propose that under verb focus, the verb head raises to Foc^0 , yielding a complex head. When this happens, low Foc^0 , *ba*, does not surface. The νP , (with the lowest copy of the verb) raises to SpecFocP , which results in the surface order seen in (23a) above.

(24) Analysis of PFD in Limbum (Becker & Nformi 2016), ex. (49)



As will be shown, our analysis of Dschang is very close to Becker and Nfomi's. However, the analysis in (24) does not directly carry over once the data set is expanded to more than direct objects in Dschang.

Verb focus in Medumba, another Bamileke language is analyzed in Kouankem & Zimmermann (2013) and Zimmermann & Kouankem (2024). Here too, the copies of the focused verb surround the direct object:

- (25) Medumba (Zimmermann & Kouankem 2024)
 Nǎná ná' lù' bí á lù'ù.
 Nana P₆ take knife FOC take
 'Nana TOOK the knife (... she did not steal it).' (ex. (3a), adapted)

Similar to Dschang, Medumba also has left peripheral and postverbal foci:

- (26) Medumba (Zimmermann & Kouankem 2024)
- a. á nǔm Nǎmí Nǎná ná' fá chàṅ lá.
 FOC to Nami Nana REM.PST give food C.DEF
 'To NAMI, Nana gave food (...and not to NGAMI)'.¹ (ex. (11), adapted)
- b. nyú fè nchóg á mè
 snake PST.NEAR bite FOC me
 'A snake bit ME (...and not NGAMI)'. (ex. (7), adapted)
- c. Nǎná ná' fá chàṅ á nǔm Nǎmí.
 Nana REM.PST give food FOC to Nami
 'Nana gave food to NAMI (...and not to NGAMI)'. (ex. (8), adapted)

They note that cases like (25) are superficially similar to what is found in Kabiye (Collins & Essizewa 2007). However, they argue that Medumba verb focus doubling does not involve a low focus position. (In their analysis, Medumba only has a left peripheral focus position.) They observe that the second instance of the verb is preceded by an *á* focus marker. Just in terms of linear order this is unexpected if the focused verb is in SpecFocP. It is also important that the focus marker *á* is used to focus other non-verbal categories like nouns and adverbs, even in the postverbal focus construction:

- (27) Medumba (Zimmermann & Kouankem 2024)
- a. Nǎná ná' lù' á bí
 Nana P₆ take FOC knife
 'Nana took a KNIFE.' (ex.(38b))

¹C.DEF is the "definite clausal determiner".

- b. Nǎná ná' zhá càŋ á ndúndú, kú bú á ndà'ndà
 Nana P₆ eat food FOC quickly, not be FOC slowly
 'Nana ate food QUICKLY, not SLOWLY.' (ex. (34))

They also point out that the second copy of the verb may occur as an infinitive (i.e., a non-verbal category) in some cases:

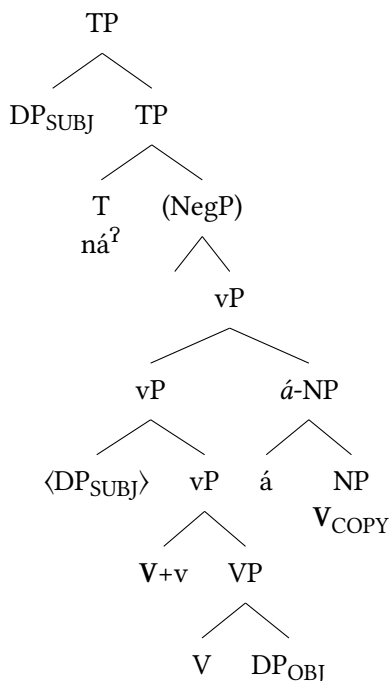
- (28) Medumba (Zimmermann & Kouankem 2024)
 Nǎná ná lù' bí á nda' nà lù'ù.
 Nana P₆ take knife FOC only INF take
 'Nana ONLY TOOK a knife.' (ex. (27))

Zimmermann & Kouankem (2024) suggest that in PFD, the second instance of the verb may not be a regular finite verb (in the cases where that verb does not carry any otherwise nominal morphology), but they leave it as an open issue. In the earlier analysis of Kouankem & Zimmermann (2013), the focus marker selects for a nominal and this is why it can occur with a clearly nominalized verb (in some cases), like an infinitive. Crucially, the focused verb need not surface in clause-final position, which is the same behavior as vP/VP adjuncts:

- (29) Medumba (Zimmermann & Kouankem 2024)
- a. Nǎná lú fá tǎŋ nǔm Nǎmi dʒù' ŋwà'nì ngàb mù á fá.
 Nana P₅ give food to Nami at school week last FOC give
 'Nana GAVE food to Nami at school last week.' (ex. (36b))
- b. Nǎná lú fá tǎŋ nǔm Nǎmi á fá dʒù' ŋwà'nì ngàb mù'.
 Nana P₅ give food to Nami FOC give at school week last
 'Nana GAVE food to Nami at school last week.' (ex. (36c))

They therefore argue that the *á*-marked copy of the verb is a right adjoined to v/VP like other v/VP adjuncts. Verb focus is brought about by copying the semantic and phonological features of the verb onto a dummy (nominalized form of the) verb that is contained in an "*á*-NP," right adjoined to vP. (See Zimmermann & Kouankem 2024 for details about the copy process.):

- (30) Analysis of PFD in Medumba (Zimmermann & Kouankem 2024 ex (49))



With this background in place let us now turn to a description of the interpretive properties of Predicate Focus Doubling (PFD) in Dschang.

3 Interpretation and morphology of Predicate Focus Doubling

In this section, we lay out the basic interpretive and morphological properties of PFD. In a nutshell, PFD is used to contrastively focus the verb. This can be seen in the dialogue in (31). The use of PFD is felicitous in (31), where ‘cut’ is contrasted with ‘cook’. (32) shows that PFD is infelicitous when the context involves focus on something which is not the verb, such as the object.

- (31) A: (tè) ù lè láà kéndòŋ-ò
Q 2SG DIST.PST cook plantain-Q
‘Did you cook the plantains?’
B: mèn lè **tyúò** kéndòŋ **tyú-ó**
1SG DIST.PST cut plantain cut-FOC
‘I CUT the plantains.’

- (32) A: *ú n-ty'úà k'éndòŋ-ò?*
 2SG.HOD.PST CONS-cut plantain-Q
 'Did you cut the plantains earlier today?'
 B: *# mén n-ty'úà ŋkëndíŋà ty'ú-ó*
 1SG.HOD.PST CONS-cut banana cut-FOC
 Intended: 'I cut the BANANAS earlier today.'

The examples in (31) and in (32) also show that PFD can be used in different tenses such as the distant past and hodiernal past. In fact, we do not know of any tense/aspect restrictions on PFD. PFD is not deployed to indicate VERUM focus, i.e., to emphasize the truth of the propositional content of a sentence. This can be seen in the contrast in felicity between the two possible responses in (33). With B1, the speaker is correcting/contesting the veracity of the proposition put forth by A. B2 shows that use of PFD in this context is infelicitous.

- (33) A: *Mérì lè tè sók sì ndjè-!è!*
 Mary DIST.PST NEG clean ground house-NEG
 'Mary didn't clean the floor!'
 B1: *ŋgân, Mérì lè sók sì ndjè!*
 no Mary DIST.PST clean ground house
 'No, Mary DID clean the floor!'
 B2: *# ŋgân, M'érì lè sók sì ndjè sók-ó!*
 no Mary DIST.PST clean ground house clean-FOC
 Intended: 'No, Mary DID clean the floor!'

Morphologically, doubling is restricted to verbal elements. The ungrammatical (34b) is an attempt to double the adverbial *zé*, which occurs in the preverbal field:

- (34) a. *Mérì lè zé n-z'à'à k'éndòŋ zá'-á*
 Mary DIST.PST again CONS-cut plantain cut-FOC
 'Mary CUT plantains again'
 b. * *Mérì lè zé n-z'à'à k'éndòŋ zè-é*
 Mary DIST.PST again CONS-cut plantain again-FOC
 Intended: 'Mary cut plantains AGAIN.'

A subset of auxiliary verbs can be doubled:

- (35) a. *mèn lè tèàlì ŋ-kàn mbàp*
 1SG DIST.PST do.early CONS-fry meat
 'I fried the meat early.'

- b. mènɲ lè tɛ̀ə̀li tɛ̀ə̀li-í ɲ-kàɲ mbàp
 1SG DIST.PST do.early do.early-FOC CONS-fry meat
 ‘I fried the meat EARLY (instead of at some other time).’
- c. mènɲ lè tɛ̀ə̀li ɲ-kàɲ mbàp tɛ̀ə̀li-í
 1SG DIST.PST do.early CONS-fry fish do.early-FOC
 ‘I fried the meat EARLY (instead of at some other time).’

It is possible, in addition, to copy verbs that are used functionally, such as those used in serial verb constructions:²

- (36) a. mènɲ lè zùk nzèt mó n-djè Shùfò
 1SG DIST.PST wipe feces child CONS-give Shufo
 ‘I cleaned the baby poop for Shufo.’
- b. mènɲ lè zùk nzèt mó {zùg-ú} n-djè Shùfò {zùg-ú}
 1SG DIST.PST wipe feces child wipe-FOC CONS-give Shufo wipe-FOC
 ‘I CLEANED the baby poop for Shufo.’
- c. mènɲ lè zùk nzèt mó n-djè Shùfò {yé-é}
 1SG DIST.PST wipe feces child CONS-give shufo give-FOC
 ‘I cleaned the baby poop FOR Shufo.’

It is not possible to double both of the verbs at the same time in the benefactive serial construction:

- (37) a. * mènɲ lè zùk nzèt mò zùg-ù n-djè Shùfò yé-é
 1SG DIST.PST wipe feces child wipe-FOC CONS-give shufo give-FOC
 Intended ‘I instead cleaned the baby poop for Shufo.’
- b. * mènɲ lè zùk nzèt mò n-djè Shùfò yé-é zùg-ú
 1SG DIST.PST wipe feces child CONS-give shufo give-FOC wipe-FOC
 Intended ‘I instead cleaned the baby poop for Shufo.’

In summary, Predicate Focus Doubling is a construction in which two copies of the same verb occur in a single clause, marking the verb for contrastive (but not verum/polarity) focus. Note that this is in line with what Zimmermann & Kouankem (2024) report for Medumba, where PFD is also used for contrastive focus.

²Curly brackets indicate optional linearization: the elements within the curly brackets may occur in either (but not both) of the positions they appear in.

4 A low focus projection in Dschang

4.1 The position of focus

In this section we argue that PFD involves a focus position that is lower than TP. That is, we argue that Dschang provides cross-linguistic support for the existence of a ν P periphery, as suggested by Belletti (2001, 2004). In our analysis, we assume, following Kayne (1994), that specifiers precede heads, heads precede complements, and movement is only to the left. The first piece of evidence for the existence of a low focus position comes from the fact that PFD seems to occur lower than T. That is, both copies of the verb occur to the right of the tense marker and NEG1, *te* in (38):

- (38) Shùfo tè lùú tɔŋ-ɔ k'ɛ̀tì tɔŋ-ɔ
 Shufo NEG1 FUT read-O.CONC book read-FOC.NEG2
 'Shufo will not READ the book.'

Note that in Dschang, the distant past precedes NEG1 (as in 13), while the distant future follows it (c.f. Nkemnji 1995 for Nweh). Thus, PFD seems to involve a domain lower than the lowest tenses.

The second piece of evidence is the fact that Dschang also has left peripheral foci, which surface preceding the subject (and tense and negation), as in (39) below, as previously noted. This is what corresponds to the expected left peripheral focus position (Aboh 1995, Rizzi 1997):

- (39) ààà kùnà ì-yì mènɲ lè tsw'è-è
 FOC pig CL-REL 1SG DIST.PST kill-REL
 'It's a pig that I killed.'

Combining the data from left peripheral and low focus yields the following template:

- (40) <Left peripheral Focus> ... S T_{PST} (NEG) T_{FUT} V1 O V2

The third piece of evidence for low focus in Dschang is that both the left peripheral focus and the PFD focus can co-occur (41):

- (41) ààà mɛ-lù è-mhì mènɲ lè zòòtì zòòtì-ì
 FOC PL-vampire CL-REL 1SG DIST.PST insult insult-FOC
 'It's vampires that I INSULTED (instead of doing something else to them).'

The fact that two distinct foci can co-occur can be made sense of if they do not compete for the same position. Putting these facts together, we conclude that there is a focus position lower than T in Dschang.

4.2 Relative ordering of V1 and V2 and other elements

We now turn to the distribution of different types of arguments and adjuncts in PFD, starting with internal arguments of the predicate, before turning to adjuncts and clausal arguments.

4.2.1 Internal arguments

Most of the examples to this point have involved transitive verbs. As in (1b) though, it is possible for intransitive verbs to double, as seen in (42) and (43).

- (42) Màngò wè'à lùtì lùtì-í.
mango this sweet sweet-FOC
'This mango is SWEET.'/'This mango is simply sweet.'

- (43) Mò lè wùì wùì-í.
child DIST.PST laugh laugh-FOC
'The child LAUGHED.'

Along the same lines, verbs of various semantic types can undergo PFD, including experiencer/psyche verbs such as 'be afraid' in (44):

- (44) Mèŋ pòk pòk-ó.
1SG afraid afraid-FOC
'...I'm AFRAID (of it).'

We have shown several instances of PFD so far. Analytically, it is important that direct objects must surface between the two copies of the verb, as the contrast in grammaticality in (45) shows:

- (45) Transitives – VOV:
a. Méri lè zá'-á kéndòŋ zá'-á.
Mary DIST.PST cut-O.CONC plantain cut-FOC
'Mary CUT plantains.'
b. *Méri lè zá'-á zá'-á kéndòŋ
Mary DIST.PST cut-O.CONC cut-FOC plantain
Intended: 'Mary CUT plantains.'

Similarly, in a double object construction, both the direct and indirect objects must surface between the two instances of the verb as in (46b); (46c) is ungrammatical.

(46) Double object constructions – V IO DO V:

- a. Shùfò lè twí-í Màfò kèt'í
 Shufo DIST.PST show-O.CONC Mafo book
 'Shufo showed Mafo the book.'
- b. Shùfò lè **twí-í** Màfò kèt'í **twì-í**
 Shufo DIST.PST show-O.CONC Mafo book show-FOC
 'Shufo **SHOWED** Mafo the book.'
- c. *Shùfò lè **twí-í** Mafo **twì-í** kèt'í
 Shufo DIST.PST show-O.CONC Mafo show-FOC book
 Intended: 'Shufo **SHOWED** Mafo the book.'

Although PFD involves two copies of the verb, they need not be strictly identical, as in (47b) and (48b), where the right edge of the second copy of the verb is altered.

- (47) a. mènɲ lè kó'ə y'ím
 1SG DIST.PST climb baobab
 'I climbed the baobab.'
- b. mènɲ lè **kó'ə y'ím kó'ə**
 1SG DIST.PST climb baobab climb-FOC
 'I **CLIMBED** the baobab instead.'
- (48) a. mènɲ lè nón-ó nè k'ò
 1SG DIST.PST lay.down-O.CONC in hammock
 'I laid down in the hammock.'
- b. mènɲ lè **nón-ó** nè k'ò **nón**
 1SG DIST.PST lay.down-O.CONC in hammock lay.down .FOC
 'I **LAI**D DOWN in the hammock instead.'

These data suggest that the second copy of the verb does indeed have a suffix when it is focused.

From the data in this section there are two things that must be accounted for: (i) direct and indirect objects must occur between V1 and V2 (ii) V2 carries a suffix. With these facts in mind, we conclude that the suffixed verb copy is the result of verb head raising (or being copied) to the low Foc⁰. To derive the surface strings, we propose a two-step derivation of PFD. In the first step, a verb bearing a focus feature undergoes head movement to Foc⁰ (see Figure 2). In the second step, *vP* undergoes movement to [Spec,FocP/ZP] (Figure 3).

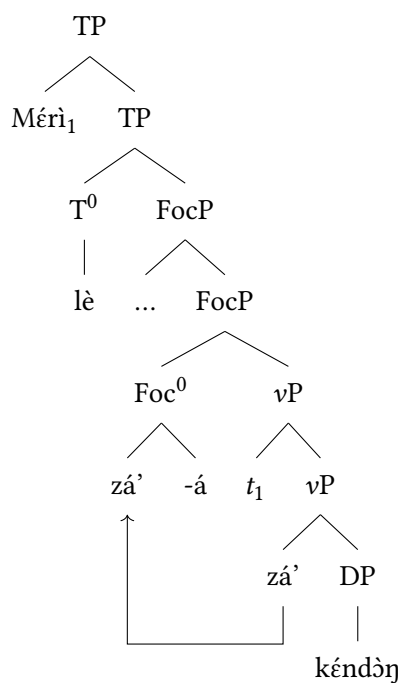


Figure 2: First step in derivation of (45)

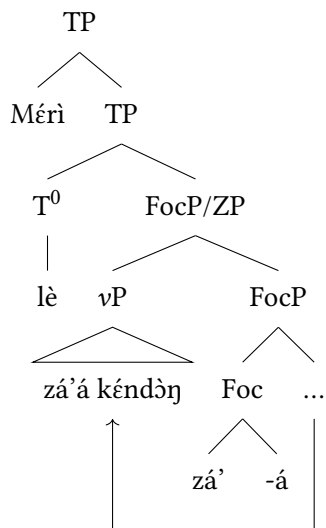


Figure 3: Second step in derivation of (45)

This captures the relative ordering of the two verbs, direct and indirect objects, and the suffix on V2, which is the head of the low focus projection. This analysis is also compatible with previous accounts of predicate focus in Nweh (Nkemnji 1995) and Shupamem (Nchare 2012). Crucially, the structure and derivations proposed also make predictions about the linearization of other elements, which we address below. We noted previously that in the PFD construction, the first instance of the verb has the same morphology and behavior as a corresponding verb in a non-PFD clause. We assume that a non-PFD clause does not involve activation of Low Focus. However, in the derivation above we have put the moved ν P in SPECFOCP/ZP. It is important to note that the derivation we have given in the trees above is “shorthand”. Since the first instance of the verb in PFD appears to be an ordinary verb, we assume that it undergoes whatever movement to whatever position a verb in the correlate non-PFD clause does. We label this position “ZP”. That is, in a PFD clause, the ν P may move through SpecFocP, but ends up in a higher position.

4.3 Prepositional phrases

Unlike internal arguments of the verb, prepositional phrases appear after V_2 as in (49):

- (49) a. mènɛ lè là-à nnà là-á [pàk Méri]
 1SG PST cook-O.CONC soup cook-FOC with Mary
 ‘I COOKED soup with Mary.’
 b. *mènɛ lè là-à nnà [pàk Méri] là-á
 1SG PST cook-O.CONC soup with Mary cook-FOC
 Intended: ‘I COOKED soup with Mary.’

On the other hand, LOCATIVE PPs freely occur before (50e) or after (50d) the second verb, or before both verbs (50c). Note, also, that locative PPs can surface following the direct object (50a, 50d, 50e) or in the preverbal field (50b, 50c). We indicate the possible positions of the PPs with curly brackets. It is not the case that the PPs can be repeated.

- (50) a. mènɛ lè fín-é mbàp {mé ntsè}
 1SG DIST.PST sell-O.CONC meat in street
 ‘I sold the meat in the street.’
 b. mènɛ lè {mé ntsè} m-fín-é mbàp
 1SG DIST.PST in street CONS-sell-O.CONC meat
 ‘I sold the meat in the street.’

- c. mènɛ lè {mé ntsè} m-fɪŋ-é mbàp fɪŋ
 1SG DIST.PST in street CONS-sell-O.CONC meat sell.FOC
 ‘I sold the meat in the street instead.’
- d. mènɛ lè fɪŋ-é mbàp fɪŋ {mé ntsè}
 1SG DIST.PST sell-O.CONC meat sell.FOC in street
 ‘I instead sold the meat in the street.’
- e. mènɛ lè fɪŋ-é mbàp {mé ntsè} fɪŋ
 1SG DIST.PST sell-O.CONC meat in street sell.FOC
 ‘I instead sold the meat in the street.’

4.4 The Position of adverbs

Postverbal MANNER ADVERBS can appear optionally before and after V_2 , as in (51).

- (51) a. mò lè wù mètù wì-í
 child DIST.PST laugh loudly laugh-FOC
 ‘The child LAUGHED loudly.’
- b. mò lè wù wì-í mètù
 child DIST.PST laugh laugh-FOC loudly
 ‘The child LAUGHED loudly.’

At least some aspectual (and manner) adverbs can surface in the preverbal field. This includes the complex phrasal adverbs corresponding to “suddenly”, as shown in (52).

- (52) a. Méri lè sà'-à ɲ-kàp
 Mary DIST.PST grab-O.CONC CL-money
 ‘Mary grabbed the money.’
- b. Méri lè sà'-à ɲ-kàp sà-ú
 Mary DIST.PST grab-O.CONC CL-money grab-FOC
 ‘Mary grabbed the money instead.’
- c. Méri lè sà'-à ɲ-kàp tà' lù'gù
 Mary DIST.PST grab-O.CONC CL-money one time
 ‘Mary suddenly grabbed the money.’
- d. Méri lè tà' lù'gù è-sà'-à ɲ-kàp.
 Mary DIST.PST one time CONS-grab-O.CONC CL-money
 ‘Mary suddenly grabbed the money.’

- e. Méri lè sà'-à ń-kàp tà' lùgù sà-ú tà' lùgù.
 Mary DIST.PST grab-O.CONC CL-money one time grab one time
 'Mary suddenly grabbed the money instead.'

Similar freedom of position is found with PP adverbs that we take to be derived from adjectives (53):

- (53) a. mò lè sòŋli [mè tjùè thù]
 child DIST.PST speak with strong head
 'The child spoke stubbornly.'
 b. mò lè sòŋli {mè tjùè thù} sòŋli-í {mè tjùè thù}
 child DIST.PST speak with strong head speak-FOC with strong head
 'The child spoke stubbornly instead.'

TEMPORAL ADVERBS pattern differently from manner adverbs and appear after V₂ as in (54):

- (54) a. mèn lè kàà kà-á zò
 1SG DIST.PST tired tired-FOC yesterday
 'I was TIRED yesterday.'
 b. *mèn lè kàà zò kà-á
 1SG DIST.PST tired yesterday tired-FOC
 Intended: 'I was TIRED yesterday.'

NOMINAL LOCATIVE ADVERBS appear before V₂ (55b) and are very degraded when they appear after V₂ (55c):

- (55) a. Méri lè gók sàà
 Mary DIST.PST work at.market
 'Mary worked at the market.'
 b. Méri lè gók sàà gòk-é
 Mary DIST.PST work at.market work-FOC
 'Mary WORKED at the market.'
 c. ?? Méri lè gók gòk-é sàà
 Mary DIST.PST work work-FOC at.market
 Intended: 'Mary WORKED at the market.'

The ADDITIVE particle sè 'also' must appear before V₂ as in (56b); it cannot appear after V₂ (56c):

- (56) a. Shùfò lè pùə sè
 Shufo DIST.PST scream also
 ‘Shufo also screamed.’
 b. Shùfò lè pùə sè pùə
 shufo DIST.PST scream also scream
 ‘Shufo also screamed instead.’
 c. *Shùfò lè pùə pùə sè
 shufo DIST.PST scream scream also
 Indended: ‘Shufo also screamed instead.’

Under our analysis, adjuncts appearing after the second verb result from a lower VP, moving to Spec,FocP, as seen in Figure 4.

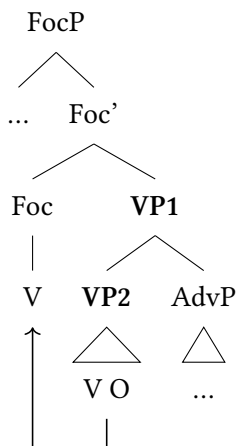


Figure 4: Derivation of adjunct position.

This analysis essentially says that the adverbial/PP adjuncts have different heights at which they can or must adjoin to VP. As a reviewer points out, if this is the case, then we should expect to find scopal or interpretive differences dependent on the attachment height of the adjunct (i.e., whether it precedes or follows the second instance of the verb). We have not been able to detect any clear differences in interpretation and we leave this prediction for future testing. It is important to note that the derivation we have suggested does not account for all of the data we have presented here, as noted by a reviewer. For example, under our proposed analysis, it's not clear how to account for the distribution of temporal adverbs or the nominal locative adverbs.

4.5 Clausal complements

In this section, we discuss different types of embedded clauses. Infinitival/control clausal complements may occur between the two copies of the verb or following the second copy (57), unlike DP complements:

- (57) a. mènɿ lè lùglì {lùgl-ì} [lè-vùè sw hè ndùn] {lùgl-ì}
 1SG DIST.PST forget forget-FOC INF-buy fish dried forget-FOC
 ‘I FORGOT to buy dried fish.’
 b. mènɿ lè fà’tì {fà’t-ì} [lè-vùè sw hè ndùn] {fà’t-ì}
 1SG DIST.PST fail fail-FOC INF-buy fish dried fail-FOC
 ‘I FAILED to buy dried fish.’

Infinitival complements of intransitive verbs also exhibit this optionality (58):

- (58) a. mènɿ káà kà-á [lè-sòɿò màtùè]
 1SG tired tired-FOC INF-pull car
 ‘I’m TIRED of driving cars.’
 b. mènɿ káà [lè-sòɿò màtùè] kà-á
 1SG tired INF-pull car tired-FOC
 ‘I’m TIRED of driving cars.’

Similarly, (at least some) adjunct clauses can also precede or follow the second verb copy, such as ‘until’ clauses (59):

- (59) a. Shùfò lè gók [t’è wò gùè]
 shufo DIST.PST work until 2SG go
 ‘Shufo worked until you left.’
 b. Shùfò lè gòk gòk-ó [t’è wò gùè]
 shufo DIST.PST work work-FOC until 2SG leave
 ‘Shufo worked instead until you left.’
 c. Shùfò lè gók [t’è wò gùè] gók-ó
 shufo DIST.PST work until 2SG leave work-FOC
 ‘Shufo worked instead until you left.’

Tensed CP complements to V are of varying degrees of strangeness or unacceptability when they intervene between the verb copies. The result is fine when the tensed CP follows the second copy of the verb in (60b) and (61b), but is degraded (60c) or ungrammatical (61c) when it precedes it.

- (60) a. Shùfò lè lyá xà wù lè zhúè mbàp
shufo DIST.PST say that 2SG DIST.PST buy meat
'Shufo said that you bought meat.'
- b. Shùfò lè lyá lyá-á xà wù lè zhúè mbàp
shufo DIST.PST say say-FOC that 2SG DIST.PST buy meat
'Shufo SAID that you bought meat.'
- c. ??Shùfò lè lyá lyá-á xà wù lè zhúè mbàp.
shufo DIST.PST say say-FOC that 2SG DIST.PST buy meat
Intended: 'Shufo said that you bought meat.'
- (61) a. Shùfò lè látí xà wù lè zhúè mbàp
shufo DIST.PST lie that 2SG DIST.PST buy meat
'Shufo lied that you bought meat.'
- b. Shùfò lè látí látí-í xà wù lè zhúè mbàp
shufo DIST.PST lie lie-FOC that 2SG DIST.PST buy meat
'Shufo lied instead that you bought meat.'
- c. *Shùfò lè láatí xà wù lè zhúè mbàp láatí-í
shufo DIST.PST lie that 2SG DIST.PST buy meat lie-FOC
Intended: 'Shufo lied instead that you bought meat.'

This pattern does not straightforwardly fall out from our analysis. The *vP* containing the TP/CP complement would move to [Spec,FocP], linearizing to the left of V2. This predicts that cases where the tensed CP occurs between the two verb copies should be acceptable. One potential solution could be that the *vP* containing the TP/CP complement *does* move to [Spec,FocP], however the TP/CP complement is able to optionally extrapose to a clause final position. We leave this as an open question, as we have not investigated CP extraposition in Dschang.

That finite clauses are extraposed is also suggested by the fact that Heavy NP shift occurs elsewhere in Dschang. It turns out that grammatically complex objects such as nouns modified by relative clauses (62) or adjectives (62) exhibit the same optionality observed with TP/CP complements; they can occur to the left or right of V2:

- (62) mən̩ lè tyúə̌ {tyú-ə̌} ŋkëndínà yì Méri lè phyi̯
1SG DIST.PST cut.down cut.down-FOC banana REL Mary DIST.PST plant
{tyú-ə̌}
cut.down-FOC
'I HARVESTED the bananas Mary planted.'

- (63) mən̩ lè tyúá {tyú-á} myè pàjá ɲkèndínà {tyú-á}
 1SG DIST.PST cut.down cut.down-FOC big ripe banana cut.down-FOC
 ‘I HARVESTED the big ripe banana.’

Since Dschang does not ordinarily allow for material to intervene between the verb and its object, the existence of Heavy NP Shift lends plausibility to the idea that other kinds of verbal complements could be right-shifted.

5 Conclusion

In this paper, we have introduced Predicate Focus Doubling in Dschang. Through our analysis of this construction, we have argued that Dschang provides strong evidence for the existence of a *vP* peripheral focus position lower than TP/IP, as argued for by Belletti (2001, 2004). This raises the issue of whether such a low focus position can be detected in other Grassfields languages, where PFD has been analyzed (e.g. Nkemnji 1995 for Nweh and Nchare 2012 for Shupamem). If this is correct, then one question that arises is whether things other than verbs can be focused in LowFocP. At this point, we have not found any cases where a non-verb can access the putative low focus position. For example, a direct object cannot be focused in low focus (64):

- (64) *mən̩ lè zhúè mbàp-á
 1SG DIST.PST buy meat-FOC
 Intended ‘I bought MEAT.’

This is unexpected, if this is simply a phrase where focus is encoded. However, at the same time, we showed previously in (18g) that the left peripheral focus is unavailable to verbs. Nothing in our analysis predicts this difference. This is thus one of the outstanding issues related to our particular analysis. It is useful to consider the issue in a larger context. In the analysis of Belletti (2004), the Low Focus position in Italian is involved in *subject* focus, in particular, the derivation of postverbal subjects. Along the same lines, Collins & Essizewa (2007) argue that postverbal *subjects* in several Bantu languages make use of a low focus position. In their description of Kabiye, Collins & Essizewa (2007) show that left peripheral focus can be used for objects, but not for subjects. The low focus position, on the other hand, can be used by subjects, objects, and verbs. Looking across languages (for which a low focus position has been posited), it is involved in the focusing of (i) subjects (Italian; Bantu) (ii) objects (Gungbe) (iii) verbs (Dschang; Limbum; Nweh) (iv) subjects, object, verbs, etc. (Agham Aboh 2007a,b; Kabiye). However,

similar restrictions are found for left peripheral focus (although much less discussed). We showed that left peripheral focus excludes verbs in Dschang, and in Kabiye it excludes subjects. The picture that emerges is quite mixed with respect to which periphery is employed for what kind of focus. Thus, there remain several outstanding questions in this area, which we leave for future research.

Abbreviations

1	first person	INF	infinitive
2	second person	IPFV	imperfective
ANIM	animate	NEG	negative
C.DEF	clausal determiner	NMLZ	nominalizer
CL	noun class	O.CONC	object concord
CONS	consecutive	PL	plural
DIST.PST	distant past	PST	past
EXPL	expletive	REM.PST	remote past
FOC	focus	Q	question particle
FUT	future	REL	relative
HOD.PST	hodiernal past	SG	singular
INANIM	inanimate		

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Cross-disciplinary approaches to Information Structure in Niger-Congo languages

This volume contains a selection of papers that were originally presented at a workshop “Cross-disciplinary approaches to Information Structure in African languages”, held in Porto-Novo, Benin in 2022. Eight papers explore information structure in Niger-Congo languages from different linguistic angles: phonetics, phonology, syntax, semantics and pragmatics.